

User Manual for GUIDE ver. 46.1*

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2 Introduction

GUIDE is an acronym for *Generalized, Unbiased, Interaction Detection and Estimation*. It is an algorithm for construction of classification and regression trees and forests. It is a descendent of the FACT (Loh and Vanichsetakul, 1988), SUPPORT

(Chaudhuri et al., 1994, 1995), QUEST (Loh and Shih, 1997), CRUISE (Kim and Loh, 2001, 2003), and LOTUS (Chan and Loh, 2004; Loh, 2006a) algorithms. GUIDE is the only classification and regression tree algorithm with all these features:

1. Unbiased variable selection for data with and without missing values
2. Unbiased importance scoring and thresholding of predictor variables
3. Automatic handling of missing values without requiring prior imputation
4. Allowance for multiple missing-value codes and missing-value flag variables
5. Optional automatic creation of missing-value indicator variables for regression
6. Periodic or cyclic variables, such as angular direction, hour of day, day of week, month of year, and seasons
7. Subgroup identification for differential treatment effects
8. Propensity score estimation
9. Linear splits for classification and regression trees
10. Kernel and nearest-neighbor node models for classification trees
11. Weighted least squares, least median of squares, logistic, quantile, Poisson, relative risk (proportional hazards), and propensity score models
12. Univariate, multivariate, censored, and longitudinal response variables
13. Piecewise polynomial, multiple, and stepwise linear regression models
14. Pairwise interaction detection at each node
15. Categorical variables may be used for splitting only, fitting only (via 0-1 dummy variables), or both in regression trees
16. Tree ensembles (bagging and forests)
17. Tree diagrams in $\text{\LaTeX}$ code
18. Predicted functions in R code

Tables 1 and 2 compare the features of GUIDE with QUEST, CRUISE, C4.5 (Quinlan, 1993), CTREE (Hothorn et al., 2006), MOB (Hothorn and Zeileis, 2015), RPART (Therneau et al., 2017)¹, and M5' (Quinlan, 1992; Witten and Frank, 2000).

The GUIDE algorithm is documented in Loh (2002) for regression trees and Loh (2009) for classification trees. Reviews of the subject may be found in Loh (2008a, 2011, 2014). Advanced features of the algorithm are reported in Chaudhuri and Loh (2002), Loh (2006b, 2008b), Kim et al. (2007), Loh et al. (2007, 2019b, 2016, 2015, 2019c), and Loh and Zhou (2021). For third-party applications of GUIDE and predecessors, see <http://www.stat.wisc.edu/~loh/apps.html>. This manual demonstrates use of the GUIDE software and interpretation of the results.

2.1 Installation

GUIDE is available free as compiled 64-bit executables for Linux, macOS, and Windows. Data and DSC files used in this manual are in the [zip file](#).

Linux: There are two executables to choose from, compiled with **gfortran** versions 11.4.0 and 13.3.0. Unzip the files with “`gunzip guide.gz`” and, if necessary, make it executable by typing “`chmod a+x guide`” in a **Terminal** window. To execute, type “`./guide`”.

macOS: There are two versions to choose from, one for Apple Arm processors (macOS Sequoia 15.4.1) and the other for Intel processors (macOS Monterey 12.7.6). Download the desired `guide.zip` file and double-click its icon to see if it opens. You will likely see a warning message saying, “Apple could not verify that guide is free of malware.” Instead of choosing to move it to Trash, choose “Done” and do this: go to System Settings > Privacy & Security and scroll down to “Security”. Look for an entry for “guide” and click “Open Anyway” and then confirm with your password. More details are given here <https://support.apple.com/guide/mac-help/open-a-mac-app-from-an-unknown-developer>. Move the `guide` file outside the Downloads folder (if that is where you have it), open a **Terminal** window, change directory to where `guide` is located, and type the command “`./guide`” to run it.

Windows: Download the file `guide.zip` and unzip it (right-click on file icon and select “Extract all”). The resulting file `guide.exe` may be placed in one of three places:

¹RPART is an implementation of CART (Breiman et al., 1984) in R. CART is a registered trademark of California Statistical Software, Inc.

Table 1: Comparison of GUIDE, QUEST, CRUISE, CART, C4.5, and CTREE classification tree algorithms. Node models: S = simple, K = kernel, LDA = linear discriminant, QDA = quadratic discriminant, N = nearest-neighbor.

	GUIDE	QUEST	CRUISE	RPART	C4.5	CTREE
Unbiased splits	Yes	Yes if no missing values	Yes	No	No	Yes if no missing values
Splits per node	2	2	≥ 2	2	2	2
Linear splits	Yes	Yes	Yes	Yes	No	No
Categorical variable splits	Subsets	Subsets	Subsets	Subsets	Atoms	Subsets
Periodic variable splits	Yes	No	No	No	No	No
Interaction tests	Yes	No	Yes	No	No	No
Class priors	Yes	Yes	Yes	Yes	No	No
Misclassification costs	Yes	Yes	Yes	Yes	No	No ^a
Case weights	No ^b	No	No	Yes	Yes	Yes ^c
Node models	S, K, N, LDA, QDA	S	S, LDA	S	S	S
Missing values in splits	Missing as observed	Node mean or mode imputation	Surrogate splits	Surrogate splits	Weights	Random splits ^d
Missing-value flag variables	Yes	No	No	No	No	No
Pruning	Yes	Yes	Yes	Yes	No	No
Tree diagrams	Text and L ^A T _E X			R	Text	R
Bagging	Yes	No	No	No	No	No
Forests	Yes	No	No	No	No	cforest
Importance scores	Yes	No	No	Yes	No	Yes

^auser defined

^bpositive weights treated as 1

^cnon-negative integer counts

^dsurrogate splits is a non-default option

Table 2: Comparison of GUIDE, RPART, M5', and MOB regression tree algorithms

	GUIDE	RPART	M5'	MOB
Unbiased splits	Yes	No	No	Yes
Linear splits	Yes	No	No	No
Interaction tests	Yes	No	No	No
Loss functions	Weighted least squares, least median of squares, logistic, quantile, Poisson, proportional hazards	Least squares, least absolute deviations	Least squares	Generalized linear models
Censored response	Yes	Yes	No	Yes
Longitudinal and multi-response	Yes	No	No	Yes
Node models	Constant, multiple, stepwise linear, polynomial, ANCOVA	Constant	Constant, stepwise	Constant, multiple linear
Variable roles	Split only, fit only, both, neither, weight, offset	Split only	Split and fit	Similar to GUIDE
Categorical variable splits	Subsets	Subsets	Singletons	Subsets
Periodic variables	Yes	No	No	No
Tree diagrams	Text and L ^A T _E X	R	PostScript	R
Sampling weights	Yes	Yes	No	No ^a
Transformations	Powers and products	No	No	Yes
Missing values in splits	Missing as observed or imputed with indicators	Surrogate splits	Mean/mode imputation	Random splits
Missing values in linear predictors	Node mean imputation & missing-value indicators	N/A	Global imputation	Omitted
Missing-value flag variables	Yes	No	No	No
Bagging & forests	Yes & yes	No & no	No & no	cforest
Importance scores	Yes	Yes	No	Yes ^b

^areplicate weights only^bfrom cforest or ctree

1. Top level of your C drive. Type “C:\guide” in a **Command Prompt** window to execute—see Section 3.1.
2. A folder that contains your data files. Type “guide” in that folder to execute.
3. A folder on your search path. Type “guide” anywhere to execute.

2.2 $\text{\LaTeX}$

GUIDE uses the public-domain software $\text{\LaTeX}$ to produce tree diagrams. The .tex files produced by GUIDE can be edited to change colors, node sizes, etc., in the trees—see [Pstricks User Guide](#).

There are two ways to produce postscript and pdf versions of the diagrams:

1. Upload the .tex file produced by GUIDE to [Overleaf](#), and compile it with the **XeLaTeX** flavor of $\text{\LaTeX}$.
2. Download and install the $\text{\LaTeX}$ software from:

Linux: [TeX Live](#)

Mac: [MacTeX](#) or [MiKTeX](#). Both include the **TeXShop** GUI app.

Windows: [MiKTeX](#). Choose [Net installer](#) under the “All downloads” tab.

There are two ways to generate pdf files from the .tex files. The following example assumes that the $\text{\LaTeX}$ file is named `diagram.tex`.

- (a) **Terminal window (simplest).** Type this single command in the **Terminal** (Linux or Mac) or **Command Prompt** (Win) window that contains `diagram.tex`:

- `xelatex diagram`

This will produce pdf file named `diagram.pdf` in the same folder.

- (b) **TeXShop, TeXworks, or TeXstudio.** Double-click `diagram.tex` to load it into one of these apps. Select **XeLaTeX** to typeset it to pdf.

In macOS, the **Preview** app can open postscript and pdf files for conversion to jpg, png, and other formats. In Windows, the same can be done with [ImageMagick](#). To insert pdf figures in MS PowerPoint or Word documents, convert them to jpg for macOS and png for Windows, or copy-and-paste them from the pdf viewer.

3 Program operation

GUIDE runs within a **terminal window** of the computer operating system.

Linux. Any terminal program will do.

macOS. The program is called **Terminal**; it is in the **Applications Folder**.

Windows. The terminal program is started from the **Start button** by choosing **All Programs → Accessories → Command Prompt**

After the terminal window is opened, change to the folder where the data and program files are stored. Mac and Windows users are unfamiliar with terminal commands may consult

<https://wiredpen.com/resources/basic-unix-commands-for-osx/>

and <https://www.geeksforgeeks.org/techtips/most-useful-cmd-commands-in-windows/>, respectively.

Do not double-click the GUIDE icon on the desktop!

3.1 Required files

GUIDE requires two text files.

Data file: This file contains the data from the training sample. Each data record consists of observations on the dependent variable, the predictor (i.e., X or independent) variables, and optional weight, missing value flag, time, offset, periodic, and event indicator (for censored responses) variables. Entries in each record are comma, space, or tab delimited (multiple spaces are treated as one space, but not for commas). A record can occupy more than one line in the file, but each record must begin on a new line.

Values of categorical variables can contain any ascii character except single and double quotation marks, which are used to enclose values that contain spaces and commas. Values can be up to 60 characters long. Class labels are truncated to 10 characters in tabular output.

A common problem among first-time users is getting the data file in proper shape. If the data are in a spreadsheet and there are **no empty cells**, export them to a **MS-DOS Comma Separated** (csv) file (the MS-DOS CSV format takes care of carriage return and line feed characters properly). If there are empty cells, a good solution is to read the spreadsheet into R (using `read.csv`

with proper specification of the `na.strings` argument), verify that the data are correctly read, and then export them to a text file using either `write.table` or `write.csv`.

Note to R users: GUIDE can optionally generate R code for the tree model and its prediction function. Because GUIDE treats "NA" (with quotes) the same as NA (without quotes), the two are treated as missing values in the R function.

DSC file: "DSC" is an abbreviation for "data specification and control." This text file provides information about the name and location of the data file, column locations and names of the variables, and their roles in the analysis. Different models may be fitted by changing the roles of the variables. An example is `fishdsc.txt` (in www.stat.wisc.edu/~loh/treeprogs/guide/datafiles.zip), whose contents are:

```
fishdata.txt
NA
2
1 species d
2 weight n
3 length1 n
4 length2 n
5 length3 n
6 heightpc n
7 widthpc n
8 sex c
```

The 1st line gives the name of the data file `fishdata.txt`, which is also in www.stat.wisc.edu/~loh/treeprogs/guide/datafiles.zip. If the file is not in the current folder, its full path must be given (e.g., "`c:\data\fishdata.txt`" for Windows users or "`~/data/fishdata.txt`" for Mac users) surrounded by matching quotes (because it contains non-alphanumeric characters). The 2nd line gives the missing value code, which can be up to 80 characters long. If it contains non-alphanumeric characters, it too must be surrounded by matching quotation marks. A missing value code **must appear** in the second line of the file even if there are no missing values in the data (in which case any character string not present among the data values can be used). The 3rd line gives the line number of the first data record in the data file. A "2" is shown here because

the variable names appear in the first line of `fishdata.txt`. If the 1st line of the data file contains the 1st record, this entry would be “1”. Blank lines in the data and DSC files are ignored. The column location, name and role of each variable comes next (in that order), with one line for each variable.

Variable names must begin with an alphabet and be not more than 60 characters long. If a name contains non-alphanumeric characters, it must be enclosed in matching single or double quotes. Spaces and the four special characters, #, %, {, and }, in a variable name are replaced by dots (periods) in the outputs. Variable names are truncated to 10 characters in tabular text output (but not in R output). Leading and trailing spaces in variable names are dropped.

The letters (lower or upper case) below are the permissible roles.

- b** Categorical variable used both for splitting and for node modeling in regression. Such variables are converted to 0-1 dummy variables when fitting models within nodes for regression. They are converted to **c** type for classification.
- c** Categorical variable used for splitting only.
- d** Dependent variable or death indicator variable. Except for longitudinal and multiple response data (Sec. 13), there can only be one **d** variable. For censored responses in proportional hazards models, it is the 0-1 event (death) indicator. For all other models, it is the response variable. It can take character string values for classification.
- e** Estimated probability variable, for logistic regression without **r** variable; see Section 15 for an example.
- f** Numerical variable used only for fitting the linear models in the nodes of the tree. It is not used for splitting the nodes and is disallowed in classification.
- i** Categorical variable internally converted to 0-1 indicator variables for fitting regression models within nodes.
- m** Missing value flag variable. Each such variable should follow immediately after a **c**, **n** or **s** variable in the DSC file. Missing value flag variables associated with any other variable type (including **b** and **p**) should be specified as **c**.
- n** Numerical variable used both for splitting the nodes and for fitting the node regression models. It is converted to type **s** in classification.
- p** Periodic (cyclic) variable, such as an angle, hour of day, day of week, or month of year. See Sec. 8 for an example.

Table 3: Predictor variable role descriptors

Type of variable	Role of variable		
	Split nodes	Fit node models	Both
Categorical	c	i	b
Numerical	s	f	n

- r** Categorical treatment (**R**x) variable used only for fitting the linear models in the nodes of the tree. It is not used for splitting the nodes.
- s** Numerical-valued variable only used for **s**plitting the nodes. It is not used as a linear predictor in in regression models. It is suitable for ordinal categorical variables if they take numerical values that reflect the orderings.
- t** **T**ime variable, either time to event for proportional hazards models or observation time for longitudinal models.
- w** **W**eight variable for weighted least squares regression or for excluding observations in the training sample from tree construction. See Sec. 21.2 for the latter. Except for longitudinal models, a record with a missing value in a **d**, **t**, or **z**-variable is automatically assigned zero weight.
- x** **E**xcluded variable. Models may be fitted to different subsets of variables by indicating excluded variables in the DSC file without editing the data file.
- z** **O**ffset variable used only in Poisson regression.

Table 3 summarizes the possible roles for predictor variables.

3.2 Running GUIDE

GUIDE uses a “restaurant” model that requires two invocations of the program. The first invocation creates an input file (similar to a restaurant order) and the second invocation reads the input file and fits the data. The first invocation is started by typing the word “guide” (lowercase) name in a terminal window and then typing “1” to answer some questions and save the answers into a file. In the following, the sign (>) is the computer prompt (not to be typed!).

```
> guide
GUIDE Classification and Regression Trees and Forests
Version 46.1 (Build date: January 23, 2026)
Compiled with GFortran 15.1.0 on macOS Tahoe 26.0.1
Copyright (c) 1997-2025 Wei-Yin Loh. All rights reserved.
Software based upon work partially supported by the U.S. Army Research Office,
National Science Foundation, National Institutes of Health,
```

Bureau of Labor Statistics, USDA Economic Research Service, and Eli Lilly.

Choose one of the following options:

- 0. Read the warranty disclaimer
- 1. Create a GUIDE input file

4 Classification: fish data

The file `fishdata.txt` give measurements on 7 predictor variables for 7 species of 159 fish. The predictor variables are weight (gm), length1 (nose to beginning of tail, cm), length2 (nose to notch of tail, cm), length3 (nose to end of tail, cm), heightpc (maximal height as percent of length3), widthpc (maximal width as percent of length3), and sex (female, male, and unknown). The species variable takes values Bream, Parkki, Perch, Pike, Roach, Smelt, and Whitefish. Further details are given in <https://search.r-project.org/CRAN/refmans/rrcov/html/fish.html>. The DSC file for this dataset is `fishdsc.txt` mentioned above. We first show how to create an input file to fit a classification tree model with univariate splits.

4.1 Univariate splits

4.1.1 Input file creation

Most of the questions prompted by GUIDE have default answers indicated by the symbol `<cr>` (“`cr`” stands for “carriage return”). For example, “`<cr>=1`” means that the answer “1” may be selected by simply hitting the carriage return once, as demonstrated several times below.

```
> guide
GUIDE Classification and Regression Trees and Forests
Version 46.1 (Build date: January 23, 2026)
Compiled with GFortran 15.2 on macOS Tahoe 26.2
Copyright (c) 1997-2025 Wei-Yin Loh. All rights reserved.
Software based upon work partially supported by the U.S. Army Research Office,
National Science Foundation, National Institutes of Health,
Bureau of Labor Statistics, USDA Economic Research Service, and Eli Lilly.
```

Choose one of the following options:

- 0. Read the warranty disclaimer
- 1. Create a GUIDE input file

Input your choice: 1

Name of batch input file: `classin.txt`

```

Input 1 for model fitting, 2 for importance or DIF scoring,
  3 for data conversion ([1:3], <cr>=1):
Name of batch output file: classout.txt
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: fishdsc.txt
Reading DSC file ...
Training sample file: fishdata.txt
Missing value code: NA
Records in data file start on line 2
6 N variables changed to S
D variable is species
Reading data file ...
Number of records in data file: 159
Length of longest entry in data file: 9
Checking for missing values ...
Finished checking
Missing values found among non-categorical variables
Recoding D values to integers
Finished recoding
Number of classes: 7
Assigning integer codes to values of 1 categorical variables
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
  Class  #Cases  Proportion
bream      35    0.22012579
parkki     11    0.06918239
perch      56    0.35220126
pike       17    0.10691824
roach      20    0.12578616
smelt      14    0.08805031
whitefish   6    0.03773585
Numbers of N, F, S, and P below exclude constant variables
  Total  #cases w/  #missing
  #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
    159         0         2        0        0        0        6
  #P-var  #M-var  #B-var  #C-var  #I-var
    0         0         0         1         0
Number of cases used for training: 159

```



```

Number of cases excluded due to 0 W or missing D variable: 0
Finished reading data file
Choose 1 for estimated priors, 2 for equal priors, 3 to input priors from a file
Input 1, 2, or 3 ([1:3], <cr>=1):
Choose 1 for unit misclassification costs, 2 to input costs from a file
Input 1 or 2 ([1:2], <cr>=1):
Default max. number of split levels: 12
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): class.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: classfit.txt
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: class.r
Input rank of top variable to split root node ([1:7], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < classin.txt

```

4.1.2 Contents of classout.txt

Following are the contents of the file `classout.txt` after running GUIDE. The selected tree is marked with two asterisks (**) in the section titled “Size and CV mean cost and SE of subtrees” below.

```

Classification tree
Pruning by cross-validation
DSC file: fishdsc.txt
Training sample file: fishdata.txt
Missing value code: NA
Records in data file start on line 2
6 N variables changed to S
D variable is species
Number of records in data file: 159
Length of longest entry in data file: 9
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Number of classes: 7
Training sample class proportions of D variable species:

```

Class	#Cases	Proportion
bream	35	0.22012579
parkki	11	0.06918239
perch	56	0.35220126
pike	17	0.10691824
roach	20	0.12578616
smelt	14	0.08805031
whitefish	6	0.03773585

Summary information for training sample of size 159

d=dependent, b=split and fit cat variable using indicator variables,
 c=split-only categorical, i=fit-only categorical (via indicators),
 s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
 m=missing-value flag variable, p=periodic variable, w=weight

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	species	d			7	
2	weight	s	5.900	1650.		2
3	length1	s	7.500	59.00		
4	length2	s	8.400	63.40		
5	length3	s	8.800	68.00		
6	heightpc	s	14.50	44.50		
7	widthpc	s	8.700	20.90		
8	sex	c			2	87

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/	#missing				
#cases	miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
159	0	2	0	0	0	6
#P-var	#M-var	#B-var	#C-var	#I-var		
0	0	0	1	0		

Number of cases used for training: 159

Number of cases excluded due to 0 W or missing D variable: 0

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Simple node models

Estimated priors

Unit misclassification costs

Univariate split highest priority

Interaction and linear splits 2nd and 3rd priorities

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 2

Ranks of variables and their 1-df chi-squared values at root node

1	0.1398E+03	heightpc
2	0.8109E+02	widthpc
3	0.6776E+02	length3
4	0.6712E+02	length2
5	0.6622E+02	length1
6	0.5275E+02	weight
7	0.1818E+02	sex

Size and CV mean cost and SE of subtrees:

Tree	#Tnodes	Mean Cost	SE(Mean)	BSE(Mean)	Median Cost	BSE(Median)
1	11	9.434E-02	2.318E-02	9.847E-03	9.375E-02	2.661E-02
2	10	9.434E-02	2.318E-02	9.847E-03	9.375E-02	2.661E-02
3**	8	7.547E-02	2.095E-02	8.597E-03	6.250E-02	7.698E-03
4	5	1.635E-01	2.933E-02	2.011E-02	1.292E-01	3.768E-02
5	4	2.390E-01	3.382E-02	3.169E-02	2.188E-01	5.079E-02
6	3	3.208E-01	3.702E-02	2.194E-02	3.125E-01	3.212E-02
7	2	4.277E-01	3.924E-02	2.459E-02	3.750E-01	4.935E-02
8	1	6.478E-01	3.788E-02	2.952E-02	6.562E-01	3.998E-02

0-SE tree based on mean is marked with * and has 8 terminal nodes

0-SE tree based on median is marked with + and has 8 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Node cost is node misclassification cost divided by number of cases in node

Node label	Total cases	Train cases	Predicted class	Node cost	Split variables	Interacting variable
1	159	159	perch	6.478E-01	heightpc	
2	113	113	perch	5.053E-01	heightpc	
4	31	31	pike	4.544E-01	weight	
8T	14	14	smelt	0.000E+00	-	
9T	17	17	pike	0.000E+00	-	
5	82	82	perch	3.191E-01	weight	
10	60	60	perch	3.850E-01	heightpc	
20T	22	22	perch	4.922E-02	-	
21	38	38	roach	5.023E-01	length2 +length3	
42T	22	22	roach	1.410E-01	length2 +heightpc	

43T	16	16	perch	0.000E+00 -
11T	22	22	perch	1.396E-01 sex
3	46	46	bream	2.425E-01 length2
6T	12	12	parkki	8.863E-02 -
7T	34	34	bream	0.000E+00 -

Number of terminal nodes of final tree: 8

Total number of nodes of final tree: 15

Second best split variable (based on curvature test) at root node is widthpc

Classification tree:

```

Node 1: heightpc <= 33.900000
  Node 2: heightpc <= 20.100000
    Node 4: weight <= 109.95000
      Node 8: smelt
      Node 4: weight > 109.95000 or NA
        Node 9: pike
        Node 2: heightpc > 20.100000 or NA
          Node 5: weight <= 452.00000 or NA
            Node 10: heightpc <= 25.250000
              Node 20: perch
              Node 10: heightpc > 25.250000 or NA
                Node 21: -0.92577064 * length3 + length2 <= -0.16303289
                  Node 42: roach
                  Node 21: -0.92577064 * length3 + length2 > -0.16303289 or NA
                    Node 43: perch
                    Node 5: weight > 452.00000
                      Node 11: perch
                      Node 1: heightpc > 33.900000 or NA
                        Node 3: length2 <= 26.150000
                          Node 6: parkki
                          Node 3: length2 > 26.150000 or NA
                            Node 7: bream

```

 Predictor means below are means of cases with no missing values.

Node 1: Intermediate node

A case goes into Node 2 if heightpc <= 33.900000

heightpc mean = 28.313208

Class	Number	Posterior
bream	35	0.2201E+00
parkki	11	0.6918E-01
perch	56	0.3522E+00
pike	17	0.1069E+00

```

roach          20  0.1258E+00
smelt          14  0.8805E-01
whitefish      6  0.3774E-01
Number of training cases misclassified = 103
Predicted class is perch
-----
Node 2: Intermediate node
A case goes into Node 4 if heightpc <= 20.100000
heightpc mean = 23.769912
  Class      Number  Posterior
bream        0  0.1376E-02
parkki       0  0.4324E-03
perch        56  0.4947E+00
pike         17  0.1502E+00
roach        20  0.1767E+00
smelt        14  0.1237E+00
whitefish     6  0.5300E-01
Number of training cases misclassified = 57
Predicted class is perch
-----
Node 4: Intermediate node
A case goes into Node 8 if weight <= 109.95000
weight mean = 399.17742
  Class      Number  Posterior
bream        0  0.1376E-02
parkki       0  0.4324E-03
perch        0  0.2201E-02
pike         17  0.5456E+00
roach        0  0.7862E-03
smelt        14  0.4493E+00
whitefish     0  0.2358E-03
Number of training cases misclassified = 14
Predicted class is pike
-----
Node 8: Terminal node
  Class      Number  Posterior
bream        0  0.1376E-02
parkki       0  0.4324E-03
perch        0  0.2201E-02
pike         0  0.6682E-03
roach        0  0.7862E-03
smelt        14  0.9943E+00
whitefish     0  0.2358E-03
Number of training cases misclassified = 0
Predicted class is smelt
-----

```

Node 9: Terminal node

Class	Number	Posterior
bream	0	0.1376E-02
parkki	0	0.4324E-03
perch	0	0.2201E-02
pike	17	0.9944E+00
roach	0	0.7862E-03
smelt	0	0.5503E-03
whitefish	0	0.2358E-03

Number of training cases misclassified = 0

Predicted class is pike

Node 5: Intermediate node

A case goes into Node 10 if weight <= 452.00000 or NA

weight mean = 341.14074

Class	Number	Posterior
bream	0	0.1376E-02
parkki	0	0.4324E-03
perch	56	0.6809E+00
pike	0	0.6682E-03
roach	20	0.2432E+00
smelt	0	0.5503E-03
whitefish	6	0.7295E-01

Number of training cases misclassified = 26

Predicted class is perch

Node 10: Intermediate node

A case goes into Node 20 if heightpc <= 25.250000

heightpc mean = 26.005000

Class	Number	Posterior
bream	0	0.1376E-02
parkki	0	0.4324E-03
perch	37	0.6150E+00
pike	0	0.6682E-03
roach	20	0.3320E+00
smelt	0	0.5503E-03
whitefish	3	0.4992E-01

Number of training cases misclassified = 23

Predicted class is perch

Node 20: Terminal node

Class	Number	Posterior
bream	0	0.1376E-02
parkki	0	0.4324E-03
perch	21	0.9508E+00
pike	0	0.6682E-03

```

roach          1 0.4596E-01
smelt          0 0.5503E-03
whitefish      0 0.2358E-03
Number of training cases misclassified = 1
Predicted class is perch
-----

```

Node 21: Intermediate node

A case goes into Node 42 if $-0.92577064 * \text{length3} + \text{length2} \leq -0.16303289$

Linear combination mean = -0.31127156

Class	Number	Posterior
bream	0	0.1376E-02
parkki	0	0.4324E-03
perch	16	0.4206E+00
pike	0	0.6682E-03
roach	19	0.4977E+00
smelt	0	0.5503E-03
whitefish	3	0.7869E-01

Number of training cases misclassified = 19

Predicted class is roach

Node 42: Terminal node

Class	Number	Posterior
bream	0	0.1376E-02
parkki	0	0.4324E-03
perch	0	0.2201E-02
pike	0	0.6682E-03
roach	19	0.8590E+00
smelt	0	0.5503E-03
whitefish	3	0.1357E+00

Number of training cases misclassified = 3

Predicted class is roach

Node 43: Terminal node

Class	Number	Posterior
bream	0	0.1376E-02
parkki	0	0.4324E-03
perch	16	0.9960E+00
pike	0	0.6682E-03
roach	0	0.7862E-03
smelt	0	0.5503E-03
whitefish	0	0.2358E-03

Number of training cases misclassified = 0

Predicted class is perch

Node 11: Terminal node

Class	Number	Posterior
-------	--------	-----------

```

bream          0 0.1376E-02
parkki         0 0.4324E-03
perch          19 0.8604E+00
pike           0 0.6682E-03
roach          0 0.7862E-03
smelt          0 0.5503E-03
whitefish      3 0.1357E+00
Number of training cases misclassified = 3
Predicted class is perch
-----
Node 3: Intermediate node
A case goes into Node 6 if length2 <= 26.150000
length2 mean = 30.056522
  Class      Number  Posterior
bream        35 0.7575E+00
parkki       11 0.2381E+00
perch         0 0.2201E-02
pike          0 0.6682E-03
roach         0 0.7862E-03
smelt         0 0.5503E-03
whitefish     0 0.2358E-03
Number of training cases misclassified = 11
Predicted class is bream
-----
Node 6: Terminal node
  Class      Number  Posterior
bream         1 0.8419E-01
parkki        11 0.9114E+00
perch          0 0.2201E-02
pike           0 0.6682E-03
roach          0 0.7862E-03
smelt          0 0.5503E-03
whitefish     0 0.2358E-03
Number of training cases misclassified = 1
Predicted class is parkki
-----
Node 7: Terminal node
  Class      Number  Posterior
bream        34 0.9951E+00
parkki         0 0.4324E-03
perch          0 0.2201E-02
pike           0 0.6682E-03
roach          0 0.7862E-03
smelt          0 0.5503E-03
whitefish     0 0.2358E-03
Number of training cases misclassified = 0

```


Predicted class is bream

Classification matrix for training sample:

Predicted class	True class					
	bream	parkki	perch	pike	roach	smelt
bream	34	0	0	0	0	0
parkki	1	11	0	0	0	0
perch	0	0	56	0	1	0
pike	0	0	0	17	0	0
roach	0	0	0	0	19	0
smelt	0	0	0	0	0	14
whitefish	0	0	0	0	0	0
Total	35	11	56	17	20	14

	whitefish
bream	0
parkki	0
perch	3
pike	0
roach	3
smelt	0
whitefish	0
Total	6

Number of cases used for tree construction: 159

Number misclassified: 8

Resubstitution estimate of mean misclassification cost: 0.50314465E-01

Observed and fitted values are stored in classfit.txt

LaTeX code for tree is in class.tex

The second last paragraph of the output file shows that 8 out of 159 observations are misclassified. The $\text{\LaTeX}$ tree diagram in Figure 1 shows that:

1. The predicted species and sample size are printed beneath each terminal node.
2. The highest posterior class probability is printed beside each terminal node. For example, nodes 7, 8, 9, and 43 have highest posterior probability 1.00, indicating that they are pure nodes (nodes containing only one species).
3. Node 5 is split on `weight` $\leq_*$ 452, with the symbol $\leq_*$ indicating that observations with missing values in `weight` go to the left branch (node 10).
4. Node 21 is a linear split on variables `length2` and `length3` and is indicated by a blue circle.

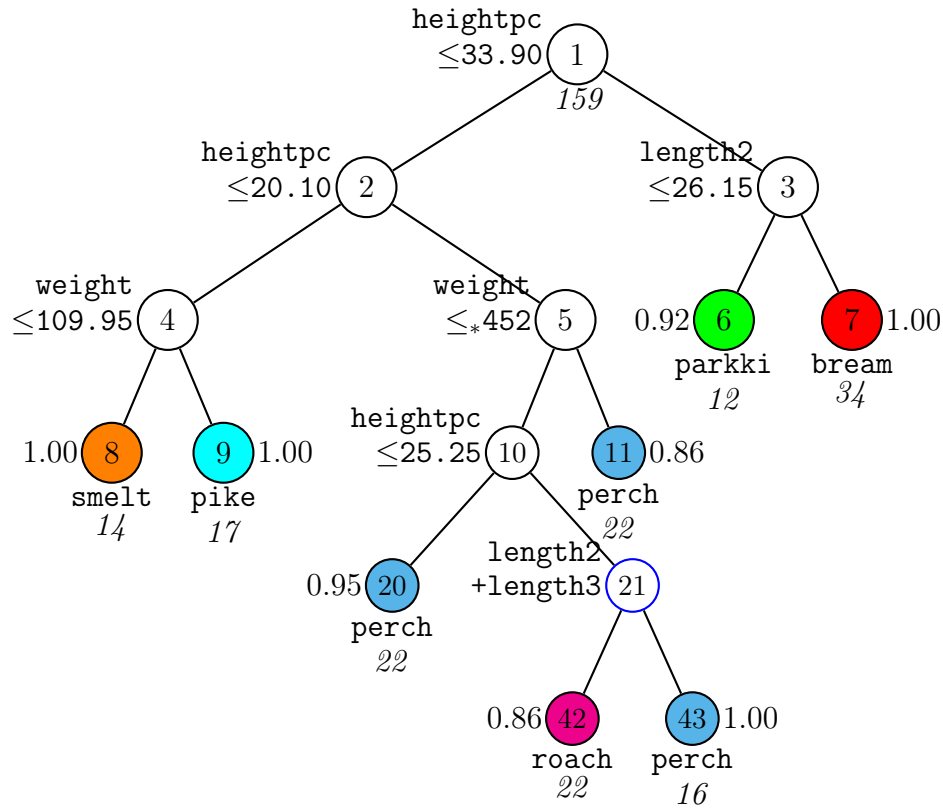


Figure 1: GUIDE v.46.1 0.250-SE classification tree for predicting **species** using estimated priors and unit misclassification costs. At each split, an observation goes to the left branch if and only if the condition is satisfied. Symbol ' $\leq_*$ ' stands for ' $\leq$ or missing'. Splits at nodes drawn with blue circles are linear splits. Predicted class and total sample size (in *italics*) printed below each terminal node; highest class posterior probability beside node. Second best split variable at root node is **widthpc**.

4.1.3 Contents of classfit.txt

Following are the top five lines of the file `classfit.txt`. They show the classification results of the first 4 observations. The column “train” shows whether an observation is used to fit the model (y, for yes) or not (n, for no). In this example, the column contains only y because all 159 observations are used in the fitting. The second column labeled “node” gives the terminal node number. In this example the first observation landed in node 6 while the other three observations went to node 7. The columns labeled “observed” and “predicted” give the true and predicted class of each observation. The remaining columns give the posterior probabilities of each observation in the seven classes.

train	node	observed	predicted	"P(bream)"	"P(parkki)"	"P(perch)"	"P(pike)"	"P(roach)"	"P(smelt)"	"P(whitefish)"
y	6	"bream"	"parkki"	0.84188E-01	0.91137E+00	0.22013E-02	0.66824E-03	0.78616E-03	0.55031E-03	0.23585E-03
y	7	"bream"	"bream"	0.99513E+00	0.43239E-03	0.22013E-02	0.66824E-03	0.78616E-03	0.55031E-03	0.23585E-03
y	7	"bream"	"bream"	0.99513E+00	0.43239E-03	0.22013E-02	0.66824E-03	0.78616E-03	0.55031E-03	0.23585E-03
y	7	"bream"	"bream"	0.99513E+00	0.43239E-03	0.22013E-02	0.66824E-03	0.78616E-03	0.55031E-03	0.23585E-03

The posterior probabilities are calculated as follows. Let J be the number of classes, N_j be the number of class j observations in the whole sample and $N = \sum_j N_j$. Let π_j be the (estimated or specified) prior probability of class j . Let $n_j(t)$ be the number of class j training samples in node t . The posterior probability of class j in t is $p_j(t) = \pi_j n_j(t) N_j^{-1} / \sum_i \pi_i n_i(t) N_i^{-1}$. If $\min_j p_j(t) = 0$, the posterior probability is redefined to be $(N p_j(t) + \pi_j) / (N + 1)$; this ensures that no probability is zero if all π_j are positive.

4.1.4 Contents of class.r

Following are the contents of `class.r` giving an R prediction function.

```

1 predicted <- function(){
2   if(!is.na(heightpc) & heightpc <= 33.9000000000 ){
3     if(!is.na(heightpc) & heightpc <= 20.1000000000 ){
4       if(!is.na(weight) & weight <= 109.9500000000 ){
5         nodeid <- 8
6         predclass <- "smelt"
7         posterior <- c( 0.13758E-02, 0.43239E-03, 0.22013E-02, 0.66824E-03, 0.78616E-03, 0.99430E+00, 0
8       } else {
9         nodeid <- 9
10        predclass <- "pike"
11        posterior <- c( 0.13758E-02, 0.43239E-03, 0.22013E-02, 0.99442E+00, 0.78616E-03, 0.55031E-03, 0
12      }
13    } else {
14      if(is.na(weight) | weight <= 452.0000000000 ){
15        if(!is.na(heightpc) & heightpc <= 25.2500000000 ){
16          nodeid <- 20
17          predclass <- "perch"
18          posterior <- c( 0.13758E-02, 0.43239E-03, 0.95078E+00, 0.66824E-03, 0.45957E-01, 0.55031E-03, 0
19        } else {
20          if(!is.na(length2) & !is.na(length3) & -0.925770636221 * length3 + length2 <= -0.16303288757
21          nodeid <- 42
22          predclass <- "roach"

```

```

23     posterior <- c( 0.13758E-02, 0.43239E-03, 0.22013E-02, 0.66824E-03, 0.85902E+00, 0.55031E-03)
24   } else {
25     nodeid <- 43
26     predclass <- "perch"
27     posterior <- c( 0.13758E-02, 0.43239E-03, 0.99595E+00, 0.66824E-03, 0.78616E-03, 0.55031E-03)
28   }
29 }
30 } else {
31   nodeid <- 11
32   predclass <- "perch"
33   posterior <- c( 0.13758E-02, 0.43239E-03, 0.86044E+00, 0.66824E-03, 0.78616E-03, 0.55031E-03, 0.23013E-03)
34 }
35 }
36 } else {
37   if(!is.na(length2) & length2 <= 26.1500000000 ){
38     nodeid <- 6
39     predclass <- "parkki"
40     posterior <- c( 0.84188E-01, 0.91137E+00, 0.22013E-02, 0.66824E-03, 0.78616E-03, 0.55031E-03, 0.23013E-03)
41   } else {
42     nodeid <- 7
43     predclass <- "breem"
44     posterior <- c( 0.99513E+00, 0.43239E-03, 0.22013E-02, 0.66824E-03, 0.78616E-03, 0.55031E-03, 0.23013E-03)
45   }
46 }
47 return(c(nodeid,predclass,posterior))
48 }
49 ## end of function
50 ##
51 ##
52 ## If desired, replace "fishdata.txt" with name of file containing new data
53 ## New file must have at least the same variables with same names
54 ## (but not necessarily the same order) as in the training data file
55 ## Missing value code is converted to NA if not already NA
56 newdata <- read.table("fishdata.txt",header=TRUE,colClasses="character")
57 ## node contains terminal node ID of each case
58 ## pred.class contains predicted class
59 ## prob contains predicted posterior probabilities
60 node <- NULL
61 pred.class <- NULL
62 prob <- NULL
63 for(i in 1:nrow(newdata)){
64   weight <- as.numeric(newdata$weight[i])
65   length2 <- as.numeric(newdata$length2[i])
66   length3 <- as.numeric(newdata$length3[i])
67   heightpc <- as.numeric(newdata$heightpc[i])
68   tmp <- predicted()
69   node <- c(node,as.numeric(tmp[1]))
70   pred.class <- rbind(pred.class,tmp[2])
71   prob <- rbind(prob,as.numeric(tmp[-c(1,2)]))
72 }

```

Figure 2 shows the partitions in the nodes of the tree. The R code for producing the plots follows.

```

1 ancestor <- function(inter,leaf){
2   n <- leaf
3   while(n > inter){ n <- floor(n/2) }

```

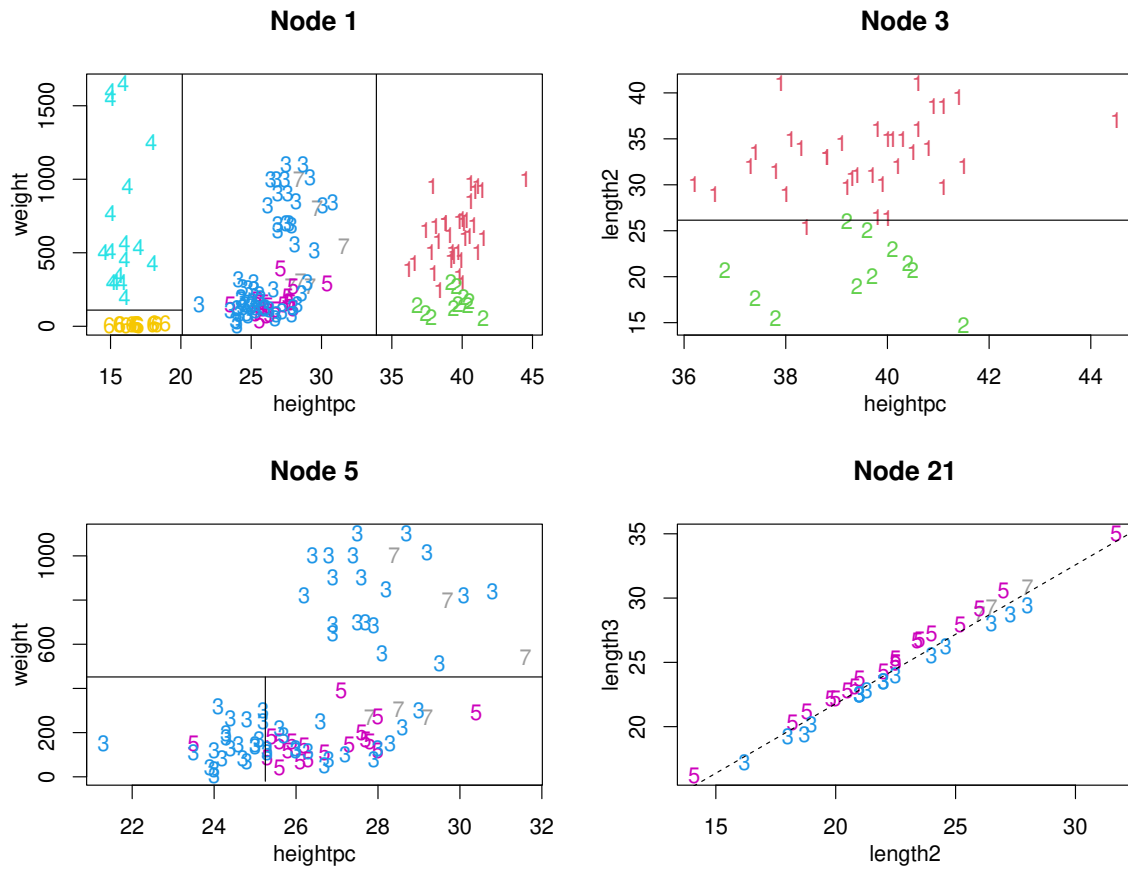


Figure 2: Data in nodes of Figure 1. 1 = Bream, 2 = Parkki, 3 = Perch, 4 = Pike, 5 = Roach, 6 = Smelt, 7 = Whitefish.

```

4   if(n == inter) TRUE else FALSE
5 }
6 i.node <- function(num,termnodes){
7   n <- length(termnodes)
8   gp <- rep(FALSE,n)
9   for(i in 1:n){ gp[i] <- ancestor(num,termnodes[i]) }
10  return(gp)
11 }
12 par(mfrow=c(2,2),mar=c(3,4,4,1),mgp=c(2,1,0),cex=1.2)
13 z <- read.table("fishdata.txt",header=TRUE)
14 var.names <- names(z)
15 z$species <- as.factor(z$species)
16 class.sym <- as.numeric(z$species)
17 class.col <- class.sym+1
18 class <- z$species
19 zf <- read.table("classfit.txt",header=TRUE)
20 node <- zf$node
21 v1.id <- 6
22 v2.id <- 2
23 x <- z[,v1.id]
24 y <- z[,v2.id]
25 gp <- rep(TRUE,length(class))
26 plot(x[gp],y[gp],xlab=var.names[v1.id],ylab=var.names[v2.id],type="n")
27 text(x[gp],y[gp],labels=class.sym[gp],col=class.col[gp])
28 abline(v=c(20.1,33.9))
29 lines(c(10,20.1),rep(109.95,2))
30 title("Node_1")
31 nodeid <- 3
32 gp <- i.node(nodeid,node)
33 v1.id <- 6
34 v2.id <- 4
35 x <- z[,v1.id]
36 y <- z[,v2.id]
37 plot(x[gp],y[gp],xlab=var.names[v1.id],ylab=var.names[v2.id],type="n")
38 text(x[gp],y[gp],labels=class.sym[gp],col=class.col[gp])
39 abline(h=26.15)
40 title("Node_3")
41 nodeid <- 5
42 gp <- i.node(nodeid,node)
43 v1.id <- 6
44 v2.id <- 2
45 x <- z[,v1.id]
46 y <- z[,v2.id]
47 plot(x[gp],y[gp],xlab=var.names[v1.id],ylab=var.names[v2.id],type="n")
48 text(x[gp],y[gp],labels=class.sym[gp],col=class.col[gp])
49 abline(h=452)

```

```
50 lines(rep(25.25,2),c(-20,452))
51 title("Node_5")
52 nodeid <- 21
53 v1.id <- 4
54 v2.id <- 5
55 x <- z[,v1.id]
56 y <- z[,v2.id]
57 gp <- i.node(nodeid,node)
58 plot(x[gp],y[gp],xlab=var.names[v1.id],ylab=var.names[v2.id],type="n")
59 text(x[gp],y[gp],labels=class.sym[gp],col=class.col[gp])
60 a <- 0.16303289/0.92577064
61 b <- 1/0.92577064
62 abline(c(a,b),lty=2)
63 title("Node_21")
```

4.2 Linear splits and discriminant models

This section shows how to construct a tree that uses linear splits and quadratic discriminant analysis (QDA) models in the nodes. As before, the default choice at each prompt (indicated by `<cr>`) can be accepted by simply hitting the “carriage return” key.

4.2.1 Input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: qda.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: qda.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1): 2
Input 1 for simple, 2 for nearest-neighbor, 3 for kernel, 4 for lda models,
    5 for qda models ([1:5], <cr>=4): 4
Input 0 to specify tree with fixed no. of nodes, 1 to prune by CV,
    2 by test sample, 3 for no pruning ([0:3], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: fishdsc.txt
Reading DSC file ...
Training sample file: fishdata.txt
Missing value code: NA
```

```
Records in data file start on line 2
D variable is species
C variables changed to B
Reading data file ...
Number of records in data file: 159
Length of longest entry in data file: 9
Checking for missing values ...
Finished checking
Missing values found among non-categorical variables
Recoding D values to integers
Finished recoding
Number of classes: 7
Assigning integer codes to values of 1 categorical variables
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...
Creating dummy variables ...
Creating missing value indicators ...
Rereading data ...
  Class  #Cases  Proportion
bream      35    0.22012579
parkki     11    0.06918239
perch      56    0.35220126
pike       17    0.10691824
roach      20    0.12578616
smelt      14    0.08805031
whitefish   6    0.03773585
Numbers of N, F, S, and P below exclude constant variables
  Total  #cases w/  #missing
  #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
    159         0         2         0         6         3         0
  #P-var  #M-var  #B-var  #C-var  #I-var
     0         0         1         0         0
Number of dummy variables (excl. missing-value indicators) created: 2
Number of missing-value indicators created: 1
Number of cases used for training: 159
Number of cases excluded due to 0 W or missing D variable: 0
Finished reading data file
Default number of cross-validations:          10
Input 1 to accept the default, 2 to change it ([1:2], <cr>=1):
Best tree may be chosen based on mean or median CV estimate
Input 1 for mean-based, 2 for median-based ([1:2], <cr>=1):
Input number of SEs for pruning ([0.00:1000.00], <cr>=0.25):
```



```
Choose 1 for estimated priors, 2 for equal priors, 3 to input priors from a file
Input 1, 2, or 3 ([1:3], <cr>=1):
Choose 1 for unit misclassification costs, 2 to input costs from a file
Input 1 or 2 ([1:2], <cr>=1):
Default maximum number of non-categorical splits at each node is 100
Choose 1 to accept the default, 2 to change it
Input 1 or 2 ([1:2], <cr>=1):
Default max. number of split levels: 15
Input 1 to accept this value, 2 to change it ([1:2], <cr>=1):
Default minimum node sample size is 5
Input 1 to use the default value, 2 to change it ([1:2], <cr>=1):
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): qda.tex
Input 1 to color terminal nodes, 2 otherwise ([1:2], <cr>=1):
Choose amount of detail in nodes of LaTeX tree diagram:
Input 0 for highest posterior, 1 for sample sizes, 2 for sample proportions,
      3 for posterior probs, 4 for misclassification cost, 5 for nothing
Input your choice ([0:5], <cr>=4):
You can store discriminant coefficients in a file
Choose 1 to skip this step, 2 to store the coefficients
Input your choice ([1:2], <cr>=2):
Input file name: qda.coefs
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: qda.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: qda.r
You can output R code for plotting data in the nodes
Input 1 to output R code
Input 2 to skip this step
Input your choice ([1:2], <cr>=1):
Input filename to storing plot data: qdaplots.r
Input file name for storing data for plotting: qdadata.txt
Input file is created!
Run GUIDE with the command: guide < qda.in
```

4.2.2 Contents of qda.out

```
Classification tree
Linear discriminant splits and QDA node models
Pruning by cross-validation
DSC file: fishdsc.txt
Training sample file: fishdata.txt
```

Missing value code: NA
 Records in data file start on line 2
 D variable is species
 C variables changed to B
 Number of records in data file: 159
 Length of longest entry in data file: 9
 Missing values found among categorical variables
 Separate categories will be created for missing categorical variables
 Missing values found among non-categorical variables
 Number of classes: 7

Training sample class proportions of D variable species:

Class	#Cases	Proportion
bream	35	0.22012579
parkki	11	0.06918239
perch	56	0.35220126
pike	17	0.10691824
roach	20	0.12578616
smelt	14	0.08805031
whitefish	6	0.03773585

Summary information for training sample of size 159
 d=dependent, b=split and fit cat variable using indicator variables,
 c=split-only categorical, i=fit-only categorical (via indicators),
 s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
 m=missing-value flag variable, p=periodic variable, w=weight

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	species	d			7	
2	weight	n	5.9000E+00	1650.		2
3	length1	n	7.500	59.00		
4	length2	n	8.400	63.40		
5	length3	n	8.800	68.00		
6	heightpc	n	14.50	44.50		
7	widthpc	n	8.700	20.90		
8	sex	b			2	87
===== Constructed variables =====						
9	sex.female	f	0.000	1.000		
10	sex.male	f	0.000	1.000		
11	weight.NA	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ miss. D	#missing	ord. vals	#X-var	#N-var	#F-var	#S-var
159	0	2	0	6	3	0	
#P-var	#M-var	#B-var	#C-var	#I-var			

0 0 1 0 0
Number of dummy variables (excl. missing-value indicators) created: 2
Number of missing-value indicators created: 1
Number of cases used for training: 159
Number of cases excluded due to 0 W or missing D variable: 0

Pruning by v-fold cross-validation, with v = 10
Selected tree is based on mean of CV estimates
Number of SE's for pruned tree: 0.2500

Estimated priors

Unit misclassification costs

Max number of splits: 100

Maximum number of split levels: 15

Minimum node sample size: 5

Size and CV mean cost and SE of subtrees:

Tree	#Tnodes	Mean Cost	SE(Mean)	BSE(Mean)	Median Cost	BSE(Median)
0	5	1.069E-01	2.451E-02	1.277E-02	1.250E-01	2.387E-02
1	4	1.069E-01	2.451E-02	1.277E-02	1.250E-01	2.387E-02
2	3	1.069E-01	2.451E-02	1.277E-02	1.250E-01	2.387E-02
3**	2	8.805E-02	2.247E-02	1.352E-02	6.250E-02	1.735E-02
4	1	1.824E-01	3.062E-02	2.438E-02	1.604E-01	3.663E-02

0-SE tree based on mean is marked with * and has 2 terminal nodes

0-SE tree based on median is marked with + and has 2 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Node cost is node misclassification cost divided by number of cases in node

Node	Total	Misclassification
label	cases	cost
1	159	1.384E-01
2T	68	0.000E+00
3T	91	2.198E-02

Number of terminal nodes of final tree: 2

Total number of nodes of final tree: 3

Classification tree with linear discriminant splits and QDA node models

```

Node 1: D1 <= 0.17486273
  Node 2: 0/68 misclassified
Node 1: D1 > 0.17486273
  Node 3: 2/91 misclassified

```

```

*****
Predictor means below are means of cases with no missing values.

```

Node 1: Intermediate node

A case goes into Node 2 if D1 <= 0.17486273 where the coefficients of D1 are the coefficients of the 1st crimcoord below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
weight	-0.10646693E-04	-0.20019881E-03	401.23503
length1	-0.41494497E-02	-0.60131669E-01	26.247170
length2	0.37732915E-02	0.68470458E-01	28.415723
heightpc	0.10100866E-01	0.30955883E-03	28.313208
widthpc	-0.54354328E-02	-0.10493626E-01	14.116352
sex.female	-0.14854441E-02	-0.30499910E-01	0.34591195
sex.male	0.64111909E-02	-0.12073413E-01	0.10691824

Class	Crim1Mean	Crim1Variance	Crim2Mean	Crim2Variance	Prior
breame	0.315985	0.203253E-03	0.179849	0.381899E-03	0.22013
parkki	0.319137	0.176877E-03	0.859815E-01	0.594026E-03	0.06918
perch	0.173082	0.198006E-03	0.113651	0.155527E-02	0.35220
pike	0.909419E-01	0.128065E-03	0.301851	0.107943E-02	0.10692
roach	0.186620	0.185920E-03	0.921213E-01	0.121835E-02	0.12579
smelt	0.114490	0.608590E-04	0.111652E-01	0.304533E-03	0.08805
whitefish	0.201284	0.552147E-04	0.143276	0.753174E-03	0.03774

Class	Correlation
breame	-0.182529
parkki	-0.363430
perch	0.641609E-01
pike	-0.204668
roach	0.206973
smelt	0.157189
whitefish	0.162032E-01

Node 2: Terminal node

Cases classified by QDA on 1st and 2nd crimcoords with coefs below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
weight	-0.32521310E-03	0.78169131E-04	372.28529
length1	-0.10887904	-0.21649848E-01	26.614706
length2	0.79213134E-01	0.39149660E-01	28.636765
length3	0.36084671E-01	-0.22898231E-01	30.607353
heightpc	0.97600350E-02	0.19359325E-01	21.158824

```
widthpc          0.27050333E-02  0.31862627E-02  13.211765
```

Class	Crim1Mean	Crim1Variance	Crim2Mean	Crim2Variance	Prior
perch	0.638530	0.284798E-02	0.426285	0.480424E-03	0.51369
pike	0.685068	0.888589E-03	0.141566	0.825493E-03	0.24911
roach	0.630274	0.364608E-04	0.388643	0.759983E-03	0.03001
smelt	0.377875	0.171544E-02	0.284848	0.612044E-03	0.20515

Class	Correlation
perch	-0.294882
pike	0.173587
roach	-0.857166
smelt	0.714333

Node 3: Terminal node

Cases classified by QDA on 1st and 2nd crimcoords with coefs below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
weight	-0.32519883E-04	0.21423970E-03	423.35393
length1	0.12447415E-01	0.22990857	25.972527
length2	0.68660090E-01	-0.27853334	28.250549
length3	-0.74942028E-01	0.41636881E-01	31.690110
widthpc	0.41454782E-02	-0.93710673E-02	14.792308

Class	Crim1Mean	Crim1Variance	Crim2Mean	Crim2Variance	Prior
bream	-0.185470	0.345736E-03	-0.656897	0.613959E-02	0.38359
parkki	-0.246289E-01	0.172909E-03	-0.511178	0.341214E-02	0.12056
perch	0.739426E-01	0.472161E-03	-0.701695	0.675896E-02	0.23153
roach	-0.313040E-01	0.567822E-03	-0.519797	0.460589E-02	0.19735
whitefish	-0.144249E-01	0.752335E-04	-0.707769	0.350085E-03	0.06576

Class	Correlation
bream	-0.253159E-01
parkki	0.951333
perch	-0.243683
roach	0.122187
whitefish	-0.870033

Classification matrix for training sample:

Predicted	True class					
class	bream	parkki	perch	pike	roach	smelt
bream	35	0	0	0	0	0
parkki	0	11	0	0	2	0
perch	0	0	56	0	0	0
pike	0	0	0	17	0	0
roach	0	0	0	0	18	0

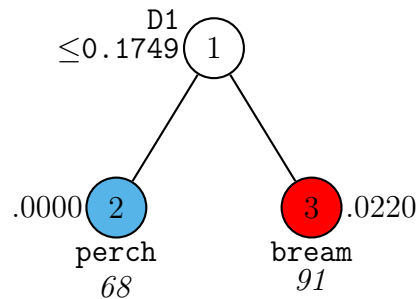


Figure 3: GUIDE v.46.1 0.250-SE classification tree for predicting `species` using linear discriminant splits and quadratic discriminant node models. At each split, an observation goes to the left branch if and only if the condition is satisfied. Class minimizing misclassification cost and sample size (in *italics*) printed below each terminal node; mean misclassification cost beside node.

smelt	0	0	0	0	0	14
whitefish	0	0	0	0	0	0
Total	35	11	56	17	20	14

	whitefish
bream	0
parkki	0
perch	0
pike	0
roach	0
smelt	0
whitefish	6
Total	6

Number of cases used for tree construction: 159

Number misclassified: 2

Resubstitution estimate of mean misclassification cost: 0.12578616E-01

Observed and fitted values are stored in `qda.fit`

LaTeX code for tree is in `qda.tex`

R code is stored in `qda.r`

Discriminant coefficients are stored in `qda.coefs`

R function for plotting data are stored in `qdaplots.r`

Plot data are stored in `qdata.txt`

Figure 3 shows that the tree has 2 terminal nodes. The results in `qda.out` show that it misclassifies only 2 observations. Figure 4 plots the data and splits in the nodes of the tree. They are produced by the R code in `qdaplots.r` whose contents are below.

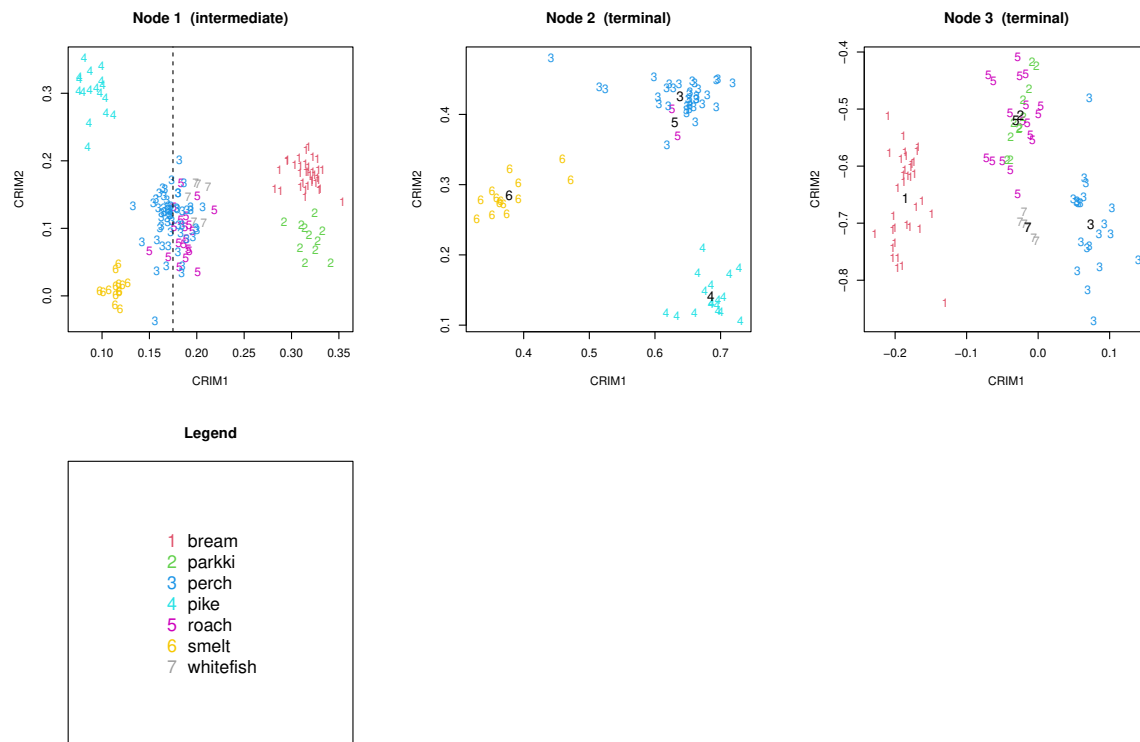


Figure 4: Data and splits in nodes of Figure 3. Class means are indicated by numbers in black.

```
1 library(MASS)
2 descendents <- function(node){
3   desc <- NULL
4   for(m in tnodes){
5     j <- m
6     while(j > 1){
7       j <- floor(j/2)
8       if(j == node) desc <- c(desc,m)
9     }
10  }
11  return(desc)
12 }
13 pdf(file='qdaplots.pdf',width=8,height=6)
14 ## postscript(file='qdaplots.ps',horizontal=TRUE)
15 par(pty='s',mfrow=c(3,4),cex=0.5,cex.main=0.8,cex.lab=0.6,cex.axis=0.6)
16 eqplot <- FALSE
17 splits <- c( 0.220278, 0.129039, 0.516425, 0.554004E-1)
18 varnames <- scan(file='qdadata.txt',what='character',nlines=1)
19 nvar <- length(varnames)
20 z <- as.data.frame(matrix(scan(file='qdadata.txt',skip=1),ncol=nvar,byrow=TRUE))
21 colnames(z) <- varnames
22 y.id <- 1 # class variable column
23 z[,y.id] <- as.factor(z[,y.id])
24 nonmissing <- !is.na(z[,y.id])
25 z <- z[nonmissing,]
26 class.codes <- unclass(z[,y.id])
27 uniq.codes <- unique(sort(class.codes))
28 pred <- read.table(file='qda.fit',header=TRUE)
29 pred <- pred[nonmissing,]
30 tnodes <- unique(sort(pred$node)) ## terminal nodes
31 coefs <- read.table(file='qda.coefs',header=TRUE)
32 inodes <- unique(sort(coefs$node))
33 inodes <- setdiff(inodes,tnodes) ## intermediate nodes
34 allnodes <- sort(c(inodes,tnodes))
35 k <- 0
36 for(node in allnodes){
37   goodplot <- TRUE
38   if(node %in% inodes){ ## intermediate node
39     terminal <- FALSE
40     d <- descendents(node)
41     if(length(d) > 0){
42       gp <- pred$node %in% d
43       k <- k+1
44     } else {
45       goodplot <- FALSE
46     }
```



```

47   } else { ## terminal node
48       terminal <- TRUE
49       gp <- pred$node == node
50       tmp <- unique(sort(z[gp,y.id]))
51       if(length(tmp) <= 1) goodplot <- FALSE
52   }
53   if(goodplot){
54       vgp <- coefs$node == node
55       xvars <- coefs$variable[vgp]
56       xmeans <- as.numeric(coefs$variablemean[vgp])
57       subdatx <- z[gp,]
58       subdatx <- subdatx[,which(names(subdatx) %in% xvars)]
59       subdaty <- z[gp,y.id]
60       for(j in 1:ncol(subdatx)){ ## impute missing values
61           tmp <- subdatx[,j]
62           tmp[is.na(tmp)] <- xmeans[j]
63           subdatx[,j] <- tmp
64       }
65       crimc <- as.matrix(coefs[vgp,3:4])
66       xmat <- as.matrix(subdatx)
67       crimx <- xmat %*% crimc
68       if(eqplot){ ## find variances
69           v1 <- 0
70           v2 <- 0
71           n <- 0
72           for(i in uniq.codes){
73               cgp <- subdaty == i
74               nc <- sum(cgp)
75               if(nc > 1){
76                   n <- n+nc
77                   v1 <- v1+(nc-1)*var(crimx[cgp,1])
78                   v2 <- v2+(nc-1)*var(crimx[cgp,2])
79               }
80           }
81           v1 <- v1/n
82           v2 <- v2/n
83           crimx[,1] <- crimx[,1]/sqrt(v1)
84           crimx[,2] <- crimx[,2]/sqrt(v2)
85           eqscplot(crimx[,1], crimx[,2], type='n',
86                   xlab='scaled_CRIM1', ylab='scaled_CRIM2')
87       } else {
88           plot(crimx[,1], crimx[,2], type='n', xlab='CRIM1', ylab='CRIM2')
89       }
90       text(crimx[,1], crimx[,2], labels=subdaty, col=1+class.codes[gp])
91       if(terminal){
92           for(i in uniq.codes){ # add class means

```

```
93         cgp <- subdaty == i
94         m1 <- mean(crimx[cgp,1])
95         m2 <- mean(crimx[cgp,2])
96         text(m1,m2,labels=as.character(i),cex=1.1)
97     }
98     if(length(allnodes) == 1){ # only 1 node
99         if(eqplot){
100             abline(v=splits/sqrt(v1),lty=2)
101         } else {
102             abline(v=splits,lty=2)
103         }
104     }
105     title(paste('Node',node,'_'(terminal)))
106 } else {
107     if(eqplot){
108         abline(v=splits[k]/sqrt(v1),lty=2)
109     } else {
110         abline(v=splits[k],lty=2)
111     }
112     title(paste('Node',node,'_'(intermediate)))
113 }
114 }
115 }
116 plot(c(-1,1),c(-1,1),type='n',xaxt='n',yaxt='n',xlab='',ylab='')
117 leg.txt <- unique(sort(pred$observed))
118 leg.pch <- as.character(uniq.codes)
119 leg.col <- unique(sort(class.codes))+1
120 legend('center',legend=leg.txt,col=leg.col,pch=leg.pch,
121 ncol=1,bty='n',cex=1.2)
122 title('Legend')
123 dev.off()
```

5 Missing-value flag variables

The Consumer Expenditure (CE) Survey is carried out by the Census Bureau for the Bureau of Labor Statistics (BLS). Conducted quarterly, the survey is a rotating panel survey that collects data on expenditures, income, and demographic characteristics of a sample of about 6000 consumer units (CUs) in the United States. After a CU is in the survey for four quarters, it is dropped and a new unit selected to replace it. The BLS defines *CU* and *reference person* of the CU as follows.

1. A CU consists of any of the following:
 - (a) All members of a particular household who are related by blood, marriage, adoption, or other legal arrangements.
 - (b) A person living alone or sharing a household with others or living as a roomer in a private home or lodging house or in permanent living quarters in a hotel or motel, but who is financially independent.
 - (c) Two or more persons living together who use their incomes to make joint expenditure decisions. Financial independence is determined by spending behavior with regard to the three major expense categories: housing, food, and other living expenses. To be considered financially independent, the respondent must provide at least two of the three major expenditure categories, either entirely or in part.
2. A reference person of the CU is the first member mentioned by the respondent when asked “What are the names of all the persons living or staying here? Start with the name of the person or one of the persons who owns or rents the home.” It is with respect to this person that the relationship of the other CU members is determined.

The data in the file `ce2021.txt` consist of 3965 observations on 550 variables. They are extracted from the second, third and fourth quarters of 2021 and the first quarter of 2022 of the Interview part of the CE survey. For the purpose of illustration and because it is not possible to link CUs between quarters, each CU in the sample is treated as unique. Table 11 in the Appendix gives the names and definitions of some of the 550 variables and their missing-value rates.

About 20% of the variables are *missing-value flags* that give the reasons for missing values. Table 4 lists the flag codes. A variable takes value NA (nonresponse) if its flag variable code is A, B, or C. The names of flag variables are typically the same as their parents, except for the addition or substitution of an underscore. For

Table 4: Codes for missing-value flag variables

A	Valid nonresponse; a response is not anticipated
B	Invalid nonresponse; nonresponse inconsistent with other data reported by CU
C	“Don’t know”, refusal, or other type of nonresponse
D	Valid unadjusted data value
T	Valid value topcoded or suppressed

example, `INTRDVX_` is the flag variable for `INTRDVX` (amount of income received from interest and dividends). In this dataset, `INTRDVX_` has no B codes and records with A codes are removed. Thirty-seven percent of the records (1478) have `INTRDVX_` = C.

A T flag code indicates that the value of a variable is “top-coded.” Top-coding is a method used by the BLS to protect the privacy of the respondents in the top 3 percent of the data. Usually, the reported values of the CUs in this group are replaced by their group mean. For example, below are the values of `AGE2` (age of spouse) and `AGE2_` in rows 112–117 of the data:

	<code>AGE2</code>	<code>AGE2_</code>
112	29	D
113	87	T
114	NA	A
115	57	D
116	87	T
117	NA	A

Respondents 113 and 116 are topcoded and have their values equal to 87, the mean of the top 3 percent of `AGE2`. See https://www.bls.gov/cex/pumd_doc.htm for names of all the variables and Loh et al. (2019b, 2020) for an analysis of an earlier dataset.

Variable `FINLWT21` is a sampling weight. For classification, GUIDE treats all observations with positive sampling weight equally; observations with non-positive weights are ignored in tree construction.

Missing-value flag variables are indicated by the letters “m” or “M” in the DSC file. To indicate to GUIDE to which variable is associated with which M variable, the latter must follow immediately after a B, C, N, P, or S variable in the DSC file. For example, the following lines from the file `ce2021class.dsc` indicate that `DIRACC_` is the flag variable for `DIRACC`, `AGE_REF_` is the flag variable for `AGE_REF`, and `INCN_NW1` is the flag variable for `INCNONW1`.

```
ce2021.txt
```

NA
2
1 DIRACC n
2 DIRACC_ m
3 AGE_REF n
4 AGE_REF_ m
5 AGE2 n
6 AGE2_ m
7 AS_COMP1 n
8 AS_COMP2 n
9 AS_COMP3 n
10 AS_COMP4 n
11 AS_COMP5 n
12 BATHRMQ n
13 BATHRMQ_ m
14 BEDROOMQ n
15 BEDR_OMQ m
16 BLS_URBN n
17 BUILDING c
18 CUTENURE c
19 EARNCOMP c
20 EDUC_REF n
21 EDUCA2 n
22 EDUCA2_ m
:
406 INTRDVX x
407 INTRDVX_ d
408 IRAB n
409 IRAB_ m
:
547 WHLFYR c
548 WHLFYR_ m
549 FFTAXOWE n
550 FSTAXOWE n

A split on an N, P, or S variable that has an associated missing-value flag variable can take several forms. For example, a split on RETSURVX (retirement, survivor, or disability pensions in past 12 months) with flag variable RETS_RVX (which takes values A, C, D, and T) can take 7 forms:

1. `RETS_RVX = A` (only A flag values go left)
2. `RETS_RVX = C` (only C flag values go left)
3. `RETSURVX = NA` (all missing values go left)
4. `RETSURVX ≤ c`
5. `RETSURVX ≤* c` (the symbol “≤*” means “≤ or is missing”)
6. `RETSURVX ≤ c` or `RETS_RVX = A`
7. `RETSURVX ≤ c` or `RETS_RVX = C`

Similarly, a split on a C variable such as `INCNONW2` that has missing-value flag variable `INCN_NW2` can take these forms (see Figure 12):

1. `INCNONW2` in S (where S is a subset of values of `INCNONW2`)
2. `INCNONW2 = NA`
3. `INCNONW2` in S or `INCN_NW2` in S^* (where S^* is a subset of flag codes)

5.1 Classification: univariate splits

This section shows how to construct a classification tree for predicting `INTRDVX_` using the DSC file `ce2021class.dsc`.

5.1.1 Input file generation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: classin.txt
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: classout.txt
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: ce2021class.dsc
Reading DSC file ...

```

```

Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX_
Reading data file ...
Number of records in data file: 3965
Length of longest entry in data file: 11
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Recoding D values to integers
Finished recoding
Number of classes: 3
Finding number of levels of M variables associated with C variables ...
Assigning integer codes to values of 48 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Finished assigning codes to 40 categorical variables
Associating missing values of N and S variables with M variable codes ...
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Smallest positive weight:  1.0725E+03
Largest positive weight:  9.3902E+04
Class #Cases      Proportion
C      1478      0.37276166
D      2431      0.61311475
T        56      0.01412358
Numbers of N, F, S, and P below exclude constant variables
      Total #cases w/ #missing
      #cases  miss. D ord. vals #X-var #N-var #F-var #S-var
      3965      0      3965      1      0      0      383
      #P-var #M-var #B-var #C-var #I-var
      0      116      0      48      0
Number of cases used for training: 3965
Number of cases excluded due to 0 W or missing D variable: 0
Finished reading data file
Warning: No interaction and linear splits; too many predictor variables

```

```

Choose 1 for estimated priors, 2 for equal priors, 3 to input priors from a file
Input 1, 2, or 3 ([1:3], <cr>=1):
Choose 1 for unit misclassification costs, 2 to input costs from a file
Input 1 or 2 ([1:2], <cr>=1):
Warning: All positive weights treated as 1
Default max. number of split levels: 12
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): class.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: classfit.txt
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: class.r
Input rank of top variable to split root node ([1:431], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < classin.txt

```

5.1.2 Contents of output file

```

Classification tree
Pruning by cross-validation
DSC file: ce2021class.dsc
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX_
Number of records in data file: 3965
Length of longest entry in data file: 11
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Number of classes: 3
Smallest and largest positive weights are 1.0725E+03 and 9.3902E+04
Training sample class proportions of D variable INTRDVX_:
Class  #Cases      Proportion
C          1478      0.37276166

```



```
D      2431      0.61311475
T       56      0.01412358
```

Summary information for training sample of size 3965

d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight

Levels of M variables are for missing values in associated variables

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	DIRACC	s	1.000	1.000		170 constant
2	DIRACC_	m			2	
3	AGE_REF	s	18.00	87.00		
4	AGE_REF_	m			0	
:						
31	FINLWT21	w	1072.	0.9390E+05		
:						
407	INTRDVX_	d			3	
:						
549	FFTAX0WE	s	-0.3368E+05	0.3997E+06		
550	FSTAX0WE	s	-3309.	0.7223E+05		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
3965	0	3965	1	0	0	378	
#P-var	#M-var	#B-var	#C-var	#I-var			
0	116	0	48	0			

Number of constant S variables: 5

Number of cases used for training: 3965

Number of cases excluded due to 0 W or missing D variable: 0

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Warning: No interaction and linear splits; too many predictor variables

Simple node models

Estimated priors

Unit misclassification costs

Warning: All positive weights treated as 1

Univariate split highest priority

No interaction splits

No linear splits

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 3

Ranks of variables and their 1-df chi-squared values at root node

1	0.2336E+03	INCLASS2
2	0.1936E+03	STATE
3	0.1519E+03	ERANKH
4	0.1350E+03	PSU
5	0.1307E+03	RETSURVX
:		
420	0.2626E-02	OTHLOAN
421	0.2542E-02	VFUELOIC
422	0.6293E-03	TEXTILCQ

Size and CV mean cost and SE of subtrees:

Tree	#Tnodes	Mean Cost	SE(Mean)	BSE(Mean)	Median Cost	BSE(Median)
1	69	3.021E-01	7.292E-03	7.727E-03	3.030E-01	1.124E-02
2	68	3.021E-01	7.292E-03	7.727E-03	3.030E-01	1.124E-02
3	67	3.021E-01	7.292E-03	7.727E-03	3.030E-01	1.124E-02
4	66	3.021E-01	7.292E-03	7.727E-03	3.030E-01	1.124E-02
5	65	3.021E-01	7.292E-03	7.727E-03	3.030E-01	1.124E-02
6	64	3.021E-01	7.292E-03	7.727E-03	3.030E-01	1.124E-02
7	63	3.021E-01	7.292E-03	7.727E-03	3.030E-01	1.124E-02
8*	60	3.021E-01	7.292E-03	7.704E-03	3.030E-01	1.139E-02
9	53	3.037E-01	7.303E-03	7.308E-03	3.030E-01	9.744E-03
10	52	3.047E-01	7.309E-03	7.311E-03	3.030E-01	1.051E-02
11	50	3.042E-01	7.306E-03	7.383E-03	3.030E-01	1.117E-02
12	48	3.052E-01	7.313E-03	7.977E-03	3.018E-01	1.223E-02
13	34	3.052E-01	7.313E-03	8.202E-03	3.018E-01	1.287E-02
14	25	3.054E-01	7.315E-03	8.156E-03	3.018E-01	1.253E-02
15	24	3.054E-01	7.315E-03	8.156E-03	3.018E-01	1.253E-02
16	23	3.044E-01	7.308E-03	8.572E-03	2.967E-01	1.316E-02
17	22	3.047E-01	7.309E-03	8.544E-03	2.980E-01	1.299E-02
18	21	3.049E-01	7.311E-03	8.659E-03	2.929E-01	1.475E-02
19**	20	3.039E-01	7.304E-03	7.919E-03	2.929E-01	1.416E-02
20	15	3.059E-01	7.318E-03	7.691E-03	3.030E-01	1.559E-02
21	13	3.057E-01	7.316E-03	7.713E-03	3.043E-01	1.599E-02
22	10	3.062E-01	7.320E-03	7.703E-03	3.131E-01	1.412E-02
23	8	3.173E-01	7.391E-03	7.984E-03	3.253E-01	1.190E-02
24	3	3.359E-01	7.501E-03	9.174E-03	3.295E-01	1.160E-02
25	2	3.417E-01	7.532E-03	1.010E-02	3.317E-01	1.779E-02
26	1	3.869E-01	7.735E-03	8.543E-03	3.851E-01	1.446E-02

0-SE tree based on mean is marked with * and has 60 terminal nodes

0-SE tree based on median is marked with + and has 20 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **
 Selected-SE tree based on mean using bootstrap SE is marked with --
 Selected-SE tree based on median and bootstrap SE is marked with ++
 ** tree same as ++ tree
 ** tree same as + tree
 ** tree same as -- tree
 ++ tree same as -- tree
 + tree same as ++ tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Node cost is node misclassification cost divided by number of cases in node

Node label	Total cases	Train cases	Predicted class	Node cost	Split variables	Interacting variable
1	3965	3965	D	3.869E-01	INCLASS2	
2	248	248	C	2.299E-01	FINCBTAX	
4T	105	105	C	2.872E-02	FINCBTAX	
5	143	143	C	3.777E-01	FINDRETX	
10	127	127	C	3.072E-01	UNISTRQ	
20T	16	16	D	6.258E-02	-	
21T	111	111	C	2.163E-01	-	
11T	16	16	D	6.258E-02	-	
3	3717	3717	D	3.613E-01	PSU	
6T	126	126	C	1.668E-01	TFOODHOP	
7	3591	3591	D	3.448E-01	STATE	
14T	1360	1360	D	2.154E-01	INCNONW2	
15	2231	2231	D	4.236E-01	RETSURVX	
30	1609	1609	D	4.201E-01	FINDRETX	
60	968	968	D	4.928E-01	INCLASS2	
120	538	538	D	4.108E-01	STOCKX	
240	507	507	D	3.984E-01	RENTEQVX	
480T	378	378	D	3.360E-01	-	
481T	129	129	C	4.187E-01	-	
241T	31	31	C	3.872E-01	-	
121	430	430	C	4.395E-01	EARNCOMP	
242	32	32	T	5.938E-01	FINCBTAX	
484T	10	10	C	1.001E-01	-	
485T	22	22	T	4.091E-01	BUILT	
243	398	398	C	4.196E-01	REFGEN	
486T	89	89	D	4.494E-01	-	
487	309	309	C	3.819E-01	REFGEN	
974T	8	8	C	2.501E-01	-	
975	301	301	C	3.854E-01	CASHCOCQ	
1950T	220	220	C	3.092E-01	-	

1951T	81	81	D	4.074E-01 -
61	641	641	D	3.105E-01 HLFBATHQ
122	350	350	D	3.514E-01 FSTAXOWE
244T	299	299	D	2.977E-01 -
245T	51	51	C	3.334E-01 -
123T	291	291	D	2.612E-01 HLFBATHQ
31	622	622	D	4.325E-01 RETSURVX
62T	86	86	C	1.397E-01 NETRENTB
63T	536	536	D	3.638E-01 STOCKYRX

Number of terminal nodes of final tree: 20

Total number of nodes of final tree: 39

Second best split variable (based on curvature test) at root node is STATE

Classification tree:

For categorical variable splits, values not in training data go to the right

Node 1: INCLASS2 = NA

Node 2: FINCBTAX <= 0.50000000

Node 4: C

Node 2: FINCBTAX > 0.50000000 or NA

Node 5: FINDRETX <= 300.00000

Node 10: UNISTRQ = "1", "2", "6", "7", "8"

Node 20: D

Node 10: UNISTRQ /= "1", "2", "6", "7", "8"

Node 21: C

Node 5: FINDRETX > 300.00000 or NA

Node 11: D

Node 1: INCLASS2 /= NA

Node 3: PSU = "S49F"

Node 6: C

Node 3: PSU /= "S49F"

Node 7: STATE = "2", "6", "10", "11", "21", "24", "25", "27", "31", "40",
"41", "47", "48", "49"

Node 14: D

Node 7: STATE /= "2", "6", "10", "11", "21", "24", "25", "27", "31", "40",
"41", "47", "48", "49"

Node 15: RETSURVX = NA & RETS_RVX = "A"

Node 30: FINDRETX <= 391.50000

Node 60: INCLASS2 <= 4.5000000

Node 120: STOCKX <= 190000.00 or STOCKX = NA & STOCKX_ = "A"

Node 240: RENTEQVX <= 1990.5000 or NA

Node 480: D

Node 240: RENTEQVX > 1990.5000

Node 481: C

Node 120: not (STOCKX <= 190000.00 or STOCKX = NA & STOCKX_ = "A")

```

Node 241: C
Node 60: INCLASS2 > 4.5000000 or NA
Node 121: EARNCOMP = "8"
Node 242: FINCBTAX <= 105777.50
Node 484: C
Node 242: FINCBTAX > 105777.50 or NA
Node 485: T
Node 121: EARNCOMP /= "8"
Node 243: REFGEN = "5"
Node 486: D
Node 243: REFGEN /= "5"
Node 487: REFGEN = "1"
Node 974: C
Node 487: REFGEN /= "1"
Node 975: CASHCOCQ <= 112.50000
Node 1950: C
Node 975: CASHCOCQ > 112.50000 or NA
Node 1951: D
Node 30: FINDRETX > 391.50000 or NA
Node 61: HLFBATHQ <= 0.50000000
Node 122: FSTAXOWE <= 10256.500
Node 244: D
Node 122: FSTAXOWE > 10256.500 or NA
Node 245: C
Node 61: HLFBATHQ > 0.50000000 or NA
Node 123: D
Node 15: not (RETSURVX = NA & RETS_RVX = "A")
Node 31: RETSURVX = NA
Node 62: C
Node 31: RETSURVX /= NA
Node 63: D

```

```

*****
Predictor means below are weighted means of cases with no missing values.

```

```

Node 1: Intermediate node
A case goes into Node 2 if INCLASS2 = NA
INCLASS2 mean = 4.4617238
Class      Number   Posterior
C           1478  0.3728E+00
D           2431  0.6131E+00
T             56  0.1412E-01
Number of training cases misclassified = 1534
Predicted class is D
-----

```

```

Node 2: Intermediate node

```

```

A case goes into Node 4 if FINCBTAX <= 0.50000000
FINCBTAX mean = 18232.969
Class      Number  Posterior
C           191  0.7701E+00
D           57  0.2299E+00
T            0  0.3561E-05
Number of training cases misclassified = 57
Predicted class is C
-----
Node 4: Terminal node
Class      Number  Posterior
C           102  0.9713E+00
D            3  0.2872E-01
T            0  0.3561E-05
Number of training cases misclassified = 3
Predicted class is C
-----
Node 5: Intermediate node
A case goes into Node 10 if FINDRETX <= 300.00000
FINDRETX mean = 1054.5743
Class      Number  Posterior
C           89  0.6223E+00
D           54  0.3777E+00
T            0  0.3561E-05
Number of training cases misclassified = 54
Predicted class is C
-----
:
-----
Node 31: Intermediate node
A case goes into Node 62 if RETSURVX = NA
RETSURVX mean = 25637.530
Class      Number  Posterior
C           259  0.4164E+00
D           353  0.5675E+00
T            10  0.1608E-01
Number of training cases misclassified = 269
Predicted class is D
-----
Node 62: Terminal node
Class      Number  Posterior
C           74  0.8603E+00
D           12  0.1397E+00
T            0  0.3561E-05
Number of training cases misclassified = 12
Predicted class is C

```

```

-----
Node 63: Terminal node
Class      Number  Posterior
C           185  0.3451E+00
D           341  0.6362E+00
T            10  0.1866E-01
Number of training cases misclassified = 195
Predicted class is D
-----

Classification matrix for training sample:
Predicted   True class
class       C         D         T
C           663       212         2
D           814      2211        41
T             1         8        13
Total       1478      2431        56

Number of cases used for tree construction: 3965
Number misclassified: 1078
Resubstitution estimate of mean misclassification cost: 0.27187894

Observed and fitted values are stored in classfit.txt
LaTeX code for tree is in class.tex
R code is stored in class.r

```

Figure 5 shows the classification tree. It has four different kinds of splits involving missing values:

Node 1. $\text{INCLASS2} = \text{NA}$ (go left if and only if INCLASS2 is missing).

Node 2. $\text{FINCBTAX} \leq 0.5$ (go left if and only if FINCBTAX is nonmissing and ≤ 0.5).

Node 15. $\text{RETS_RVX} = \text{A}$ (go left if and only if RETSURVX is missing with flag variable $\text{RETS_RVX} = \text{A}$).

Node 240. $\text{RENTEQVX} \leq_* 1990.5$ (go left if and only if $\text{RENTEQVX} = \text{NA}$ or ≤ 1990.5).

The top several lines of the file of fitted values `class.fit` are given below. The posterior probabilities of predicting class T are very low (see Section 4.1.3 for the calculation of the posterior probabilities).

train	node	observed	predicted	"P(C)"	"P(D)"	"P(T)"
y	6	"C"	"C"	0.83322E+00	0.16678E+00	0.35612E-05

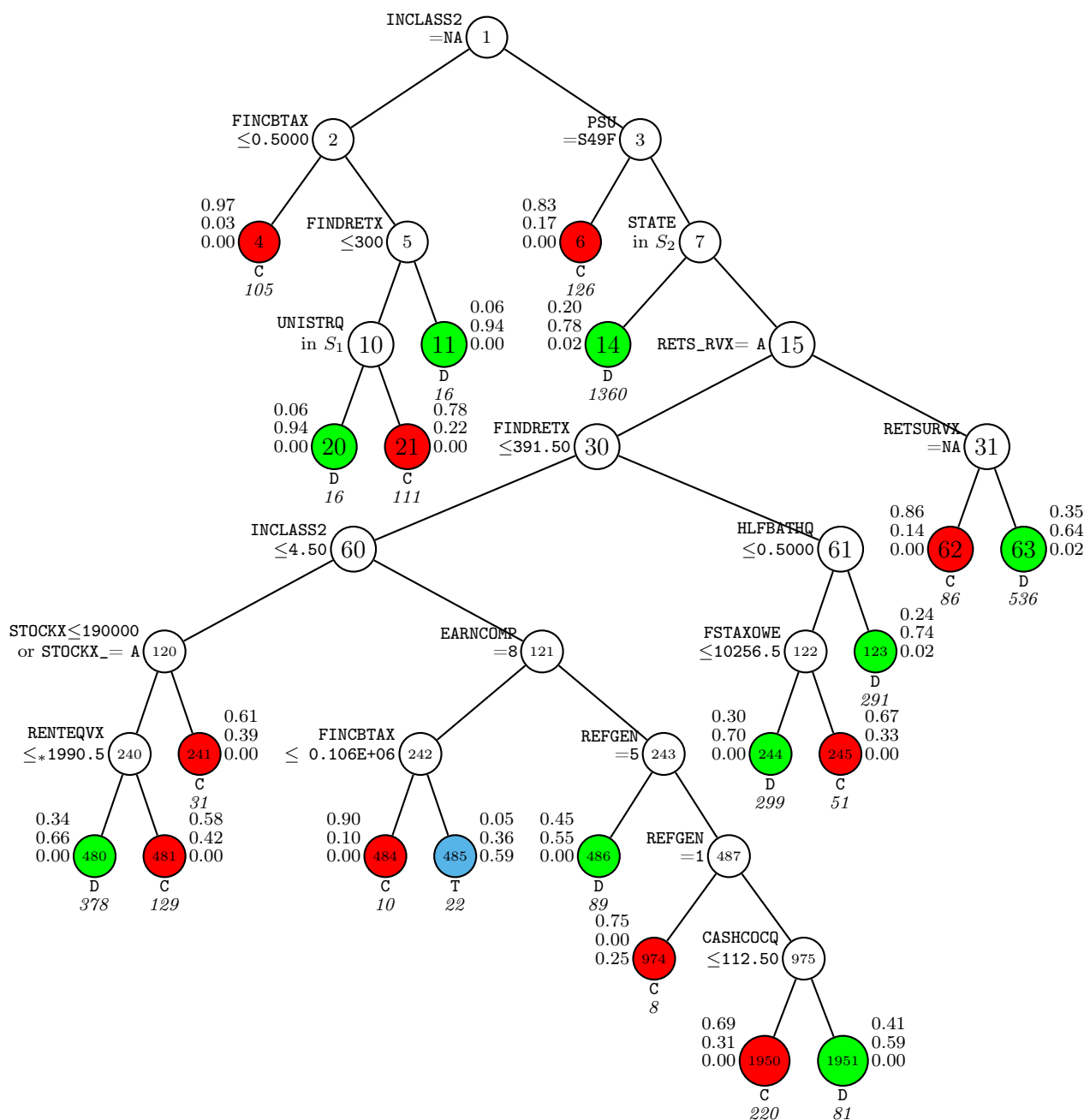


Figure 5: GUIDE v.46.1 0.250-SE classification tree for predicting INTRDVX_ using estimated priors and unit misclassification costs. At each split, an observation goes to the left branch if and only if the condition is satisfied. Symbol ‘ $\leq^*$ ’ stands for ‘ $\leq$ or missing’. $S_1 = \{1, 2, 6, 7, 8\}$. $S_2 = \{2, 6, 10, 11, 21, 24, 25, 27, 31, 40, 41, 47, 48, 49\}$. Predicted class and sample size (in *italics*) printed below each terminal node; class posterior probabilities for INTRDVX_ = C, D, and T, respectively, beside node. Second best split variable at root node is STATE.

y	14	"D"	"D"	0.19779E+00	0.78456E+00	0.17647E-01
y	14	"D"	"D"	0.19779E+00	0.78456E+00	0.17647E-01
y	1950	"D"	"C"	0.69083E+00	0.30917E+00	0.35612E-05
y	14	"D"	"D"	0.19779E+00	0.78456E+00	0.17647E-01
y	123	"T"	"D"	0.23711E+00	0.73883E+00	0.24055E-01

5.2 Classification: linear trees

Following are results from fitting a classification tree with linear splits and LDA node models. C and M variables are automatically converted to B.

```

Classification tree
Linear discriminant splits and LDA node models
Pruning by cross-validation
DSC file: ce2021class.dsc
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
D variable is INTRDVX_
M variables changed to B
C variables changed to B
Number of records in data file: 3965
Length of longest entry in data file: 11
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Number of classes: 3
Smallest and largest positive weights are 1.0725E+03 and 9.3902E+04
Training sample class proportions of D variable INTRDVX_:
Class  #Cases    Proportion
C      1478      0.37276166
D      2431      0.61311475
T        56      0.01412358

```

```

Summary information for training sample of size 3965
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight

```

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	DIRACC	n	1.0000E+00	1.000		170 constant
2	DIRACC_	b			3	

3	AGE_REF	n	18.00	87.00		
4	AGE_REF_	b			2	
5	AGE2	n	2.1000E+01	87.00		1734
6	AGE2_	b			3	
:						
31	FINLWT21	w	1072.	0.9390E+05		
:						
407	INTRDVX_	d			3	
:						
549	FFTAXOWE	n	-0.3368E+05	0.3997E+06		
550	FSTAXOWE	n	-3309.	0.7223E+05		
===== Constructed variables =====						
551	DIRACC_.C	f	0.000	1.000		
552	DIRACC_.D	f	0.000	1.000		
553	AGE_REF_.T	f	0.000	1.000		
554	AGE2_.D	f	0.000	1.000		
555	AGE2_.T	f	0.000	1.000		
:						
1100	WHOLIFX.NA	f	0.000	1.000		
1101	VEHICTAX.NA	f	0.000	1.000		
1102	CREDYR.NA	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/	#missing					
#cases	miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var	
3965	0	3965	1	378	552	0	
#P-var	#M-var	#B-var	#C-var	#I-var			
0	0	164	0	0			

Number of constant N variables: 5

Number of dummy variables (excl. missing-value indicators) created: 467

Number of missing-value indicators created: 85

Number of cases used for training: 3965

Number of cases excluded due to 0 W or missing D variable: 0

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Estimated priors

Unit misclassification costs

Warning: All positive weights treated as 1

Max number of splits: 100

Maximum number of split levels: 15

Minimum node sample size: 30

crimcoords switched in node 2

crimcoords switched in node 5

crimcoords switched in node 10
 crimcoords switched in node 6
 crimcoords switched in node 7

Size and CV mean cost and SE of subtrees:

Tree	#Tnodes	Mean Cost	SE(Mean)	BSE(Mean)	Median Cost	BSE(Median)
1	25	3.972E-01	7.771E-03	1.915E-02	3.934E-01	1.785E-02
2	24	3.985E-01	7.775E-03	1.863E-02	3.934E-01	1.752E-02
3	23	3.945E-01	7.762E-03	1.922E-02	3.859E-01	2.171E-02
4	22	3.876E-01	7.737E-03	2.062E-02	3.733E-01	2.616E-02
5	21	3.854E-01	7.729E-03	2.106E-02	3.690E-01	2.800E-02
6	18	3.834E-01	7.721E-03	2.107E-02	3.615E-01	2.667E-02
7	15	4.005E-01	7.782E-03	3.370E-02	3.577E-01	3.041E-02
8	14	4.005E-01	7.782E-03	3.370E-02	3.577E-01	3.041E-02
9	13	3.950E-01	7.763E-03	3.540E-02	3.564E-01	3.171E-02
10	4	3.602E-01	7.624E-03	3.371E-02	3.140E-01	3.704E-02
11**	2	3.011E-01	7.285E-03	1.040E-02	2.992E-01	1.294E-02
12	1	3.291E-01	7.462E-03	8.841E-03	3.317E-01	1.334E-02

0-SE tree based on mean is marked with * and has 2 terminal nodes

0-SE tree based on median is marked with + and has 2 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Node cost is node misclassification cost divided by number of cases in node

Node	Total	Misclassification
label	cases	cost
1	3965	3.120E-01
2T	3664	2.604E-01
3T	301	2.924E-01

Number of terminal nodes of final tree: 2

Total number of nodes of final tree: 3

Classification tree with linear discriminant splits and LDA node models

Node 1: D1 <= 0.36376700E-1

Node 2: 954/3664 misclassified

Node 1: D1 > 0.36376700E-1

Node 3: 88/301 misclassified

 Predictor means below are weighted means of cases with no missing values.

Node 1: Intermediate node

A case goes into Node 2 if $D1 \leq 0.36376700E-1$ where the coefficients of $D1$ are the coefficients of the 1st crimcoord below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
AGE_REF	0.10286356E-03	-0.18305945E-03	56.815873
AGE2	-0.55263465E-05	-0.17796121E-03	56.569604
BATHRMQ	-0.66238493E-03	-0.23729633E-03	1.9788369
BEDROOMQ	0.26475570E-02	-0.14139684E-02	3.0598017
EDUC_REF	-0.14127559E-02	0.37877657E-02	14.419353
:			
RETSRVBX.NA	-0.24148305E-03	0.41288727E-03	0.98644714
STOCKYRX.NA	-0.39704159E-03	-0.47029904E-03	0.93379120
STOCKX.NA	-0.43097163E-03	-0.36438702E-03	0.92629161
WHOLIFX.NA	0.31332909E-06	-0.17989280E-03	0.97664602

Class	Crim1Mean	Crim1Variance	Crim2Mean	Crim2Variance	Prior
C	0.180418E-01	0.201764E-03	-0.657768E-02	0.217229E-03	0.37276
D	0.127579E-01	0.201764E-03	0.492010E-02	0.217229E-03	0.61311
T	0.710481E-01	0.201764E-03	0.185411E-01	0.217229E-03	0.01412

Class	Correlation
C	-0.218710
D	0.328716E-01
T	0.870809

 Node 2: Terminal node

Cases classified by LDA on 1st and 2nd crimcoords with coefs below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
AGE_REF	0.96285117E-04	-0.63857323E-04	56.278127
EDUC_REF	-0.19041907E-04	0.14995258E-02	14.406429
FINDRETX	-0.47640375E-06	0.30569910E-06	3503.0889
:			
STOCKYRX.NA	0.50317660E-02	0.30502394E-03	0.93767194
STOCKX.NA	-0.56818615E-03	-0.53271423E-03	0.92981836

Class	Crim1Mean	Crim1Variance	Crim2Mean	Crim2Variance	Prior
C	0.258611E-01	0.239888E-03	0.470209E-02	0.218749E-03	0.36081
D	0.242641E-01	0.239888E-03	0.199238E-01	0.218749E-03	0.63783
T	0.179619	0.239888E-03	0.409829E-01	0.218749E-03	0.00136

Class	Correlation
C	-0.541558E-01
D	0.518177E-02

T 0.954405

Node 3: Terminal node

Cases classified by LDA on 1st and 2nd crimcoords with coefs below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
BLS_URBN	0.33425912E-01	0.11807432	1.0367633
FINCBTAX	-0.16920898E-06	-0.14394214E-06	206064.08
INC_RANK	0.42300178E-01	-0.25649083E-01	0.69192227
ERANKHM	-0.71334772E-02	-0.25523355E-01	0.74976596
FS_MTHI	0.25232937E-01	0.38869046E-01	3.8260458
STUDFINX	-0.88087983E-03	0.12671364E-02	12.736634
FINCBT_X.T	0.46683415E-01	0.43507913E-01	0.52302645
INC__ANK.T	0.46683415E-01	0.43507913E-01	0.52302645
ERANKH_.D	0.14007054E-01	-0.60120585E-02	0.72058900
LIQU_YRX.D	-0.15443567E-01	-0.63917957E-01	0.11509276
LIQUIDX_.C	-0.25385359E-01	0.25380838E-01	0.10418600
LIQUIDX_.D	0.36041174E-01	-0.92514583E-01	0.12576118
RETS_RVX.C	-0.29421341E-01	0.49024608E-01	0.13415560
RETS_VBX.B	0.15085335E-01	-0.15294843E-01	0.97117048E-01
INCLASS2.NA	-0.14007054E-01	0.60120585E-02	0.27941100
IRAX.NA	0.62440841E-03	0.34600390E-01	0.87740311
IRAYRX.NA	0.20033036E-01	0.25966634E-01	0.88171064
LIQUDYRX.NA	0.12205883E-01	-0.88538824E-01	0.85571789
LIQUIDX.NA	-0.46188132E-01	-0.58461879E-01	0.83981802
RETSURVX.NA	0.24330200E-01	0.70294665E-01	0.84697755

Class	Crim1Mean	Crim1Variance	Crim2Mean	Crim2Variance	Prior
C	0.138806	0.184603E-02	0.250541	0.273223E-02	0.51827
D	0.192089	0.184603E-02	0.205875	0.273223E-02	0.31229
T	0.240629	0.184603E-02	0.274041	0.273223E-02	0.16944

Class	Correlation
C	-0.253259
D	0.189470
T	0.457696

Classification matrix for training sample:

Predicted	True class		
class	C	D	T
C	703	234	0
D	764	2184	20
T	11	13	36
Total	1478	2431	56

Number of cases used for tree construction: 3965

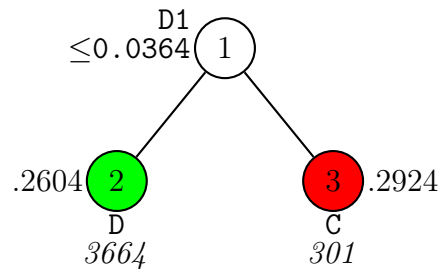


Figure 6: GUIDE v.46.1 0.250-SE classification tree for predicting INTRDVX_ using linear discriminant splits and linear discriminant node models. At each split, an observation goes to the left branch if and only if the condition is satisfied. Class minimizing misclassification cost and sample size (in *italics*) printed below each terminal node; mean misclassification cost beside node.

Number misclassified: 1042

Resubstitution estimate of mean misclassification cost: 0.26279950

Observed and fitted values are stored in lda.fit

LaTeX code for tree is in lda.tex

R code is stored in lda.r

Discriminant coefficients are stored in lda.coefs

R function for plotting data are stored in ldaplot.r

Plot data are stored in ldadata.txt

Figure 6 shows the tree and Figure 7 shows the data in the nodes.

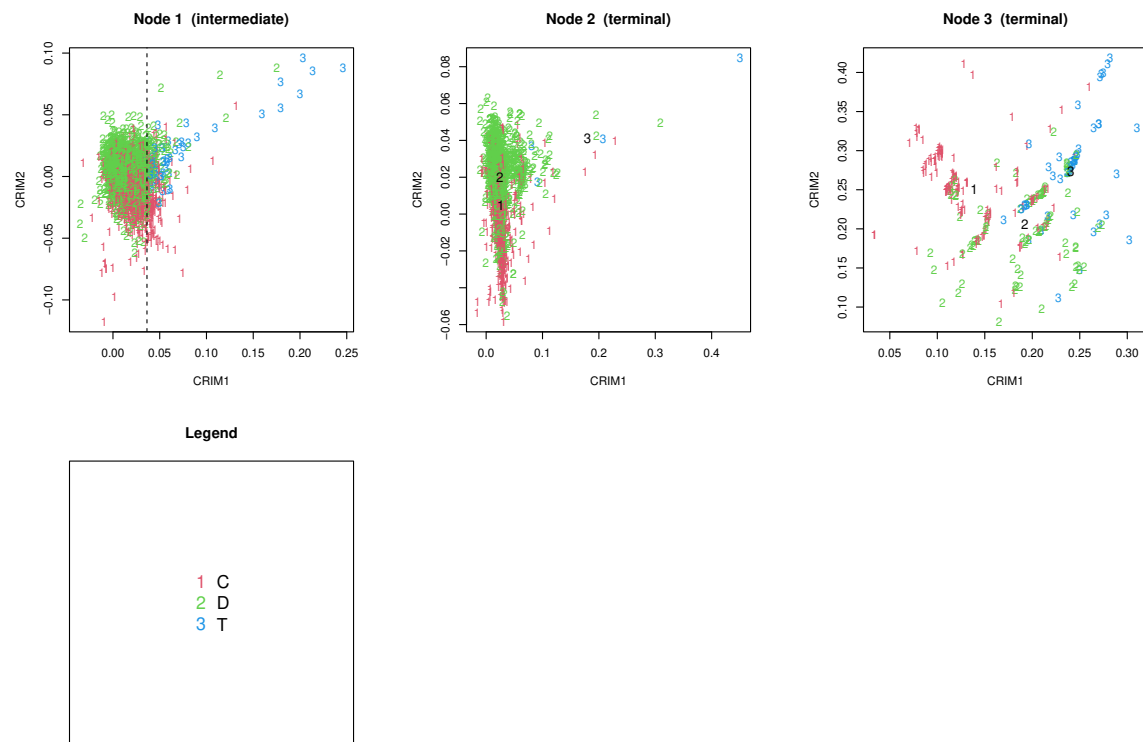


Figure 7: Data and splits in nodes of Figure 6. Class means are indicated by numbers in black.

6 Least squares regression

GUIDE can fit least-squares (LS), quantile, Poisson, proportional hazards, and least-median-of-squares (LMS) regression tree models. We illustrate least squares and quantile models with the CE data, using INTRDVX as the dependent (d) variable and excluding (x) its flag INTRDVX_. The DSC file is ce2021reg.dsc, which sets FINLWT21 as a weight (w) variable.

6.1 Piecewise constant

6.1.1 Input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: cons.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: cons.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1):
Input 1 for least squares, 2 least median of squares ([1:2], <cr>=1):
Choose complexity of model to use at each node:
Choose 0 for stepwise linear regression (recommended for prediction)
Choose 1 for multiple regression
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
0: stepwise linear, 1: multiple linear, 2: best simple polynomial, 3: constant,
4: best simple stepwise ANCOVA ([0:4], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: ce2021reg.dsc
Reading DSC file ...
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX
```



```

Reading data file ...
Number of records in data file: 3965
Length of longest entry in data file: 11
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Finding number of levels of M variables associated with C variables ...
Assigning integer codes to values of 48 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Finished assigning codes to 40 categorical variables
Associating missing values of N and S variables with M variable codes ...
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Smallest positive weight: 1.0725E+03
Largest positive weight: 9.3902E+04
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      3965      1478      3965      1      0      0      383
      #P-var  #M-var  #B-var  #C-var  #I-var
      0      116      0      48      0
Weight variable FINLWT21 in column: 31
Number of cases used for training: 2487
Number of cases excluded due to 0 W or missing D variable: 1478
Finished reading data file
Input 1 for unweighted, 2 for weighted error estimates during pruning ([1:2], <cr>=2):
Warning: No interaction and linear splits; too many predictor variables
Default max. number of split levels: 12
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2): 1
Input file name to store LaTeX code (use .tex as suffix): cons.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: cons.fit
You can output an R function for predicting future observations

```

```

Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: cons.r
Input rank of top variable to split root node ([1:431], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < cons.in

```

6.1.2 Contents of cons.out

```

Least squares regression tree
Pruning by cross-validation
DSC file: ce2021reg.dsc
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX
Piecewise constant model
Number of records in data file: 3965
Length of longest entry in data file: 11
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Smallest and largest positive weights are 1.0725E+03 and 9.3902E+04

Summary information for training sample of size 2487 (excluding observations
with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight
Levels of M variables are for missing values in associated variables

```

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	DIRACC	s	1.000	1.000		125 constant
2	DIRACC_	m			2	
3	AGE_REF	s	19.00	87.00		
4	AGE_REF_	m			0	
:						
31	FINLWT21	w	1072.	0.9390E+05		

```

:
404 FSMPFRMX s -0.1160E+06 0.7703E+06
405 FSMP_RMX m 0
406 INTRDVX d 1.000 0.1413E+06
:
547 WHLFYR c 1 2487
548 WHLFYR_ m 1
549 FFTAXOWE s -0.3368E+05 0.3380E+06
550 FSTAXOWE s -3074. 0.5654E+05

```

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/	#missing					
#cases	miss.	D	ord. vals	#X-var	#N-var	#F-var	#S-var
3965	1478		3965	1	0	0	376
#P-var	#M-var	#B-var	#C-var	#I-var			
0	116	0	48	0			

Number of constant S variables: 7

Weight variable FINLWT21 in column: 31

Number of cases used for training: 2487

Number of cases excluded due to 0 W or missing D variable: 1478

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Weighted error estimates used for pruning

Warning: No interaction and linear splits; too many predictor variables

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 12

Minimum node sample size: 2

Ranks of variables and their 1-df chi-squared values at root node

```

1 0.8297E+02 REFGEN
2 0.8111E+02 AGE_REF
3 0.7066E+02 INCNONW1
:
416 0.3569E-02 TGASMOTC
417 0.1029E-03 MAINRPPQ

```

Size and CV MSE and SE of subtrees:

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	62	6.106E+12	7.688E+11	4.771E+11	5.779E+12	7.257E+11
2	61	6.106E+12	7.688E+11	4.771E+11	5.779E+12	7.257E+11
:						
35	10	6.103E+12	7.677E+11	4.945E+11	5.796E+12	7.342E+11

36**	9	5.915E+12	7.601E+11	4.635E+11	5.668E+12	6.327E+11
37	8	6.433E+12	8.423E+11	5.350E+11	6.000E+12	7.476E+11
38	5	6.573E+12	8.685E+11	6.476E+11	6.000E+12	9.316E+11
39	4	7.614E+12	9.359E+11	7.113E+11	8.502E+12	1.107E+12
40	1	8.287E+12	1.032E+12	6.955E+11	8.542E+12	7.418E+11

0-SE tree based on mean is marked with * and has 9 terminal nodes

0-SE tree based on median is marked with + and has 9 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is weighted mean of INTRDVX in the node

Cases fit give the number of cases used to fit node

MSE is residual sum of squares divided by number of cases in node

Node label	Total cases	Cases fit	Matrix rank	Node D-mean	Node MSE	Split variable
1	2487	2487	1	5.131E+03	8.287E+12	REFGEN
2T	573	573	1	9.804E+02	1.105E+12	PSU
3	1914	1914	1	6.476E+03	1.027E+13	INC_RANK
6T	1345	1345	1	2.823E+03	1.657E+12	REF_RACE
7	569	569	1	1.498E+04	2.828E+13	EARNCOMP
14	75	75	1	5.278E+04	7.423E+13	RETSURV
28T	46	46	1	2.764E+04	4.076E+13	-
29T	29	29	1	8.586E+04	7.873E+13	-
15	494	494	1	9.170E+03	1.560E+13	FFTAXOWE
30T	247	247	1	2.647E+03	3.560E+12	UNISTRQ
31	247	247	1	1.570E+04	2.573E+13	AGE2
62T	156	156	1	8.036E+03	1.274E+13	LIQDYRBX
63	91	91	1	3.030E+04	4.131E+13	BATHRMQ
126	87	87	1	2.564E+04	3.295E+13	STATE
252T	42	42	1	4.145E+04	5.339E+13	-
253T	45	45	1	5.032E+03	8.537E+11	STATE
127T	4	4	1	1.243E+05	8.997E+12	-

Number of terminal nodes of final tree: 9

Total number of nodes of final tree: 17

Second best split variable (based on curvature test) at root node is AGE_REF

Regression tree:

For categorical variable splits, values not in training data go to the right

```

Node 1: REFGEN = "5"
  Node 2: INTRDVX-mean = 980.35292
Node 1: REFGEN /= "5"
  Node 3: INC_RANK <= 0.84018625
    Node 6: INTRDVX-mean = 2822.6445
  Node 3: INC_RANK > 0.84018625 or NA
    Node 7: EARNCOMP = "8"
      Node 14: RETSURV = "1"
        Node 28: INTRDVX-mean = 27641.282
      Node 14: RETSURV /= "1"
        Node 29: INTRDVX-mean = 85859.276
    Node 7: EARNCOMP /= "8"
      Node 15: FFTAXOWE <= 27769.500
        Node 30: INTRDVX-mean = 2646.5367
      Node 15: FFTAXOWE > 27769.500 or NA
        Node 31: AGE2 <= 56.500000 or NA
          Node 62: INTRDVX-mean = 8036.3341
        Node 31: AGE2 > 56.500000
          Node 63: BATHRMQ <= 4.5000000
            Node 126: STATE = "6", "17", "21", "24", "27", "29", "32", "37", "42"
              Node 252: INTRDVX-mean = 41453.995
            Node 126: STATE /= "6", "17", "21", "24", "27", "29", "32", "37", "42"
              Node 253: INTRDVX-mean = 5032.1141
          Node 63: BATHRMQ > 4.5000000 or NA
            Node 127: INTRDVX-mean = 124323.81

```

Predictor means below are weighted means of cases with no missing values.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if REFGEN = "5"

REFGEN mode = "3"

Coefficients of weighted least squares regression function and weighted means:

Regressor	Coefficient	t-stat	p-value
-----------	-------------	--------	---------

Constant	5131.	12.43	0.3014-313
----------	-------	-------	------------

Node 2: Terminal node

Coefficients of weighted least squares regression function and weighted means:

Regressor	Coefficient	t-stat	p-value
Constant	980.4	3.232	0.3014-313

INTRDVX mean = 980.353

Node 3: Intermediate node

A case goes into Node 6 if INC_RANK <= 0.84018625

INC_RANK mean = 0.64479638

Node 6: Terminal node

Coefficients of weighted least squares regression function and weighted means:

Regressor	Coefficient	t-stat	p-value
Constant	2823.	11.12	0.3014-313

INTRDVX mean = 2822.64

Node 7: Intermediate node

A case goes into Node 14 if EARNCOMP = "8"

EARNCOMP mode = "2"

Node 14: Intermediate node

A case goes into Node 28 if RETSURV = "1"

RETSURV mode = "1"

Node 28: Terminal node

Coefficients of weighted least squares regression function and weighted means:

Regressor	Coefficient	t-stat	p-value
Constant	0.2764E+05	4.116	0.3014-313

INTRDVX mean = 27641.3

Node 29: Terminal node

Coefficients of weighted least squares regression function and weighted means:

Regressor	Coefficient	t-stat	p-value
Constant	0.8586E+05	7.878	0.3014-313

INTRDVX mean = 85859.3

Node 15: Intermediate node

A case goes into Node 30 if FFTAXOWE <= 27769.500

FFTAXOWE mean = 36491.278

Node 30: Terminal node

Coefficients of weighted least squares regression function and weighted means:

Regressor	Coefficient	t-stat	p-value
Constant	2647.	3.066	0.3014-313

```

INTRDVX mean = 2646.54
-----
Node 31: Intermediate node
A case goes into Node 62 if AGE2 <= 56.500000 or NA
AGE2 mean = 54.155118
-----
Node 62: Terminal node
Coefficients of weighted least squares regression function and weighted means:
Regressor    Coefficient  t-stat    p-value
Constant      8036.      3.982     0.3014-313
INTRDVX mean = 8036.33
-----
Node 63: Intermediate node
A case goes into Node 126 if BATHRMQ <= 4.5000000
BATHRMQ mean = 2.8097992
-----
Node 126: Intermediate node
A case goes into Node 252 if STATE = "6", "17", "21", "24", "27", "29", "32", "37", "42"
STATE mode = "42"
-----
Node 252: Terminal node
Coefficients of weighted least squares regression function and weighted means:
Regressor    Coefficient  t-stat    p-value
Constant      0.4145E+05   5.568     0.3014-313
INTRDVX mean = 41454.0
-----
Node 253: Terminal node
Coefficients of weighted least squares regression function and weighted means:
Regressor    Coefficient  t-stat    p-value
Constant      5032.       4.335     0.3014-313
INTRDVX mean = 5032.11
-----
Node 127: Terminal node
Coefficients of weighted least squares regression function and weighted means:
Regressor    Coefficient  t-stat    p-value
Constant      0.1243E+06  14.02     0.3014-313
INTRDVX mean = 124324.
-----
Proportion of variance (R-squared) explained by tree model: 0.4174

Observed and fitted values are stored in cons.fit
LaTeX code for tree is in cons.tex
R code is stored in cons.r

```

In the above results, the pruned tree is marked with two asterisks (tree #36).

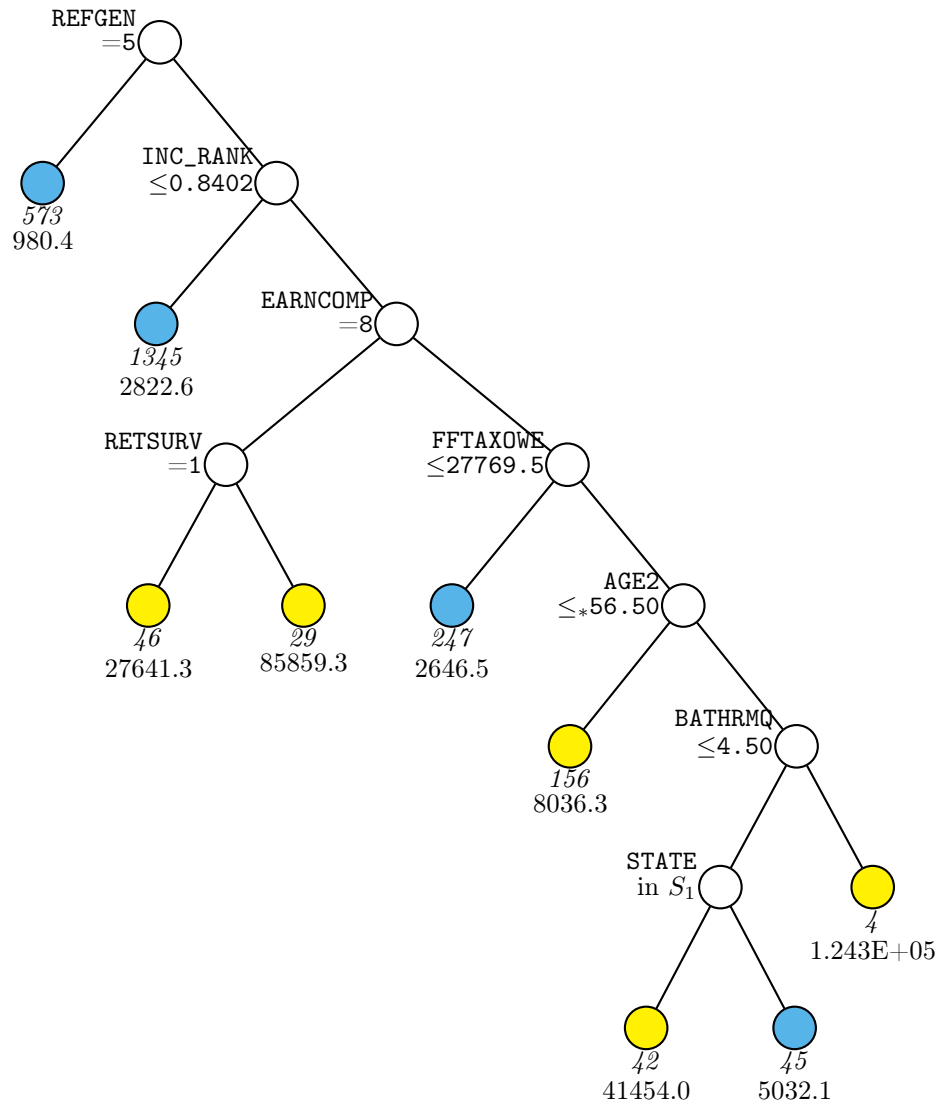


Figure 8: GUIDE v.46.1 0.250-SE piecewise-constant weighted least-squares regression tree for predicting INTRDVX. At each split, an observation goes to the left branch if and only if the condition is satisfied. Symbol ' $\leq_*$ ' stands for ' $\leq$ or missing'. $S_1 = \{6, 17, 21, 24, 27, 29, 32, 37, 42\}$. Sample size (in *italics*) and weighted mean of INTRDVX printed below nodes. Terminal nodes with means above and below value of 5130.6 at root node are painted yellow and skyblue respectively. Second best split variable at root node is AGE_REF.

It has 9 terminal nodes and a cross-validation estimate of prediction mean squared error of 5.915E+12. Figure 8 shows the tree. The first split is on `REFGEN=5`, meaning that millennials go to node 2, which is terminal. The first 7 entries of `cons.fit` below show that the 1st observation, for which `INTRDVX` is missing (the letter “n” in the first column indicates that it is not used to train the model), belongs to node 6 and has a predicted value of \$2822.64.

train	node	observed	predicted
n	6	NA	2822.64
y	6	1087.00	2822.64
y	6	1000.00	2822.64
y	6	300.000	2822.64
y	2	10.0000	980.353
y	252	141304.	41454.0
y	6	55.0000	2822.64

6.1.3 Population mean estimation

Predicted values from the regression tree may be used to estimate population means. Let w_i denote the sampling weight (`FINLWT21`) and S_1 , S_2 denote the sets of observations nonmissing and missing y_i (`INTRDVX`). Then an estimate of the population mean of `INTRDVX` is

$$\left(\sum_{k \in S_1 \cup S_2} w_k \right)^{-1} \left(\sum_{i \in S_1} w_i y_i + \sum_{j \in S_2} w_j \hat{y}_j \right) \quad (1)$$

where $\hat{y}_j$ denotes the predicted value of `INTRDVX`. The R code below gives an estimated population mean `INTRDVX` of 4750.376. See [Loh et al. \(2019b\)](#) for a similar analysis of an earlier data set.

```
1 data <- read.table("ce2021.txt",header=TRUE)
2 y <- data$INTRDVX
3 w <- data$FINLWT21
4 fitted <- read.table("cons.fit",header=TRUE)
5 pred <- fitted$predicted
6 S1 <- !is.na(fitted$observed)
7 S2 <- is.na(fitted$observed)
8 popmean <- (sum(w[S1]*y[S1])+sum(w[S2]*pred[S2]))/sum(w)
```

6.2 Piecewise simple polynomial

GUIDE can also fit a simple polynomial regression model in each node of the form

$$y = \beta_0 + \sum_{k=1}^p \beta_k x^k + \epsilon \quad (2)$$

where p is the degree of polynomial desired and x is selected from the set of **n** and **f** variables. The variable x is the one among all **n** and **f** variables that yields the smallest sum of squared residuals. Variable x can vary from node to node. If there are missing values in the x variable, GUIDE fits two separate models to the data in the node: model (2) to the observations with complete values in x and y and a constant ($y = \beta_0 + \epsilon$) to those with missing values in x . This is equivalent to imputing missing x values with a constant c and adding the missing-value indicator $I(x = \text{NA})$ as linear predictor:

$$y = \beta_0 + \sum_{k=1}^p \beta_k x_1^k + \beta_2 x_2 + \epsilon$$

where $x_1 = xI(x \neq \text{NA}) + cI(x = \text{NA})$ and $x_2 = I(x = \text{NA})$. The predicted values are independent of c but the least-squares estimates of the β coefficients are not.

Truncation note: Extrapolation can adversely affect the prediction accuracy of parametric models. To guard against extrapolation, GUIDE has several options to truncate the predicted values, with the default being to truncate the predicted values if they fall outside the range of the observed values. The option of no truncation is available as well. Default truncation is used in this manual.

6.2.1 Input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: lin.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: lin.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
```

```

Input choice ([1:7], <cr>=1):
Input 1 for least squares, 2 least median of squares ([1:2], <cr>=1):
Choose complexity of model to use at each node:
Choose 0 for stepwise linear regression (recommended for prediction)
Choose 1 for multiple regression
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
0: stepwise linear, 1: multiple linear, 2: best simple polynomial, 3: constant,
4: best simple stepwise ANCOVA ([0:4], <cr>=3): 2
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: ce2021reg.dsc
Reading DSC file ...
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
D variable is INTRDVX
Reading data file ...
Number of records in data file: 3965
Length of longest entry in data file: 11
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Finding number of levels of M variables associated with C variables ...
Assigning integer codes to values of 48 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Finished assigning codes to 40 categorical variables
Associating missing values of N and S variables with M variable codes ...
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Smallest positive weight: 1.0725E+03
Largest positive weight: 9.3902E+04
Numbers of N, F, S, and P below exclude constant variables
  Total  #cases w/  #missing
#cases   miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
  3965      1478      3965      1      383      0      0

```

```

      #P-var  #M-var  #B-var  #C-var  #I-var
        0      116      0      48      0
Weight variable FINLWT21 in column: 31
Number of cases used for training: 2487
Number of cases excluded due to 0 W or missing D variable: 1478
Finished reading data file
Constant fitted to cases with missing values in regressor variables
Input 1 for unweighted, 2 for weighted error estimates during pruning ([1:2], <cr>=2):
Warning: No interaction and linear splits; too many predictor variables
Default max. number of split levels: 12
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): lin.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save regressor names in a file, 1 otherwise ([1:2], <cr>=2):
Input file name: lin.var
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: lin.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: lin.r
Input rank of top variable to split root node ([1:431], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < lin.in

```

6.2.2 Partial output

```

Size and CV MSE and SE of subtrees:

```

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	36	6.941E+12	8.475E+11	9.177E+11	5.842E+12	1.280E+12
2	35	6.941E+12	8.475E+11	9.177E+11	5.842E+12	1.280E+12
:						
12	22	6.861E+12	8.491E+11	1.085E+12	5.115E+12	1.436E+12
13+	21	6.861E+12	8.491E+11	1.085E+12	5.115E+12	1.436E+12
14	18	6.874E+12	8.491E+11	1.084E+12	5.206E+12	1.434E+12
15	17	6.833E+12	8.484E+11	1.086E+12	5.166E+12	1.452E+12
16	16	6.714E+12	8.414E+11	1.061E+12	5.195E+12	1.303E+12
17	10	6.597E+12	8.385E+11	1.018E+12	5.157E+12	1.347E+12
18	9	6.432E+12	8.279E+11	1.060E+12	5.154E+12	1.439E+12
19**	8	6.144E+12	7.700E+11	8.069E+11	5.239E+12	1.583E+12

20	7	6.430E+12	8.188E+11	7.088E+11	5.868E+12	1.326E+12
21	5	7.158E+12	8.931E+11	7.326E+11	7.966E+12	1.153E+12
22	4	7.397E+12	9.042E+11	7.487E+11	8.079E+12	1.044E+12
23	1	8.303E+12	1.011E+12	7.002E+11	8.514E+12	8.534E+11

0-SE tree based on mean is marked with * and has 8 terminal nodes

0-SE tree based on median is marked with + and has 21 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

** tree same as ++ tree

** tree same as -- tree

++ tree same as -- tree

* tree same as ** tree

* tree same as ++ tree

* tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is weighted mean of INTRDVX in the node

Cases fit give the number of cases used to fit node

MSE and R² are based on all cases in node

Node label	Total cases	Cases fit	Matrix rank	Node D-mean	Node MSE	Node R ²	Split variable	Other variables
1	2487	209	2	5.131E+03	7.702E+12	0.0710	CUTENURE	+STOCKYRX
2	855	855	2	8.856E+03	1.219E+13	0.1292	FJSSDEDX	+FINCBTAX
4	578	578	2	9.919E+03	1.137E+13	0.2914	FINCBTAX	+FINCBTAX
8T	500	500	2	3.863E+03	1.323E+12	0.3043	INC_RANK	+ALCBEVCQ
9	78	47	2	5.014E+04	4.793E+13	0.2919	RETSURV	-RETSURVX
18T	47	47	2	2.613E+04	2.689E+13	0.2996	- +FULOILCQ	
19T	31	7	2	8.229E+04	3.579E+13	0.5077	- -ROYESTX	
5T	277	277	2	6.780E+03	4.930E+12	0.4882	PERINSCQ	+ETOTALC
3	1632	1087	2	3.221E+03	4.540E+12	0.1032	RENTEQVX	+RENTEQVX
6T	1520	1520	2	1.846E+03	1.564E+12	0.0959	PSU +VELECTRC	
7	112	112	2	2.163E+04	3.560E+13	0.1590	FOODCQ	+JFS_AMT
14	58	58	2	2.970E+04	2.765E+13	0.5220	FINCBTAX	+PERINSCQ
28T	26	26	2	1.943E+04	7.830E+12	0.8283	- +APPARCQ	
29T	32	32	2	3.722E+04	1.910E+13	0.7186	- +HEALTHCQ	
15T	54	54	2	1.262E+04	1.537E+13	0.3474	HLTHINPQ	+JFS_AMT

Number of terminal nodes of final tree: 8

Total number of nodes of final tree: 15

Second best split variable (based on curvature test) at root node is REFGEN

Regression tree:

For categorical variable splits, values not in training data go to the right

```

Node 1: CUTENURE = "2", "6"
  Node 2: FJSSDEDX <= 2720.0000
    Node 4: FINCBTAX <= 114750.50
      Node 8: INTRDVX-mean = 3863.4422
    Node 4: FINCBTAX > 114750.50 or NA
      Node 9: RETSURV = "1"
        Node 18: INTRDVX-mean = 26127.783
      Node 9: RETSURV /= "1"
        Node 19: INTRDVX-mean = 82288.430
  Node 2: FJSSDEDX > 2720.0000 or NA
    Node 5: INTRDVX-mean = 6780.2396
Node 1: CUTENURE /= "2", "6"
  Node 3: RENTEQVX <= 3625.0000 or NA
    Node 6: INTRDVX-mean = 1845.8461
  Node 3: RENTEQVX > 3625.0000
    Node 7: FOODCQ <= 1029.1667
      Node 14: FINCBTAX <= 244530.00
        Node 28: INTRDVX-mean = 19427.633
      Node 14: FINCBTAX > 244530.00 or NA
        Node 29: INTRDVX-mean = 37216.426
    Node 7: FOODCQ > 1029.1667 or NA
      Node 15: INTRDVX-mean = 12623.057

```

Predictor means below are weighted means of cases with no missing values.

Regression coefficients are computed from the complete cases.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if CUTENURE = "2", "6"

CUTENURE mode = "1"

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	3167.	2.042	0.4246E-01			

```

STOCKYRX      0.1749E-01   11.42       0.000       0.000       0.3617E+06   0.5450E+07
If regressor has missing values, predicted value = 4720.7960
Predicted values truncated at 1.00000 & 141304.
-----

```

```

Node 2: Intermediate node
A case goes into Node 4 if FJSSDEDX <= 2720.0000
FJSSDEDX mean = 3273.8110
-----

```

```

Node 4: Intermediate node
A case goes into Node 8 if FINCBTAX <= 114750.50
FINCBTAX mean = 60208.956
-----

```

```

Node 8: Terminal node
Coefficients of weighted least squares regression function and weighted means:
Regressor    Coefficient  t-stat    p-value    Minimum    Mean    Maximum
Constant      2892.        8.287    0.9637E-15
ALCBVCQ       19.81       14.76    0.1403E-15    0.000    49.04    4670.
If regressor has missing values, predicted value = 3863.4422
Predicted values truncated at 1.00000 & 141304.
-----

```

```

Node 9: Intermediate node
A case goes into Node 18 if RETSURV = "1"
RETSURV mode = "1"
-----

```

```

:
Node 29: Terminal node
Coefficients of weighted least squares regression function and weighted means:
Regressor    Coefficient  t-stat    p-value    Minimum    Mean    Maximum
Constant      0.1557E+05   2.856    0.7707E-02
HEALTHCQ      83.68       8.753    0.9246E-09    0.000    258.7    1470.
If regressor has missing values, predicted value = 37216.426
Predicted values truncated at 1.00000 & 141304.
-----

```

```

Node 15: Terminal node
Coefficients of weighted least squares regression function and weighted means:
Regressor    Coefficient  t-stat    p-value    Minimum    Mean    Maximum
Constant      9035.        2.534    0.1432E-01
JFS_AMT       93.41       5.262    0.2738E-05    0.000    38.41    1050.
If regressor has missing values, predicted value = 12623.057
Predicted values truncated at 1.00000 & 141304.
-----

```

```

Proportion of variance (R-squared) explained by tree model: 0.6047

```

```

Observed and fitted values are stored in lin.fit
Regressor names and coefficients are stored in lin.var
LaTeX code for tree is in lin.tex

```

Figure 9 shows the pruned tree, which has 8 terminal nodes and a cross-validation estimate of prediction mean squared error of 6.142E+12. Below each terminal node are printed the sample size (in *italics*), the sample mean of INTRDVX and the signed simple linear predictor, with the sign being that of the slope coefficient. Nodes with mean of the *d* variable above and below the mean at the root node are colored yellow and purple, respectively.

6.2.3 Plots of data

Figure 10 shows plots of the data and fitted regression lines in the terminal nodes of the tree. The plots are drawn using the R code in Figure 11, which reads the files `lin.fit` and `lin.var`. The contents of the latter are below. The first row is a header line. Each subsequent row gives the terminal node number, predictor variable name, intercept and slope of the regression line, and lower and upper truncation limits on the predicted values (the defaults are the global minimum and maximum observed values of the dependent variable).

node	variable	beta0	beta1	lower	upper
8	ALCBEVCQ	2892.	19.81	1.000	0.1413E+6
18	FULOILCQ	0.2096E+5	181.5	1.000	0.1413E+6
19	ROYESTX	0.1215E+6	-0.5119	1.000	0.1413E+6
5	ETOTALC	-953.9	0.7695	1.000	0.1413E+6
6	VELECTRC	1875.	166.1	1.000	0.1413E+6
14	DMSXCCPQ	0.1346E+5	112.6	1.000	0.1413E+6
15	ECARTKUC	7371.	302.2	1.000	0.1413E+6

6.3 Stepwise linear

Besides piecewise constant and best simple polynomial, GUIDE can fit a multiple linear (where all *n* and *f* variables are used as regressors) or a stepwise linear (where forward and backward selection is used to select a subset of regressors) regression model at each node. Quite often, these models have higher prediction accuracy, as hinted by the cross-validation estimates of MSE in the output.

For stepwise regression, missing values in each *x* variable are imputed with the weighted mean of *x* in the node and a stepwise linear regression model is fitted to the *y* variable, using the imputed *x* variables and their missing-value indicators. The name of the indicator of *x* is denoted by “*x*.NA”, where $x.NA = I(x = NA)$.

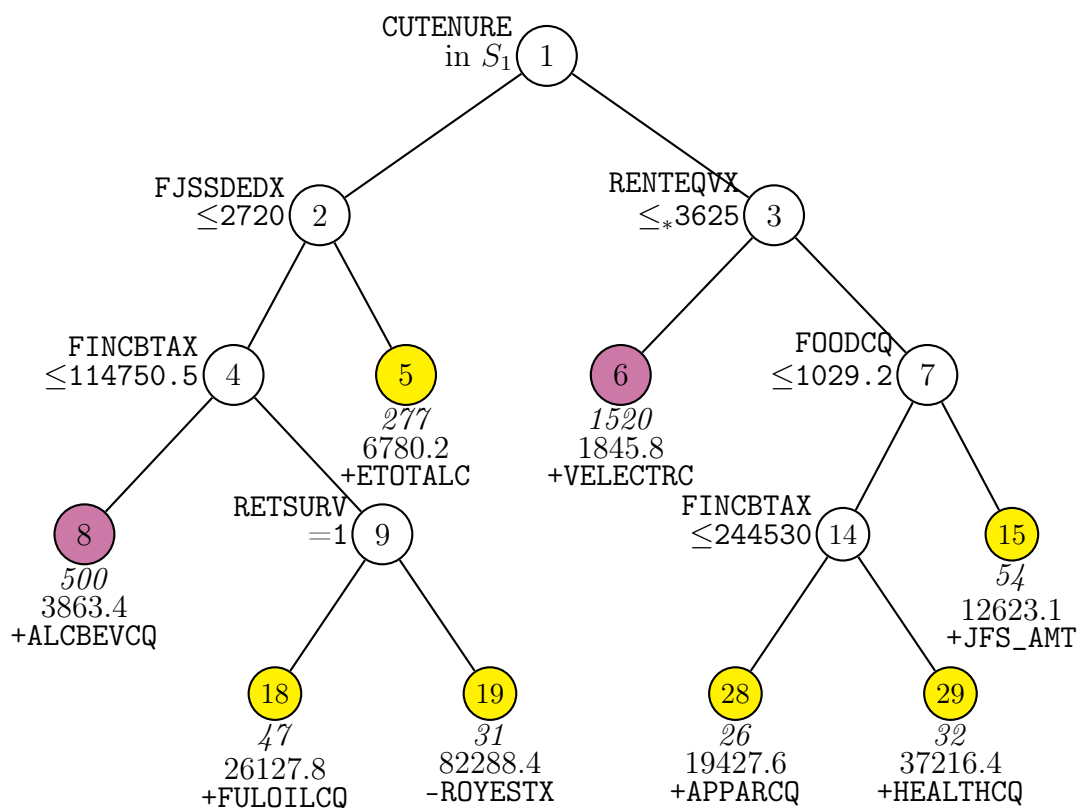


Figure 9: GUIDE v.46.1 0.250-SE piecewise simple linear weighted least-squares regression tree (constant fitted to incomplete cases in terminal nodes) for predicting INTRDVX. At each split, an observation goes to the left branch if and only if the condition is satisfied. Symbol ' $\leq^*$ ' stands for ' $\leq$ or missing'. $S_1 = \{2, 6\}$. Sample size (in *italics*), weighted mean of INTRDVX, and name of regressor (with sign of slope) printed below nodes. Terminal nodes with means above and below value of 5130.6 at root node are painted **yellow** and **purple** respectively. Second best split variable at root node is REFGEN.

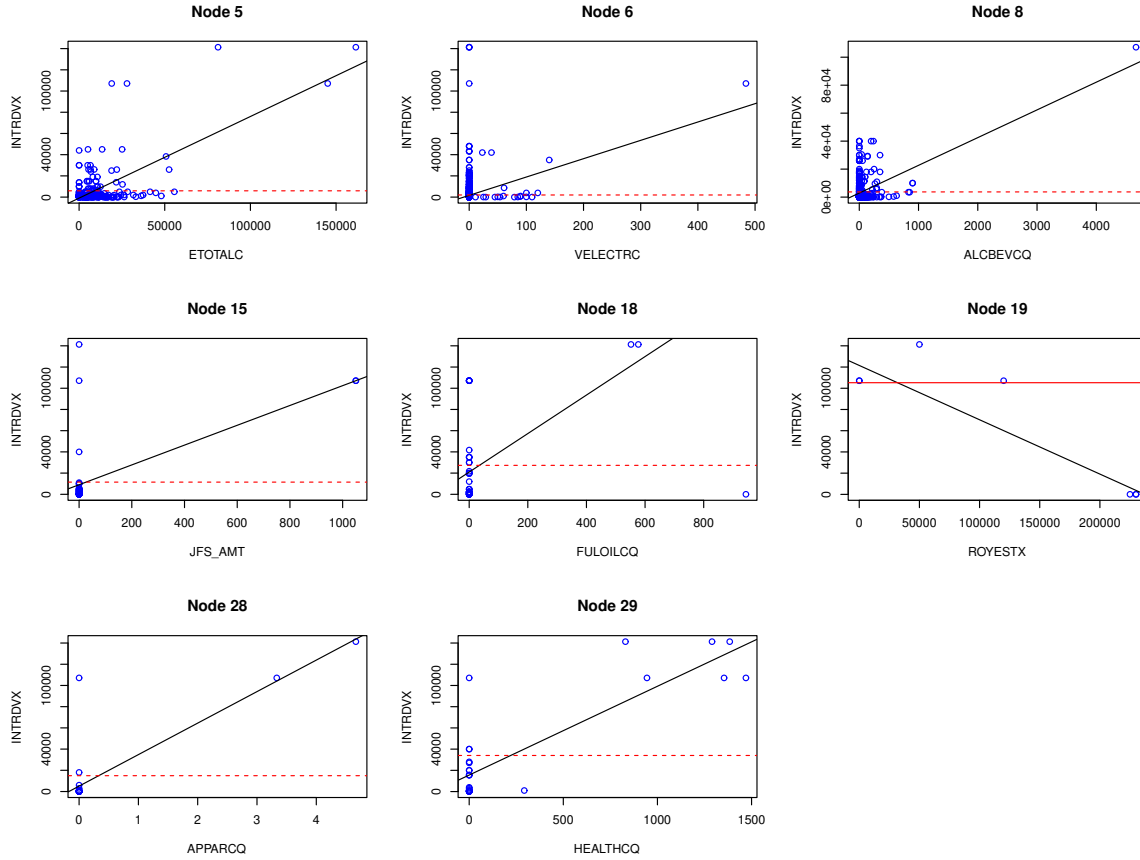


Figure 10: Data and regression lines in terminal nodes of tree in Figure 9. If there are missing values in the regressor, a solid red line marks their $\bar{d}$ mean. If there are no missing values, a dashed red line marks the $\bar{d}$ mean of all points in the node.

```
1 z <- read.table("ce2021.txt",header=TRUE)
2 par(mfrow=c(3,3))
3 z1 <- read.table("lin.fit",header=TRUE)
4 z2 <- read.table("lin.var",header=TRUE)
5 nodes <- unique(sort(z1$node))
6 y <- z$INTRDVX
7 for(n in nodes){
8     gp <- z1$node == n & z1$train == "y"
9     vrow <- z2$node == n
10    b0 <- z2$beta0[vrow]
11    b1 <- z2$beta1[vrow]
12    reg <- z2$variable[vrow]
13    k <- which(names(z) %in% reg)
14    x <- z[,k]
15    plot(y[gp] ~ x[gp],xlab=reg,ylab="INTRDVX",col="blue")
16    abline(c(b0,b1))
17    nomiss <- z1$node == n & z1$train == "y" & !is.na(x)
18    if(sum(nomiss) < sum(gp)){
19        miss <- z1$node == n & z1$train == "y" & is.na(x)
20        abline(h=mean(y[miss]),col="red",lty=1)
21    } else {
22        abline(h=mean(y[gp]),col="red",lty=2)
23    }
24    title(paste("Node",n))
25 }
```

Figure 11: R code for Figure 10

6.3.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: step.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: step.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1):
Input 1 for least squares, 2 least median of squares ([1:2], <cr>=1):
Choose complexity of model to use at each node:
Choose 0 for stepwise linear regression (recommended for prediction)
Choose 1 for multiple regression
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
0: stepwise linear, 1: multiple linear, 2: best simple polynomial, 3: constant,
4: best simple stepwise ANCOVA ([0:4], <cr>=3): 0
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input 1 for univariate splits, 2 for univariate+linear splits ([1:2], <cr>=2):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: ce2021reg.dsc
Reading DSC file ...
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
D variable is INTRDVX
M variables changed to B
Reading data file ...
Number of records in data file: 3965
Length of longest entry in data file: 11
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 164 categorical variables
Finished assigning codes to 10 categorical variables

```

```

Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Finished assigning codes to 40 categorical variables
Finished assigning codes to 50 categorical variables
Finished assigning codes to 60 categorical variables
Finished assigning codes to 70 categorical variables
Finished assigning codes to 80 categorical variables
Finished assigning codes to 90 categorical variables
Finished assigning codes to 100 categorical variables
Finished assigning codes to 110 categorical variables
Finished assigning codes to 120 categorical variables
Finished assigning codes to 130 categorical variables
Finished assigning codes to 140 categorical variables
Finished assigning codes to 150 categorical variables
Finished assigning codes to 160 categorical variables
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...
Creating dummy variables ...
Creating missing value indicators ...
Rereading data ...
Smallest positive weight:  1.0725E+03
Largest positive weight:   9.3902E+04
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      3965      1478      3965      1      383      309      0
      #P-var  #M-var  #B-var  #C-var  #I-var
      0        0      116      48      0
Number of dummy variables (excl. missing-value indicators) created: 224
Number of missing-value indicators created: 85
Weight variable FINLWT21 in column: 31
Number of cases used for training: 2487
Number of cases excluded due to 0 W or missing D variable: 1478
Finished reading data file
Input 1 for unweighted, 2 for weighted error estimates during pruning ([1:2], <cr>=2):
Warning: No interaction and linear splits; too many predictor variables
Default max. number of split levels: 8
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): step.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values

```

```

Input your choice ([1:3], <cr>=1):
Input 2 to save regressor names in a file, 1 otherwise ([1:2], <cr>=2):
Input file name: step.reg
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: step.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: step.r
Input rank of top variable to split root node ([1:856], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < step.in

```

6.3.2 Results

The tree has no splits because the fitted regression model in the root node has a R^2 value very close to 1.0. The regression coefficients from `step.out` are below.

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.2316E+05	876.2	0.3625E-16			
FINCBTAX	1.000	0.3081E+05	0.6113E-16	-0.1050E+06	0.1160E+06	0.1039E+07
FJSSDEDX	-0.1466E-03	-0.9102	0.3628	0.000	6354.	0.4366E+05
FRRETIRX	-1.000	-0.2400E+05	0.1970E-15	0.000	0.1071E+05	0.1116E+06
FSALARYX	-1.000	-0.2841E+05	0.8374E-16	0.000	0.8172E+05	0.7645E+06
FSSIX	-1.000	-3698.	0.1310E-15	0.000	83.02	0.3600E+05
OTHRINCX	-1.000	-6139.	0.6918E-16	150.0	0.2064E+05	0.1041E+06
PERSLT18	-0.3120	-0.7659	0.4438	0.000	0.3957	7.000
DMSXCCPQ	0.7173E-03	1.021	0.3076	-2408.	144.6	0.1527E+05
CASHCOPQ	-0.6518E-05	-0.7545E-01	0.9399	0.000	802.6	0.1200E+06
EENTRMTC	-0.7989E-04	-0.2645	0.7914	0.000	376.7	0.3022E+05
FSMPFRMX	-1.000	-0.2671E+05	0.5637E-16	-0.1160E+06	6641.	0.7703E+06
JFS_AMT	-0.9994	-1843.	0.3068E-15	0.000	64.43	9600.
NETRENTX	-1.000	-0.1885E+05	0.1083E-15	-0.1402E+05	0.1462E+05	0.1589E+06
NETRNTBX	-1.000	-3706.	0.1756E-15	-2400.	9325.	0.7130E+05
OTHREGBX	-1.000	-2468.	0.000	488.0	6380.	0.4200E+05
OTHREGX	-0.9998	-9576.	0.000	100.0	0.1281E+05	0.8288E+05
RETSURVX	-1.000	-0.2268E+05	0.4449E-16	134.0	0.2762E+05	0.1739E+06
RETSRVBX	-1.000	-5520.	0.000	3500.	0.2850E+05	0.6200E+05
ROYESTBX	-1.001	-244.1	0.2051E-15	200.0	7464.	0.6000E+05
ROYESTB	5.369	0.2390	0.8112	1.000	2.461	12.00
ROYESTX	-1.000	-0.1609E+05	0.000	5.000	0.4599E+05	0.2300E+06
FINCBT_X.T	0.4732	0.1146	0.9088	0.000	0.1044	1.000
FSAL_RYX.T	0.8994	0.1913	0.8483	0.000	0.4732E-01	1.000

```

OTHR_NCX.T      -0.5737      -0.3731E-01    0.9702      0.000      0.3928E-02    1.000
WELF_REX.C      -0.1096E+05    -323.0        0.000      0.000      0.9226E-04    1.000
WELF_REX.D      -915.6        -131.3        0.2932E-15  0.000      0.2086E-02    1.000
FSMP_RMX.T      -2.382        -0.4181       0.6759      0.000      0.1027E-01    1.000
NETR_NTB.D      -9325.        -1538.        0.2245E-15  0.000      0.2754E-02    1.000
NETR_NTX.T      -0.8048       -0.1449       0.8848      0.000      0.6320E-02    1.000
OTHR_GBX.D      -6380.        -1647.        0.9177E-16  0.000      0.6767E-02    1.000
OTHREGX_.T      -15.27        -1.556        0.1197      0.000      0.2762E-02    1.000
RETS_RVX.T      -4.974        -0.8233       0.4104      0.000      0.1413E-01    1.000
ROYE_TBX.D      -7463.        -1150.        0.000      0.000      0.2422E-02    1.000
ROYESTX_.T      -0.8850       -0.7034E-01   0.9439      0.000      0.4268E-02    1.000
OTHRINCX.NA     0.2064E+05    7103.        0.4358E-16  0.000      0.9715        1.000
NETRENTX.NA     0.1462E+05    0.1090E+05   0.9388E-16  0.000      0.9211        1.000
OTHREGX.NA     0.1281E+05    0.1070E+05   0.1820E-15  0.000      0.9028        1.000
RETSURVX.NA     0.2762E+05    0.2160E+05   0.000      0.000      0.7632        1.000
RETSRVBX.NA     0.2850E+05    5780.        0.1534E-15  0.000      0.9949        1.000
ROYESTX.NA     0.4599E+05    0.1798E+05   0.000      0.000      0.9594        1.000
INTRDVX mean = 5130.60
Predicted values truncated at 1.00000 & 141304.
-----
Proportion of variance (R-squared) explained by tree model: 1.0000

Observed and fitted values are stored in step.fit
Regressor names and coefficients are stored in step.reg
LaTeX code for tree is in step.tex

```

The names of the selected predictor variables can also be read from the output file `step.reg`, which gives, on each line, the terminal node number, lower and upper truncation values, and the names of the variables selected by the stepwise regression model in the node.

7 Quantile regression: CE data

GUIDE can build piecewise-constant and piecewise-linear quantile regression models. First we show how to build a piecewise-constant 0.90-quantile regression model.

7.1 Piecewise constant: one quantile

7.1.1 Input file creation

0. Read the warranty disclaimer
1. Create a GUIDE input file

```

Input your choice: 1
Name of batch input file: quantcon.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: quantcon.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 2
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple polynomial, 3: constant ([1:3], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input 1 for 1 quantile, 2 for 2 quantiles ([1:2], <cr>=1):
Input quantile probability ([0.00:1.00], <cr>=0.50): 0.90
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: ce2021reg.dsc
Reading DSC file ...
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX
Reading data file ...
Number of records in data file: 3965
Length of longest entry in data file: 11
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Finding number of levels of M variables associated with C variables ...
Assigning integer codes to values of 48 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Finished assigning codes to 40 categorical variables
Associating missing values of N and S variables with M variable codes ...

```



```

Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Smallest positive weight:  1.0725E+03
Largest positive weight:  9.3902E+04
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      3965    1478    3965      1      0      0      383
      #P-var  #M-var  #B-var  #C-var  #I-var
      0       116     0       48     0
Number of cases used for training: 2487
Number of cases excluded due to 0 W or missing D variable: 1478
Finished reading data file
Input 1 for unweighted, 2 for weighted error estimates during pruning ([1:2], <cr>=2):
Warning: No interaction and linear splits; too many predictor variables
Warning: All positive weights treated as 1
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): quantcon.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: quantcon.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: quantcon.r
Input rank of top variable to split root node ([1:431], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < quantcon.in

```

Contents of quantcon.out

```

Quantile regression tree with quantile probability 0.9000
Pruning by cross-validation
DSC file: ce2021reg.dsc
Training sample file: ce2021.txt

```

Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX
Piecewise constant model
Number of records in data file: 3965
Length of longest entry in data file: 11
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Smallest and largest positive weights are 1.0725E+03 and 9.3902E+04

Summary information for training sample of size 2487 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight
Levels of M variables are for missing values in associated variables

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	DIRACC	s	1.000	1.000		125 constant
2	DIRACC_	m			2	
3	AGE_REF	s	19.00	87.00		
4	AGE_REF_	m			0	
5	AGE2	s	21.00	87.00		1092
6	AGE2_	m			1	
:						
31	FINLWT21	w	1072.	0.9390E+05		
:						
406	INTRDVX	d	1.000	0.1413E+06		
:						
547	WHLFYR	c			1	2487
548	WHLFYR_	m			1	
549	FFTAX0WE	s	-0.3368E+05	0.3380E+06		
550	FSTAX0WE	s	-3074.	0.5654E+05		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss. D	ord. vals 3965	#X-var 1	#N-var 0	#F-var 0	#S-var 376
3965	1478						
#P-var	#M-var	#B-var	#C-var	#I-var			
0	116	0	48	0			

7.1 Piecewise constant: one quantile 7 QUANTILE REGRESSION: CE DATA

Number of constant S variables: 7
 Number of cases used for training: 2487
 Number of cases excluded due to 0 W or missing D variable: 1478

Pruning by v-fold cross-validation, with v = 10
 Selected tree is based on mean of CV estimates
 Number of SE's for pruned tree: 0.2500

Weighted error estimates used for pruning
 Warning: No interaction and linear splits; too many predictor variables
 Warning: All positive weights treated as 1
 No nodewise interaction tests
 Split values for N and S variables based on exhaustive search
 Max number of interaction splits: 100
 Maximum number of split levels: 15
 Minimum node sample size: 2

Ranks of variables and their 1-df chi-squared values at root node

1	0.6943E+02	CUTENURE
2	0.6324E+02	REFGEN
3	0.5982E+02	RENTEQVX
4	0.5957E+02	AGE_REF
5	0.5754E+02	AGE2
6	0.5689E+02	STOCKX
7	0.5598E+02	INCNONW1
8	0.5547E+02	NO_EARNR
:		
413	0.1064E-02	ECARTKNP
414	0.4983E-03	FULOILPQ
415	0.3478E-03	TOTHFARP
416	0.4489E-04	MAJAPPPQ
417	0.3034E-04	OTHLNYR

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	48	5.948E+07	4.764E+06	4.889E+06	5.743E+07	7.105E+06
2	47	5.948E+07	4.764E+06	4.889E+06	5.743E+07	7.105E+06
3	46	5.946E+07	4.763E+06	4.878E+06	5.743E+07	7.105E+06
4	45	5.946E+07	4.763E+06	4.879E+06	5.743E+07	7.112E+06
5	42	5.946E+07	4.763E+06	4.879E+06	5.743E+07	7.112E+06
6	41	5.941E+07	4.761E+06	4.867E+06	5.749E+07	7.120E+06
7*	39	5.941E+07	4.761E+06	4.871E+06	5.749E+07	7.122E+06
8	37	5.953E+07	4.761E+06	4.886E+06	5.772E+07	7.145E+06
:						
18	26	5.964E+07	4.775E+06	4.803E+06	5.790E+07	6.953E+06
19	25	5.984E+07	4.773E+06	4.802E+06	5.790E+07	6.960E+06
20**	24	6.021E+07	4.807E+06	4.653E+06	5.738E+07	6.891E+06

21	23	6.171E+07	4.997E+06	4.587E+06	5.720E+07	6.362E+06
22	21	6.171E+07	4.997E+06	4.587E+06	5.720E+07	6.362E+06
23	20	6.140E+07	4.984E+06	4.660E+06	5.720E+07	6.656E+06
24	17	6.087E+07	4.957E+06	4.440E+06	5.720E+07	5.580E+06
25	15	6.103E+07	4.964E+06	4.385E+06	5.717E+07	5.761E+06
26	14	6.078E+07	4.980E+06	4.519E+06	5.717E+07	5.861E+06
27	13	6.109E+07	5.026E+06	4.483E+06	5.769E+07	6.522E+06
28	12	6.155E+07	5.042E+06	4.334E+06	5.769E+07	6.263E+06
29	10	6.155E+07	5.042E+06	4.334E+06	5.769E+07	6.263E+06
30++	9	6.147E+07	5.029E+06	4.341E+06	5.708E+07	6.351E+06
31	7	6.580E+07	5.409E+06	5.027E+06	6.057E+07	8.897E+06
32	5	7.073E+07	5.668E+06	6.656E+06	6.916E+07	1.257E+07
33	4	7.094E+07	5.845E+06	6.872E+06	7.191E+07	1.356E+07
34	3	7.561E+07	5.755E+06	5.616E+06	6.904E+07	8.622E+06
35	1	9.107E+07	7.613E+06	5.017E+06	9.297E+07	5.472E+06

0-SE tree based on mean is marked with * and has 39 terminal nodes

0-SE tree based on median is marked with + and has 9 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

** tree same as -- tree

+ tree same as ++ tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-quant is quantile of INTRDVX in the node

Cases fit give the number of cases used to fit node

Node label	Total cases	Cases fit	Matrix rank	Node D-quant	Split variable	Other variables
1	2487	2487	1	9.800E+03	CUTENURE	
2	872	872	1	2.500E+04	INC_RANK	
4	664	664	1	1.200E+04	RNTAPYCQ	
8	660	660	1	1.000E+04	INC_RANK	
16T	333	333	1	5.361E+03	-	
17	327	327	1	1.800E+04	TTRANPRP	
34	249	249	1	1.124E+04	AGE2	
68	209	209	1	7.500E+03	TGASMOTC	
136T	207	207	1	7.139E+03	-	
137T	2	2	1	1.071E+05	-	
69T	40	40	1	2.700E+04	-	
35T	78	78	1	3.000E+04	-	
9T	4	4	1	4.800E+04	-	
5	208	208	1	1.071E+05	RETPENPQ	

10	64	64	1	1.413E+05	PROPTXCQ
20T	47	47	1	1.413E+05	PROPTXCQ
21T	17	17	1	1.071E+05	-
11	144	144	1	3.000E+04	EENTMSCC
22	121	121	1	1.500E+04	STATE
44T	17	17	1	4.400E+04	-
45T	104	104	1	5.600E+03	-
23T	23	23	1	1.071E+05	-
3	1615	1615	1	4.200E+03	FFTAXOWE
6	1374	1374	1	3.000E+03	REF_RACE
12T	28	28	1	1.500E+04	FLRCVRPQ
13	1346	1346	1	2.400E+03	STOCKYRX
26	1341	1341	1	2.400E+03	AGE_REF
52	990	990	1	1.000E+03	STATE
104T	388	388	1	3.000E+03	FSSIX
105T	602	602	1	4.500E+02	OCCUCOD2
53	351	351	1	8.161E+03	OTHHEXCQ
106T	176	176	1	4.200E+03	EDUC_REF
107	175	175	1	1.400E+04	OTHSTYRB
214T	173	173	1	1.400E+04	-
215T	2	2	1	1.071E+05	-
27T	5	5	1	1.413E+05	-
7	241	241	1	2.123E+04	REF_RACE
14T	4	4	1	1.413E+05	-
15	237	237	1	2.000E+04	LIQUDYRB
30	231	231	1	1.550E+04	AGE2
60T	187	187	1	6.000E+03	ROYESTX
61	44	44	1	1.071E+05	INCNONW2
122	38	38	1	4.200E+04	VNATLGAC
244T	35	35	1	2.355E+04	-
245T	3	3	1	1.071E+05	-
123T	6	6	1	1.071E+05	-
31T	6	6	1	1.071E+05	-

Warning: tree very large, omitting node numbers in LaTeX file

Number of terminal nodes of final tree: 24

Total number of nodes of final tree: 47

Second best split variable (based on curvature test) at root node is REFGEN

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: CUTENURE = "2", "5"

Node 2: INC_RANK <= 0.81810030

Node 4: RNTAPYCQ <= 1158.3333

Node 8: INC_RANK <= 0.49406705

```

Node 16: INTRDVX sample quantile = 5361.0000
Node 8: INC_RANK > 0.49406705 or NA
Node 17: TTRANPRP <= 20.500000
Node 34: AGE2 <= 76.500000 or NA
Node 68: TGASMOTC <= 660.50000
Node 136: INTRDVX sample quantile = 7139.0000
Node 68: TGASMOTC > 660.50000 or NA
Node 137: INTRDVX sample quantile = 107121.00
Node 34: AGE2 > 76.500000
Node 69: INTRDVX sample quantile = 27000.000
Node 17: TTRANPRP > 20.500000 or NA
Node 35: INTRDVX sample quantile = 30000.000
Node 4: RNTAPYCQ > 1158.3333 or NA
Node 9: INTRDVX sample quantile = 48000.000
Node 2: INC_RANK > 0.81810030 or NA
Node 5: RETPENPQ <= 90.250000
Node 10: PROPTXCQ <= 875.00000
Node 20: INTRDVX sample quantile = 141304.00
Node 10: PROPTXCQ > 875.00000 or NA
Node 21: INTRDVX sample quantile = 107121.00
Node 5: RETPENPQ > 90.250000 or NA
Node 11: EENTMSCC <= 66.666500
Node 22: STATE = "1", "17", "29", "37", "46"
Node 44: INTRDVX sample quantile = 44000.000
Node 22: STATE /= "1", "17", "29", "37", "46"
Node 45: INTRDVX sample quantile = 5600.0000
Node 11: EENTMSCC > 66.666500 or NA
Node 23: INTRDVX sample quantile = 107121.00
Node 1: CUTENURE /= "2", "5"
Node 3: FFTAXOWE <= 31377.000
Node 6: REF_RACE = "3"
Node 12: INTRDVX sample quantile = 15000.000
Node 6: REF_RACE /= "3"
Node 13: STOCKYRX <= 875000.00 or NA
Node 26: AGE_REF <= 64.500000
Node 52: STATE = "1", "2", "6", "8", "9", "10", "12", "15", "21",
"22", "36", "41", "45", "48", "53"
Node 104: INTRDVX sample quantile = 3000.0000
Node 52: STATE /= "1", "2", "6", "8", "9", "10", "12", "15", "21",
"22", "36", "41", "45", "48", "53"
Node 105: INTRDVX sample quantile = 450.00000
Node 26: AGE_REF > 64.500000 or NA
Node 53: OTHHEXCQ <= 16.666650
Node 106: INTRDVX sample quantile = 4200.0000
Node 53: OTHHEXCQ > 16.666650 or NA
Node 107: OTHSTYRB = NA & OTHS_YRB = "A"

```

7.1 Piecewise constant: one quantile 7 QUANTILE REGRESSION: CE DATA

```
Node 214: INTRDVX sample quantile = 14000.000
Node 107: not (OTHSTYRB = NA & OTHS_YRB = "A")
Node 215: INTRDVX sample quantile = 107121.00
Node 13: STOCKYRX > 875000.00
Node 27: INTRDVX sample quantile = 141304.00
Node 3: FFTAXOWE > 31377.000 or NA
Node 7: REF_RACE = "6"
Node 14: INTRDVX sample quantile = 141304.00
Node 7: REF_RACE /= "6"
Node 15: LIQUDYRB = NA & LIQU_YRB = "A"
Node 30: AGE2 <= 57.500000 or NA
Node 60: INTRDVX sample quantile = 6000.0000
Node 30: AGE2 > 57.500000
Node 61: INCNONW2 = "2"
      or (INCNONW2 = NA & INCN_NW2 = "A")
Node 122: VNATLGAC <= 11.000000
Node 244: INTRDVX sample quantile = 23554.000
Node 122: VNATLGAC > 11.000000 or NA
Node 245: INTRDVX sample quantile = 107121.00
Node 61: INCNONW2 /= "2"
      & not (INCNONW2 = NA & INCN_NW2 = "A")
Node 123: INTRDVX sample quantile = 107121.00
Node 15: not (LIQUDYRB = NA & LIQU_YRB = "A")
Node 31: INTRDVX sample quantile = 107121.00
```

Predictor means below are weighted means of cases with no missing values.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if CUTENURE = "2", "5"

CUTENURE mode = "1"

Predicted quantile = 9800.00

Node 2: Intermediate node

A case goes into Node 4 if INC_RANK <= 0.81810030

```

INC_RANK mean = 0.59625137
-----
Node 4: Intermediate node
A case goes into Node 8 if RNTAPYCQ <= 1158.3333
RNTAPYCQ mean = 11.380252
-----
:
Node 60: Terminal node
Predicted quantile = 6000.00
-----
Node 61: Intermediate node
A case goes into Node 122 if INCNONW2 = "2"
    or INCNONW2 = NA & INCN_NW2 = "A"
    INCN_NW2 mode = "A"
-----
Node 122: Intermediate node
A case goes into Node 244 if VNATLGAC <= 11.000000
VNATLGAC mean = 3.4919712
-----
Node 244: Terminal node
Predicted quantile = 23554.0
-----
Node 245: Terminal node
Predicted quantile = 107121.
-----
Node 123: Terminal node
Predicted quantile = 107121.
-----
Node 31: Terminal node
Predicted quantile = 107121.
-----
Observed and fitted values are stored in quantcon.fit
LaTeX code for tree is in quantcon.tex

```

Figure 12 shows the quantile regression tree. The sample size (in *italics*) and 0.90-quantile are given beneath each terminal node.

7.2 Best simple linear

We demonstrate this with a linear 0.90-quantile regression tree.

7.2.1 Input file creation

0. Read the warranty disclaimer

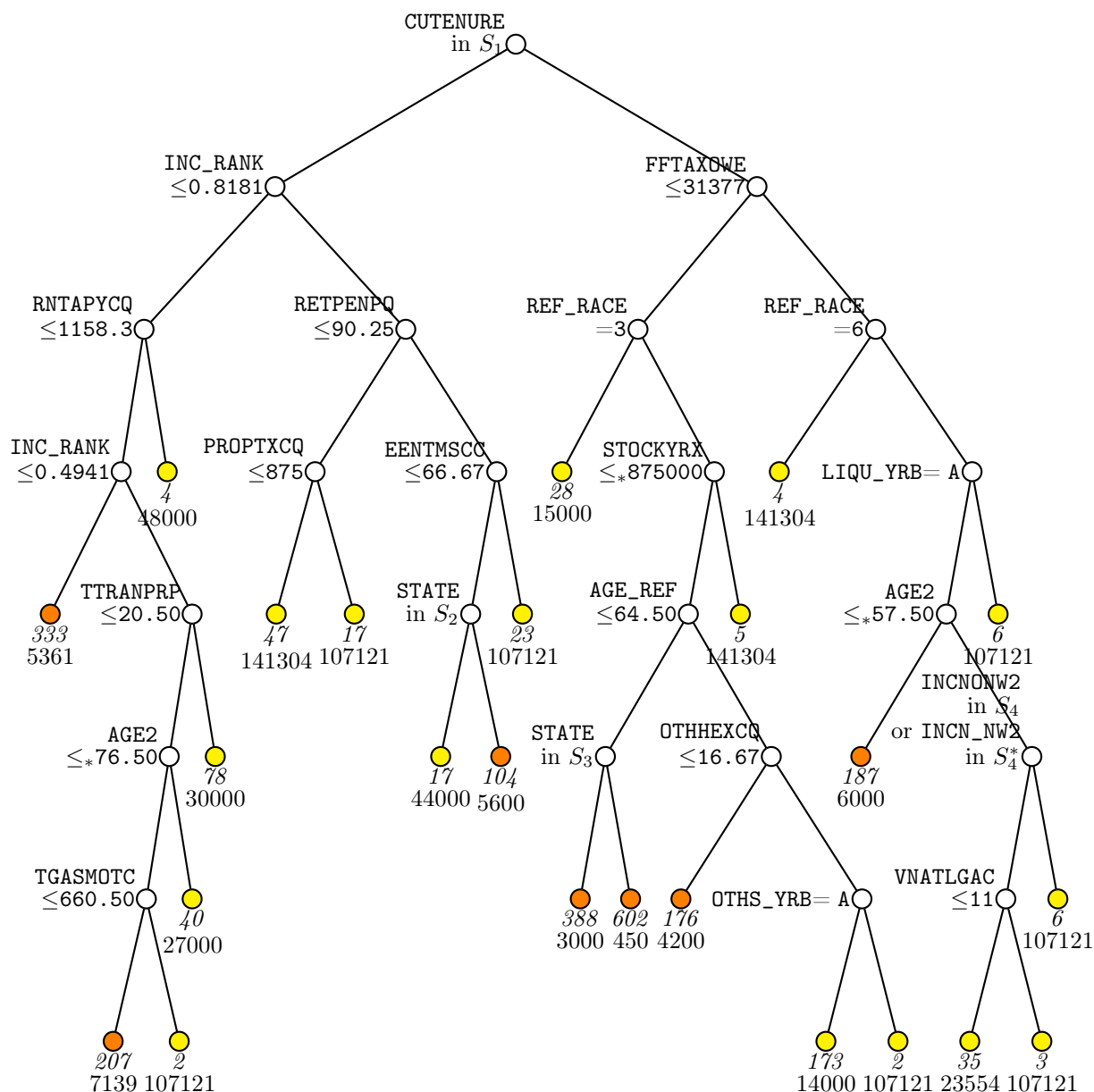


Figure 12: GUIDE v.46.1 0.250-SE piecewise-constant 0.900-quantile regression tree for predicting INTRDVX. At each split, an observation goes to the left branch if and only if the condition is satisfied. Symbol ' $\leq$ ' stands for ' $\leq$ or missing'. $S_1 = \{2, 5\}$. $S_2 = \{1, 17, 29, 37, 46\}$. $S_3 = \{1, 2, 6, 8, 9, 10, 12, 15, 21, 22, 36, 41, 45, 48, 53\}$. $S_4 = \{2\}$; $S_4^* = \{A\}$. Sample size (in *italics*) and 0.900-quantile of INTRDVX printed below nodes. Terminal nodes with quantiles above and below value of 9800 at root node are painted yellow and orange respectively. Second best split variable at root node is REFGEN.

```

1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: quantlin.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: quantlin.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 2
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple polynomial, 3: constant ([1:3], <cr>=3): 2
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input quantile probability ([0.00:1.00], <cr>=0.50): 0.90
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: ce2021reg.dsc
Reading DSC file ...
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
D variable is INTRDVX
Reading data file ...
Number of records in data file: 3965
Length of longest entry in data file: 11
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Finding number of levels of M variables associated with C variables ...
Assigning integer codes to values of 48 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Finished assigning codes to 40 categorical variables
Associating missing values of N and S variables with M variable codes ...
Re-checking data ...

```

```

Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Smallest positive weight: 1.0725E+03
Largest positive weight: 9.3902E+04
Numbers of N, F, S, and P below exclude constant variables
      Total #cases w/ #missing
      #cases miss. D ord. vals #X-var #N-var #F-var #S-var
      3965      1478      3965      1      383      0      0
      #P-var #M-var #B-var #C-var #I-var
      0      116      0      48      0
Number of cases used for training: 2487
Number of cases excluded due to 0 W or missing D variable: 1478
Finished reading data file
Constant fitted to cases with missing values in regressor variables
Input 1 for unweighted, 2 for weighted error estimates during pruning ([1:2], <cr>=2):
Warning: No interaction and linear splits; too many predictor variables
Warning: All positive weights treated as 1
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): quantlin.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save regressor names in a file, 1 otherwise ([1:2], <cr>=2):
Input file name: quantlin.reg
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: quantlin.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: quantlin.r
Input rank of top variable to split root node ([1:431], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < quantlin.in

```

Contents of quantlin.out

```

Quantile regression tree with quantile probability 0.9000
No truncation of predicted values

```

Pruning by cross-validation
 DSC file: ce2021reg.dsc
 Training sample file: ce2021.txt
 Missing value code: NA
 Records in data file start on line 2
 Number of M variables associated with C variables: 19
 D variable is INTRDVX
 Piecewise simple linear or constant model
 Powers are dropped if they are not significant at level 1.0000
 Number of records in data file: 3965
 Length of longest entry in data file: 11
 Missing values found in D variable
 Missing values found among categorical variables
 Separate categories will be created for missing categorical variables
 Missing values found among non-categorical variables
 Smallest and largest positive weights are 1.0725E+03 and 9.3902E+04

Summary information for training sample of size 2487 (excluding observations
 with non-positive weight or missing values in d, e, t, r or z variables)
 d=dependent, b=split and fit cat variable using indicator variables,
 c=split-only categorical, i=fit-only categorical (via indicators),
 s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
 m=missing-value flag variable, p=periodic variable, w=weight
 Levels of M variables are for missing values in associated variables

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	DIRACC	n	1.0000E+00	1.000		125 constant
2	DIRACC_	m			2	
3	AGE_REF	n	19.00	87.00		
4	AGE_REF_	m			0	
5	AGE2	n	2.1000E+01	87.00		1092
6	AGE2_	m			1	
:						
31	FINLWT21	w	1072.	0.9390E+05		
:						
406	INTRDVX	d	1.000	0.1413E+06		
:						
547	WHLFYR	c			1	2487
548	WHLFYR_	m			1	
549	FFTAX0WE	n	-0.3368E+05	0.3380E+06		
550	FSTAX0WE	n	-3074.	0.5654E+05		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/	#missing				
#cases	miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var

```

      3965      1478      3965      1      376      0      0
    #P-var  #M-var  #B-var  #C-var  #I-var
      0      116      0      48      0

```

Number of constant N variables: 7

Number of cases used for training: 2487

Number of cases excluded due to 0 W or missing D variable: 1478

Constant fitted to cases with missing values in regressor variables

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Weighted error estimates used for pruning

Warning: No interaction and linear splits; too many predictor variables

Warning: All positive weights treated as 1

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 25

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	32	6.034E+07	5.099E+06	4.652E+06	6.263E+07	8.243E+06
2	30	6.033E+07	5.099E+06	4.648E+06	6.263E+07	8.246E+06
3	29	6.030E+07	5.099E+06	4.656E+06	6.259E+07	8.243E+06
:						
19	11	5.915E+07	4.948E+06	3.769E+06	6.056E+07	4.819E+06
20**	10	5.872E+07	4.905E+06	3.882E+06	5.921E+07	5.297E+06
21	9	6.000E+07	4.936E+06	4.054E+06	6.137E+07	6.008E+06
22++	8	6.037E+07	4.486E+06	3.640E+06	5.969E+07	6.153E+06
23	7	6.087E+07	4.695E+06	4.010E+06	6.498E+07	6.932E+06
24	6	6.420E+07	5.043E+06	5.316E+06	6.498E+07	7.856E+06
25	4	7.040E+07	5.888E+06	5.816E+06	7.232E+07	9.671E+06
26	3	7.352E+07	5.991E+06	6.546E+06	6.965E+07	1.071E+07
27	1	8.361E+07	6.413E+06	3.673E+06	8.229E+07	5.259E+06

0-SE tree based on mean is marked with * and has 10 terminal nodes

0-SE tree based on median is marked with + and has 10 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree same as + tree

** tree same as + tree

** tree same as -- tree

* tree same as ** tree

* tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-quant is quantile of INTRDVX in the node

Cases fit give the number of cases used to fit node

Node label	Total cases	Cases fit	Matrix rank	Node D-quant	Split variable	Other variables
1	2487	1939	2	9.800E+03	MRTINTPQ	
2	1371	1371	2	1.400E+04	FINCBTAX	
4	916	916	2	7.000E+03	STOCKX	
8T	874	874	2	6.000E+03	STATE	
9T	42	42	2	3.000E+04	-	
5	455	320	2	4.500E+04	RETPENPQ	
10	117	117	2	1.071E+05	RETSURV	
20T	67	20	2	1.071E+05	-	
21T	50	50	2	1.413E+05	-	
11	338	338	2	1.500E+04	RENTEQVX	
22T	305	305	2	6.000E+03	RETSURVX	
23T	33	33	2	1.071E+05	-	
3	1116	19	2	5.000E+03	STOCKYRX	
6	1077	19	2	4.400E+03	FFTAXOWE	
12T	930	7	2	3.000E+03	REFGEN	
13	147	147	2	2.123E+04	FDXMAPCQ	
26T	72	72	2	1.071E+05	RENTEQVX	
27T	75	75	2	1.000E+04	-	
7T	39	39	2	1.071E+05	-	

Number of terminal nodes of final tree: 10

Total number of nodes of final tree: 19

Second best split variable (based on curvature test) at root node is EMRTPNOP

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: MRTINTPQ <= 1.5000000

Node 2: FINCBTAX <= 95076.000

Node 4: STOCKX <= 125000.00 or STOCKX = NA & STOCKX_ = "A"

Node 8: INTRDVX sample quantile = 6000.0000

Node 4: not (STOCKX <= 125000.00 or STOCKX = NA & STOCKX_ = "A")

Node 9: INTRDVX sample quantile = 30000.000

Node 2: FINCBTAX > 95076.000 or NA

Node 5: RETPENPQ <= 571.83330

Node 10: RETSURV = "1"

Node 20: INTRDVX sample quantile = 107121.00

```

Node 10: RETSURV /= "1"
Node 21: INTRDVX sample quantile = 141304.00
Node 5: RETPENPQ > 571.83330 or NA
Node 11: RENTEQVX <= 3073.0000 or NA
Node 22: INTRDVX sample quantile = 6000.0000
Node 11: RENTEQVX > 3073.0000
Node 23: INTRDVX sample quantile = 107121.00
Node 1: MRTINTPQ > 1.5000000 or NA
Node 3: STOCKYRX <= 135000.00 or STOCKYRX = NA & STOC_YRX = "A"
Node 6: FFTAXOWE <= 40686.000
Node 12: INTRDVX sample quantile = 3000.0000
Node 6: FFTAXOWE > 40686.000 or NA
Node 13: FDXMAPCQ <= 238.33335
Node 26: INTRDVX sample quantile = 107121.00
Node 13: FDXMAPCQ > 238.33335 or NA
Node 27: INTRDVX sample quantile = 10000.000
Node 3: not (STOCKYRX <= 135000.00 or STOCKYRX = NA & STOC_YRX = "A")
Node 7: INTRDVX sample quantile = 107121.00

```

```

*****
Predictor means below are weighted means of cases with no missing values.
Regression coefficients are computed from the complete cases.

```

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in *Design and Analysis of Subgroups with Biopharmaceutical Applications*, Springer, pp.147-165.

```

Node 1: Intermediate node
A case goes into Node 2 if MRTINTPQ <= 1.5000000
MRTINTPQ mean = 619.33599
Coefficients of quantile regression function:
Regressor    Coefficient Minimum      Mean      Maximum
Constant     -5793.
RENTEQVX      9.896         65.00      2252.      6302.
If regressor has missing values, predicted quantile = 3000.00
-----

```

```

Node 2: Intermediate node
A case goes into Node 4 if FINCBTAX <= 95076.000
FINCBTAX mean = 89898.898

```

```

-----
Node 4: Intermediate node
A case goes into Node 8 if STOCKX <= 125000.00 or STOCKX_ = "A"
STOCKX mean = 324021.34
-----
Node 8: Terminal node
Coefficients of quantile regression function:
Regressor   Coefficient Minimum      Mean      Maximum
Constant    1969.
FRRETIRM    0.2330      0.000      0.1471E+05  0.7378E+05
If regressor has missing values, predicted quantile = 6000.00
-----
Node 9: Terminal node
Coefficients of quantile regression function:
Regressor   Coefficient Minimum      Mean      Maximum
Constant    -6275.
INC_RANK    0.6613E+05  0.9591E-01  0.4320      0.7292
If regressor has missing values, predicted quantile = 30000.0
-----
Node 5: Intermediate node
A case goes into Node 10 if RETPENPQ <= 571.83330
RETPENPQ mean = 2829.5709
-----
Node 10: Intermediate node
A case goes into Node 20 if RETSURV = "1"
RETSURV mode = "1"
-----
Node 20: Terminal node
Coefficients of quantile regression function:
Regressor   Coefficient Minimum      Mean      Maximum
Constant    0.1071E+06
CREDITX     -14.01      0.000      833.7      7648.
If regressor has missing values, predicted quantile = 40000.0
-----
Node 21: Terminal node
Coefficients of quantile regression function:
Regressor   Coefficient Minimum      Mean      Maximum
Constant    -0.6379E+05
INCLASS2    0.3418E+05  5.000      5.601      6.000
If regressor has missing values, predicted quantile = 141304.
-----
Node 11: Intermediate node
A case goes into Node 22 if RENTEQVX <= 3073.0000 or NA
RENTQVX mean = 2523.9099
-----
Node 22: Terminal node

```


Coefficients of quantile regression function:

Regressor	Coefficient	Minimum	Mean	Maximum
Constant	4200.			
PROPTXCQ	32.49	0.000	227.1	3168.

If regressor has missing values, predicted quantile = 6000.00

Node 23: Terminal node

Coefficients of quantile regression function:

Regressor	Coefficient	Minimum	Mean	Maximum
Constant	0.3182E+05			
TALCBEVC	376.5	0.000	46.91	448.0

If regressor has missing values, predicted quantile = 107121.

Node 3: Intermediate node

A case goes into Node 6 if STOCKYRX <= 135000.00 or STOC_YRX = "A"

STOCKYRX mean = 277744.06

Node 6: Intermediate node

A case goes into Node 12 if FFTAXOWE <= 40686.000

FFTAXOWE mean = 17090.349

Node 12: Terminal node

Coefficients of quantile regression function:

Regressor	Coefficient	Minimum	Mean	Maximum
Constant	-4692.			
WHLFYRBX	3.195	1500.	0.1386E+05	0.3500E+05

If regressor has missing values, predicted quantile = 3000.00

Node 13: Intermediate node

A case goes into Node 26 if FDXMAPCQ <= 238.33335

FDXMAPCQ mean = 510.41115

Node 26: Terminal node

Coefficients of quantile regression function:

Regressor	Coefficient	Minimum	Mean	Maximum
Constant	0.1550E+05			
OWNDWECQ	28.30	0.000	450.6	7125.

If regressor has missing values, predicted quantile = 107121.

Node 27: Terminal node

Coefficients of quantile regression function:

Regressor	Coefficient	Minimum	Mean	Maximum
Constant	5000.			
MISCPQ	78.55	0.000	121.7	1412.

If regressor has missing values, predicted quantile = 10000.0

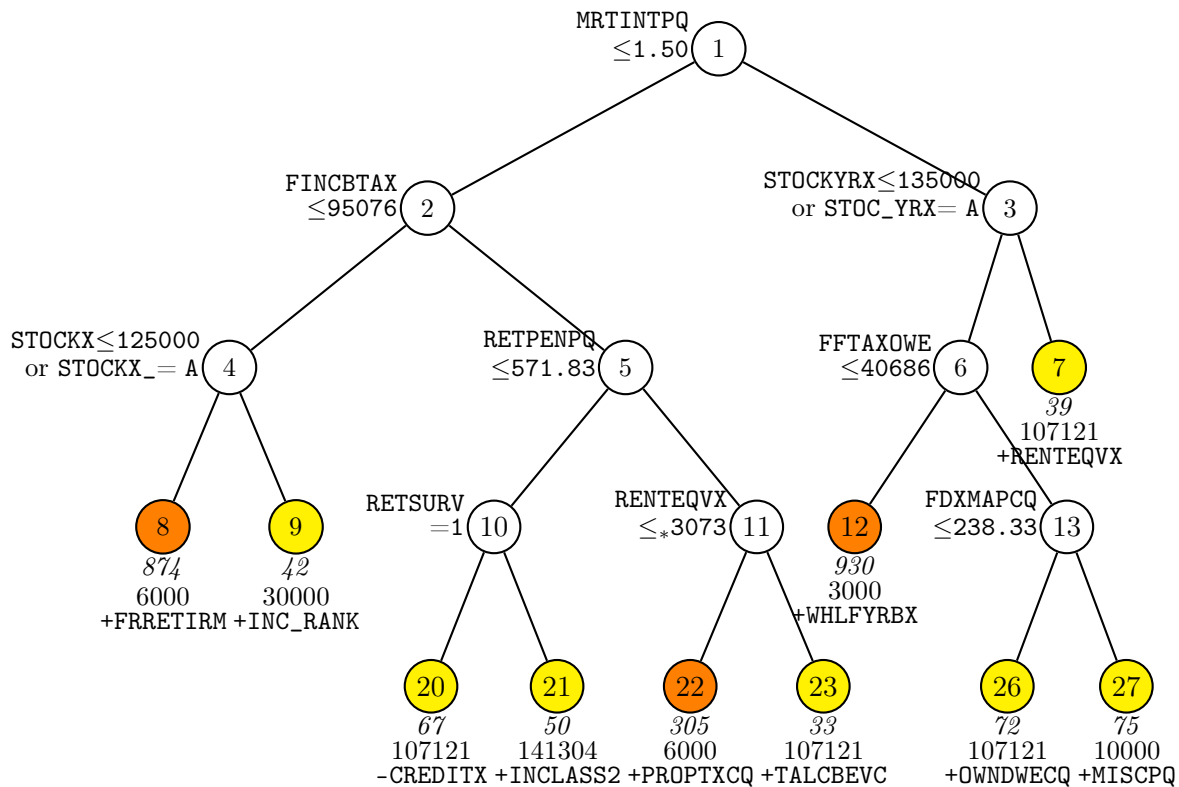


Figure 13: GUIDE v.46.1 0.250-SE piecewise simple linear 0.900-quantile regression tree (constant fitted to incomplete cases in terminal nodes) for predicting INTRDVX. At each split, an observation goes to the left branch if and only if the condition is satisfied. Symbol ' $\leq_*$ ' stands for ' $\leq$ or missing'. Sample size (in *italics*), 0.900-quantile of INTRDVX, and sign and name of best regressor printed below nodes. Terminal nodes with quantiles above and below value of 9800 at root node are painted yellow and orange respectively. Second best split variable at root node is EMRTPNOP.

Node 7: Terminal node

Coefficients of quantile regression function:

Regressor	Coefficient	Minimum	Mean	Maximum
Constant	-0.2434E+05			
RENTEQVX	26.32	925.0	3338.	6294.

If regressor has missing values, predicted quantile = 107121.

Observed and fitted values are stored in quantlin.fit

Regressor names and coefficients are stored in quantlin.reg

LaTeX code for tree is in quantlin.tex

Figure 13 shows the 0.90-quantile simple linear regression tree.

7.3 Two quantiles: checking variance heterogeneity

Checking variance homogeneity in the residuals is a standard practice in fitting regression models. Here we show how GUIDE can do this by constructing a quantile regression tree models for the 25th and 75th quantiles simultaneously.

7.3.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: twoquant.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: twoquant.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 2
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple polynomial, 3: constant ([1:3], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1): 2
Input 1 for 1 quantile, 2 for 2 quantiles ([1:2], <cr>=1): 2
Input 1st quantile probability ([0.00:1.00], <cr>=0.25):
Input 2nd quantile probability ([0.00:1.00], <cr>=0.75):
Input 1 for interaction tests, 2 to skip them ([1:2], <cr>=1):
Input 0 to get tree with fixed no. of nodes, 1 to prune by CV, 2 for no pruning ([0:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: ce2021reg.dsc
Reading DSC file ...
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX
Reading data file ...
Number of records in data file: 3965

```

```

Length of longest entry in data file: 11
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Finding number of levels of M variables associated with C variables ...
Assigning integer codes to values of 48 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Finished assigning codes to 40 categorical variables
Associating missing values of N and S variables with M variable codes ...
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Smallest positive weight: 1.0725E+03
Largest positive weight: 9.3902E+04
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      3965      1478      3965      1      0      0      383
      #P-var  #M-var  #B-var  #C-var  #I-var
      0      116      0      48      0
Number of cases used for training: 2487
Number of cases excluded due to 0 W or missing D variable: 1478
Finished reading data file
Default number of cross-validations: 10
Input 1 to accept the default, 2 to change it ([1:2], <cr>=1):
Best tree may be chosen based on mean or median CV estimate
Input 1 for mean-based, 2 for median-based ([1:2], <cr>=1):
Input number of SEs for pruning ([0.00:1000.00], <cr>=0.25):
Input 1 for unweighted, 2 for weighted error estimates during pruning ([1:2], <cr>=2):
Warning: No interaction and linear splits; too many predictor variables
Warning: All positive weights treated as 1
Default max. number of split levels: 15
Input 1 to accept this value, 2 to change it ([1:2], <cr>=1):
Default minimum node sample size is 2
Input 1 to use the default value, 2 to change it ([1:2], <cr>=1):
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): twoquant.tex
Choose a color for the terminal nodes:

```

```

(1) white
(2) lightgray
(3) aqua
(4) skyblue
(5) lime
(6) yellow
(7) red
(8) mauve
(9) green
(10) orange
(11) cyan
Input your choice ([1:11], <cr>=8):
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: twoquant.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: twoquant.r
Input rank of top variable to split root node ([1:431], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < twoquant.in

```

7.3.2 Output file

```

Dual-quantile regression tree with 0.2500 and 0.7500 quantiles
Pruning by cross-validation
DSC file: ce2021reg.dsc
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX
Piecewise constant model
Number of records in data file: 3965
Length of longest entry in data file: 11
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables

```

Missing values found among non-categorical variables
 Smallest and largest positive weights are 1.0725E+03 and 9.3902E+04

Summary information for training sample of size 2487 (excluding observations
 with non-positive weight or missing values in d, e, t, r or z variables)
 d=dependent, b=split and fit cat variable using indicator variables,
 c=split-only categorical, i=fit-only categorical (via indicators),
 s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
 m=missing-value flag variable, p=periodic variable, w=weight
 Levels of M variables are for missing values in associated variables

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	DIRACC	s	1.000	1.000		125 constant
2	DIRACC_	m			2	
3	AGE_REF	s	19.00	87.00		
4	AGE_REF_	m			0	
5	AGE2	s	21.00	87.00		1092
6	AGE2_	m			1	
:						
31	FINLWT21	w	1072.	0.9390E+05		
:						
406	INTRDVX	d	1.000	0.1413E+06		
:						
549	FFTAX0WE	s	-0.3368E+05	0.3380E+06		
550	FSTAX0WE	s	-3074.	0.5654E+05		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	miss. D	#missing ord. vals	#X-var	#N-var	#F-var	#S-var
3965	1478	3965	1	0	0	376	
#P-var	#M-var	#B-var	#C-var	#I-var			
0	116	0	48	0			

Number of constant S variables: 7

Number of cases used for training: 2487

Number of cases excluded due to 0 W or missing D variable: 1478

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Weighted error estimates used for pruning

Warning: No interaction and linear splits; too many predictor variables

Warning: All positive weights treated as 1

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 2

Ranks of variables and their 1-df chi-squared values at root node

```

1  0.1744E+03  STATE
2  0.1192E+03  FINCBTAX
3  0.1135E+03  INC_RANK
:
415 0.5618E-03  TFEESADC
416 0.4974E-03  INC_HRS2
417 0.4468E-03  OTHLNRYX

```

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	68	1.078E+08	7.583E+06	5.419E+06	1.098E+08	8.082E+06
2	66	1.078E+08	7.583E+06	5.419E+06	1.098E+08	8.082E+06
:						
11	51	1.078E+08	7.584E+06	5.417E+06	1.097E+08	8.074E+06
12+	50	1.078E+08	7.584E+06	5.417E+06	1.097E+08	8.074E+06
13	49	1.078E+08	7.584E+06	5.417E+06	1.097E+08	8.078E+06
:						
31	13	1.077E+08	7.636E+06	5.331E+06	1.118E+08	7.930E+06
32++	11	1.059E+08	7.642E+06	5.547E+06	1.107E+08	7.791E+06
33	10	1.065E+08	7.705E+06	5.541E+06	1.128E+08	8.121E+06
34--	7	1.058E+08	7.671E+06	5.251E+06	1.128E+08	7.954E+06
35	5	1.084E+08	8.069E+06	5.362E+06	1.134E+08	7.313E+06
36**	4	1.074E+08	7.820E+06	4.994E+06	1.128E+08	6.979E+06
37	1	1.164E+08	9.292E+06	5.964E+06	1.190E+08	7.350E+06

0-SE tree based on mean is marked with * and has 7 terminal nodes

0-SE tree based on median is marked with + and has 50 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Cases fit give the number of cases used to fit node

Column labeled 'Split variable' gives median if node is terminal

Node	Total	Cases	Matrix	Node	Split	Other
label	cases	fit	rank	median	variable	variables
1	2487	2487	1	1.500E+01	STATE	
2T	420	420	1	5.000E+00	2.500E+02	FFTAXOWE

3	2067	2067	1	2.000E+01	INC_RANK	
6T	1554	1554	1	1.500E+01	1.500E+03	CUTENURE
7	513	513	1	1.200E+02	INCNONW2	
14T	417	417	1	1.000E+02	3.000E+03	STATE
15T	96	96	1	4.000E+02	1.071E+05	OCCUCOD1

Number of terminal nodes of final tree: 4

Total number of nodes of final tree: 7

Second best split variable (based on curvature test) at root node is FINCBTAX

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: STATE = "19", "24", "31", "40", "45", "49", "51", "54"

Node 2: INTRDVX sample quantiles = 5.0000000, 250.00000

Node 1: STATE /= "19", "24", "31", "40", "45", "49", "51", "54"

Node 3: INC_RANK <= 0.87960485

Node 6: INTRDVX sample quantiles = 15.000000, 1500.0000

Node 3: INC_RANK > 0.87960485 or NA

Node 7: INCNONW2 = NA & INCN_NW2 = "A"

Node 14: INTRDVX sample quantiles = 100.00000, 3000.0000

Node 7: not (INCNONW2 = NA & INCN_NW2 = "A")

Node 15: INTRDVX sample quantiles = 400.00000, 107121.00

Predictor means below are weighted means of cases with no missing values.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.

2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.

3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if STATE = "19", "24", "31", "40", "45", "49", "51", "54"

STATE mode = "6"

Sample 0.250-quantile, 0.750-quantile, and median:

1.5000E+01	1.9300E+03	1.7400E+02
------------	------------	------------

Node 2: Terminal node

Sample 0.250-quantile, 0.750-quantile, and median:


```

      5.0000E+00      2.5000E+02      2.0000E+01
-----
Node 3: Intermediate node
A case goes into Node 6 if INC_RANK <= 0.87960485
INC_RANK mean = 0.65419245
-----
Node 6: Terminal node
Sample 0.250-quantile, 0.750-quantile, and median:
      1.5000E+01      1.5000E+03      1.5000E+02
-----
Node 7: Intermediate node
A case goes into Node 14 if INCNONW2 = NA & INCN_NW2 = "A"
INCN_NW2 mode = "A"
-----
Node 14: Terminal node
Sample 0.250-quantile, 0.750-quantile, and median:
      1.0000E+02      3.0000E+03      5.0000E+02
-----
Node 15: Terminal node
Sample 0.250-quantile, 0.750-quantile, and median:
      4.0000E+02      1.0712E+05      9.5000E+03
-----
Observed and fitted values are stored in twoquant.fit
LaTeX code for tree is in twoquant.tex

```

Figure 14 shows the tree. Beneath each terminal node are three numbers. The first (in *italics*) is the node sample size. The other two are the sample 0.75 and 0.25-quantiles in the node. The large differences in inter-quartile range in the nodes indicates substantial variance heterogeneity.

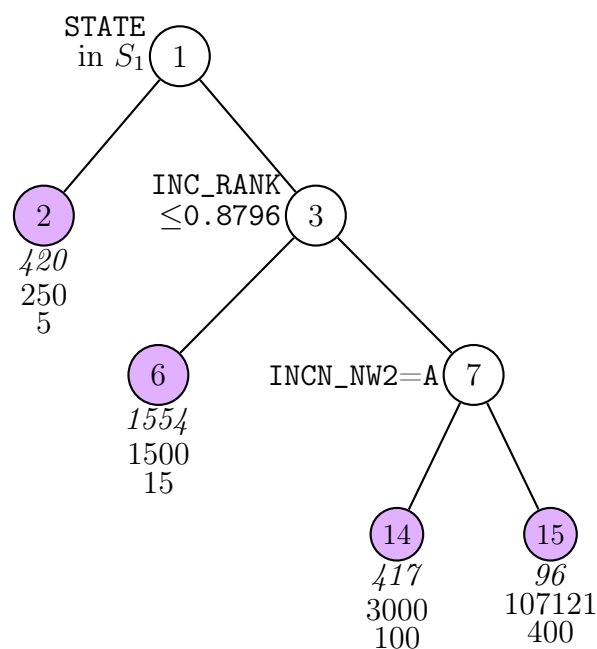


Figure 14: GUIDE v.46.1 0.250-SE piecewise-constant 0.250 and 0.750-quantile regression tree for predicting INTRDVX. At each split, an observation goes to the left branch if and only if the condition is satisfied. $S_1 = \{19, 24, 31, 40, 45, 49, 51, 54\}$. Sample size (in *italics*) and sample 0.750 and 0.250-quantiles of INTRDVX printed below nodes. Second best split variable at root node is FINCBTAX.

8 Periodic variables: NHTSA data

Periodic variables that have a cyclic property, such as angular measurements, hour of day, day of week, and month of year, can be designated as **P** variables in the DSC file. There can be multiple **P** variables in the same data set. Unlike the other types of variables, each line in the DSC file containing a **P** variable must have the value of its period (e.g., 360 for angular measurements, 24 for hour of day, 7 for day of week, and 12 for month of year) immediately after **P** on the same line. GUIDE does not allow **P** variables to have missing-value flag (**M**) variables.

The National Highway Traffic Safety Administration (NHTSA) has been conducting vehicle crash tests since 1972. Data from 3310 crash tests are in the file `nhtsadatam.txt` (see www-nrd.nhtsa.dot.gov/database/veh/ for more information). Variable **HIC** (head injury criterion) is a measure of severity of head injury. Experts believe that $HIC > 999$ is life threatening. Table 5 gives the definitions of the variables appearing in the models below. Besides missing values, there are many variables with illogical values (such as negative values for diameter). To identify these values, we adopt the strategy in the CE data of creating a missing-value flag variable for each variable having illogical values, with the flags being **A**, **B**, and **D** for validly missing, illogical, and valid response, respectively. The data also contain some angular variables (with periods of 360 degrees and for which 0 degrees indicates straight-ahead or head-on) that are defined as **P**.

8.1 Regression

The DSC file for regression is `nhtsadsc.txt`; its contents follow.

```
nhtsadatam.txt
NA
2
1 BARRIG c
2 BARSHP b
3 BARANG p 360
4 BARDIA n
5 OCCWT n
6 OCCWT_ m
7 DUMSIZ c
8 HH n
9 HH_ m
10 HW n
11 HW_ m
12 HR n
```

13 HR_ m
14 HS_ n
15 HS_ m
16 CD_ n
17 CD_ m
18 CS_ n
19 CS_ m
20 AD_ n
21 AD_ m
22 HD_ n
23 HD_ m
24 KD_ n
25 KD_ m
26 HB_ n
27 HB_ m
28 NB_ n
29 NB_ m
30 CB_ n
31 CB_ m
32 KB_ n
33 SEPOSN_ c
34 HIC_ d
35 TKSURF_ c
36 TKCOND_ c
37 CLSSPD_ n
38 CLSSPD_ m
39 IMPANG_ p 360
40 OFFSET_ n
41 IMPPNT_ n
42 MAKED_ c
43 MODEL_ c
44 YEAR_ n
45 BODY_ c
46 ENGINE_ c
47 ENGDSP_ n
48 ENGDSP_ m
49 TRANSM_ c
50 VEHTWT_ n
51 VEHTWT_ m
52 CURBWT_ n
53 WHLBAS_ n
54 WHLBAS_ m
55 VEHLEN_ n
56 VEHLEN_ m
57 VEHWID_ n
58 VEHWID_ m

59 VEHCG n
60 VEHCG_ m
61 COLMEC c
62 BX1 n
63 BX1_ m
64 BX2 n
65 BX2_ m
66 BX3 n
67 BX3_ m
68 BX4 n
69 BX4_ m
70 BX5 n
71 BX5_ m
72 BX6 n
73 BX6_ m
74 BX7 n
75 BX7_ m
76 BX8 n
77 BX8_ m
78 BX9 n
79 BX9_ m
80 BX10 n
81 BX10_ m
82 BX11 n
83 BX11_ m
84 BX12 n
85 BX12_ m
86 BX13 n
87 BX13_ m
88 BX14 n
89 BX14_ m
90 BX15 n
91 BX15_ m
92 BX16 n
93 BX16_ m
94 BX17 n
95 BX17_ m
96 BX18 n
97 BX18_ m
98 BX19 n
99 BX19_ m
100 BX20 n
101 BX20_ m
102 BX21 n
103 BX21_ m
104 VEHSPD n

Table 5: Some variable definitions for NHTSA data

Variable	Meaning
BARSHP	barrier shape (21 values)
BX2	distance from rear surface of vehicle to front of engine (mm)
BX5	distance from rear surface of vehicle to upper leading edge of left door (mm)
BX8	distance from rear surface of vehicle to upper trailing edge of right door (mm)
BX12	distance from rear surface of vehicle to bottom of a post of right side (mm)
COLMEC	steering column collapse mechanism (9 values)
ENGDSP	engine displacement (liters)
IMPANG	impact angle (clockwise with 0 degrees being straight ahead)
OCCAGE	dummy occupant age
PDOF	principal direction of force (degrees)
TRANSM	transmission type (9 values)
VEHTWT	vehicle test weight (kg)
VEHSPD	vehicle speed (km/h)
VEHWID	vehicle width (mm)
WHLBAS	wheel base (mm)
YEAR	vehicle model year (1972–2017)

```

105 VEHSPD_ m
106 CRBANG p 360
107 PDOF p 360
108 CARANG p 360
109 VEHOR p 360
110 RSTFRT c
111 HIC2 x
112 estHIC2 x

```

We show the results of fitting a piecewise-linear regression tree here.

8.1.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: lin.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: lin.out

```

```

Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
  7=binary logistic regression.
Input choice ([1:7], <cr>=1):
Input 1 for least squares, 2 least median of squares ([1:2], <cr>=1):
Choose complexity of model to use at each node:
Choose 0 for stepwise linear regression (recommended for prediction)
Choose 1 for multiple regression
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
0: stepwise linear, 1: multiple linear, 2: best simple polynomial, 3: constant,
4: best simple stepwise ANCOVA ([0:4], <cr>=3): 2
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: nhtsadsc.txt
Reading DSC file ...
Training sample file: nhtsadatam.txt
Missing value code: NA
Records in data file start on line 2
Warning: B variables changed to C
D variable is HIC
Reading data file ...
Number of records in data file: 3310
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 11 categorical variables
Finished assigning codes to 10 categorical variables
Associating missing values of N, P and S variables with M variable codes ...
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
#cases   miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var

```

3310	34	3310	5	48	0	0
#P-var	#M-var	#B-var	#C-var	#I-var		
6	42	0	11	0		

No weight variable in data file
 Number of cases used for training: 3276
 Number of cases excluded due to 0 W or missing D variable: 34
 Finished reading data file
 Constant fitted to cases with missing values in regressor variables
 Default max. number of split levels: 15
 Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
 Input file name to store LaTeX code (use .tex as suffix): lin.tex
 You can store the variables and/or values used to split and fit in a file
 Choose 1 to skip this step, 2 to store split and fit variables,
 3 to store split variables and their values
 Input your choice ([1:3], <cr>=1):
 Input 2 to save regressor names in a file, 1 otherwise ([1:2], <cr>=2):
 Input file name: lin.reg
 Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
 Input name of file to store node ID and fitted value of each case: lin.fit
 You can output an R function for predicting future observations
 Input 1 to skip this step
 Input 2 to output an R function that applies to the training data
 Input 3 to output an R function that applies to a new data set
 Input your choice ([1:3], <cr>=2):
 Input filename to write R function: lin.r
 Input rank of top variable to split root node ([1:65], <cr>=1):
 Input file is created!
 Run GUIDE with the command: guide < lin.in

8.1.2 Results

Least squares regression tree
 Predictions truncated at global min. and max. of D sample values
 Pruning by cross-validation
 DSC file: nhtsadsc.txt
 Training sample file: nhtsadatam.txt
 Missing value code: NA
 Records in data file start on line 2
 Warning: B variables changed to C
 D variable is HIC
 Piecewise simple linear or constant model
 Powers are dropped if they are not significant at level 0.0500
 Number of records in data file: 3310
 Length of longest entry in data file: 19
 Missing values found in D variable

Missing values found among categorical variables

Separate categories will be created for missing categorical variables

Missing values found among non-categorical variables

Summary information for training sample of size 3276 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)

d=dependent, b=split and fit cat variable using indicator variables,

c=split-only categorical, i=fit-only categorical (via indicators),

s=split-only numerical, n=split and fit numerical, f=fit-only numerical,

m=missing-value flag variable, p=periodic variable, w=weight

Levels of M variables are for missing values in associated variables

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	BARRIG	c			3	
2	BARSHP	c			21	
3	BARANG	p	0.000	330.0	360	14
4	BARDIA	n	1.9100E+02	1000.		2807
:						
33	SEPOSN	c			5	81
34	HIC	d	0.000	0.1225E+05		
35	TKSURF	c			5	80
:						
101	BX20_	m			2	
102	BX21	n	5.4000E+01	0.1085E+05		964
103	BX21_	m			2	
104	VEHSPD	n	3.0000E-01	99.10		6
105	VEHSPD_	m			2	
106	CRBANG	p	0.000	315.0	360	24
107	PDQF	p	0.000	345.0	360	23
108	CARANG	p	0.000	99.00	360	991
109	VEHOR	p	0.000	90.00	360	995
110	RSTFRT	c			3	

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	miss. D	#missing ord. vals	#X-var	#N-var	#F-var	#S-var
3310	34		3310	5	48	0	0
#P-var	#M-var	#B-var	#C-var	#I-var			
6	42	0	11	0			

No weight variable in data file

Number of cases used for training: 3276

Number of cases excluded due to 0 W or missing D variable: 34

Constant fitted to cases with missing values in regressor variables

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates
 Number of SE's for pruned tree: 0.2500

Nodewise interaction tests on all variables
 Split values for N and S variables based on exhaustive search
 Max number of interaction splits: 100
 Maximum number of split levels: 15
 Minimum node sample size: 33
 Size and CV MSE and SE of subtrees:

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	77	4.136E+05	7.785E+04	7.934E+04	3.550E+05	8.140E+04
2	75	4.136E+05	7.785E+04	7.934E+04	3.550E+05	8.140E+04
:						
46	14	4.327E+05	8.921E+04	1.012E+05	3.435E+05	7.641E+04
47+	12	4.317E+05	8.921E+04	1.012E+05	3.435E+05	7.611E+04
48	11	3.852E+05	7.886E+04	9.474E+04	3.439E+05	4.904E+04
49	10	3.872E+05	7.917E+04	9.452E+04	3.439E+05	4.783E+04
50**	6	3.558E+05	6.574E+04	5.578E+04	3.448E+05	3.927E+04
51	1	3.750E+05	6.843E+04	5.388E+04	4.032E+05	4.736E+04

0-SE tree based on mean is marked with * and has 6 terminal nodes
 0-SE tree based on median is marked with + and has 12 terminal nodes
 Selected-SE tree based on mean using naive SE is marked with **
 Selected-SE tree based on mean using bootstrap SE is marked with --
 Selected-SE tree based on median and bootstrap SE is marked with ++
 ** tree same as ++ tree
 ** tree same as -- tree
 ++ tree same as -- tree
 * tree same as ** tree
 * tree same as ++ tree
 * tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is mean of HIC in the node
 Cases fit give the number of cases used to fit node
 MSE and R² are based on all cases in node

Node label	Total cases	Cases fit	Matrix rank	Node D-mean	Node MSE	Node R ²	Split variable	Other variables
1	3276	3272	2	5.127E+02	3.743E+05	0.1007	BARSHP -YEAR	
2T	40	32	2	1.606E+02	1.337E+04	0.3122	- +HW	
3	3236	3232	2	5.170E+02	3.749E+05	0.1066	BARSHP -YEAR	
6T	34	31	2	2.749E+02	5.098E+04	0.2716	- -VEHSPD	
7	3202	3199	2	5.196E+02	3.771E+05	0.1080	BARSHP -YEAR	

14	320	206	2	3.440E+02	4.708E+05	0.4316	BX4	-IMPPNT
28T	65	7	2	6.285E+02	4.322E+05	0.7203	-	-BARDIA
29T	255	171	2	2.715E+02	2.911E+05	0.5381	SEPOSN	-IMPPNT
15	2882	2879	2	5.391E+02	3.315E+05	0.1147	CLSSPD	-YEAR
30T	1292	9	2	4.440E+02	5.638E+05	0.0267	HS	-CB
31T	1590	1590	2	6.164E+02	1.141E+05	0.4156	TRANSM	-YEAR

Number of terminal nodes of final tree: 6

Total number of nodes of final tree: 11

Second best split variable (based on curvature test) at root node is IMPANG

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: BARSHP = "488", "EOL", "GRL", "MBR", "OTH", "ROR"

Node 2: HIC-mean = 160.57500

Node 1: BARSHP /= "488", "EOL", "GRL", "MBR", "OTH", "ROR"

Node 3: BARSHP = "128", "SGN", "UNK", "US1"

Node 6: HIC-mean = 274.94118

Node 3: BARSHP /= "128", "SGN", "UNK", "US1"

Node 7: BARSHP = "134", "EOB", "FAB", "IAT", "LUM"

Node 14: BX4 <= 2536.5000 or BX4 = NA & BX4_ = "A"

Node 28: HIC-mean = 628.50769

Node 14: not (BX4 <= 2536.5000 or BX4 = NA & BX4_ = "A")

Node 29: HIC-mean = 271.47843

Node 7: BARSHP /= "134", "EOB", "FAB", "IAT", "LUM"

Node 15: CLSSPD <= 55.450000

Node 30: HIC-mean = 443.96749

Node 15: CLSSPD > 55.450000 or NA

Node 31: HIC-mean = 616.36541

Predictor means below are means of cases with no missing values.

Regression coefficients are computed from the complete cases.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.

2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.

3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if BARSHP = "488", "EOL", "GRL", "MBR", "OTH", "ROR"
BARSHP mode = "LCB"

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.3893E+05	19.40	0.1488E-14			
YEAR	-19.21	-19.14	0.9685E-15	1972.	2000.	2017.

If regressor has missing values, predicted value = 471.00000

Predicted values truncated at 0.00000 & 12246.0

Node 2: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	-84.23	-0.7127	0.4815			
HW	0.5093	2.369	0.2447E-01	414.0	540.9	787.0

If regressor has missing values, predicted value = 37.875000

Predicted values truncated at 0.00000 & 12246.0

Node 3: Intermediate node

A case goes into Node 6 if BARSHP = "128", "SGN", "UNK", "US1"
BARSHP mode = "LCB"

Node 6: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	767.7	4.103	0.3026E-03			
VEHSPD	-7.404	-2.561	0.1590E-01	40.20	63.02	98.00

If regressor has missing values, predicted value = 5.3333333

Predicted values truncated at 0.00000 & 12246.0

Node 7: Intermediate node

A case goes into Node 14 if BARSHP = "134", "EOB", "FAB", "IAT", "LUM"
BARSHP mode = "LCB"

Node 14: Intermediate node

A case goes into Node 28 if BX4 <= 2536.5000 or BX4_ = "A"
BX4 mean = 3273.8683

Node 28: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.5577E+06	3.041	0.2873E-01			
BARDIA	-2557.	-3.027	0.2917E-01	216.0	217.1	218.0

If regressor has missing values, predicted value = 411.89655

Predicted values truncated at 0.00000 & 12246.0

Node 29: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	145.6	3.044	0.2708E-02			
IMPPNT	-20.71	-15.03	0.000	-203.0	-5.936	0.000

If regressor has missing values, predicted value = 277.47619
 Predicted values truncated at 0.00000 & 12246.0

 Node 15: Intermediate node

A case goes into Node 30 if CLSSPD <= 55.450000
 CLSSPD mean = 49.475944

 Node 30: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.1004E+05	14.75	0.1575E-05			
CB	-9.554	-13.47	0.2915E-05	537.0	948.6	1000.

If regressor has missing values, predicted value = 440.23461
 Predicted values truncated at 0.00000 & 12246.0

 Node 31: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.5474E+05	33.99	0.1594E-15			
YEAR	-27.07	-33.60	0.000	1974.	1999.	2017.

If regressor has missing values, predicted value = 616.36541
 Predicted values truncated at 0.00000 & 12246.0

 Proportion of variance (R-squared) explained by tree model: 0.2578

Observed and fitted values are stored in lin.fit

Regressor names and coefficients are stored in lin.reg

LaTeX code for tree is in lin.tex

The piecewise-linear regression tree is shown in Figure 15.

8.2 Classification

8.2.1 Univariate splits

Below are the results of fitting a classification tree with univariate splits to HIC2 which takes value 1 if HIC > 999 and value 0 otherwise. The DSC file is nhtsahic2.dsc.

Classification tree

Pruning by cross-validation

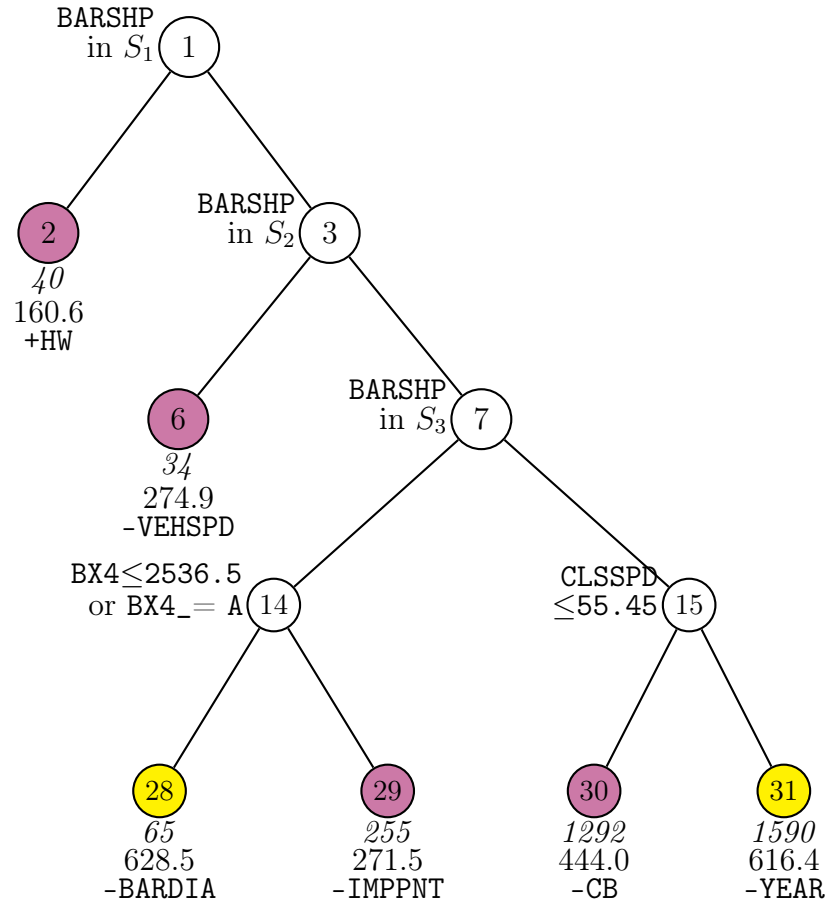


Figure 15: GUIDE v.46.1 0.250-SE piecewise simple linear least-squares regression tree (constant fitted to incomplete cases in terminal nodes) for predicting HIC. At each split, an observation goes to the left branch if and only if the condition is satisfied. $S_1 = \{488, \text{EOL}, \text{GRL}, \text{MBR}, \text{OTH}, \text{ROR}\}$. $S_2 = \{128, \text{SGN}, \text{UNK}, \text{US1}\}$. $S_3 = \{134, \text{EOB}, \text{FAB}, \text{IAT}, \text{LUM}\}$. Sample size (in *italics*), mean of HIC, and name of regressor (with sign of slope) printed below nodes. Terminal nodes with means above and below value of 512.7 at root node are painted yellow and purple respectively. Second best split variable at root node is IMPANG.

```

DSC file: nhtsahic2.dsc
Training sample file: nhtsadatam.txt
Missing value code: NA
Records in data file start on line 2
48 N variables changed to S
Warning: B variables changed to C
D variable is HIC2
Number of records in data file: 3310
Length of longest entry in data file: 19
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Number of classes: 2
Training sample class proportions of D variable HIC2:
Class #Cases      Proportion
0      2999      0.91544567
1       277      0.08455433

```

Summary information for training sample of size 3276 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight
Levels of M variables are for missing values in associated variables

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	BARRIG	c			3	
2	BARSHP	c			21	
3	BARANG	p	0.000	330.0	360	14
4	BARDIA	s	191.0	1000.		2807
5	OCCWT	s	72.00	83.00		3265
6	OCCWT_	m			2	
:						
106	CRBANG	p	0.000	315.0	360	24
107	PDOF	p	0.000	345.0	360	23
108	CARANG	p	0.000	99.00	360	991
109	VEHOR	p	0.000	90.00	360	995
110	RSTFRT	c			3	
111	HIC2	d			2	

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/	#missing				
#cases	miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var

```

      3310      34      3310      5      0      0      48
    #P-var  #M-var  #B-var  #C-var  #I-var
          6      42      0      11      0

```

Number of cases used for training: 3276

Number of cases excluded due to 0 W or missing D variable: 34

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Simple node models

Estimated priors

Misclassification cost matrix from file costmatrix.txt:

```

Predicted   True class
           0           1
0           0.000E+00  1.000E+01
1           1.000E+00  0.000E+00

```

Class Altered prior

0 0.5198

1 0.4802

Univariate split highest priority

Interaction and linear splits 2nd and 3rd priorities

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 12

Minimum node sample size: 3

Ranks of variables and their 1-df chi-squared values at root node

```

  1  0.4697E+03  COLMEC
  2  0.3441E+03  YEAR
  3  0.3169E+03  RSTFRT
  4  0.2902E+03  HS
  :
64  0.8370E+00  BARDIA
65  0.1921E-01  IMPANG

```

Size and CV mean cost and SE of subtrees:

Tree	#Tnodes	Mean Cost	SE(Mean)	BSE(Mean)	Median Cost	BSE(Median)
1	51	3.480E-01	2.721E-02	1.915E-02	3.582E-01	1.993E-02
2	50	3.480E-01	2.721E-02	1.915E-02	3.582E-01	1.993E-02
:						
23	23	2.576E-01	2.047E-02	2.144E-02	2.470E-01	3.640E-02
24*	22	2.561E-01	2.046E-02	2.365E-02	2.424E-01	3.951E-02
25	20	2.582E-01	2.047E-02	2.371E-02	2.500E-01	3.756E-02
26	11	2.616E-01	2.069E-02	2.143E-02	2.530E-01	3.319E-02
27**	10	2.592E-01	2.027E-02	1.991E-02	2.546E-01	2.753E-02

28	9	2.637E-01	2.069E-02	2.090E-02	2.473E-01	3.482E-02
29++	8	2.625E-01	2.049E-02	2.150E-02	2.443E-01	3.918E-02
30	7	2.643E-01	2.029E-02	2.509E-02	2.550E-01	4.066E-02
31	6	2.842E-01	2.076E-02	2.261E-02	2.855E-01	3.314E-02
32	5	2.824E-01	1.951E-02	2.107E-02	2.886E-01	3.030E-02
33	4	3.092E-01	2.163E-02	2.774E-02	2.805E-01	2.103E-02
34	2	3.700E-01	2.630E-02	2.480E-02	3.572E-01	2.321E-02
35	1	8.455E-01	4.861E-02	4.656E-03	8.537E-01	7.280E-03

0-SE tree based on mean is marked with * and has 22 terminal nodes

0-SE tree based on median is marked with + and has 22 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree same as + tree

** tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Node cost is node misclassification cost divided by number of cases in node

Node label	Total cases	Train cases	Predicted class	Node cost	Split variables	Interacting variable
1	3276	3276	0	8.369E-01	COLMEC	
2	2596	2596	0	2.707E-01	RSTFRT	
4	602	602	0	6.576E-01	BARSHP	
8T	64	64	1	5.877E-01	BODY	
9T	538	538	0	2.576E-01	CRBANG	
5	1994	1994	0	1.539E-01	IMPANG	
10	74	74	1	8.292E-01	YEAR	
20T	50	50	1	7.522E-01	HW	
21T	24	24	0	0.000E+00	-	
11T	1920	1920	0	9.794E-02	IMPPNT	
3	680	680	1	6.899E-01	RSTFRT	
6	598	598	1	6.554E-01	BARSHP	
12T	510	510	1	6.074E-01	TKSURF	
13	88	88	0	5.623E-01	CURBWT	
26T	81	81	0	2.444E-01	YEAR +VEHWID	
27T	7	7	1	5.656E-01	BX19	
7	82	82	0	4.828E-01	PDOF	
14T	79	79	0	2.506E-01	TKSURF	
15T	3	3	1	3.299E-01	-	

Number of terminal nodes of final tree: 10

Total number of nodes of final tree: 19

Second best split variable (based on curvature test) at root node is YEAR

Classification tree:

For categorical variable splits, values not in training data go to the right

Node 1: COLMEC = "BWU", "NA", "NAP", "UNK"

Node 2: RSTFRT = "0"

Node 4: BARSHP = "LCB"

Node 8: 1

Node 4: BARSHP /= "LCB"

Node 9: 0

Node 2: RSTFRT /= "0"

Node 5: IMPANG in [269, 287]

Node 10: YEAR <= 2010.5000

Node 20: 1

Node 10: YEAR > 2010.5000 or NA

Node 21: 0

Node 5: IMPANG not in [269, 287] or NA

Node 11: 0

Node 1: COLMEC /= "BWU", "NA", "NAP", "UNK"

Node 3: RSTFRT = "0"

Node 6: BARSHP = "FLB", "LCB", "LUM"

Node 12: 1

Node 6: BARSHP /= "FLB", "LCB", "LUM"

Node 13: CURBWT <= 1529.0000 or NA

Node 26: 0

Node 13: CURBWT > 1529.0000

Node 27: 1

Node 3: RSTFRT /= "0"

Node 7: PDOF in [-1, 1]

Node 14: 0

Node 7: PDOF not in [-1, 1] or NA

Node 15: 1

Predictor means below are means of cases with no missing values.

Node 1: Intermediate node

A case goes into Node 2 if COLMEC = "BWU", "NA", "NAP", "UNK"

COLMEC mode = "UNK"

Class	Number	Posterior
0	2999	0.9154E+00
1	277	0.8455E-01

Misclassification cost = 2770

Predicted class is 0

```

Node 2: Intermediate node
A case goes into Node 4 if RSTFRT = "0"
RSTFRT mode = "1"
Class      Number  Posterior
0          2525  0.9727E+00
1           71  0.2735E-01
Misclassification cost =                      710
Predicted class is 0
-----

```

```

Node 4: Intermediate node
A case goes into Node 8 if BARSHP = "LCB"
BARSHP mode = "POL"
Class      Number  Posterior
0          562  0.9336E+00
1          40  0.6645E-01
Misclassification cost =                      400
Predicted class is 0
-----

```

```

:
Node 7: Intermediate node
A case goes into Node 14 if PDOF in [-1, 1]
PDOF mean = 5.1219512
Class      Number  Posterior
0          78  0.9512E+00
1           4  0.4878E-01
Misclassification cost =                      40
Predicted class is 0
-----

```

```

Node 14: Terminal node
Class      Number  Posterior
0          77  0.9747E+00
1           2  0.2532E-01
Misclassification cost =                      20
Predicted class is 0
-----

```

```

Node 15: Terminal node
Class      Number  Posterior
0           1  0.3333E+00
1           2  0.6667E+00
Misclassification cost =                      1
Predicted class is 1
-----

```

Classification matrix for training sample:

Predicted	True class	
class	0	1
0		
1		

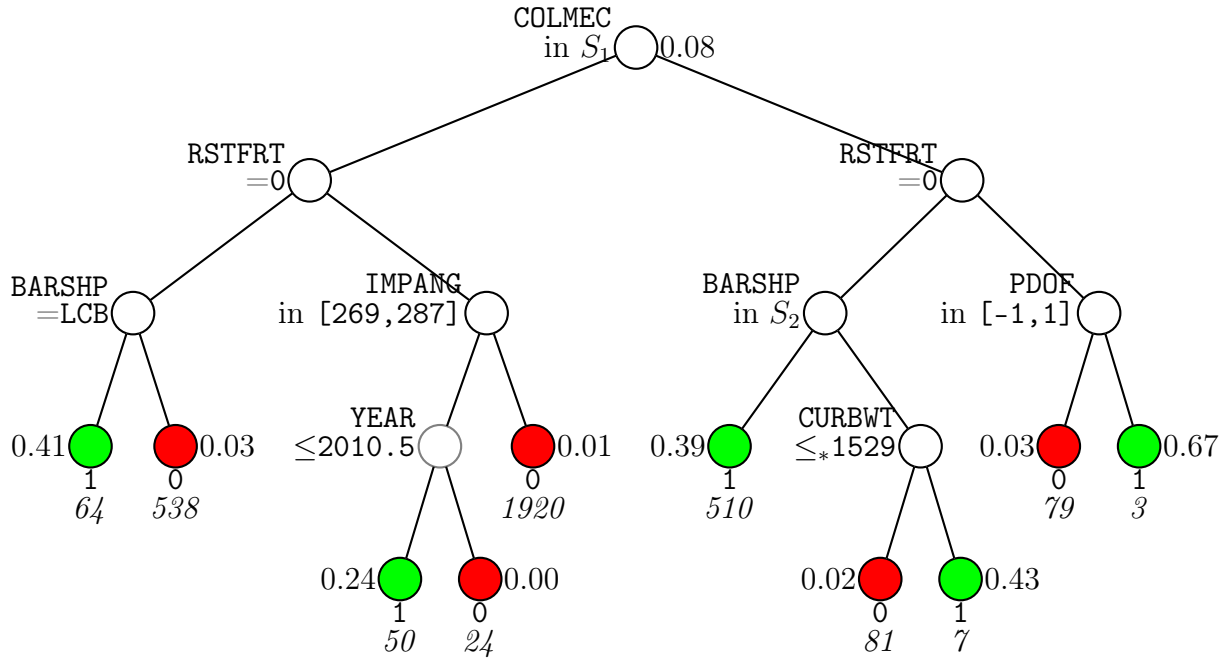


Figure 16: GUIDE v.45.2 0.250-SE classification tree for predicting HIC2 using estimated priors and specified misclassification costs. At each split, an observation goes to the left branch if and only if the condition is satisfied. Symbol ' $\leq_*$ ' stands for ' $\leq$ or missing'. $S_1 = \{\text{BWU, NA, NAP, UNK}\}$. $S_2 = \{\text{FLB, LCB, LUM}\}$. Predicted class and sample size (in *italics*) printed below each terminal node; class sample proportion for HIC2 = 1 beside node. Second best split variable at root node is YEAR.

0	2605	37
1	394	240
Total	2999	277

Number of cases used for tree construction: 3276

Number misclassified: 431

Resubstitution estimate of mean misclassification cost: 0.23321123

Observed and fitted values are stored in class10.fit

LaTeX code for tree is in class10.tex

8.2.2 Linear splits

To use P variables for classification, we need to convert their values to sines and cosines. The data after conversion are in the file `sincosdata.txt`. The created variables have names beginning with “s.” (for `sine`) and “c.” (for `cosine`). For example, the values of the variable `s.BARANG` are $\sin(2\pi \times \text{BARANG}/360)$. The contents of the DSC file `sincosdschic2.txt` for this task are given below.

```
sincosdata.txt
NA
2
1 BARRIG b
2 BARSHP b
3 BARANG p 360
4 BARDIA n
5 OCCWT n
6 OCCWT_ m
7 DUMSIZ b
8 HH n
9 HH_ m
10 HW n
11 HW_ m
12 HR n
13 HR_ m
14 HS n
15 HS_ m
16 CD n
17 CD_ m
18 CS n
19 CS_ m
20 AD n
21 AD_ m
22 HD n
23 HD_ m
24 KD n
25 KD_ m
26 HB n
27 HB_ m
28 NB n
29 NB_ m
30 CB n
31 CB_ m
32 KB n
33 SEPOSN b
34 HIC x
35 TKSURF b
```

36 TKCOND b
37 CLSSPD n
38 CLSSPD_ m
39 IMPANG p 360
40 OFFSET n
41 IMPPNT n
42 YEAR n
43 BODY b
44 ENGINE b
45 ENGDSP n
46 ENGDSP_ m
47 TRANSM b
48 VEHTWT n
49 VEHTWT_ m
50 CURBWT n
51 WHLBAS n
52 WHLBAS_ m
53 VEHLEN n
54 VEHLEN_ m
55 VEHWID n
56 VEHWID_ m
57 VEHCG n
58 VEHCG_ m
59 COLMEC b
60 BX1 n
61 BX1_ m
62 BX2 n
63 BX2_ m
64 BX3 n
65 BX3_ m
66 BX4 n
67 BX4_ m
68 BX5 n
69 BX5_ m
70 BX6 n
71 BX6_ m
72 BX7 n
73 BX7_ m
74 BX8 n
75 BX8_ m
76 BX9 n
77 BX9_ m
78 BX10 n
79 BX10_ m
80 BX11 n
81 BX11_ m

```
82 BX12 n
83 BX12_ m
84 BX13 n
85 BX13_ m
86 BX14 n
87 BX14_ m
88 BX15 n
89 BX15_ m
90 BX16 n
91 BX16_ m
92 BX17 n
93 BX17_ m
94 BX18 n
95 BX18_ m
96 BX19 n
97 BX19_ m
98 BX20 n
99 BX20_ m
100 BX21 n
101 BX21_ m
102 VEHSPD n
103 VEHSPD_ m
104 CRBANG p 360
105 PDOF p 360
106 CARANG p 360
107 VEHOR p 360
108 RSTFRT b
109 HIC2 d
110 HIC3 x
111 s.BARANG n
112 c.BARANG n
113 s.IMPANG n
114 c.IMPANG n
115 s.CRBANG n
116 c.CRBANG n
117 s.PDOF n
118 c.PDOF n
119 s.CARANG n
120 c.CARANG n
121 s.VEHOR n
122 c.VEHOR n
```

8.2.3 Results

The output using LDA splits and LDA node models follow.

```

Classification tree
Linear discriminant splits and LDA node models
Pruning by cross-validation
DSC file: sincosdscHIC2.txt
Training sample file: sincosdata.txt
Missing value code: NA
Records in data file start on line 2
Warning: P variables excluded from LDA and QDA models
D variable is HIC2
M variables changed to B
Number of records in data file: 3310
Length of longest entry in data file: 20
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Number of classes: 2
Training sample class proportions of D variable HIC2:
Class #Cases      Proportion
0      2999      0.91544567
1       277      0.08455433

```

Summary information for training sample of size 3276 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	BARRIG	b			3	
2	BARSHP	b			21	
4	BARDIA	n	1.9100E+02	1000.		2807
:						
108	RSTFRT	b			3	
109	HIC2	d			2	
111	s.BARANG	n	-5.0000E-01	1.000		14
112	c.BARANG	n	6.1232E-17	1.000		14
113	s.IMPANG	n	-1.0000E+00	1.000		4
114	c.IMPANG	n	-1.0000E+00	1.000		4
115	s.CRBANG	n	-1.0000E+00	1.000		24
116	c.CRBANG	n	-9.3969E-01	1.000		24
117	s.PDOF	n	-1.0000E+00	1.000		23
118	c.PDOF	n	-1.0000E+00	1.000		23
119	s.CARANG	n	0.0000E+00	0.9877		991


```

120 c.CARANG      n -1.5643E-01  1.000      991
121 s.VEHOR       n  0.0000E+00  1.000      995
122 c.VEHOR       n  6.1232E-17  1.000      995
===== Constructed variables =====
123 BARRIG.R      f  0.000      1.000
124 BARRIG.U      f  0.000      1.000
125 BARSHP.134    f  0.000      1.000
126 BARSHP.488    f  0.000      1.000
:
354 BX21.NA      f  0.000      1.000
355 VEHSPD.NA     f  0.000      1.000
356 s.BARANG.NA   f  0.000      1.000
357 c.BARANG.NA   f  0.000      1.000
358 s.IMPANG.NA   f  0.000      1.000
359 c.IMPANG.NA   f  0.000      1.000
360 s.CRBANG.NA   f  0.000      1.000
361 c.CRBANG.NA   f  0.000      1.000
362 s.PDOF.NA     f  0.000      1.000
363 c.PDOF.NA     f  0.000      1.000
364 s.CARANG.NA   f  0.000      1.000
365 c.CARANG.NA   f  0.000      1.000
366 s.VEHOR.NA    f  0.000      1.000
367 c.VEHOR.NA    f  0.000      1.000

```

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/	#missing					
#cases	miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var	
3310	34	3310	8	60	244	0	
#P-var	#M-var	#B-var	#C-var	#I-var			
0	0	53	0	0			

Number of constant F variables: 1

Number of dummy variables (excl. missing-value indicators) created: 185

Number of missing-value indicators created: 60

Number of cases used for training: 3276

Number of cases excluded due to 0 W or missing D variable: 34

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Estimated priors

Unit misclassification costs

Max number of splits: 100

Maximum number of split levels: 15

Minimum node sample size: 29

Size and CV mean cost and SE of subtrees:

Tree	#Tnodes	Mean Cost	SE(Mean)	BSE(Mean)	Median Cost	BSE(Median)
1	23	9.646E-02	5.158E-03	3.529E-03	9.604E-02	2.221E-03
2	20	9.646E-02	5.158E-03	3.529E-03	9.604E-02	2.221E-03
3	19	9.188E-02	5.047E-03	2.306E-03	9.007E-02	3.784E-03
4	18	9.096E-02	5.024E-03	1.852E-03	9.146E-02	2.930E-03
5	15	9.005E-02	5.001E-03	2.758E-03	9.007E-02	3.124E-03
6	14	8.974E-02	4.994E-03	3.018E-03	9.007E-02	3.178E-03
7	9	8.791E-02	4.947E-03	2.959E-03	8.855E-02	3.878E-03
8**	6	8.333E-02	4.829E-03	2.240E-03	8.396E-02	4.838E-03
9++	2	8.730E-02	4.932E-03	3.248E-03	8.397E-02	3.987E-03
10	1	9.707E-02	5.172E-03	3.415E-03	9.756E-02	4.450E-03

0-SE tree based on mean is marked with * and has 6 terminal nodes

0-SE tree based on median is marked with + and has 6 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree same as + tree

** tree same as + tree

** tree same as -- tree

* tree same as ** tree

* tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Node cost is node misclassification cost divided by number of cases in node

Node	Total	Misclassification
label	cases	cost
1	3276	8.489E-02
2	3183	8.271E-02
4	2473	9.925E-02
8	148	3.611E-01
16T	82	2.293E-01
17T	66	2.999E-01
9	2325	8.173E-02
18T	253	0.000E+00
19T	2072	8.741E-02
5T	710	2.509E-02
3T	93	0.000E+00

Number of terminal nodes of final tree: 6

Total number of nodes of final tree: 11

Classification tree with linear discriminant splits and LDA node models

```

Node 1: D1 <= 0.44816965E-01
Node 2: D2 <= 1.8490250
Node 4: D4 <= -0.36315036E-02
Node 8: D8 <= -0.64715070
Node 16: 19/82 misclassified
Node 8: D8 > -0.64715070
Node 17: 20/66 misclassified
Node 4: D4 > -0.36315036E-02
Node 9: D9 <= -2.9160169
Node 18: 0/253 misclassified
Node 9: D9 > -2.9160169
Node 19: 183/2072 misclassified
Node 2: D2 > 1.8490250
Node 5: 18/710 misclassified
Node 1: D1 > 0.44816965E-01
Node 3: 0/93 misclassified

```

Predictor means below are means of cases with no missing values.

Node 1: Intermediate node

A case goes into Node 2 if D1 <= 0.44816965E-01 where the coefficients of D1 are the coefficients of the 1st crimcoord below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
HH	-0.26082179E-04	-0.27612200E-04	360.68138
HW	0.20159169E-04	0.32088560E-04	601.14091
HR	-0.16556541E-04	-0.15234824E-04	223.20486
:			
HD.NA	-0.17015129E-01	-0.59529266E-02	0.60439560E-01
IMPPNT.NA	0.59105812E-02	0.24570562E-02	0.51678877
CURBWT.NA	-0.15752245E-02	-0.12535642E-02	0.87118437
BX19.NA	-0.74896802E-02	0.24272996E-02	0.16483516
BX20.NA	0.10360932E-01	-0.42393704E-02	0.16452991
s.CARANG.NA	-0.22888994E-02	0.96496985E-03	0.30250305
c.CARANG.NA	-0.22888994E-02	0.96496985E-03	0.30250305
s.VEHOR.NA	0.16756736E-02	-0.11686222E-02	0.30372405
c.VEHOR.NA	0.16756736E-02	-0.11686222E-02	0.30372405

Class	Crim1Mean	Crim1Variance	Prior
0	0.291610E-01	0.230109E-03	0.91545
1	-0.202517E-02	0.230109E-03	0.08455

Node 2: Intermediate node

A case goes into Node 4 if D2 <= 1.8490250 where the coefficients of D2 are the coefficients of the 1st crimcoord below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
HH	-0.18283492E-04	0.37090012E-04	361.65711
HW	0.97458529E-05	-0.28266264E-04	599.65925
HR	-0.11262926E-04	0.56739314E-04	222.45743
:			
IMPPNT.NA	0.70009329E-02	0.18940052E-02	0.52466227
CURBWT.NA	0.14010492E-02	-0.28062937E-02	0.87998743
BX1.NA	0.62704176E-02	0.43320642E-02	0.15519950
BX19.NA	-0.74606039E-02	0.24567888E-01	0.15739868
BX20.NA	0.55646220E-02	-0.31285166E-01	0.15708451
s.CARANG.NA	-0.34155275E-02	0.18553556E-02	0.30788564
c.CARANG.NA	-0.34155275E-02	0.18553556E-02	0.30788564
s.VEHOR.NA	0.42417812E-02	-0.44439240E-02	0.30914232
c.VEHOR.NA	0.42417812E-02	-0.44439240E-02	0.30914232

Class	Crim1Mean	Crim1Variance	Prior
0	1.83870	0.236742E-03	0.91298
1	1.80745	0.236742E-03	0.08702

Node 4: Intermediate node

A case goes into Node 8 if D4 <= -0.36315036E-02 where the coefficients of D4 are the coefficients of the 1st crimcoord below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
HW	0.68509332E-05	-0.23218602E-04	584.71009
HR	-0.19941354E-04	-0.52771962E-04	217.38686
HS	0.50748455E-04	-0.10019163E-03	297.49170
CS	-0.42544670E-04	-0.87227140E-05	304.67124
:			
RSTFRT.1	0.15745298E-01	0.55253049E-02	0.54346947
HS.NA	-0.23738869E-02	0.45846082E-02	0.73999191E-01
HD.NA	-0.16121582E-02	-0.84382252E-02	0.73594824E-01
IMPPNT.NA	0.30091921E-02	-0.45152172E-02	0.58592802
CURBWT.NA	-0.68622555E-02	-0.29935730E-02	0.91872220
BX19.NA	-0.13671335E-01	0.18293443E-01	0.16215123
BX20.NA	0.18542804E-01	-0.21064303E-01	0.16174687

Class	Crim1Mean	Crim1Variance	Prior
0	0.348710E-01	0.307940E-03	0.89001
1	0.344502E-02	0.307940E-03	0.10999

Node 8: Intermediate node

A case goes into Node 16 if D8 <= -0.64715070 where the coefficients of D8 are the coefficients of the 1st crimcoord below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
CD	-0.11617182E-02	0.11154176E-02	558.12687
CS.NA	0.20856652	0.21722404	0.81081081E-01

Class	Crim1Mean	Crim1Variance	Prior
0	-0.603613	0.628197E-02	0.41892
1	-0.651562	0.628197E-02	0.58108

Node 16: Terminal node

Cases classified by LDA on 1st crimcoord with coefs below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
TKSURF.DOS	0.51266023	-0.50000000	0.24390244E-01
ENGINE.S6IF	0.51266023	0.50000000	0.24390244E-01

Class	Crim1Mean	Crim1Variance	Prior
0	0.891583E-01	0.108556E-01	0.28049
1	0.00000	0.108556E-01	0.71951

Node 17: Terminal node

Cases classified by LDA on 1st crimcoord with coefs below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
BARDIA	-0.32814153	0.51703301	216.66667
TRANSM.MF	0.25421953	0.16134364	0.21212121

Class	Crim1Mean	Crim1Variance	Prior
0	-71.0825	0.133512E-01	0.59091
1	-70.9870	0.133512E-01	0.40909

Node 9: Intermediate node

A case goes into Node 18 if D9 <= -2.9160169 where the coefficients of D9 are the coefficients of the 1st crimcoord below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
HW	0.88247264E-05	0.91092030E-04	588.62545
HR	0.17950110E-04	-0.68764149E-05	220.03499
HS	-0.57206862E-04	0.72666912E-04	302.03050
:			
IMPPNT.NA	-0.96667873E-02	0.19714749E-01	0.56387097
BX1.NA	-0.16071609E-01	0.95530151E-02	0.15741935
BX19.NA	-0.57664540E-02	0.11290297E-02	0.16043011
BX20.NA	0.17985659E-01	0.22360619E-03	0.16000000

Class	Crim1Mean	Crim1Variance	Prior
0	-2.89873	0.380789E-03	0.92000
1	-2.87276	0.380789E-03	0.08000

Node 18: Terminal node

Cases classified by LDA on 1st crimcoord with coefs below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
DUMSIZ.UN	0.61621885E-01	-0.17497988	0.39525692E-02

HH_.B	0.24110964	-0.22997355	0.11857708E-01
HW_.D	-0.72232482E-02	-0.39995400E-01	0.98418972
HR_.D	-0.49885955E-01	0.11998620	0.98814229
:			
ENGDSP.NA	-0.25563576E-01	0.80589233E-15	0.15810277E-01
WHLBAS.NA	0.18959178E-01	-0.14998275E-01	0.79051383E-02
VEHLEN.NA	0.18959178E-01	-0.14998275E-01	0.79051383E-02
VEHWID.NA	-0.29613193E-01	0.36394333E-15	0.15810277E-01
VEHCG.NA	-0.14013121	0.70661542E-15	0.15810277E-01

Class	Crim1Mean	Crim1Variance	Prior
0	0.450076	0.398406E-02	0.99209
1	0.450076	0.398406E-02	0.00791

Node 19: Terminal node

Cases classified by LDA on 1st crimcoord with coeffs below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
HW	-0.12968685E-04	0.28863139E-04	581.90505
HR	-0.20887869E-04	-0.12800171E-03	217.46098
HS	0.59987332E-04	0.31142132E-03	295.26527
CS	-0.18317599E-04	-0.12019715E-03	307.01234
:			
BX20_.D	0.25809426E-01	-0.91565862E-02	0.83928571
IMPPNT.NA	0.89226448E-02	0.52268840E-02	0.62113900
BX1.NA	0.26774011E-01	-0.43044841E-02	0.15781853
BX19.NA	0.54807095E-02	-0.78635445E-02	0.16119691
BX20.NA	-0.25809426E-01	0.91565862E-02	0.16071429

Class	Crim1Mean	Crim1Variance	Prior
0	3.37049	0.434208E-03	0.91120
1	3.34592	0.434208E-03	0.08880

Node 5: Terminal node

Cases classified by LDA on 1st crimcoord with coeffs below:

Variable	Crimcoord1 coef	Crimcoord2 coef	Variable mean
ENGINE.4CIF	0.23089070	-0.24340008	0.12676056E-01
ENGINE.OTHR	0.26092209	0.24158366	0.11267606E-01

Class	Crim1Mean	Crim1Variance	Prior
0	0.521075E-02	0.135158E-02	0.99296
1	0.983626E-01	0.135158E-02	0.00704

Node 3: Terminal node

Pure node

All cases classified as "0"

Classification matrix for training sample:

Predicted	True class	
class	0	1
0	2956	199
1	43	78
Total	2999	277

Number of cases used for tree construction: 3276

Number misclassified: 242

Resubstitution estimate of mean misclassification cost: 0.73870574E-01

Observed and fitted values are stored in lda.fit

LaTeX code for tree is in lda.tex

R code is stored in lda.r

Discriminant coefficients are stored in lda.coefs

R function for plotting data are stored in ldaplots.r

Plot data are stored in ldadata.txt

The LDA classification tree is shown in Figure 17 and the data plots in Figure 18.

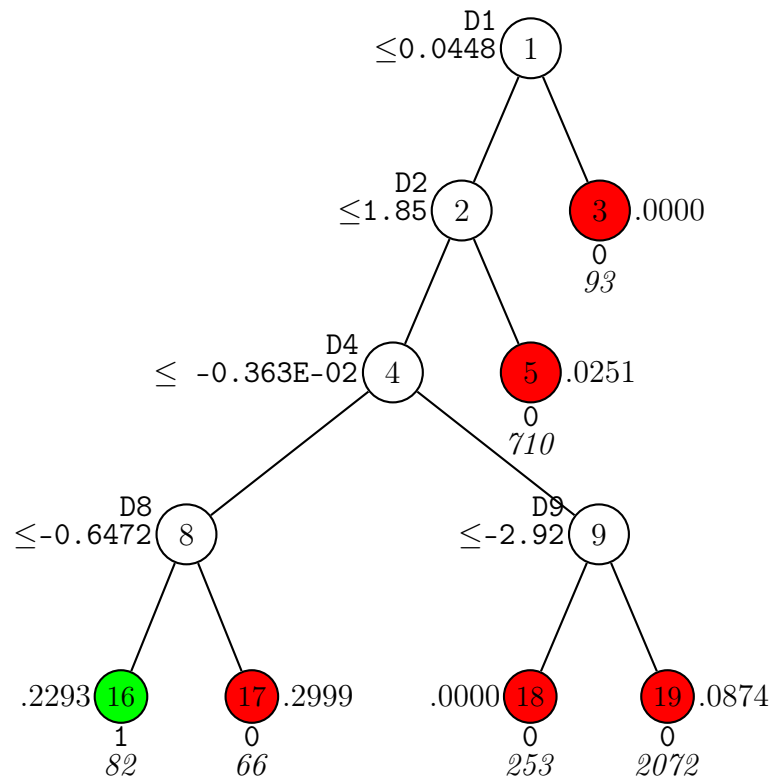


Figure 17: GUIDE v.46.1 0.250-SE classification tree for predicting HIC2 using linear discriminant splits and linear discriminant node models. At each split, an observation goes to the left branch if and only if the condition is satisfied. Class minimizing misclassification cost and sample size (in *italics*) printed below each terminal node; mean misclassification cost beside node.

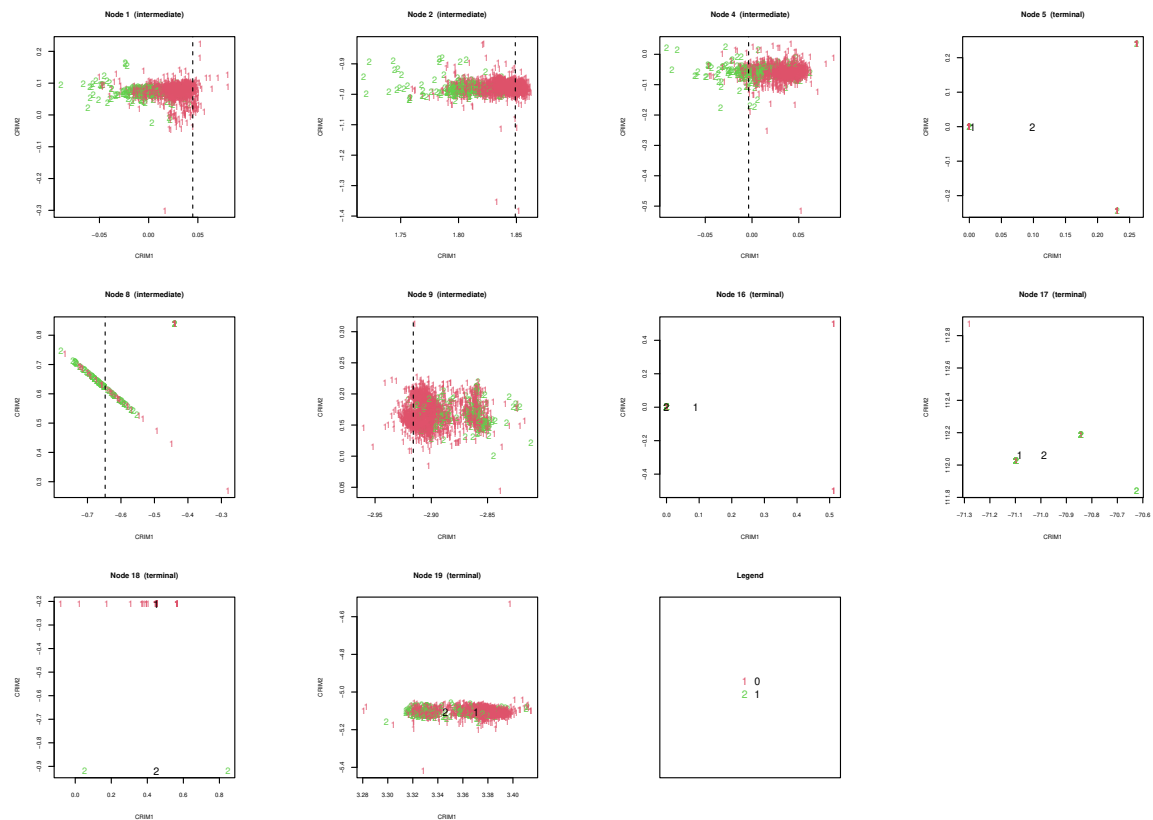


Figure 18: Data and splits in nodes of tree in Figure 17.

9 Poisson regression: solder data

We use a data set on printed circuit board soldering to show how GUIDE fits Poisson regression models. The data were analyzed in [Chambers and Hastie \(1992\)](#) and are given in `solder.dat`. The DSC file `solder.dsc` uses the `b` descriptor for the 5 categorical variables:

```
solder.dat
NA
1
1, skips, d
2, opening, b
3, solder, b
4, mask, b
5, padtype, b
6, panel, b
```

9.1 Piecewise-constant: solder data

9.1.1 Input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: cons.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: cons.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 3
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple polynomial, 3: constant ([1:3], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: solder.dsc
Reading DSC file ...
```

```

Training sample file: solder.dat
Missing value code: NA
Records in data file start on line 1
Warning: B variables changed to C
D variable is skips
Reading data file ...
Number of records in data file: 720
Length of longest entry in data file: 6
Checking for missing values ...
Finished checking
Assigning integer codes to values of 5 categorical variables
Re-checking data ...
Assigning codes to missing values, if any ...
Data checks complete
Number of cases with positive D values: 478
Rereading data ...
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
        720         0         0        0        0        0        0
      #P-var  #M-var  #B-var  #C-var  #I-var
         0         0         0        5        0
No offset variable in data file.
Number of cases used for training: 720
Finished reading data file
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): cons.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: cons.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: cons.r
Input rank of top variable to split root node ([1:5], <cr>=1):
Input file is created!

```

The tree is shown in Figure 19, which is quite large. One way to reduce the size of the tree is to fit a more complex Poisson regression model in each node.

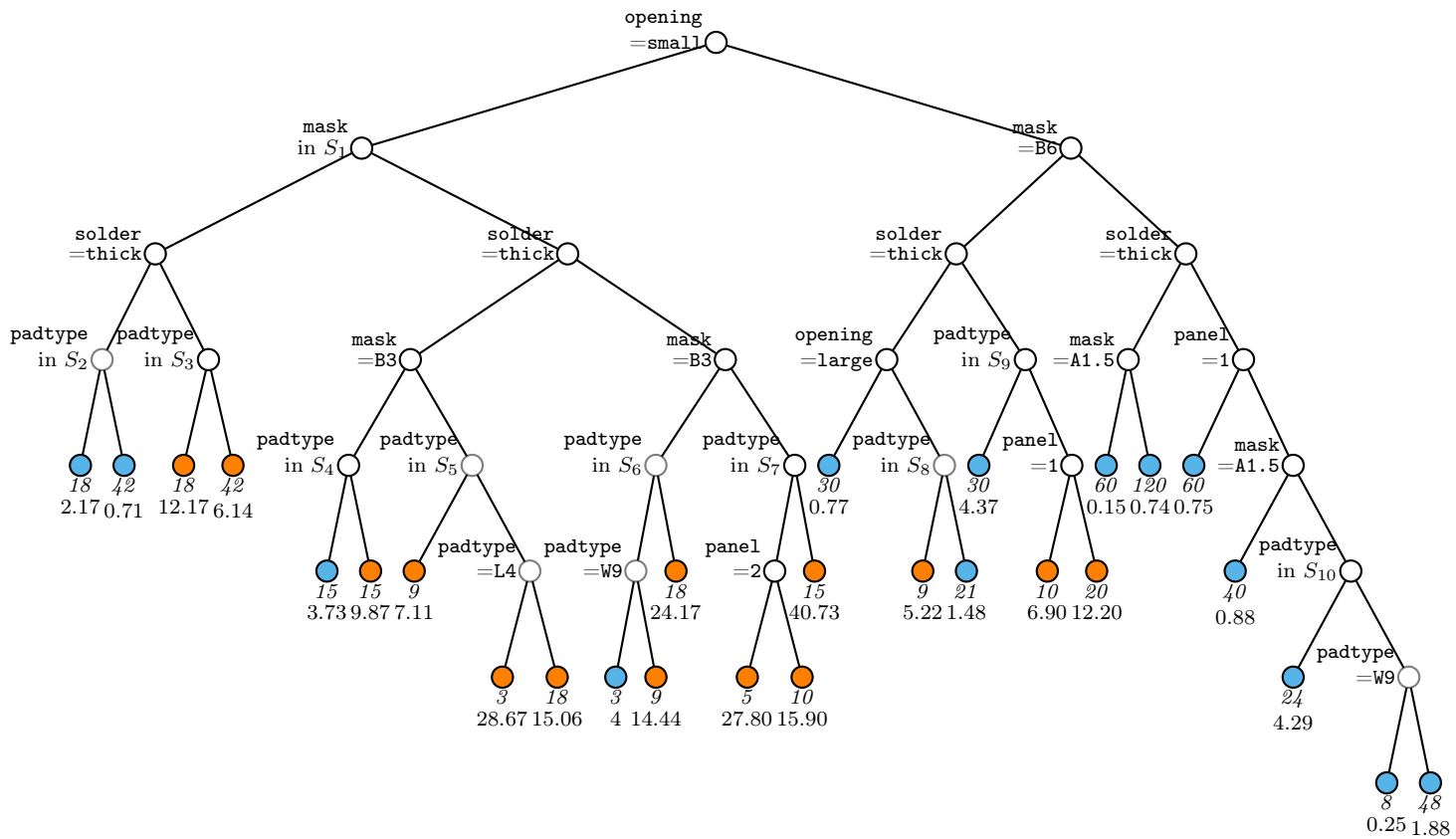


Figure 19: GUIDE v.46.1 0.250-SE piecewise-constant Poisson regression tree for predicting **skips**. At each split, an observation goes to the left branch if and only if the condition is satisfied. $S_1 = \{A1.5, A3\}$. $S_2 = \{D7, L4, L8\}$. $S_3 = \{D4, D7, L4\}$. $S_4 = \{D6, L6, L9, W4, W9\}$. $S_5 = \{L6, L9, W9\}$. $S_6 = \{L6, L7, L9, W9\}$. $S_7 = \{L6, L7, L8, L9, W9\}$. $S_8 = \{D7, L4, L8\}$. $S_9 = \{D6, L6, L7, L9, W9\}$. $S_{10} = \{D4, D7, L4\}$. Splits at nodes drawn with gray circles are not statistically significant. Sample size (in *italics*) and mean of **skips** printed below nodes. Terminal nodes with means above and below value of 4.97 at root node are painted orange and skyblue respectively. Second best split variable at root node is **mask**.

9.2 Multiple linear: solder data

Now we construct a tree where each node is fitted with a Poisson model containing only the main effects. This is where the “B” descriptor in `solder.dsc` is for.

9.2.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: mul.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: mul.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 3
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple polynomial, 3: constant ([1:3], <cr>=3): 1
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: solder.dsc
Reading DSC file ...
Training sample file: solder.dat
Missing value code: NA
Records in data file start on line 1
D variable is skips
Reading data file ...
Number of records in data file: 720
Length of longest entry in data file: 6
Checking for missing values ...
Finished checking
Assigning integer codes to values of 5 categorical variables
Re-checking data ...
Assigning codes to missing values, if any ...
Data checks complete
Number of cases with positive D values: 478
GUIDE will try to create the variables in the description file.
```

```

If it is unsuccessful, please create the columns yourself...
Creating dummy variables ...
Rereading data ...
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      720      0      0      0      0      0      17      0
      #P-var  #M-var  #B-var  #C-var  #I-var
      0      0      5      0      0
Number of dummy variables created: 17
No offset variable in data file.
Number of cases used for training: 720
Finished reading data file
Constant fitted to cases with missing values in regressor variables
Default max. number of split levels: 8
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): mul.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: mul.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: mul.r
Input rank of top variable to split root node ([1:22], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < mul.in

```

9.2.2 Contents of mul.out

```

Poisson regression tree
No truncation of predicted values
Pruning by cross-validation
DSC file: solder.dsc
Training sample file: solder.dat
Missing value code: NA
Records in data file start on line 1
D variable is skips
Piecewise linear model
Number of records in data file: 720
Length of longest entry in data file: 6

```

Number of cases with positive D values: 478

Summary information for training sample of size 720

d=dependent, b=split and fit cat variable using indicator variables,
 c=split-only categorical, i=fit-only categorical (via indicators),
 s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
 m=missing-value flag variable, p=periodic variable, w=weight,
 z=offset variable

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	skips	d	0.000	48.00		
2	opening	b			3	
3	solder	b			2	
4	mask	b			4	
5	padtype	b			10	
6	panel	b			3	
===== Constructed variables =====						
7	opening.medium	f	0.000	1.000		
8	opening.small	f	0.000	1.000		
9	solder.thin	f	0.000	1.000		
10	mask.A3	f	0.000	1.000		
11	mask.B3	f	0.000	1.000		
12	mask.B6	f	0.000	1.000		
13	padtype.D6	f	0.000	1.000		
14	padtype.D7	f	0.000	1.000		
15	padtype.L4	f	0.000	1.000		
16	padtype.L6	f	0.000	1.000		
17	padtype.L7	f	0.000	1.000		
18	padtype.L8	f	0.000	1.000		
19	padtype.L9	f	0.000	1.000		
20	padtype.W4	f	0.000	1.000		
21	padtype.W9	f	0.000	1.000		
22	panel.2	f	0.000	1.000		
23	panel.3	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
720	0	0	0	0	0	17	0
#P-var	#M-var	#B-var	#C-var	#I-var			
0	0	5	0	0			

Number of dummy variables created: 17

No offset variable in data file.

Number of cases used for training: 720

Constant fitted to cases with missing values in regressor variables
 Pruning by v-fold cross-validation, with v = 10
 Selected tree is based on mean of CV estimates
 Number of SE's for pruned tree: 0.2500

Nodewise interaction tests on all variables
 Split values for N and S variables based on exhaustive search
 Max number of interaction splits: 100
 Maximum number of split levels: 8
 Minimum node sample size: 7
 Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	55	2.849E+00	1.872E-01	2.033E-01	2.772E+00	2.508E-01
2	53	2.849E+00	1.872E-01	2.033E-01	2.772E+00	2.508E-01
3	52	2.849E+00	1.872E-01	2.033E-01	2.772E+00	2.508E-01
:						
36	4	1.488E+00	8.070E-02	8.672E-02	1.449E+00	7.036E-02
37**	3	1.457E+00	7.447E-02	9.380E-02	1.343E+00	7.680E-02
38	2	1.527E+00	7.949E-02	9.597E-02	1.455E+00	6.790E-02
39	1	1.660E+00	8.239E-02	7.060E-02	1.651E+00	7.689E-02

0-SE tree based on mean is marked with * and has 3 terminal nodes
 0-SE tree based on median is marked with + and has 3 terminal nodes
 Selected-SE tree based on mean using naive SE is marked with **
 Selected-SE tree based on mean using bootstrap SE is marked with --
 Selected-SE tree based on median and bootstrap SE is marked with ++
 * tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is mean of skips in the node

Cases fit give the number of cases used to fit node

Deviance is mean residual deviance for all cases in node

Node label	Total cases	Cases fit	Matrix rank	Node D-mean	Node deviance	Split variable	Other variables
1	720	720	18	4.965E+00	1.610E+00	solder	
2T	360	360	17	2.481E+00	1.279E+00	mask	
3	360	360	17	7.450E+00	1.628E+00	opening	:mask
6T	120	120	15	1.636E+01	1.367E+00	padtype	
7T	240	240	16	2.996E+00	1.403E+00	mask	

Number of terminal nodes of final tree: 3

Total number of nodes of final tree: 5

Second best split variable (based on curvature test) at root node is opening

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: solder = "thick"

Node 2: skips sample mean = 2.4805556

Node 1: solder /= "thick"

Node 3: opening = "small"

Node 6: skips sample mean = 16.358333

Node 3: opening /= "small"

Node 7: skips sample mean = 2.9958333

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.

2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.

3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if solder = "thick"

solder mode = "thick"

Coefficients of regression function for log mean:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	-1.220	-12.81	0.6614E-15			
mask.A3	0.4282	5.674	0.2043E-07	0.000	0.2500	1.000
mask.B3	1.202	17.95	0.3926E-15	0.000	0.2500	1.000
mask.B6	1.866	29.58	0.000	0.000	0.2500	1.000
opening.medium	0.2585	3.884	0.1126E-03	0.000	0.3333	1.000
opening.small	1.893	35.31	0.7867E-15	0.000	0.3333	1.000
padtype.D6	-0.3687	-5.164	0.3144E-06	0.000	0.1000	1.000
padtype.D7	-0.9844E-01	-1.487	0.1374	0.000	0.1000	1.000
padtype.L4	0.2624	4.321	0.1774E-04	0.000	0.1000	1.000
padtype.L6	-0.6685	-8.525	0.000	0.000	0.1000	1.000
padtype.L7	-0.4902	-6.619	0.7177E-10	0.000	0.1000	1.000
padtype.L8	-0.2712	-3.907	0.1023E-03	0.000	0.1000	1.000
padtype.L9	-0.6365	-8.203	0.1231E-14	0.000	0.1000	1.000
padtype.W4	-0.1100	-1.657	0.9804E-01	0.000	0.1000	1.000
padtype.W9	-1.438	-13.80	0.000	0.000	0.1000	1.000
panel.2	0.3335	7.929	0.8472E-14	0.000	0.3333	1.000

```

panel.3          0.2544      5.947      0.4318E-08      0.000      0.3333      1.000
solder.thin      1.100      28.46      0.000      0.000      0.5000      1.000
If regressor has missing values, predicted value = 1.6024692
-----

```

Node 2: Terminal node

Coefficients of regression function for log mean:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	-2.431	-10.68	0.000			
mask.A3	0.4670	2.373	0.1820E-01	0.000	0.2500	1.000
mask.B3	1.831	11.01	0.000	0.000	0.2500	1.000
mask.B6	2.520	15.71	0.000	0.000	0.2500	1.000
opening.medium	0.8641	5.567	0.5228E-07	0.000	0.3333	1.000
opening.small	2.465	18.18	0.000	0.000	0.3333	1.000
padtype.D6	-0.3238	-2.034	0.4274E-01	0.000	0.1000	1.000
padtype.D7	0.1201	0.8480	0.3970	0.000	0.1000	1.000
padtype.L4	0.6985	5.534	0.6221E-07	0.000	0.1000	1.000
padtype.L6	-0.4002	-2.458	0.1448E-01	0.000	0.1000	1.000
padtype.L7	0.4167E-01	0.2887	0.7730	0.000	0.1000	1.000
padtype.L8	0.1481	1.052	0.2936	0.000	0.1000	1.000
padtype.L9	-0.5921	-3.426	0.6877E-03	0.000	0.1000	1.000
padtype.W4	-0.5466E-01	-0.3696	0.7119	0.000	0.1000	1.000
padtype.W9	-1.324	-5.886	0.9394E-08	0.000	0.1000	1.000
panel.2	0.2224	2.718	0.6895E-02	0.000	0.3333	1.000
panel.3	0.6825E-01	0.8049	0.4214	0.000	0.3333	1.000
solder.thin	0.000	0.000	1.000	0.000	0.000	0.000

If regressor has missing values, predicted value = 0.90848255

Node 3: Intermediate node

A case goes into Node 6 if opening = "small"
opening mode = "large"

Node 6: Terminal node

Coefficients of regression function for log mean:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	2.080	21.50	0.000			
mask.A3	0.3085	3.329	0.1202E-02	0.000	0.2500	1.000
mask.B3	1.050	12.84	0.000	0.000	0.2500	1.000
mask.B6	1.504	19.34	0.000	0.000	0.2500	1.000
opening.medium	0.000	0.000	1.000	0.000	0.000	0.000
opening.small	0.000	0.000	1.000	1.000	1.000	1.000
padtype.D6	-0.2534	-2.788	0.6302E-02	0.000	0.1000	1.000
padtype.D7	-0.1476	-1.671	0.9763E-01	0.000	0.1000	1.000
padtype.L4	0.8309E-01	0.9980	0.3206	0.000	0.1000	1.000
padtype.L6	-0.7187	-6.847	0.4730E-09	0.000	0.1000	1.000
padtype.L7	-0.6473	-6.315	0.6560E-08	0.000	0.1000	1.000
padtype.L8	-0.4255	-4.452	0.2127E-04	0.000	0.1000	1.000

```

padtype.L9      -0.6404      -6.262      0.8418E-08      0.000      0.1000      1.000
padtype.W4      -0.8668E-01  -0.9978      0.3207      0.000      0.1000      1.000
padtype.W9      -1.376      -10.29      0.000      0.000      0.1000      1.000
panel.2         0.3070      5.470      0.3070E-06      0.000      0.3333      1.000
panel.3         0.1850      3.210      0.1762E-02      0.000      0.3333      1.000
solder.thin     0.000      0.000      1.000      1.000      1.000      1.000
If regressor has missing values, predicted value = 2.7947375
-----
Node 7: Terminal node
Coefficients of regression function for log mean:
Regressor      Coefficient  t-stat      p-value      Minimum      Mean      Maximum
Constant       -0.3711      -1.947      0.5284E-01
mask.A3         0.8061      4.546      0.8965E-05      0.000      0.2500      1.000
mask.B3         1.008      5.849      0.1735E-07      0.000      0.2500      1.000
mask.B6         2.267      14.64      0.2731E-15      0.000      0.2500      1.000
opening.medium  0.1030      1.379      0.1692      0.000      0.5000      1.000
opening.small   0.000      0.000      1.000      0.000      0.000      0.000
padtype.D6      -0.7995      -4.649      0.5709E-05      0.000      0.1000      1.000
padtype.D7      -0.1915      -1.345      0.1800      0.000      0.1000      1.000
padtype.L4       0.2065      1.601      0.1108      0.000      0.1000      1.000
padtype.L6      -0.8201      -4.735      0.3894E-05      0.000      0.1000      1.000
padtype.L7      -0.7595      -4.477      0.1206E-04      0.000      0.1000      1.000
padtype.L8      -0.3606      -2.413      0.1662E-01      0.000      0.1000      1.000
padtype.L9      -0.6660      -4.051      0.7039E-04      0.000      0.1000      1.000
padtype.W4      -0.2254      -1.568      0.1183      0.000      0.1000      1.000
padtype.W9      -1.747      -7.027      0.2514E-10      0.000      0.1000      1.000
panel.2         0.5841      5.732      0.3190E-07      0.000      0.3333      1.000
panel.3         0.6931      6.931      0.4388E-10      0.000      0.3333      1.000
solder.thin     0.000      0.000      1.000      1.000      1.000      1.000
If regressor has missing values, predicted value = 1.0972224
-----
Observed and fitted values are stored in mul.fit
LaTeX code for tree is in mul.tex

```

Figure 20 shows the tree, which is much shorter than that in Figure 19. Node 3 is drawn in brown color to indicate that the split there is due to an interaction between two variables (opening and mask).

9.3 Offset variable: lung cancer data

We use a data set from an epidemiological study of the effect of public drinking water on cancer mortality in Missouri (Choi et al., 2005). The data file `lungcancer.txt` gives the number of deaths (`deaths`) from lung cancer among 115 counties (`county`) during the period 1972–1981 for both sexes (`sex`) and four age groups (`agegp`): 45–54, 55–64, 65–74, and over 75. The DSC file `lungcancer.dsc` below lists the variables

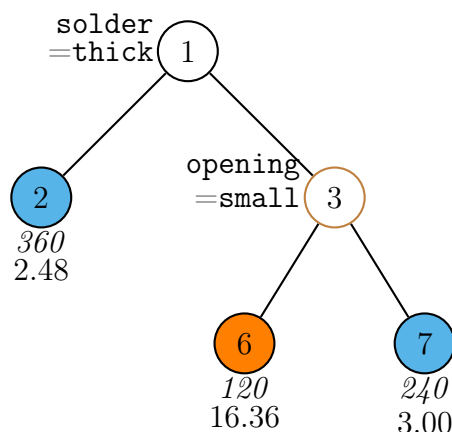


Figure 20: GUIDE v.46.1 0.250-SE multiple linear Poisson regression tree (constant fitted to incomplete cases in terminal nodes) for predicting `skips`. At each split, an observation goes to the left branch if and only if the condition is satisfied. Splits at nodes drawn with **brown** circles are interaction splits. Sample size (in *italics*) and mean of `skips` printed below nodes. Terminal nodes with means above and below value of 4.97 at root node are painted **orange** and **skyblue** respectively. Second best split variable at root node is `opening`.

together with the county population (`pop`) and the natural log of `pop` (`logpop`). The latter is specified as `z` to serve as an offset variable and `pop` is excluded (`x`) from the analysis. The contents of `lungcancer.dsc` are:

```

lungcancer.txt
NA
1
1 county c
2 sex b
3 agegp c
4 deaths d
5 pop x
6 logpop z

```

Our goal is to construct a Poisson regression tree for the gender-specific rate of lung cancer deaths, where rate is the expected number of deaths in a county divided by its population size for each gender. That is, letting μ denote the expected number of gender-specific deaths in a county, we fit this model in each node of the tree:

$$\log(\mu/\text{pop}) = \beta_0 + \beta_1 I(\text{sex} = \text{M}).$$

This is achieved by fitting a linear Poisson regression model with `sex` as `b` so that its dummy indicator variable serves as a linear predictor in the Poisson node models.

9.3.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: poi.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: poi.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 3
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple polynomial, 3: constant ([1:3], <cr>=3): 1
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: lungcancer.dsc
Reading DSC file ...
Training sample file: lungcancer.txt
Missing value code: NA
Records in data file start on line 1
D variable is deaths
Reading data file ...
Number of records in data file: 920
Length of longest entry in data file: 8
Checking for missing values ...
Finished checking
Assigning integer codes to values of 3 categorical variables
Re-checking data ...
Assigning codes to missing values, if any ...
Data checks complete
Number of cases with positive D values: 869
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...

```

```

Creating dummy variables ...
Rereading data ...
Numbers of N, F, S, and P below exclude constant variables
      Total #cases w/ #missing
      #cases miss. D ord. vals #X-var #N-var #F-var #S-var
      920      0      0      0      1      0      1      0
      #P-var #M-var #B-var #C-var #I-var
      0      0      1      2      0
Number of dummy variables created: 1
Offset variable in column: 6
Number of cases used for training: 920
Finished reading data file
Constant fitted to cases with missing values in regressor variables
Default max. number of split levels: 8
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): poi.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: poi.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: poi.r
Input rank of top variable to split root node ([1:4], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < poi.in

```

9.3.2 Results

```

Poisson regression tree
No truncation of predicted values
Pruning by cross-validation
DSC file: lungcancer.dsc
Training sample file: lungcancer.txt
Missing value code: NA
Records in data file start on line 1
D variable is deaths
Piecewise linear model
Number of records in data file: 920
Length of longest entry in data file: 8

```

Number of cases with positive D values: 869

Summary information for training sample of size 920

d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight,
z=offset variable

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	county	c			115	
2	sex	b			2	
3	agegp	c			4	
4	deaths	d	0.000	1046.		
6	logpop	z	4.828	10.96		
===== Constructed variables =====						
7	sex.M	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	miss. D	#missing ord. vals	#X-var	#N-var	#F-var	#S-var
920	0	0	1	0	1	0	
#P-var	#M-var	#B-var	#C-var	#I-var			
0	0	1	2	0			

Number of dummy variables created: 1

Offset variable in column 6

Number of cases used for training: 920

Constant fitted to cases with missing values in regressor variables

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Nodewise interaction tests on all variables

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 8

Minimum node sample size: 7

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	55	2.680E+00	2.855E-01	1.856E-01	2.609E+00	3.165E-01
2	54	2.680E+00	2.855E-01	1.856E-01	2.609E+00	3.165E-01
:						
44	4	2.329E+00	3.284E-01	2.704E-01	2.126E+00	3.242E-01
45**	3	2.255E+00	3.279E-01	2.717E-01	1.954E+00	2.711E-01

46	2	4.702E+00	8.054E-01	4.866E-01	4.153E+00	6.629E-01
47	1	9.431E+00	1.420E+00	9.674E-01	9.043E+00	9.329E-01

0-SE tree based on mean is marked with * and has 3 terminal nodes
 0-SE tree based on median is marked with + and has 3 terminal nodes
 Selected-SE tree based on mean using naive SE is marked with **
 Selected-SE tree based on mean using bootstrap SE is marked with --
 Selected-SE tree based on median and bootstrap SE is marked with ++
 * tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Rate is mean of Y/exp(offset)

Cases fit give the number of cases used to fit node

Deviance is mean residual deviance for all cases in node

Node label	Total cases	Cases fit	Matrix rank	Node rate	Node deviance	Split variable	Other variables
1	920	920	2	1.382E-02	9.179E+00	agegp	
2T	230	230	2	5.493E-03	1.863E+00	county	
3	690	690	2	1.763E-02	4.357E+00	agegp	
6T	230	230	2	1.339E-02	3.003E+00	county	
7T	460	460	2	2.093E-02	1.802E+00	agegp	

Number of terminal nodes of final tree: 3

Total number of nodes of final tree: 5

Second best split variable (based on curvature test) at root node is sex

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: agegp = "45-54"

Node 2: deaths sample rate = 0.54928582E-2

Node 1: agegp /= "45-54"

Node 3: agegp = "55-64"

Node 6: deaths sample rate = 0.13389777E-1

Node 3: agegp /= "55-64"

Node 7: deaths sample rate = 0.20932715E-1

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", Statistics in Medicine, v.35, 4837-4855.

2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.

3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if agegp = "45-54"

agegp mode = "45-54"

Coefficients of regression function for log expected rate:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	-5.172	-366.9	0.000			
sex.M	1.437	89.64	0.000	0.000	0.5000	1.000

Node mean for offset variable = 6.727

If regressor has missing values, predicted rate = 0.13824405E-1

Node 2: Terminal node

Coefficients of regression function for log expected rate:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	-5.834	-161.5	0.3750E-15			
sex.M	1.038	24.44	0.1308E-15	0.000	0.5000	1.000

Node mean for offset variable = 6.857

If regressor has missing values, predicted rate = 0.54928582E-2

Node 3: Intermediate node

A case goes into Node 6 if agegp = "55-64"

agegp mode = "55-64"

Node 6: Terminal node

Coefficients of regression function for log expected rate:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	-5.117	-199.8	0.000			
sex.M	1.285	43.87	0.000	0.000	0.5000	1.000

Node mean for offset variable = 6.920

If regressor has missing values, predicted rate = 0.13389777E-1

Node 7: Terminal node

Coefficients of regression function for log expected rate:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	-4.907	-256.9	0.000			
sex.M	1.714	79.68	0.000	0.000	0.5000	1.000

Node mean for offset variable = 6.567

If regressor has missing values, predicted rate = 0.20932715E-1

Observed and fitted values are stored in poi.fit

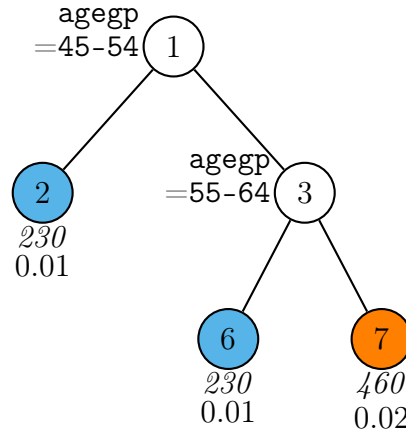


Figure 21: GUIDE v.46.1 0.250-SE multiple linear Poisson regression tree (constant fitted to incomplete cases in terminal nodes) for predicting rate of **deaths**. At each split, an observation goes to the left branch if and only if the condition is satisfied. Sample size (in *italics*) and sample rate printed below nodes. Terminal nodes with rates above and below value of 0.014 at root node are painted **orange** and **skyblue** respectively. Second best split variable at root node is **sex**.

LaTeX code for tree is in `poi.tex`

The results show that the death rate increases with age and that the rate for males is consistently higher than that for females. The tree diagram is given in Figure 21.

10 Censored response: RHC data

Doctors believe that direct measurement of cardiac function by right heart catheterization (RHC) for some critically ill patients yields better outcomes. Relative risk of death is higher in elderly and patients with acute myocardial infarction who received RHC. Benefit of RHC has not been demonstrated in a randomized clinical trial, because physicians refuse to allow their patients to be randomized. Treatment selection is confounded with patient factors that are also related to outcomes, e.g., patients with low blood pressure are more likely to get RHC, and such patients are also more likely to die. Data consist of observations on more than 60 variables for 5735 patients from 5 medical centers over 5 years (Connors et al., 1996). Response

Table 6: Demographics & outcomes for RHC data [#missing in brackets]

Variable	Definition
swang1	Right heart catheterization (RHC) [0]
age	Age in years [0]
sex	Sex (female/male) [0]
wtkilo1	Weight in kilograms [515]
edu	Years of Education [0]
race	Race [0]
income	Income bracket (<11k, 11–25k, 25–50k, >50k) [0]
ninsclas	Medical insurance (Medicaid, Medicare, Medicare & Medicaid, no insurance, private, private & Medicare) [0]
t3d30	Days from admission to death within 30 days [0]
dth30	Death indicator for t3d30 (0=no, 1=yes) [0]
survtime	Days from admission to death or last contact day [0]
death	Death indicator for survtime (0=no, 1=yes) [0]
transhx	Transfer (> 24 hours) from another hospital (no/yes) [0]

variables are **t3d30** (censored 30-day survival time) and **survtime** (days till death or last contact). Variable **dth30** is 1 if death occurs within 30 days of hospital admission and 0 otherwise; **death** is 1 if the subject eventually dies and 0 if death is unknown. Variable **swang1** takes values “RHC” and “NoRHC”, indicating whether or not a patient received RHC. Other variables are given in Tables 6–8.

GUIDE can fit a proportional hazards model to the censored survival time if the event indicator **death** is specified as “D” and **survtime** as “T”. The DSC file is **rhcdsc2.txt** whose contents follow.

```
rhcdsc2.txt
NA
2
1 X x
2 cat1 c
3 cat2 c
4 ca c
5 sadmdte x
6 dschdte x
7 dthdte x
8 lstctdte x
9 death d
10 cardiohx c
11 chfhx c
12 dementhx c
```

Table 7: Disease variables for RHC data [#missing in brackets]

Variable	Definition
cat1	Primary disease category (9 levels) [0]
cat2	Secondary disease category (6 levels) [2798]
ca	Cancer (3 levels) [0]
card	Cardiovascular diagnosis [0]
gastr	Gastrointestinal diagnosis [0]
hema	Hematologic diagnosis [0]
meta	Metabolic diagnosis [0]
neuro	Neurological diagnosis [0]
ortho	Orthopedic diagnosis [0]
renal	Renal diagnosis [0]
resp	Respiratory diagnosis [0]
seps	Sepsis diagnosis [0]
trauma	Trauma diagnosis [0]

Table 8: Medical history for RHC data [#missing in brackets]

Variable	Definition
amihx	Definite myocardial infarction (no/yes) [0]
cardiohx	Acute MI, peripheral vascular disease, severe cardiovascular symptoms [0]
chfhx	Congestive heart failure (no/yes) [0]
chrpulhx	Chronic or severe pulmonary disease (no/yes) [0]
dementhx	Dementia, stroke or cerebral infarction, Parkinson's disease (no/yes) [0]
gibledhx	Upper GI bleeding (no/yes) [0]
liverhx	Cirrhosis, hepatic failure (no/yes) [0]
malighx	Solid tumor, metastatic disease, chronic leukemia/myeloma, acute leukemia, lymphoma (no/yes) [0]
immunhx	Immunosuppression, organ transplant, HIV positivity, diabetes mellitus, connective tissue disease (no/yes) [0]
psychhx	Psychiatric history, active psychosis or severe depression (no/yes) [0]
renalhx	Chronic renal disease, chronic hemodialysis or peritoneal dialysis (no/yes) [0]

Table 9: Admission variables for RHC data [#missing in brackets]

Variable	Definition
alb1	Albumin [0]
bili1	Bilirubin [0]
crea1	Serum creatinine [0]
hema1	Hematocrit [0]
hrt1	Heart rate [159]
meanbp1	Mean blood pressure [80]
pot1	Serum potassium [0]
pao2	Partial pressure of arterial oxygen [0]
fio2	Fraction of inspired oxygen [0]
pafi1	$\text{PaO}_2 / (0.01 * \text{FiO}_2)$ [0]
paco21	Partial pressure of arterial carbon dioxide [0]
ph1	Serum ph [0]
resp1	Respiration rate [136]
scoma1	Glasgow coma score [0]
sod1	Serum sodium [0]
temp1	Temperature (Celsius) [0]
urin1	Urine output [3028]
wb1c1	White blood cell count [0]
aps1	APACHE III score ignoring coma [0]
adld3p	Katz Activities of Daily Living Scale [3016]
das2d3pc	DASI (Duke Activity Status Index) [0]
dnr1	DNR (do-not-resuscitate) status [0]
surv2md1	Estimated probability of 2-month survival [0]

13 psychhx c
14 chrpulhx c
15 renalhx c
16 liverhx c
17 gibledhx c
18 malighx c
19 immunhx c
20 transhx c
21 amihx c
22 age n
23 sex c
24 edu n
25 surv2md1 n
26 das2d3pc n
27 t3d30 x
28 dth30 x
29 aps1 n
30 scoma1 n
31 meanbp1 n
32 wblc1 n
33 hrt1 n
34 resp1 n
35 temp1 n
36 pafi1 n
37 alb1 n
38 hema1 n
39 bili1 n
40 crea1 n
41 sod1 n
42 pot1 n
43 paco21 n
44 ph1 n
45 swang1 c
46 wtkilo1 n
47 dnr1 c
48 ninsclas c
49 resp c
50 card c
51 neuro c
52 gastr c
53 renal c
54 meta c
55 hema c
56 seps c
57 trauma c
58 ortho c

```

59 adld3p n
60 urin1 n
61 race c
62 income c
63 ptid x
64 survtime t

```

10.1 Proportional hazards

GUIDE has two options for modeling censored response data. The first is a piecewise Cox proportional hazards model.

Let the survival time of a subject be U with probability density $f(u)$ and distribution function $F(u)$. The survival probability function is $S(u) = P(U > u) = 1 - F(u)$ and the hazard rate (instantaneous rate of death) at time u is $\lambda(u) = f(u)/S(u)$. Let U_i and C_i be survival and censoring times of subject i . Let $Y_i = \min(U_i, C_i)$ be the observed censored survival time and let $\delta_i = I(U_i < C_i)$ denote the event indicator. The proportional hazards model assumes that $\lambda(u, \mathbf{x}) = \lambda_0(u) \exp(\beta' \mathbf{x})$, where $\lambda_0(u)$ is an unknown baseline hazard function. Unlike other regression tree methods for survival data, $\lambda_0(u)$ is the same for all terminal nodes of a GUIDE tree.

10.1.1 Input file generation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: censored.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: censored.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 4
Input 1 for proportional hazards, 2 for restricted mean event time ([1:2], <cr>=1):
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple linear in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple linear, 3: constant ([1:3], <cr>=3):

```

```

Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: rhcdsc2.txt
Reading DSC file ...
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
23 N variables changed to S
D variable is death
Reading data file ...
Number of records in data file: 5735
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 31 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Finished processing 5000 of 5735 observations
Data checks complete
Smallest uncensored survtime: 2.0000
Number of complete cases excluding censored T < smallest uncensored T: 5735
Number of cases used to compute baseline hazard: 5735
Number of cases with D=1 and T >= smallest uncensored: 3722
Creating missing value indicators ...
Rereading data ...
Largest uncensored and censored survtime with positive weight: 1943.0000 1351.0000
Numbers of N, F, S, and P below exclude constant variables
      Total #cases w/ #missing
      #cases  miss. D ord. vals #X-var #N-var #F-var #S-var
      5735      0      5157      8      0      0      23
      #P-var #M-var #B-var #C-var #I-var
      0      0      0      31      0
Survival time variable in column: 64
Event indicator variable in column: 9
Proportion uncensored among nonmissing T and D variables: .649
Number of cases used for training: 5735
Number of cases excluded due to 0 W or missing D or T variables: 0
Finished reading data file
Default max. number of split levels: 15

```



```

Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): censored.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: censored.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: censored.r
Input rank of top variable to split root node ([1:54], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < censored.in

```

10.1.2 Output file

```

Proportional hazards regression tree for censored response
Pruning by cross-validation
DSC file: rhcdsc2.txt
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
23 N variables changed to S
D variable is death
Piecewise constant model
Number of records in data file: 5735
Length of longest entry in data file: 19
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Number of complete cases excluding censored T < smallest uncensored T: 5735
Number of cases used to compute baseline hazard: 5735
Number of cases with D=1 and T >= smallest uncensored: 3722
Smallest uncensored survtime: 2.0000
Largest uncensored and censored survtime with positive weight: 1943.0000 1351.0000

Summary information for training sample of size 5735
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight,
t=survival time variable

```

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
2	cat1	c			9	
3	cat2	c			6	4535
4	ca	c			3	
9	death	d	0.000	1.000		
10	cardiohx	c			2	
11	chfhx	c			2	
12	dementhx	c			2	
13	psychhx	c			2	
14	chrpulhx	c			2	
15	renalhx	c			2	
16	liverhx	c			2	
17	gibledhx	c			2	
18	malighx	c			2	
19	immunhx	c			2	
20	transhx	c			2	
21	amihx	c			2	
22	age	s	18.04	101.8		
23	sex	c			2	
24	edu	s	0.000	30.00		
25	surv2md1	s	0.000	0.9620		
26	das2d3pc	s	11.00	33.00		
29	aps1	s	3.000	147.0		
30	scoma1	s	0.000	100.0		
31	meanbp1	s	10.00	259.0		80
32	wblc1	s	0.000	192.0		
33	hrt1	s	8.000	250.0		159
34	resp1	s	2.000	100.0		136
35	temp1	s	27.00	43.00		
36	pafi1	s	11.60	937.5		
37	alb1	s	0.3000	29.00		
38	hema1	s	2.000	66.19		
39	bili1	s	0.9999E-01	58.20		
40	crea1	s	0.9999E-01	25.10		
41	sod1	s	101.0	178.0		
42	pot1	s	1.100	11.90		
43	paco21	s	1.000	156.0		
44	ph1	s	6.579	7.770		
45	swang1	c			2	
46	wtkilo1	s	19.50	244.0		515
47	dnr1	c			2	
48	ninsclas	c			6	
49	resp	c			2	
50	card	c			2	

```

51 neuro      c                      2
52 gastr      c                      2
53 renal      c                      2
54 meta       c                      2
55 hema       c                      2
56 seps       c                      2
57 trauma     c                      2
58 ortho      c                      2
59 adld3p     s      0.000      7.000      4296
60 urin1      s      0.000      9000.      3028
61 race       c                      3
62 income     c                      4
64 survtime   t      2.000      1943.
===== Constructed variables =====
65 lnbasehaz  z      -3.818      2.038

```

Numbers of N, F, S, and P below exclude constant variables

```

Total #cases w/ #missing
#cases miss. D ord. vals #X-var #N-var #F-var #S-var
5735      0      5157      8      0      0      23
#P-var #M-var #B-var #C-var #I-var
0      0      0      31      0

```

Survival time variable in column: 64

Event indicator variable in column: 9

Proportion uncensored among nonmissing T and D variables: 0.649

Number of cases used for training: 5735

Number of cases excluded due to 0 W or missing D or T variables: 0

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Nodewise interaction tests on all variables

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 5

Number of iterations for fitting: 20

Ranks of variables and their 1-df chi-squared values at root node

```

1  0.7537E+03  surv2md1
2  0.3238E+03  adld3p
3  0.2324E+03  cat1
4  0.2262E+03  aps1
:
51 0.4567E-01  meta
52 0.2905E-01  chfhx

```

```

53 0.1966E-01 cardiohx
54 0.1463E-01 chrpulhx

```

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	693	1.897E+00	4.509E-02	3.758E-02	1.879E+00	5.321E-02
2	692	1.897E+00	4.509E-02	3.758E-02	1.879E+00	5.321E-02
:						
420	11	1.254E+00	1.908E-02	1.498E-02	1.238E+00	2.194E-02
421**	10	1.242E+00	1.732E-02	1.342E-02	1.227E+00	1.880E-02
422	8	1.250E+00	1.701E-02	1.140E-02	1.235E+00	1.777E-02
423	7	1.257E+00	1.701E-02	1.238E-02	1.243E+00	2.317E-02
424	6	1.267E+00	1.698E-02	1.215E-02	1.257E+00	1.838E-02
425	5	1.283E+00	1.726E-02	1.261E-02	1.281E+00	1.891E-02
426	3	1.291E+00	1.700E-02	1.253E-02	1.281E+00	2.092E-02
427	2	1.332E+00	1.684E-02	1.207E-02	1.329E+00	1.682E-02
428	1	1.455E+00	1.624E-02	6.159E-03	1.451E+00	9.894E-03

0-SE tree based on mean is marked with * and has 10 terminal nodes

0-SE tree based on median is marked with + and has 10 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Cases fit give the number of cases used to fit node

Deviance is mean residual deviance for all cases in node

Node	Total	Cases	Matrix	Median	Node	Split	Other
label	cases	fit	rank	survtime	deviance	variable	variables
1	5735	5735	1	1.920E+02	1.455E+00	surv2md1	
2	2157	2157	1	2.400E+01	1.494E+00	adld3p	
4	1924	1924	1	1.900E+01	1.523E+00	surv2md1	
8T	705	705	1	1.100E+01	1.431E+00	cat1	
9	1219	1219	1	2.800E+01	1.487E+00	dnr1	
18T	1025	1025	1	3.700E+01	1.423E+00	psychhx	:surv2md1
19T	194	194	1	8.500E+00	1.427E+00	liverhx	
5T	233	233	1	1.950E+02	9.305E-01	ca	
3	3578	3578	1	3.290E+02	1.223E+00	surv2md1	
6	1806	1806	1	2.270E+02	1.345E+00	adld3p	
12	1364	1364	1	1.270E+02	1.455E+00	dnr1	
24T	1213	1213	1	1.680E+02	1.410E+00	das2d3pc	
25T	151	151	1	2.500E+01	1.589E+00	dementhx	

13T	442	442	1	3.750E+02	8.581E-01	das2d3pc
7	1772	1772	1	4.030E+02	1.022E+00	chfhx
14	1279	1279	1	4.410E+02	1.039E+00	das2d3pc
28T	861	861	1	3.640E+02	1.071E+00	cat1
29T	418	418	1	5.470E+02	8.881E-01	surv2md1
15T	493	493	1	3.730E+02	9.358E-01	surv2md1

Number of terminal nodes of final tree: 10

Total number of nodes of final tree: 19

Second best split variable (based on curvature test) at root node is adld3p

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: surv2md1 <= 0.56398225

Node 2: adld3p = NA

Node 4: surv2md1 <= 0.35699725

Node 8: Median survival time = 11.000000

Node 4: surv2md1 > 0.35699725 or NA

Node 9: dnr1 = "No"

Node 18: Median survival time = 37.000000

Node 9: dnr1 /= "No"

Node 19: Median survival time = 8.5000000

Node 2: adld3p /= NA

Node 5: Median survival time = 195.00000

Node 1: surv2md1 > 0.56398225 or NA

Node 3: surv2md1 <= 0.71695924

Node 6: adld3p = NA

Node 12: dnr1 = "No"

Node 24: Median survival time = 168.00000

Node 12: dnr1 /= "No"

Node 25: Median survival time = 25.000000

Node 6: adld3p /= NA

Node 13: Median survival time = 375.00000

Node 3: surv2md1 > 0.71695924 or NA

Node 7: chfhx = "0"

Node 14: das2d3pc <= 24.798830

Node 28: Median survival time = 364.00000

Node 14: das2d3pc > 24.798830 or NA

Node 29: Median survival time = 547.00000

Node 7: chfhx /= "0"

Node 15: Median survival time = 373.00000

Predictor means below are means of cases with no missing values.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if surv2md1 <= 0.56398225

surv2md1 mean = 0.59245008

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value
Constant	0.000		

Predicted log-relative hazard = 192.00000

Node 2: Intermediate node

A case goes into Node 4 if adld3p = NA

adld3p mean = 1.3605150

Node 4: Intermediate node

A case goes into Node 8 if surv2md1 <= 0.35699725

surv2md1 mean = 0.38119025

Node 8: Terminal node

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value
Constant	1.011		

Predicted log-relative hazard = 11.000000

Node 9: Intermediate node

A case goes into Node 18 if dnr1 = "No"

dnr1 mode = "No"

Node 18: Terminal node

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value
Constant	0.3648		

Predicted log-relative hazard = 37.000000

Node 19: Terminal node

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value

```

Constant      1.168
Predicted log-relative hazard = 8.5000000
-----
Node 5: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value
Constant      -0.2260E-02
Predicted log-relative hazard = 195.00000
-----
Node 3: Intermediate node
A case goes into Node 6 if surv2md1 <= 0.71695924
surv2md1 mean = 0.71756975
-----
Node 6: Intermediate node
A case goes into Node 12 if adld3p = NA
adld3p mean = 1.1606335
-----
Node 12: Intermediate node
A case goes into Node 24 if dnr1 = "No"
dnr1 mode = "No"
-----
Node 24: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value
Constant      -0.1478E-01
Predicted log-relative hazard = 168.00000
-----
Node 25: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value
Constant      0.5762
Predicted log-relative hazard = 25.000000
-----
Node 13: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value
Constant      -0.5115
Predicted log-relative hazard = 375.00000
-----
Node 7: Intermediate node
A case goes into Node 14 if chfhx = "0"
chfhx mode = "0"
-----
Node 14: Intermediate node
A case goes into Node 28 if das2d3pc <= 24.798830
das2d3pc mean = 21.932716

```

```

-----
Node 28: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor   Coefficient  t-stat      p-value
Constant    -0.5874
Predicted log-relative hazard = 364.00000
-----
Node 29: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor   Coefficient  t-stat      p-value
Constant    -1.261
Predicted log-relative hazard = 547.00000
-----
Node 15: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor   Coefficient  t-stat      p-value
Constant    -0.4073
Predicted log-relative hazard = 373.00000
-----
Observed and fitted values are stored in censored.fit
LaTeX code for tree is in censored.tex

```

Figure 22 shows the fitted model. Sample size (in *italics*), relative hazard, and median survival time are printed below each terminal node. The top lines of the file `censored.fit` are:

train	node	observed	event	logbasecumhaz	survivalprob	mediansurvtime
y	13	0.240000E+003	n	-0.256265E+000	0.628727E+000	0.375000E+003
y	15	0.450000E+002	y	-0.799746E+000	0.741494E+000	0.373000E+003
y	8	0.317000E+003	n	-0.495237E-001	0.731318E-001	0.110000E+002
y	18	0.370000E+002	y	-0.884460E+000	0.551714E+000	0.370000E+002
y	19	0.200000E+001	y	-0.400645E+001	0.943169E+000	0.850000E+001

The columns give the following information:

train: equals `y` if observation is used for model fitting; equals `n` if not used.

node: terminal node label of observation.

observed: observed survival time (`t` variable in DSC file).

event: equals `y` if **observed** is uncensored (`d=1`); equals `n` if censored (`d=0`).

logbasecumhaz: log of the estimated baseline cumulative hazard function $\log \Lambda_0(t) = \log \int_0^t \lambda_0(u) du$ at observed time t .

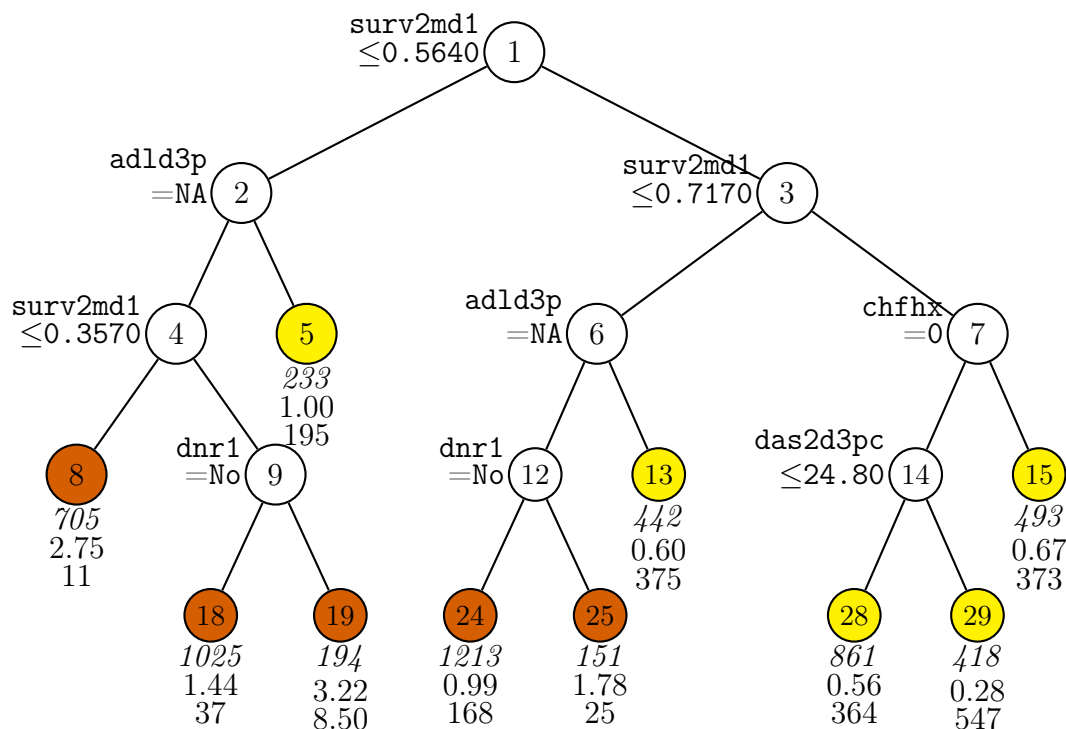


Figure 22: GUIDE v.46.1 0.250-SE piecewise-constant proportional hazards regression tree for `survtime`. At each split, an observation goes to the left branch if and only if the condition is satisfied. Sample size (in *italics*), relative hazard, and median survival time printed below nodes. Terminal nodes with median survival times above and below 192 (median at root node) are painted yellow and vermillion respectively. Second best split variable at root node is `adld3p`.

Kaplan–Meier survival curves

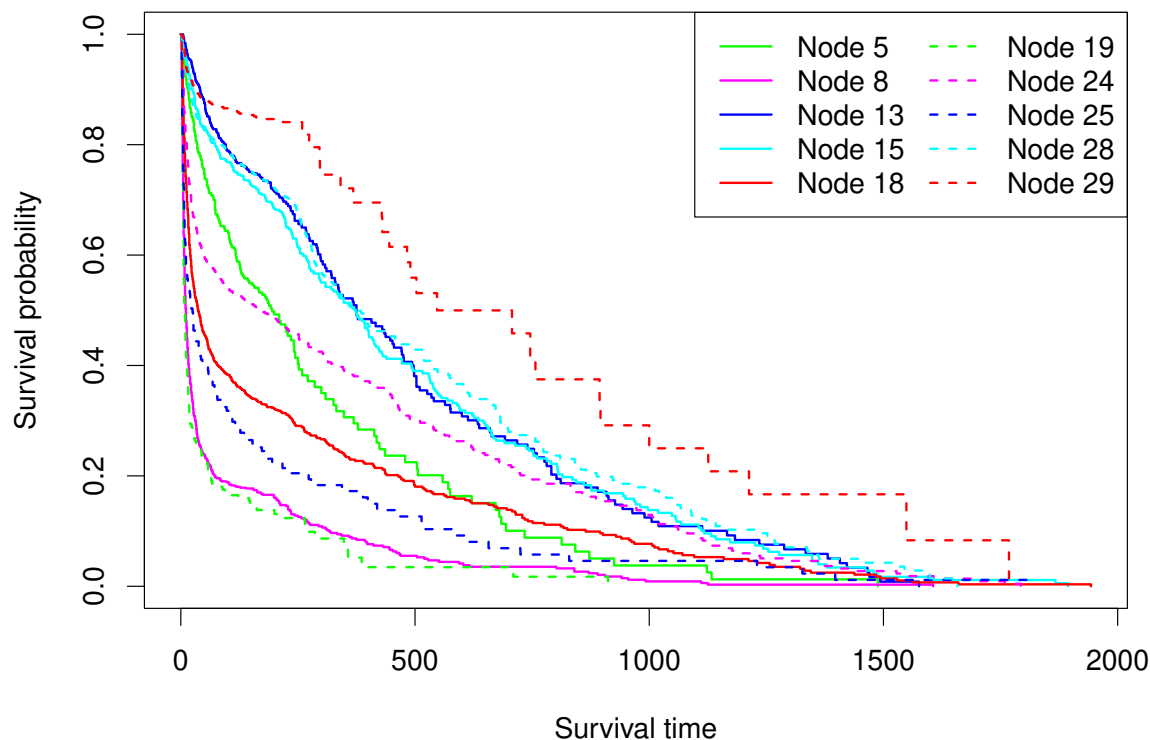


Figure 23: Kaplan-Meier survival curves for data in terminal nodes of Figure 22

survivalprob: probability that the subject survives up to observed time t . For the first subject, this is

$$\begin{aligned}
 \exp\{-\Lambda_0(t) \exp(\beta' \mathbf{x})\} &= \exp\{-\exp(\beta_0 + \text{logbasecumhaz})\} \\
 &= \exp(-\exp(-0.514911594896 - 0.261185)) \\
 &= 0.6311581
 \end{aligned}$$

where $t = 240$ and $\beta_0 = -0.514911594896$ is the constant term in the node (`censored.r` gives β_0 to higher precision than `censored.out`).

mediansurvtime: median survival time among observations in node estimated from Kaplan-Meier survival function. A trailing plus (+) sign indicates estimate is censored.

Figure 23 plots the estimated survival curves in the terminal nodes of the tree. The plot is produced by the following R code.

```

library(survival)
z0 <- read.table("rhcddata.txt",header=TRUE)
z1 <- read.table("censored.fit",header=TRUE)
nodenum <- unique(sort(z1$node))
leg.txt <- paste("Node",nodenum)
leg.col <- c("green","magenta","blue","cyan","red")
leg.lty <- rep(c(1,2),c(5,5))
fit <- survfit(Surv(z0$survtime,z0$death) ~ z1$node, conf.type="none")
plot(fit,mark.time=FALSE,xlab="Survival time",ylab="Survival probability",
     col=leg.col,lwd=2,lty=leg.lty)
title("Kaplan-Meier survival curves")
legend("topright",legend=leg.txt,lty=leg.lty,col=leg.col,lwd=2,ncol=2)

```

10.2 Restricted mean event time

The mean survival time is not estimable if there is censoring. But given a pre-specified time point τ , the restricted mean survival time $\mu(X) = E(Y|X)$ is estimable, where $Y = \min(U, C, \tau)$ and X is a covariate vector ([Andersen et al., 2004](#); [Chen and Tsiatis, 2001](#); [Tian et al., 2014](#)). GUIDE has an option to fit a *restricted event time model* to each node of the tree such that $\mu(X)$ is linear in the covariates.

10.2.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: rest.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: rest.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 4
Input 1 for proportional hazards, 2 for restricted mean event time ([1:2], <cr>=1): 2
Choose complexity of model to use at each node:
Choose 0 for stepwise linear regression (recommended for prediction)
Choose 1 for multiple regression
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)

```

```

0: stepwise linear, 1: multiple linear, 2: best simple polynomial, 3: constant,
4: best simple stepwise ANCOVA ([0:4], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: rhcdsc2.txt
Reading DSC file ...
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
23 N variables changed to S
D variable is death
Reading data file ...
Number of records in data file: 5735
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 31 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Finished processing 5000 of 5735 observations
Data checks complete
Creating missing value indicators ...
Rereading data ...
Largest uncensored and censored survtime with positive weight: 1943.0000 1351.0000
Smallest observed uncensored time is 2.0000
Largest observed censored or uncensored time is 1943.0000
Input restriction on event time ([2.00:1943.00], <cr>=972.00):

Numbers of N, F, S, and P below exclude constant variables
      Total #cases w/ #missing
      #cases  miss. D ord. vals #X-var #N-var #F-var #S-var
      5735      0      5157      8      0      0      23
      #P-var #M-var #B-var #C-var #I-var
      0      0      0      31      0

No weight variable in data file
Number of cases used for training: 3732
Number of cases excluded due to 0 W or missing D variable: 2003
Finished reading data file
Default max. number of split levels: 15

```

```

Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): rest.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: rest.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: rest.r
Input rank of top variable to split root node ([1:54], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < rest.in

```

10.2.2 Contents of rest.out

```

Restricted mean event time regression tree
Pruning by cross-validation
DSC file: rhcdsc2.txt
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
23 N variables changed to S
D variable is death
Piecewise constant model
Number of records in data file: 5735
Length of longest entry in data file: 19
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Smallest uncensored survtime: 2.0000
Largest uncensored and censored survtime with positive weight: 1943.0000 1351.0000
Interval for restricted mean event time is from 0 to 972.

Summary information for training sample of size 3732 (excluding observations
  with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight
                                     #Codes/

```

Column	Name		Minimum	Maximum	Levels/ Periods	#Missing
2	cat1	c			9	
3	cat2	c			6	2807
4	ca	c			3	
9	death	d	0.000	1.000		
10	cardiohx	c			2	
:						
60	urin1	s	0.000	9000.		2093
61	race	c			3	
62	income	c			4	
64	survtime	t	2.000	1943.		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
5735	0	5157	8	0	0	23	
#P-var	#M-var	#B-var	#C-var	#I-var			
0	0	0	31	0			

No weight variable in data file

Number of cases used for training: 3732

Number of cases excluded due to 0 W or missing D variable: 2003

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Nodewise interaction tests on all variables

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 3

Ranks of variables and their 1-df chi-squared values at root node

1	0.1868E+03	adld3p
2	0.1629E+03	surv2md1
3	0.1122E+03	cat1
4	0.6234E+02	aps1
:		
52	0.2165E+00	race
53	0.1196E+00	amihx
54	0.6209E-01	income

Size and CV MSE and SE of subtrees:

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	771	1.417E+05	5.391E+03	2.630E+03	1.392E+05	4.541E+03
2	770	1.417E+05	5.391E+03	2.630E+03	1.392E+05	4.541E+03

3	769	1.417E+05	5.391E+03	2.630E+03	1.392E+05	4.541E+03
:						
504	8	1.134E+05	3.719E+03	1.992E+03	1.110E+05	2.521E+03
505	7	1.112E+05	3.530E+03	1.623E+03	1.109E+05	2.809E+03
506--	6	1.084E+05	3.219E+03	1.489E+03	1.092E+05	1.373E+03
507**	4	1.091E+05	3.046E+03	1.501E+03	1.097E+05	2.588E+03
508++	3	1.097E+05	3.048E+03	1.409E+03	1.093E+05	1.989E+03
509	2	1.102E+05	3.057E+03	1.510E+03	1.101E+05	2.162E+03
510	1	1.225E+05	3.100E+03	2.805E+02	1.225E+05	4.687E+02

0-SE tree based on mean is marked with * and has 6 terminal nodes

0-SE tree based on median is marked with + and has 6 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree same as + tree

* tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is weighted mean of death in the node

Cases fit give the number of cases used to fit node

MSE is residual sum of squares divided by number of cases in node

Node label	Total cases	Cases fit	Matrix rank	Node D-mean	Node MSE	Split variable	Interacting variable
1	3732	3732	1	3.144E+02	1.800E+05	adld3p	
2T	664	664	1	4.685E+02	2.273E+05	surv2md1	
3	3068	3068	1	2.647E+02	1.556E+05	surv2md1	
6T	1262	1262	1	1.607E+02	8.878E+04	dnr1	
7	1806	1806	1	3.225E+02	1.880E+05	urin1	
14T	800	800	1	2.032E+02	8.072E+04	adld3p	
15T	1006	1006	1	4.004E+02	2.479E+05	surv2md1	

Number of terminal nodes of final tree: 4

Total number of nodes of final tree: 7

Second best split variable (based on curvature test) at root node is surv2md1

Regression tree:

Node 1: adld3p <= 5.5000000

Node 2: survtime-mean = 468.46294

Node 1: adld3p > 5.5000000 or NA

Node 3: surv2md1 <= 0.49098337

Node 6: survtime-mean = 160.70095

```

Node 3: surv2md1 > 0.49098337 or NA
Node 7: urin1 <= 7212.5000
Node 14: survtime-mean = 203.19504
Node 7: urin1 > 7212.5000 or NA
Node 15: survtime-mean = 400.36912

```

```

*****
Predictor means below are means of cases with no missing values.

```

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

```

Node 1: Intermediate node
A case goes into Node 2 if adld3p <= 5.5000000
adld3p mean = 1.2733830
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat      p-value
Constant      314.4        45.27      0.3032-313
-----

```

```

Node 2: Terminal node
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat      p-value
Constant      468.5        25.32      0.3032-313
survtime mean = 468.463
-----

```

```

Node 3: Intermediate node
A case goes into Node 6 if surv2md1 <= 0.49098337
surv2md1 mean = 0.54259828
-----

```

```

Node 6: Terminal node
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat      p-value
Constant      160.7        19.16      0.3032-313
survtime mean = 160.701
-----

```

```

Node 7: Intermediate node
A case goes into Node 14 if urin1 <= 7212.5000
urin1 mean = 1998.7301

```

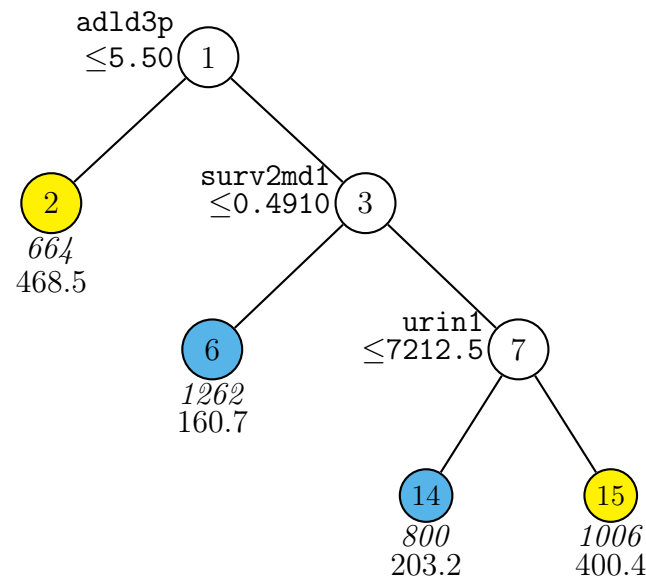



Figure 24: GUIDE v.46.1 0.250-SE piecewise-constant regression tree for mean `survtime` restricted to less than 972.000. At each split, an observation goes to the left branch if and only if the condition is satisfied. Sample size (in *italics*) and restricted mean of `survtime` printed below nodes. Terminal nodes with means above and below value of 314.4 at root node are painted yellow and skyblue respectively. Second best split variable at root node is `surv2md1`.

```

-----
Node 14: Terminal node
Coefficients of least squares regression function:
Regressor   Coefficient  t-stat    p-value
Constant    203.2           20.23     0.3032-313
survtime mean = 203.195
-----

```

```

Node 15: Terminal node
Coefficients of least squares regression function:
Regressor   Coefficient  t-stat    p-value
Constant    400.4        25.50     0.3032-313
survtime mean = 400.369
-----

```

```

Observed and fitted values are stored in rest.fit
LaTeX code for tree is in rest.tex

```

Figure 24 shows the restricted mean event time tree.

11 Randomized treatments

Causal effects of treatments are best studied in a randomized trial where the treatments are assigned randomly to subjects. The goal is to show that one treatment is more efficacious than another across all subjects. If this determination is not achieved, a secondary goal may be to search for subgroups of subjects with differential treatment effects.

There are two types of covariates for identification of subgroups with differential treatment effects. A *prognostic* variable is a clinical or biologic characteristic that provides information on the likely outcome of the disease in an untreated individual (e.g., patient age, family history, disease stage, and prior therapy). A *predictive* variable is one that provides information on the likely benefit from the treatment. **Predictive variables can be used to identify subgroups of patients who are most likely to benefit from a given therapy.** In general, prognostic variables define the effects of patient or tumor characteristics on the patient outcome, whereas predictive variables define the effect of treatment on the tumor (Italiano, 2011). Accordingly, GUIDE has two options, called **Gi** and **Gs**. **Gi** is more sensitive to predictive variables and **Gs** tends to be equally sensitive to prognostic and predictive variables (Loh et al., 2015).

11.1 Multiple treatment arms: CAPE data

We first demonstrate this on a data set from a three-armed randomized controlled experiment to find out whether two interventions (DVD or Phone) are more efficacious than a control at promoting mammography screening. The relevant data and DSC files are `cape.txt` and `cape.dsc`. Note that the three treatment levels (contained in the treatment (R) variable `group`) are assumed to be categorical (i.e., nominal valued). See Loh et al. (2016) for more information on the data.

Because the response variable (`resp6`) is 0-1 (0=no, 1=yes), we use least-squares regression with `resp6` designated as the dependent variable `D` or `d` in the DSC file. The treatment variable (`group`) is designated as `R` or `r` (for “Rx”).

11.1.1 Input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: gi.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1):
```

```

Name of batch output file: gi.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
  7=binary logistic regression.
Input choice ([1:7], <cr>=1):
Input 1 for least squares, 2 least median of squares ([1:2], <cr>=1):
Choose complexity of model to use at each node:
Choose 0 for stepwise linear regression (recommended for prediction)
Choose 1 for multiple regression
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
0: stepwise linear, 1: multiple linear, 2: best simple polynomial, 3: constant,
4: best simple stepwise ANCOVA ([0:4], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: cape.dsc
Reading DSC file ...
Training sample file: cape.txt
Missing value code: NA
Records in data file start on line 1
R variable present
21 N variables changed to S
Warning: model changed to linear in treatment
D variable is resp6
Reading data file ...
Number of records in data file: 1681
Length of longest entry in data file: 25
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 18 categorical variables
Finished assigning codes to 10 categorical variables
Treatment (R) variable is group with values "Control", "DVD", and "Phone"
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...

```

```

Choose a subgroup identification method:
1 = Prognostic priority (Gs)
2 = Predictive priority (Gi)
Input your choice: ([1:2], <cr>=2):
Creating dummy variables ...
Creating missing value indicators ...
Rereading data ...
Proportion of training sample for each level of group
"Control"      0.3278
  "DVD"        0.3309
  "Phone"      0.3413
    Total  #cases w/  #missing
    #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
    1681      43      84      1      0      0      21
    #P-var  #M-var  #B-var  #C-var  #I-var  #R-var
      0      0      0      17      0      1
No weight variable in data file
Number of cases used for training: 1638
Number of split variables: 38
Number of dummy variables created: 2
Number of cases excluded due to 0 W or missing D or R variables: 43
Finished reading data file
Default max. number of split levels: 14
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): gi.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: gi.fit
Input 2 to write R function for predicting new cases, 1 otherwise ([1:2], <cr>=2):
Input file name: gi.r
Input rank of top variable to split root node ([1:41], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < gi.in

```

11.1.2 Contents of gi.out

```

Least squares regression tree
Pruning by cross-validation
DSC file: cape.dsc
Training sample file: cape.txt
Missing value code: NA
Records in data file start on line 1
R variable present

```

11.1 Multiple treatment arms: CAPE data 11 RANDOMIZED TREATMENTS

```

21 N variables changed to S
Warning: model changed to linear in treatment
D variable is resp6
Piecewise linear model
Number of records in data file: 1681
Length of longest entry in data file: 24
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Treatment (R) variable is group with values "Control", "DVD", and "Phone"
Proportion of training sample for each level of group
"Control"    0.3278
"DVD"        0.3309
"Phone"      0.3413

```

Summary information for training sample of size 1638 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	resp6	d	0.000	1.000		
3	group	r			3	
4	age	s	41.00	75.00		1
5	educyrs	s	2.000	20.00		
6	collegeormore	c			2	
:						
37	susc	s	5.000	25.00		
38	fear	s	8.000	40.00		
39	fatal	s	11.00	42.00		
40	know	s	1.000	7.000		
41	stage	c			4	
===== Constructed variables =====						
42	group.DVD	f	0.000	1.000		
43	group.Phone	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
1681	43	84	1	0	2	21	
#P-var	#M-var	#B-var	#C-var	#I-var	#R-var		
0	0	0	17	0	1		

11.1 Multiple treatment arms: CAPE data 11 RANDOMIZED TREATMENTS

Number of dummy variables created: 2
No weight variable in data file
Number of cases used for training: 1638
Number of cases excluded due to 0 W or missing D or R variables: 43

Predictive priority (Gi)
Pruning by v-fold cross-validation, with v = 10
Selected tree is based on mean of CV estimates
Number of SE's for pruned tree: 0.2500

No nodewise interaction tests
Split values for N and S variables based on exhaustive search
Max number of interaction splits: 100
Maximum number of split levels: 8
Minimum node sample size: 8
Minimum fraction of cases per treatment at each node: 0.066
Size and CV MSE and SE of subtrees:

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	16	2.561E-01	4.759E-03	4.872E-03	2.580E-01	7.191E-03
2	15	2.560E-01	4.748E-03	4.807E-03	2.580E-01	7.191E-03
3	14	2.561E-01	4.732E-03	4.732E-03	2.580E-01	7.247E-03
4	13	2.560E-01	4.725E-03	4.717E-03	2.580E-01	7.188E-03
5	11	2.560E-01	4.692E-03	4.769E-03	2.580E-01	7.191E-03
6	9	2.530E-01	4.434E-03	4.209E-03	2.537E-01	5.715E-03
7	8	2.528E-01	4.310E-03	3.948E-03	2.537E-01	5.991E-03
8	7	2.424E-01	3.624E-03	4.306E-03	2.415E-01	5.952E-03
9*	5	2.408E-01	3.157E-03	2.616E-03	2.410E-01	3.743E-03
10**	1	2.414E-01	2.372E-03	5.044E-04	2.410E-01	6.719E-04

0-SE tree based on mean is marked with * and has 5 terminal nodes
0-SE tree based on median is marked with + and has 1 terminal nodes
Selected-SE tree based on mean using naive SE is marked with **
Selected-SE tree based on mean using bootstrap SE is marked with --
Selected-SE tree based on median and bootstrap SE is marked with ++
** tree same as ++ tree
** tree same as + tree
** tree same as -- tree
++ tree same as -- tree
+ tree same as ++ tree

WARNING: tree based on mean CV estimate of error has no splits
Choosing smallest nontrivial tree with no larger CV error estimate

Structure of final tree. Each terminal node is marked with a T.

D-mean is mean of resp6 in the node

11.1 Multiple treatment arms: CAPE data 11 RANDOMIZED TREATMENTS

Cases fit give the number of cases used to fit node
MSE and R² are based on all cases in node

Node label	Total cases	Cases fit	Matrix rank	Node D-mean	Node MSE	Node R ²	Split variable	Other variables
1	1638	1638	3	4.035E-01	2.410E-01	0.0006	sf12gh	
2	903	903	3	3.732E-01	2.336E-01	0.0046	know	
4	703	703	3	3.898E-01	2.384E-01	0.0018	educyrs	
8	543	543	3	3.720E-01	2.324E-01	0.0105	yearmam	
16T	427	427	3	2.998E-01	2.091E-01	0.0107	-	
17T	116	116	3	6.379E-01	2.248E-01	0.0518	-	
9T	160	160	3	4.500E-01	2.387E-01	0.0535	know	
5T	200	200	3	3.150E-01	2.039E-01	0.0693	fear	
3T	735	735	3	4.408E-01	2.455E-01	0.0081	sf12sf	

Number of terminal nodes of final tree: 5

Total number of nodes of final tree: 9

Second best split variable (based on curvature test) at root node is know

Regression tree:

Node 1: sf12gh <= 72.500000

Node 2: know <= 6.5000000

Node 4: educyrs <= 15.500000

Node 8: yearmam <= 3.5000000

Node 16: resp6-mean = 0.29976581

Node 8: yearmam > 3.5000000 or NA

Node 17: resp6-mean = 0.63793103

Node 4: educyrs > 15.500000 or NA

Node 9: resp6-mean = 0.45000000

Node 2: know > 6.5000000 or NA

Node 5: resp6-mean = 0.31500000

Node 1: sf12gh > 72.500000 or NA

Node 3: resp6-mean = 0.44081633

Predictor means below are means of cases with no missing values.

Regression coefficients are computed from the complete cases.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", Statistics in Medicine, v.35, 4837-4855.

2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", Statistics in Medicine, v.38, 545-557.

11.1 Multiple treatment arms: CAPE data 11 RANDOMIZED TREATMENTS

3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification",
in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if sf12gh $\leq$ 72.500000

sf12gh mean = 65.921856

group.DVD	-0.7366E-02	-0.2465	0.8054	0.000	0.3309	1.000
-----------	-------------	---------	--------	-------	--------	-------

group.Phone	0.2188E-01	0.7378	0.4608	0.000	0.3413	1.000
-------------	------------	--------	--------	-------	--------	-------

No truncation of predicted values

Node 2: Intermediate node

A case goes into Node 4 if know $\leq$ 6.5000000

know mean = 5.6087154

Node 4: Intermediate node

A case goes into Node 8 if educyrs $\leq$ 15.500000

educyrs mean = 13.800853

Node 8: Intermediate node

A case goes into Node 16 if yearmam $\leq$ 3.5000000

yearmam mean = 2.0055249

Node 16: Terminal node

Coefficients of least squares regression function:

group.DVD	-0.9843E-01	-1.790	0.7419E-01	0.000	0.3489	1.000
-----------	-------------	--------	------------	-------	--------	-------

group.Phone	0.2237E-02	0.4068E-01	0.9676	0.000	0.3489	1.000
-------------	------------	------------	--------	-------	--------	-------

resp6 mean = 0.299766

No truncation of predicted values

Node 17: Terminal node

Coefficients of least squares regression function:

group.DVD	0.1154	1.037	0.3019	0.000	0.3362	1.000
-----------	--------	-------	--------	-------	--------	-------

group.Phone	0.2674	2.458	0.1550E-01	0.000	0.3707	1.000
-------------	--------	-------	------------	-------	--------	-------

resp6 mean = 0.637931

No truncation of predicted values

Node 9: Terminal node

Coefficients of least squares regression function:

group.DVD	0.2366	2.611	0.9889E-02	0.000	0.3250	1.000
-----------	--------	-------	------------	-------	--------	-------

group.Phone	-0.2165E-01	-0.2244	0.8227	0.000	0.2625	1.000
-------------	-------------	---------	--------	-------	--------	-------

resp6 mean = 0.450000

No truncation of predicted values

Node 5: Terminal node

Coefficients of least squares regression function:

group.DVD	0.2883	3.791	0.1993E-03	0.000	0.3500	1.000
-----------	--------	-------	------------	-------	--------	-------


```

group.Phone      0.1050      1.321      0.1882      0.000      0.2950      1.000
resp6 mean = 0.315000
No truncation of predicted values
-----
Node 3: Terminal node
Coefficients of least squares regression function:
group.DVD      -0.1101      -2.407      0.1634E-01      0.000      0.3156      1.000
group.Phone    -0.3832E-01  -0.8659      0.3868      0.000      0.3619      1.000
resp6 mean = 0.440816
No truncation of predicted values
-----
Number of times Li-Martin approximation used = 3
Proportion of variance (R-squared) explained by tree model: 0.0579

Observed and fitted values are stored in gi.fit
LaTeX code for tree is in gi.tex

```

Figure 25 shows the tree diagram. The tree has 5 terminal nodes (subgroups) and the results for each terminal node give the treatment effects of DVD and Phone versus Control, which is the first treatment level in alphabetical order.

11.2 Censored response: prop. hazards

We now consider a randomized controlled breast cancer trial where the response variable is a censored survival time (Schmoor et al., 1996). The data are in the file `cancerdata.txt`; they are included in the `TH.data` R package (Hothorn, 2017) as well. In the DSC file `cancerdsc.txt` below, the treatment variable is hormone therapy, `horTh`. The variable `time` is (censored) time to recurrence of cancer and the event indicator `event` = 1 if the cancer recurred and = 0 if it did not. Ordinal predictor variables may be designated as “n” or “s” (with this option of no linear prognostic control, n variables are automatically changed to s when the program executes). See Loh et al. (2019a, 2016, 2015, 2019c) and Loh and Zhou (2020) for further analysis of the data.

```

cancerdata.txt
NA
1
1 horTh r
2 age n
3 menostat c
4 tsize n
5 tgrade c
6 pnodes n
7 progrec n

```

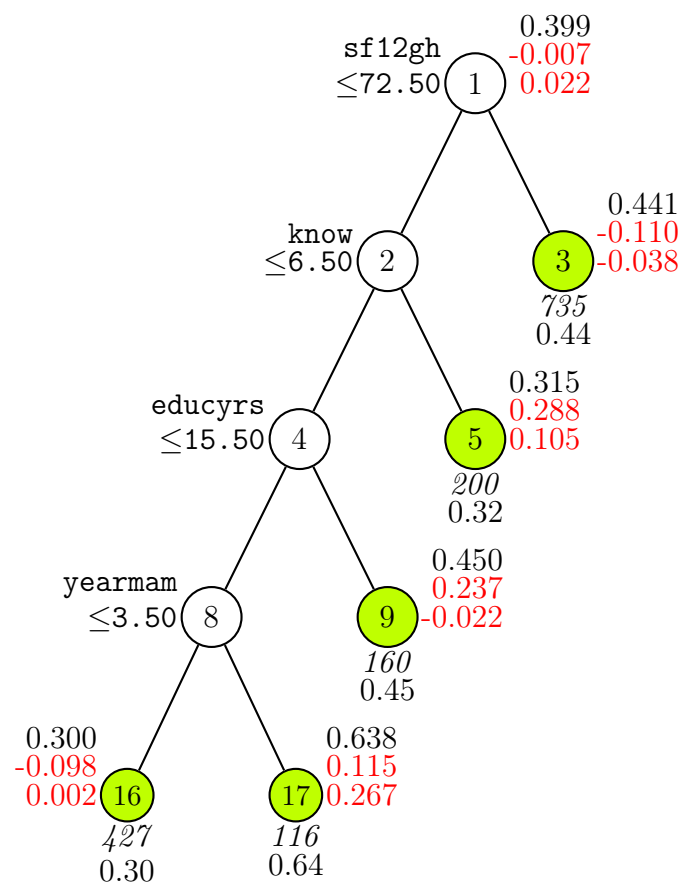


Figure 25: GUIDE v.46.1 0.000-SE (0.250-SE has no splits) least-squares regression tree using Gi option for dependent variable `resp6` and treatment variable `group` without adjustment for linear prognostic effects. At each split, an observation goes to the left branch if and only if the condition is satisfied. Sample size (in *italics*) and mean of `resp6` printed below nodes. `resp6` mean for `group` reference level `Control` followed by `group` effects (in red) of levels DVD, Phone (relative to `Control`) beside nodes. Second best split variable at root node is `know`.

```

8 estrec n
9 time t
10 event d

```

11.2.1 Without linear prognostic control

The simplest model only uses the covariates to split the intermediate nodes; terminal nodes are fitted with treatment means.

Input file generation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: ph-gi.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1):
Name of batch output file: ph-gi.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
  7=binary logistic regression.
Input choice ([1:7], <cr>=1): 4
Input 1 for proportional hazards, 2 for restricted mean event time ([1:2], <cr>=1):
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple linear in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple linear, 3: constant ([1:3], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: cancerdsc.txt
Reading DSC file ...
Training sample file: cancerdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
6 N variables changed to S
Warning: model changed to linear in treatment
D variable is death
Reading data file ...

```

```

Number of records in data file: 686
Length of longest entry in data file: 4
Checking for missing values ...
Finished checking
Assigning integer codes to values of 2 categorical variables
Treatment (R) variable is horTh with values "no" and "yes"
Re-checking data ...
Assigning codes to missing values, if any ...
Data checks complete
Smallest uncensored time: 72.0000
Number of cases dropped due to missing D or T or censored T < smallest uncensored T: 14
Number of complete cases excluding censored T < smallest uncensored T: 672
Number of cases used to compute baseline hazard: 672
Number of cases with D=1 and T >= smallest uncensored: 299
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...
Choose a subgroup identification method:
1 = Prognostic priority (Gs)
2 = Predictive priority (Gi)
Input your choice: ([1:2], <cr>=2):
Creating dummy variables ...
Rereading data ...
Largest uncensored and censored time by horTh
  "no"      2456.0000    2563.0000
  "yes"     2372.0000    2659.0000
Proportion of training sample for each level of horTh
  "no"      0.6399
  "yes"     0.3601
    Total  #cases w/  #missing
    #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      686         0         0        0        0        0        6
    #P-var  #M-var  #B-var  #C-var  #I-var  #R-var
        0         0         0         1         0         1
Survival time variable in column: 9
Event indicator variable in column: 10
Proportion uncensored among nonmissing T and D variables: .445
Number of cases used for training: 672
Number of split variables: 7
Number of dummy variables created: 1
Finished reading data file
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): ph-gi.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values

```

```

Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: ph-gi.fit
Input 2 to write R function for predicting new cases, 1 otherwise ([1:2], <cr>=2):
Input file name: ph-gi.r
Input rank of top variable to split root node ([1:9], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < ph-gi.in

```

Results The contents of ph-gi.out follow.

```

Proportional hazards regression tree for censored response
Pruning by cross-validation
DSC file: cancerdsc.txt
Training sample file: cancerdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
6 N variables changed to S
Warning: model changed to linear in treatment
D variable is death
Piecewise linear model
Number of records in data file: 686
Length of longest entry in data file: 4
Treatment (R) variable is horTh with values "no" and "yes"
Number of cases dropped due to missing D or T or censored T < smallest uncensored T: 14
Number of complete cases excluding censored T < smallest uncensored T: 672
Number of cases used to compute baseline hazard: 672
Number of cases with D=1 and T >= smallest uncensored: 299
Smallest uncensored time: 72.0000
Largest uncensored and censored time by horTh
  horTh      Uncensored      Censored
  "no"      2456.0000      2563.0000
  "yes"      2372.0000      2659.0000
Proportion of training sample for each level of horTh
  "no"      0.6399
  "yes"      0.3601

Summary information for training sample of size 672 (excluding observations
  with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight,
t=survival time variable

```

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	horTh	r			2	
2	age	s	21.00	80.00		
3	menostat	c			2	
4	tsize	s	3.000	120.0		
5	tgrade	s	1.000	3.000		
6	pnodes	s	1.000	51.00		
7	progre	s	0.000	2380.		
8	estrec	s	0.000	1144.		
9	time	t	72.00	2659.		
10	death	d	0.000	1.000		
===== Constructed variables =====						
11	lnbasehaz	z	-6.510	0.5887E-01		
12	horTh.yes	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss.	D	ord. vals	#X-var	#N-var	#F-var	#S-var
686	0	0	0	0	0	0	1	6
#P-var	#M-var	#B-var	#C-var	#I-var	#R-var			
0	0	0	1	0	1			

Number of dummy variables created: 1

Survival time variable in column: 9

Event indicator variable in column: 10

Proportion uncensored among nonmissing T and D variables: 0.445

Number of cases used for training: 672

Constant fitted to cases with missing values in regressor variables

Predictive priority (Gi)

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 8

Minimum node sample size: 6

Minimum fraction of cases per treatment at each node: 0.072

Number of iterations for fitting: 20

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	18	1.571E+00	7.736E-02	8.051E-02	1.474E+00	1.007E-01
2	17	1.570E+00	7.758E-02	8.031E-02	1.482E+00	1.007E-01

3	16	1.562E+00	7.713E-02	8.018E-02	1.480E+00	9.509E-02
4	14	1.562E+00	7.713E-02	8.018E-02	1.480E+00	9.509E-02
5	13	1.559E+00	7.702E-02	7.846E-02	1.480E+00	9.493E-02
6	12	1.560E+00	7.699E-02	7.817E-02	1.480E+00	9.478E-02
7	11	1.550E+00	7.681E-02	7.878E-02	1.457E+00	1.003E-01
8	10	1.547E+00	7.680E-02	7.906E-02	1.457E+00	9.958E-02
9	9	1.521E+00	7.628E-02	7.936E-02	1.410E+00	1.003E-01
10	8	1.490E+00	7.465E-02	8.255E-02	1.424E+00	9.213E-02
11	6	1.463E+00	6.426E-02	6.453E-02	1.425E+00	8.040E-02
12	4	1.418E+00	5.585E-02	3.363E-02	1.443E+00	4.561E-02
13**	2	1.398E+00	5.064E-02	1.949E-02	1.400E+00	2.803E-02
14	1	1.435E+00	5.100E-02	1.066E-02	1.446E+00	1.482E-02

0-SE tree based on mean is marked with * and has 2 terminal nodes

0-SE tree based on median is marked with + and has 2 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Cases fit give the number of cases used to fit node

Deviance is mean residual deviance for all cases in node

Node	Total	Cases	Matrix	Median	Node	Split
label	cases	fit	rank	survtime	deviance	variable
1	672	672	1	1.807E+03	1.431E+00	progrec
2T	274	274	1	1.140E+03	1.601E+00	estrec
3T	398	398	1	2.286E+03	1.188E+00	menostat

Number of terminal nodes of final tree: 2

Total number of nodes of final tree: 3

Second best split variable (based on curvature test) at root node is estrec

Regression tree:

Node 1: progrec <= 21.500000

Node 2: Median survival time = 1140.0000

Node 1: progrec > 21.500000 or NA

Node 3: Median survival time = 2286.0000

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if progrec <= 21.500000

progrec mean = 110.91518

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.000					
horTh.yes	-0.3654	-2.933	0.3471E-02	0.000	0.3601	1.000

Predicted log-relative hazard = -0.30206062E-02

Node 2: Terminal node

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.3729					
horTh.yes	-0.1140	-0.6871	0.4926	0.000	0.3613	1.000

Predicted log-relative hazard = 0.45682185

Node 3: Terminal node

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	-0.2596					
horTh.yes	-0.6453	-3.375	0.8098E-03	0.000	0.3593	1.000

Predicted log-relative hazard = -0.34487497

Observed and fitted values are stored in ph-gi.fit

LaTeX code for tree is in phi-gi.tex

Let $\lambda(u, \mathbf{x})$ denote the hazard function at time u and predictor values $\mathbf{x}$ and let $\lambda_0(u)$ denote the baseline hazard function. The results in `ph-gi.out` show that the fitted proportional hazards model is

$$\begin{aligned} \lambda(u, \mathbf{x}) = & \lambda_0(u) [\exp\{\hat{\beta}_1 + \hat{\gamma}_1 I(\text{horTh} = \text{yes})\} I(\text{progrec} \leq 21.5) \\ & + \exp\{\hat{\beta}_2 + \hat{\gamma}_2 I(\text{horTh} = \text{yes})\} I(\text{progrec} > 21.5)] \end{aligned}$$

with $\hat{\beta}_1 = 0.37292$, $\hat{\gamma}_1 = -0.11404$, $\hat{\beta}_2 = -0.25964$, and $\hat{\gamma}_2 = -0.64531$.

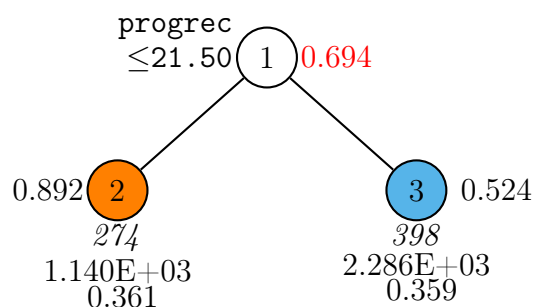


Figure 26: GUIDE v.45.0 0.250-SE proportional hazards regression tree using Gi option for time and event indicator **death** and treatment variable **horTh** without adjustment for linear prognostic effects. Constants fitted to incomplete cases in terminal nodes. At each split, an observation goes to the left branch if and only if the condition is satisfied. Treatment **horTh** hazard ratio of level **yes** to level **no** beside nodes. Sample size (in *italics*), median survival time, and proportion of **horTh** = **yes** printed below nodes. Terminal nodes with treatment hazard ratio above and below 0.694 (ratio at root node) are painted orange and skyblue respectively. Second best split variable at root node is **estrec**.

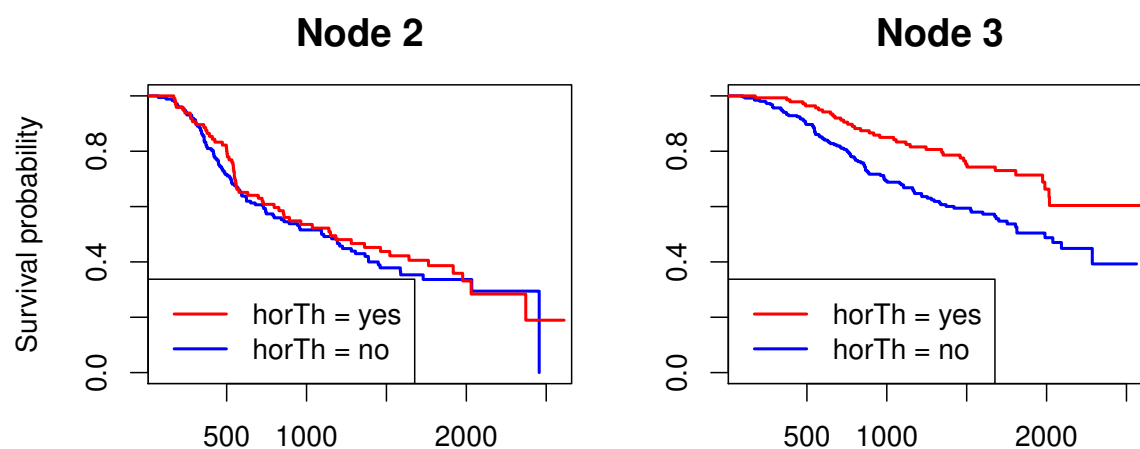


Figure 27: Estimated survival probability functions for breast cancer data

Figure 26 shows the tree diagram. The numbers beside each terminal node are relative hazards of `horTh = yes` versus `no`, namely, $\exp(\hat{\gamma}_1) = \exp(-0.11404) = 0.8922223$ for node 2 and $\exp(\hat{\gamma}_2) = \exp(-0.64531) = 0.5244999$ for node 3. Figure 27 shows Kaplan-Meier survival functions of the data in the terminal nodes. The plots are produced by the following R code.

```
library(survival)
z <- read.table("cancerdata.txt",header=TRUE)
leg.txt <- c("horTh = yes","horTh = no")
leg.col <- c("red","blue")
leg.lty <- 1:2
xr <- range(z$time)
zg <- read.table("ph-gi.fit",header=TRUE)
nodes <- zg$node
uniq.gp <- unique(sort(nodes))
plotted <- FALSE
for(g in uniq.gp){
  gp <- nodes == g
  y <- z$time[gp]
  stat <- z$death[gp]
  treat <- z$horTh[gp]
  fit <- survfit(Surv(y,stat) ~ treat, conf.type="none")
  if(plotted){
    plot(fit,xlim=xr,mark.time=FALSE,xlab="",ylab="",col=c("blue","red"),lwd=2)
  } else {
    plot(fit,xlim=xr,mark.time=FALSE,xlab="",ylab="Survival probability",
         col=c("blue","red"),lwd=2)
    plotted <- TRUE
  }
  title(paste("Node",g))
  legend("bottomleft",legend=leg.txt,lty=1,col=leg.col,lwd=2)
}
```

Estimated relative risks and survival probabilities The file `ph-gi.fit` gives the terminal node number, observed survival time, event indicator (`y`=uncensored, `n`=censored), log baseline cumulative hazard, survival probability, median survival time, and treatment effect (regression coefficient of treatment indicator) of each observation in the training sample (`cancerdata.txt`). The results for the first few observations are shown below.

train	node	observed	event	logbasecumhaz	survivalprob	mediansurvtime	horTh.yes
y	3	1814.00	y	-0.335623	0.576131	2286.00	-0.645311
y	3	2018.00	y	-0.210308	0.720485	2286.00	-0.645311
y	3	712.000	y	-1.28452	0.894065	2286.00	-0.645311

y	3	1807.00	y	-0.358191	0.753697	2286.00	-0.645311
y	3	772.000	y	-1.16232	0.785652	2286.00	-0.645311
y	2	448.000	y	-2.08322	0.834592	1140.00	-0.114042
y	3	2172.00	n	-0.121866	0.698971	2286.00	-0.645311

11.2.2 Simple linear prognostic control

To reduce or eliminate confounding between treatment and covariate variables, it may be desirable to adjust for the effects of the latter by fitting a regression model that allows for the linear effects of one or more prognostic variables in each node (Loh et al., 2019c). This is done by choosing the “simple linear” or the “multiple linear” option and specifying each potential linear predictor as “n” in the DSC file (no change is needed in `cancerdsc.txt`). First we show how to choose the simple linear model, where a single prognostic variable is used as regressor in each node. There are two options: the **Gi** (default) option is more sensitive to detecting *predictive* variables while the **Gs** option is equally sensitive to detecting *prognostic* variables—see Loh et al. (2015) for definitions.

Input file generation for Gi method

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: lin-gi.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1):
Name of batch output file: lin-gi.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
1=linear, 2=quantile, 3=Poisson, 4=censored response,
5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
7=binary logistic regression.
Input choice ([1:7], <cr>=1): 4
Input 1 for proportional hazards, 2 for restricted mean event time ([1:2], <cr>=1):
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple linear in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple linear, 3: constant ([1:3], <cr>=3): 2
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: cancerdsc.txt
```

```

Reading DSC file ...
Training sample file: cancerdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
D variable is death
Reading data file ...
Number of records in data file: 686
Length of longest entry in data file: 4
Checking for missing values ...
Finished checking
Assigning integer codes to values of 2 categorical variables
Treatment (R) variable is horTh with values "no" and "yes"
Re-checking data ...
Assigning codes to missing values, if any ...
Data checks complete
Smallest uncensored time: 72.0000
Number of cases dropped due to missing D or T or censored T < smallest uncensored T: 14
Number of complete cases excluding censored T < smallest uncensored T: 672
Number of cases used to compute baseline hazard: 672
Number of cases with D=1 and T >= smallest uncensored: 299
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...
Choose a subgroup identification method:
1 = Prognostic priority (Gs)
2 = Predictive priority (Gi)
Input your choice: ([1:2], <cr>=2):
Creating dummy variables ...
Rereading data ...
Largest uncensored and censored time by horTh
  "no"      2456.0000    2563.0000
  "yes"     2372.0000    2659.0000
Proportion of training sample for each level of horTh
  "no"      0.6399
  "yes"     0.3601
    Total #cases w/ #missing
    #cases miss. D ord. vals #X-var #N-var #F-var #S-var
      686      0      0      0      6      0      0
    #P-var #M-var #B-var #C-var #I-var #R-var
      0      0      0      1      0      1
Survival time variable in column: 9
Event indicator variable in column: 10
Proportion uncensored among nonmissing T and D variables: .445
Number of cases used for training: 672
Number of split variables: 7
Number of dummy variables created: 1

```

```

Finished reading data file
Warning: missing regressor values imputed with node means
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): lin-gi.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save regressor names in a file, 1 otherwise ([1:2], <cr>=2):
Input file name: lin-gi.reg
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: lin-gi.fit
Input 2 to write R function for predicting new cases, 1 otherwise ([1:2], <cr>=2):
Input file name: lin-gi.r
Input rank of top variable to split root node ([1:9], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < lin-gi.in

```

Results for Gi method The following output shows that the pruned tree is trivial with no splits and that the variable `pnodes` is the best simple linear predictor.

```

Proportional hazards regression tree for censored response
No truncation of predicted values
Pruning by cross-validation
DSC file: cancerdsc.txt
Training sample file: cancerdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
D variable is death
Piecewise simple linear or constant model
Powers are dropped if they are not significant at level 1.0000
Number of records in data file: 686
Length of longest entry in data file: 4
Treatment (R) variable is horTh with values "no" and "yes"
Number of cases dropped due to missing D or T or censored T < smallest uncensored T: 14
Number of complete cases excluding censored T < smallest uncensored T: 672
Number of cases used to compute baseline hazard: 672
Number of cases with D=1 and T >= smallest uncensored: 299
Smallest uncensored time: 72.0000
Largest uncensored and censored time by horTh
  horTh      Uncensored      Censored
  "no"      2456.0000      2563.0000

```

```

"yes"      2372.0000    2659.0000
Proportion of training sample for each level of horTh
"no"       0.6399
"yes"      0.3601

```

Summary information for training sample of size 672 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight,
t=survival time variable

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	horTh	r			2	
2	age	n	21.00	80.00		
3	menostat	c			2	
4	tsize	n	3.000	120.0		
5	tgrade	n	1.000	3.000		
6	pnodes	n	1.000	51.00		
7	progre	n	0.000	2380.		
8	estrec	n	0.000	1144.		
9	time	t	72.00	2659.		
10	death	d	0.000	1.000		
===== Constructed variables =====						
11	lnbasehaz	z	-6.510	0.5887E-01		
12	horTh.yes	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
686	0	0	0	6	1	0	
#P-var	#M-var	#B-var	#C-var	#I-var	#R-var		
0	0	0	1	0	1		

Number of dummy variables created: 1

Survival time variable in column: 9

Event indicator variable in column: 10

Proportion uncensored among nonmissing T and D variables: 0.445

Number of cases used for training: 672

Warning: missing regressor values imputed with node means

Predictive priority (Gi)

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 7

Minimum fraction of cases per treatment at each node: 0.072

Number of iterations for fitting: 20

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	18	1.247E+07	1.219E+07	1.214E+07	2.720E+00	3.919E+06
2	17	1.247E+07	1.219E+07	1.214E+07	2.720E+00	3.919E+06
3	15	1.247E+07	1.219E+07	1.214E+07	2.720E+00	3.919E+06
4	14	1.247E+07	1.219E+07	1.214E+07	2.701E+00	3.919E+06
5	9	1.247E+07	1.219E+07	1.214E+07	2.616E+00	3.919E+06
6	8	2.741E+05	2.739E+05	2.591E+05	2.484E+00	6.230E-01
7	7	2.741E+05	2.739E+05	2.591E+05	1.936E+00	6.723E-01
8	6	2.741E+05	2.739E+05	2.591E+05	1.542E+00	1.931E-01
9++	2	1.370E+00	7.295E-02	5.276E-02	1.320E+00	3.197E-02
10**	1	1.355E+00	5.363E-02	2.719E-02	1.330E+00	2.698E-02

0-SE tree based on mean is marked with * and has 1 terminal node

0-SE tree based on median is marked with + and has 2 terminal node

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

** tree same as -- tree

+ tree same as ++ tree

* tree same as ** tree

* tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Cases fit give the number of cases used to fit node

Deviance is mean residual deviance for all cases in node

Node	Total	Cases	Matrix	Median	Node	Split
label	cases	fit	rank	survtime	deviance	variable
1T	672	672	3	1.807E+03	1.343E+00	estrec

Best split at root node is estrec <= 4.5000

Number of terminal nodes of final tree: 1

Total number of nodes of final tree: 1

Regression tree:

Node 1: Median survival time = 1807.0000

Node 1: Terminal node

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.000					
pnodes	0.5630E-01	8.575	0.000	1.000	4.987	51.00
horTh.yes	-0.3465	-2.778	0.5627E-02	0.000	0.3601	1.000

Observed and fitted values are stored in lin-gi.fit

Regressor names and coefficients are stored in lin-gi.reg

LaTeX code for tree is in lin-gi.tex

The file `lin-gi.reg` reports the selected regressor in each terminal node of the tree (there is only one node here):

node bestvar

1 pnodes

Input file generation for Gs method

0. Read the warranty disclaimer

1. Create a GUIDE input file

Input your choice: 1

Name of batch input file: `lin-gs.in`

Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data conversion ([1:3], <cr>=1):

Name of batch output file: `lin-gs.out`

Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):

Input 1 for classification, 2 for regression, 3 for propensity score tree

Input your choice ([1:3], <cr>=1): 2

Choose type of regression model:

1=linear, 2=quantile, 3=Poisson, 4=censored response,

5=multiresponse or itemresponse, 6=longitudinal data (with T variables),

7=binary logistic regression.

Input choice ([1:7], <cr>=1): 4

Input 1 for proportional hazards, 2 for restricted mean event time ([1:2], <cr>=1):

Choose complexity of model to use at each node:

Choose 1 for multiple regression (recommended for prediction)

Choose 2 for best simple linear in one N or F variable

Choose 3 for constant fit (recommended for interpretability or if there is an R variable)


```

1: multiple linear, 2: best simple linear, 3: constant ([1:3], <cr>=3): 2
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: cancerdsc.txt
Reading DSC file ...
Training sample file: cancerdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
D variable is death
Reading data file ...
Number of records in data file: 686
Length of longest entry in data file: 4
Checking for missing values ...
Finished checking
Assigning integer codes to values of 2 categorical variables
Treatment (R) variable is horTh with values "no" and "yes"
Re-checking data ...
Assigning codes to missing values, if any ...
Data checks complete
Smallest uncensored time: 72.0000
Number of cases dropped due to missing D or T or censored T < smallest uncensored T: 14
Number of complete cases excluding censored T < smallest uncensored T: 672
Number of cases used to compute baseline hazard: 672
Number of cases with D=1 and T >= smallest uncensored: 299
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...
Choose a subgroup identification method:
1 = Prognostic priority (Gs)
2 = Predictive priority (Gi)
Input your choice: ([1:2], <cr>=2): 1
Creating dummy variables ...
Rereading data ...
Largest uncensored and censored time by horTh
  "no"      2456.0000    2563.0000
  "yes"     2372.0000    2659.0000
Proportion of training sample for each level of horTh
  "no"      0.6399
  "yes"     0.3601
  Total  #cases w/  #missing
  #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
    686         0         0        0        6        0        0
  #P-var  #M-var  #B-var  #C-var  #I-var  #R-var
    0         0         0        1        0        1
Survival time variable in column: 9
Event indicator variable in column: 10

```

```

Proportion uncensored among nonmissing T and D variables: .445
Number of cases used for training: 672
Number of split variables: 7
Number of dummy variables created: 1
Finished reading data file
Warning: missing regressor values imputed with node means
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): lin-gs.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save regressor names in a file, 1 otherwise ([1:2], <cr>=2):
Input file name: lin-gs.reg
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: lin-gs.fit
Input 2 to write R function for predicting new cases, 1 otherwise ([1:2], <cr>=2):
Input file name: lin-gs.r
Input rank of top variable to split root node ([1:9], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < lin-gs.in

```

Results for Gs method The Gs method gives a tree with three terminal nodes.

```

Proportional hazards regression tree for censored response
No truncation of predicted values
Pruning by cross-validation
DSC file: cancerdsc.txt
Training sample file: cancerdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
D variable is death
Piecewise simple linear or constant model
Powers are dropped if they are not significant at level 1.0000
Number of records in data file: 686
Length of longest entry in data file: 4
Treatment (R) variable is horTh with values "no" and "yes"
Number of cases dropped due to missing D or T or censored T < smallest uncensored T: 14
Number of complete cases excluding censored T < smallest uncensored T: 672
Number of cases used to compute baseline hazard: 672
Number of cases with D=1 and T >= smallest uncensored: 299
Smallest uncensored time: 72.0000
Largest uncensored and censored time by horTh

```

```

horTh      Uncensored      Censored
"no"       2456.0000      2563.0000
"yes"      2372.0000      2659.0000
Proportion of training sample for each level of horTh
"no"       0.6399
"yes"      0.3601

```

Summary information for training sample of size 672 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight,
t=survival time variable

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	horTh	r			2	
2	age	n	21.00	80.00		
3	menostat	c			2	
4	tsize	n	3.000	120.0		
5	tgrade	n	1.000	3.000		
6	pnodes	n	1.000	51.00		
7	progre	n	0.000	2380.		
8	estrec	n	0.000	1144.		
9	time	t	72.00	2659.		
10	death	d	0.000	1.000		
===== Constructed variables =====						
11	lnbasehaz	z	-6.510	0.5887E-01		
12	horTh.yes	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
686	0	0	0	6	1	0	
#P-var	#M-var	#B-var	#C-var	#I-var	#R-var		
0	0	0	1	0	1		

Number of dummy variables created: 1

Survival time variable in column: 9

Event indicator variable in column: 10

Proportion uncensored among nonmissing T and D variables: 0.445

Number of cases used for training: 672

Warning: missing regressor values imputed with node means

Prognostic priority (Gs)

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 7

Minimum fraction of cases per treatment at each node: 0.072

Number of iterations for fitting: 20

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	14	1.940E+00	1.773E-01	1.115E-01	1.892E+00	1.131E-01
2	13	1.962E+00	1.774E-01	1.183E-01	1.892E+00	1.364E-01
3	12	1.960E+00	1.774E-01	1.180E-01	1.892E+00	1.342E-01
4	9	1.809E+00	1.056E-01	1.040E-01	1.790E+00	1.197E-01
5	8	1.722E+00	1.007E-01	1.006E-01	1.681E+00	1.295E-01
6	7	1.666E+00	1.006E-01	1.034E-01	1.599E+00	1.750E-01
7	6	1.565E+00	9.305E-02	1.036E-01	1.424E+00	9.816E-02
8	4	1.441E+00	6.851E-02	5.627E-02	1.424E+00	7.813E-02
9**	3	1.336E+00	5.196E-02	3.403E-02	1.289E+00	3.960E-02
10	2	1.362E+00	5.631E-02	3.638E-02	1.314E+00	5.650E-02
11	1	1.383E+00	5.502E-02	2.787E-02	1.359E+00	2.776E-02

0-SE tree based on mean is marked with * and has 3 terminal nodes

0-SE tree based on median is marked with + and has 3 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Cases fit give the number of cases used to fit node

Deviance is mean residual deviance for all cases in node

Node	Total	Cases	Matrix	Median	Node	Split
label	cases	fit	rank	survtime	deviance	variable
1	672	672	3	1.807E+03	1.371E+00	pnodes
2	370	370	3	2.659E+03+	1.092E+00	age
4T	142	142	3	2.563E+03+	9.548E-01	tsize
5T	228	228	3	2.030E+03	1.044E+00	tgrade
3T	302	302	3	9.830E+02	1.552E+00	progre

Number of terminal nodes of final tree: 3

Total number of nodes of final tree: 5
 Second best split variable (based on curvature test) at root node is progrec

Regression tree:

Node 1: pnodes <= 3.5000000
 Node 2: age <= 49.500000
 Node 4: Median survival time = 2563.0000+
 Node 2: age > 49.500000 or NA
 Node 5: Median survival time = 2030.0000
 Node 1: pnodes > 3.5000000 or NA
 Node 3: Median survival time = 983.00000

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if pnodes <= 3.5000000

pnodes mean = 4.9866071

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.000					
pnodes	0.5725E-01	8.744	0.000	1.000	4.987	51.00
horTh.yes	-0.3528	-2.828	0.4823E-02	0.000	0.3601	1.000

Node 2: Intermediate node

A case goes into Node 4 if age <= 49.500000

age mean = 53.235135

Node 4: Terminal node

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	5.162					
age	-0.1344	-5.463	0.2096E-06	21.00	43.00	49.00
horTh.yes	-0.7981	-1.502	0.1353	0.000	0.1690	1.000

Node 5: Terminal node

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	0.3737					
progre	-0.3152E-02	-2.547	0.1152E-01	0.000	112.1	1490.
horTh.yes	-0.6723	-2.877	0.4400E-02	0.000	0.4474	1.000

Node 3: Terminal node

Coefficients of log-relative hazard function (relative to baseline hazard):

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	1.039					
progre	-0.2870E-02	-4.036	0.6925E-04	0.000	105.2	2380.
horTh.yes	-0.3303	-2.112	0.3549E-01	0.000	0.3841	1.000

Observed and fitted values are stored in lin-gs.fit

Regressor names and coefficients are stored in lin-gs.reg

LaTeX code for tree is in lin-gs.tex

The tree is shown in Figure 28. It does not display the linear predictor selected at each terminal node. This information is given in the file lin-gs.out or, more conveniently, in tabular form in lin-gs.reg as shown below.

```
node bestvar
4 age
5 progre
3 progre
```

11.3 Censored response: restricted mean

Besides a proportional hazards tree, GUIDE can also fit a tree to estimate the restricted mean survival time in each node (Chen and Tsiatis, 2001; Tian et al., 2014). This section shows how this is carried out. The time restriction may be changed by the user during when the input file is created.

11.3.1 Without linear prognostic control

The piecewise-constant Gi tree has no splits when the restricted mean option is chosen.

Input file generation for Gi method

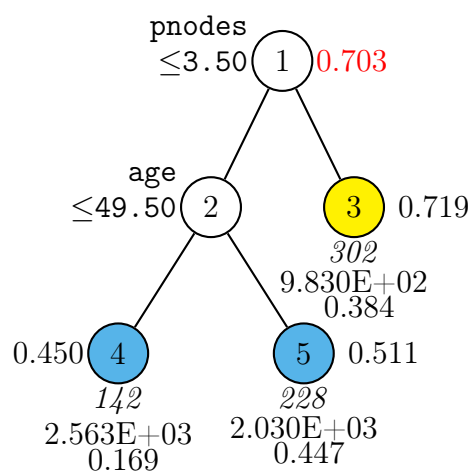


Figure 28: GUIDE v.46.1 0.250-SE proportional hazards regression tree using Gs option for `time` and event indicator `death` and treatment variable `horTh` with adjustment for simple linear prognostic effects. Missing regressor values imputed with node means. At each split, an observation goes to the left branch if and only if the condition is satisfied. Sample size (in *italics*), median survival time, and proportion of `horTh` = `yes` printed below nodes. Treatment `horTh` hazard ratio of level `yes` to `no` beside nodes. Terminal nodes with treatment hazard ratio above and below **0.703** (ratio at root node) are painted **yellow** and **skyblue** respectively. Second best split variable at root node is `progre`.

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: rest-gi.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1):
Name of batch output file: rest-gi.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
  7=binary logistic regression.
Input choice ([1:7], <cr>=1): 4
Input 1 for proportional hazards, 2 for restricted mean event time ([1:2], <cr>=1): 2
Choose complexity of model to use at each node:
Choose 0 for stepwise linear regression (recommended for prediction)
Choose 1 for multiple regression
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
0: stepwise linear, 1: multiple linear, 2: best simple polynomial, 3: constant,
4: best simple stepwise ANCOVA ([0:4], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: cancerdsc.txt
Reading DSC file ...
Training sample file: cancerdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
6 N variables changed to S
Warning: model changed to linear in treatment
D variable is death
Reading data file ...
Number of records in data file: 686
Length of longest entry in data file: 4
Checking for missing values ...
Finished checking
Assigning integer codes to values of 2 categorical variables
Treatment (R) variable is horTh with values "no" and "yes"
Re-checking data ...
Assigning codes to missing values, if any ...
Data checks complete
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...
```



```

Choose a subgroup identification method:
1 = Prognostic priority (Gs)
2 = Predictive priority (Gi)
Input your choice: ([1:2], <cr>=2):
Creating dummy variables ...
Rereading data ...
Largest uncensored and censored time by horTh
      "no"      2456.0000      2563.0000
      "yes"     2372.0000      2659.0000
Smallest observed uncensored time is 72.0000
Largest observed censored or uncensored time is 2659.0000
Input restriction on event time ([72.00:2659.00], <cr>=1222.00):
Proportion of training sample for each level of horTh
      "no"      0.6360
      "yes"     0.3640
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      686      0      0      0      0      0      0      6
      #P-var  #M-var  #B-var  #C-var  #I-var  #R-var
      0      0      0      1      0      1
No weight variable in data file
Number of cases used for training: 533
Number of split variables: 7
Number of dummy variables created: 1
Finished reading data file
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): rest-gi.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: rest-gi.fit
Input 2 to write R function for predicting new cases, 1 otherwise ([1:2], <cr>=2):
Input file name: rest-gi.r
Input rank of top variable to split root node ([1:9], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < rest-gi.in

```

Results for Gi method

```

Restricted mean event time regression tree
Pruning by cross-validation
DSC file: cancerdsc.txt

```

```

Training sample file: cancerdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
6 N variables changed to S
Warning: model changed to linear in treatment
D variable is death
Piecewise linear model
Number of records in data file: 686
Length of longest entry in data file: 4
Treatment (R) variable is horTh with values "no" and "yes"
Smallest uncensored time: 72.0000
Largest uncensored and censored time by horTh
  horTh      Uncensored      Censored
  "no"      2456.0000      2563.0000
  "yes"      2372.0000      2659.0000
Interval for restricted mean event time is from 0 to 1222.
Proportion of training sample for each level of horTh
  "no"      0.6360
  "yes"      0.3640

```

Summary information for training sample of size 533 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	horTh	r			2	
2	age	s	21.00	80.00		
3	menostat	c			2	
4	tsize	s	3.000	120.0		
5	tgrade	s	1.000	3.000		
6	pnodes	s	1.000	36.00		
7	progrec	s	0.000	1490.		
8	estrec	s	0.000	1091.		
9	time	t	72.00	2659.		
10	death	d	0.000	1.000		
===== Constructed variables =====						
11	horTh.yes	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
-------	---------------------	---------------------	-----------	--------	--------	--------	--------

```

        686          0          0          0          0          1          6
      #P-var  #M-var  #B-var  #C-var  #I-var  #R-var
        0          0          0          1          0          1

```

Number of dummy variables created: 1
 No weight variable in data file
 Number of cases used for training: 533

Constant fitted to cases with missing values in regressor variables
 Predictive priority (Gi) using restricted mean event time
 Pruning by v-fold cross-validation, with v = 10
 Selected tree is based on mean of CV estimates
 Number of SE's for pruned tree: 0.2500

No nodewise interaction tests
 Split values for N and S variables based on exhaustive search
 Max number of interaction splits: 100
 Maximum number of split levels: 8
 Minimum node sample size: 6
 Minimum fraction of cases per treatment at each node: 0.073
 Size and CV MSE and SE of subtrees:

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	10	4.514E+05	2.208E+04	1.033E+04	4.444E+05	1.310E+04
2	8	4.514E+05	2.208E+04	1.033E+04	4.444E+05	1.310E+04
3	6	4.514E+05	2.208E+04	1.033E+04	4.444E+05	1.310E+04
4	5	4.488E+05	2.182E+04	9.798E+03	4.433E+05	1.329E+04
5	2	4.408E+05	2.079E+04	8.011E+03	4.414E+05	1.228E+04
6**	1	4.338E+05	1.732E+04	6.012E+03	4.385E+05	7.335E+03

0-SE tree based on mean is marked with * and has 1 terminal node
 0-SE tree based on median is marked with + and has 1 terminal node
 Selected-SE tree based on mean using naive SE is marked with **
 Selected-SE tree based on mean using bootstrap SE is marked with --
 Selected-SE tree based on median and bootstrap SE is marked with ++
 * tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is weighted mean of death in the node
 Cases fit give the number of cases used to fit node
 MSE and R² are based on all cases in node

Node	Total	Cases	Matrix	Node	Node	Node	Split	Other
label	cases	fit	rank	D-mean	MSE	R ²	variable	variables
1T	533	533	2	9.873E+02	1.519E+05	0.0106	estrec	

Best split at root node is estrec <= 8.5000

```
Number of terminal nodes of final tree: 1
Total number of nodes of final tree: 1
```

```
Regression tree:
```

```
Node 1: terminal
```

```
*****
```

```
Node 1: Terminal node
```

```
Coefficients of least squares regression function:
```

```
horTh.yes      73.85      2.385      0.1744E-01      0.000      0.3591      1.000
```

```
time mean = 987.273
```

```
No truncation of predicted values
```

```
-----
```

```
Observed and fitted values are stored in rest-gi.fit
```

```
LaTeX code for tree is in rest-gi.tex
```

Results for Gs method The piecewise-constant Gs tree has one split, as shown below.

```
Restricted mean event time regression tree
```

```
Pruning by cross-validation
```

```
DSC file: cancerdsc.txt
```

```
Training sample file: cancerdata.txt
```

```
Missing value code: NA
```

```
Records in data file start on line 2
```

```
R variable present
```

```
6 N variables changed to S
```

```
Warning: model changed to linear in treatment
```

```
D variable is death
```

```
Piecewise linear model
```

```
Number of records in data file: 686
```

```
Length of longest entry in data file: 4
```

```
Treatment (R) variable is horTh with values "no" and "yes"
```

```
Smallest uncensored time: 72.0000
```

```
Largest uncensored and censored time by horTh
```

horTh	Uncensored	Censored
"no"	2456.0000	2563.0000
"yes"	2372.0000	2659.0000

```
Interval for restricted mean event time is from 0 to 1222.
```

```
Proportion of training sample for each level of horTh
```

```
"no"      0.6360
```

"yes" 0.3640

Summary information for training sample of size 533 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
 d=dependent, b=split and fit cat variable using indicator variables,
 c=split-only categorical, i=fit-only categorical (via indicators),
 s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
 m=missing-value flag variable, p=periodic variable, w=weight

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	horTh	r			2	
2	age	s	21.00	80.00		
3	menostat	c			2	
4	tsize	s	3.000	120.0		
5	tgrade	s	1.000	3.000		
6	pnodes	s	1.000	36.00		
7	progre	s	0.000	1490.		
8	estrec	s	0.000	1091.		
9	time	t	72.00	2659.		
10	death	d	0.000	1.000		
===== Constructed variables =====						
11	horTh.yes	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
686	0	0	0	0	0	1	6
#P-var	#M-var	#B-var	#C-var	#I-var	#R-var		
0	0	0	1	0	1		

Number of dummy variables created: 1

No weight variable in data file

Number of cases used for training: 533

Constant fitted to cases with missing values in regressor variables

Prognostic priority (Gs) using restricted mean event time

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 8

Minimum node sample size: 6

Minimum fraction of cases per treatment at each node: 0.073

Size and CV MSE and SE of subtrees:

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	15	4.129E+05	2.146E+04	1.382E+04	4.201E+05	2.357E+04
2	14	4.129E+05	2.146E+04	1.382E+04	4.201E+05	2.357E+04
3	13	4.129E+05	2.146E+04	1.382E+04	4.201E+05	2.357E+04
4	11	4.129E+05	2.146E+04	1.382E+04	4.201E+05	2.357E+04
5	10	4.127E+05	2.146E+04	1.387E+04	4.201E+05	2.374E+04
6	8	4.123E+05	2.145E+04	1.405E+04	4.201E+05	2.375E+04
7	6	4.118E+05	2.136E+04	1.404E+04	4.178E+05	2.348E+04
8	5	4.090E+05	2.127E+04	1.396E+04	4.020E+05	2.373E+04
9	4	4.094E+05	2.122E+04	1.393E+04	4.039E+05	2.327E+04
10	3	4.028E+05	2.096E+04	1.376E+04	4.023E+05	1.971E+04
11**	2	3.798E+05	1.852E+04	1.523E+04	3.804E+05	1.576E+04
12	1	4.338E+05	1.732E+04	6.012E+03	4.385E+05	7.335E+03

0-SE tree based on mean is marked with * and has 2 terminal nodes

0-SE tree based on median is marked with + and has 2 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is weighted mean of death in the node

Cases fit give the number of cases used to fit node

MSE and R² are based on all cases in node

Node	Total	Cases	Matrix	Node	Node	Node	Split	Other
label	cases	fit	rank	D-mean	MSE	R ²	variable	variables
1	533	533	2	9.873E+02	1.519E+05	0.0106	pnodes	
2T	332	332	2	1.073E+03	1.048E+05	0.0129	estrec	
3T	201	201	2	8.312E+02	1.842E+05	0.0174	progrec	

Number of terminal nodes of final tree: 2

Total number of nodes of final tree: 3

Second best split variable (based on curvature test) at root node is progrec

Regression tree:

Node 1: pnodes <= 4.5000000

Node 2: terminal

Node 1: pnodes > 4.5000000 or NA

Node 3: terminal

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.

2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.

3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if pnodes $\leq$ 4.5000000

pnodes mean = 4.8475943

horTh.yes	73.85	2.385	0.1744E-01	0.000	0.3591	1.000
-----------	-------	-------	------------	-------	--------	-------

No truncation of predicted values

Node 2: Terminal node

Coefficients of least squares regression function:

horTh.yes	66.83	2.074	0.3884E-01	0.000	0.3483	1.000
-----------	-------	-------	------------	-------	--------	-------

time mean = 1072.91

No truncation of predicted values

Node 3: Terminal node

Coefficients of least squares regression function:

horTh.yes	106.5	1.879	0.6164E-01	0.000	0.3786	1.000
-----------	-------	-------	------------	-------	--------	-------

time mean = 831.171

No truncation of predicted values

Observed and fitted values are stored in rest-gs.fit

LaTeX code for tree is in rest-gs.tex

11.3.2 With linear prognostic control

A trivial tree is obtained for both the Gi and Gs methods if a linear regressor is included in each node.

12 Nonrandomized treatments: RHC data

For the RHC data, GUIDE can fit a tree model to find subgroups where the treatment (represented by variable swang1) is beneficial or not for survival. This is done by

specifying the treatment variable as “r” and the event variable **death** (1=die, 0=not die) as “d” in the DSC file `rhcdsc3.txt` below.

```
rhcdsc3.txt
NA
2
1 X x
2 cat1 c
3 cat2 c
4 ca c
5 sadmdte x
6 dschdte x
7 dthdte x
8 lstctdte x
9 death d
10 cardiohx c
11 chfhx c
12 dementhx c
13 psychhx c
14 chrpulhx c
15 renalhx c
16 liverhx c
17 giblethx c
18 malighx c
19 immunhx c
20 transhx c
21 amihx c
22 age n
23 sex c
24 edu n
25 surv2md1 n
26 das2d3pc n
27 t3d30 x
28 dth30 x
29 aps1 n
30 scoma1 n
31 meanbp1 n
32 wblc1 n
33 hrt1 n
34 resp1 n
35 temp1 n
36 pafi1 n
37 alb1 n
38 hema1 n
39 bili1 n
40 crea1 n
41 sod1 n
```



```

42 pot1 n
43 paco21 n
44 ph1 n
45 swang1 r
46 wtkilo1 n
47 dnr1 c
48 ninsclas c
49 resp c
50 card c
51 neuro c
52 gastr c
53 renal c
54 meta c
55 hema c
56 seps c
57 trauma c
58 ortho c
59 adld3p n
60 urin1 n
61 race c
62 income c
63 ptid x
64 survtime t

```

12.1 Proportional hazards

GUIDE can fit models with the Gi or Gs options. The Gi option is designed to be sensitive to detect *predictive* variables (variables that have interactions with the treatment variable) while Gs option is equally sensitive to such variables as well as *prognostic* variables (those that have an effect on the outcome irrespective of the treatment). See [Loh et al. \(2015\)](#) for details.

12.1.1 Gi option

Gi input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: surv-gi.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1):
Name of batch output file: surv-gi.out

```

12.1 Proportional hazards12 NONRANDOMIZED TREATMENTS: RHC DATA

```
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
  7=binary logistic regression.
Input choice ([1:7], <cr>=1): 4
Input 1 for proportional hazards, 2 for restricted mean event time ([1:2], <cr>=1):
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple linear in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple linear, 3: constant ([1:3], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: rhcdsc3.txt
Reading DSC file ...
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
23 N variables changed to S
Warning: model changed to linear in treatment
D variable is death
Reading data file ...
Number of records in data file: 5735
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 31 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Treatment (R) variable is swang1 with values "NoRHC" and "RHC"
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Finished processing 5000 of 5735 observations
Data checks complete
Smallest uncensored survtime: 2.0000
Number of complete cases excluding censored T < smallest uncensored T: 5735
Number of cases used to compute baseline hazard: 5735
```

12.1 Proportional hazards12 NONRANDOMIZED TREATMENTS: RHC DATA

```
Number of cases with D=1 and T >= smallest uncensored: 3722
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...
Choose a subgroup identification method:
1 = Prognostic priority (Gs)
2 = Predictive priority (Gi)
Input your choice: ([1:2], <cr>=2):
Creating dummy variables ...
Creating missing value indicators ...
Rereading data ...
Largest uncensored and censored survtime by swang1
"NorHC"      1867.0000    1243.0000
"RHC"        1943.0000    1351.0000
Proportion of training sample for each level of swang1
"NorHC"      0.6192
"RHC"        0.3808
  Total  #cases w/  #missing
#cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
  5735      0      5157      8      0      0      23
#P-var  #M-var  #B-var  #C-var  #I-var  #R-var
    0      0      0      30      0      1
Survival time variable in column: 64
Event indicator variable in column: 9
Proportion uncensored among nonmissing T and D variables: .649
Number of cases used for training: 5735
Number of split variables: 53
Number of dummy variables created: 1
Number of cases excluded due to 0 W or missing D, T or R variables: 0
Finished reading data file
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): surv-gi.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: surv-gi.fit
Input 2 to write R function for predicting new cases, 1 otherwise ([1:2], <cr>=2):
Input file name: surv-gi.r
Input rank of top variable to split root node ([1:55], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < surv-gi.in
```

Contents of surv-gi.out

12.1 Proportional hazards12 NONRANDOMIZED TREATMENTS: RHC DATA

```

Proportional hazards regression tree for censored response
Pruning by cross-validation
DSC file: rhcdsc3.txt
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
23 N variables changed to S
Warning: model changed to linear in treatment
D variable is death
Piecewise linear model
Number of records in data file: 5735
Length of longest entry in data file: 19
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Treatment (R) variable is swang1 with values "NoRHC" and "RHC"
Number of complete cases excluding censored T < smallest uncensored T: 5735
Number of cases used to compute baseline hazard: 5735
Number of cases with D=1 and T >= smallest uncensored: 3722
Smallest uncensored survtime: 2.0000
Largest uncensored and censored survtime by swang1
  swang1      Uncensored      Censored
  "NoRHC"      1867.0000      1243.0000
  "RHC"        1943.0000      1351.0000
Proportion of training sample for each level of swang1
  "NoRHC"      0.6192
  "RHC"        0.3808

```

Summary information for training sample of size 5735
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight,
t=survival time variable

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
2	cat1	c			9	
3	cat2	c			6	4535
4	ca	c			3	
9	death	d	0.000	1.000		
10	cardiohx	c			2	
11	chfhx	c			2	
:						
56	seps	c			2	

12.1 Proportional hazards12 NONRANDOMIZED TREATMENTS: RHC DATA

```

57 trauma      c                      2
58 ortho       c                      2
59 adld3p      s      0.000      7.000      4296
60 urin1       s      0.000      9000.      3028
61 race        c                      3
62 income      c                      4
64 survtime    t      2.000      1943.
===== Constructed variables =====
65 lnbasehaz0  z      -3.818      2.038
66 swang1.RHC  f      0.000      1.000

```

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/	#missing					
#cases	miss.	D	ord. vals	#X-var	#N-var	#F-var	#S-var
5735	0		5157	8	0	1	23
#P-var	#M-var	#B-var	#C-var	#I-var	#R-var		
0	0	0	30	0	1		

Number of dummy variables created: 1

Survival time variable in column: 64

Event indicator variable in column: 9

Proportion uncensored among nonmissing T and D variables: 0.649

Number of cases used for training: 5735

Number of cases excluded due to 0 W or missing D, T or R variables: 0

Predictive priority (Gi)

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 8

Minimum node sample size: 7

Minimum fraction of cases per treatment at each node: 0.076

Number of iterations for fitting: 20

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	21	1.401E+00	2.992E-02	3.401E-02	1.353E+00	3.843E-02
2	20	1.396E+00	2.973E-02	3.530E-02	1.353E+00	4.072E-02
3	19	1.387E+00	2.888E-02	3.537E-02	1.353E+00	3.181E-02
4	17	1.359E+00	2.024E-02	1.939E-02	1.352E+00	1.744E-02
5	15	1.356E+00	2.002E-02	1.931E-02	1.349E+00	1.559E-02
6	11	1.354E+00	1.976E-02	1.946E-02	1.337E+00	1.324E-02
7	10	1.353E+00	1.974E-02	1.937E-02	1.338E+00	1.235E-02
8	9	1.340E+00	1.626E-02	9.638E-03	1.336E+00	9.990E-03

12.1 Proportional hazards12 NONRANDOMIZED TREATMENTS: RHC DATA

9	8	1.338E+00	1.624E-02	1.003E-02	1.337E+00	9.677E-03
10	7	1.338E+00	1.621E-02	1.009E-02	1.337E+00	9.949E-03
11--	6	1.327E+00	1.561E-02	1.001E-02	1.329E+00	1.023E-02
12**	5	1.330E+00	1.552E-02	8.802E-03	1.330E+00	1.108E-02
13	1	1.367E+00	1.526E-02	6.317E-03	1.358E+00	9.980E-03

0-SE tree based on mean is marked with * and has 6 terminal nodes

0-SE tree based on median is marked with + and has 6 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree same as + tree

** tree same as ++ tree

* tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Cases fit give the number of cases used to fit node

Deviance is mean residual deviance for all cases in node

Node label	Total cases	Cases fit	Matrix rank	Median survtime	Node deviance	Split variable
1	5735	5735	1	1.920E+02	1.367E+00	ph1
2	1411	1411	1	1.150E+02	1.454E+00	cat2
4T	1307	1307	1	1.570E+02	1.416E+00	paco21
5T	104	104	1	1.400E+01	1.636E+00	malighx
3	4324	4324	1	2.070E+02	1.334E+00	resp1
6	3341	3341	1	2.200E+02	1.333E+00	paco21
12T	687	687	1	6.900E+01	1.531E+00	income
13T	2654	2654	1	2.390E+02	1.265E+00	paco21
7T	983	983	1	1.640E+02	1.319E+00	hrt1

Number of terminal nodes of final tree: 5

Total number of nodes of final tree: 9

Second best split variable (based on curvature test) at root node is resp1

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: ph1 <= 7.3344730

Node 2: cat2 = "MOSF w/Sepsis", "NA"

Node 4: Median survival time = 157.00000

Node 2: cat2 /= "MOSF w/Sepsis", "NA"

Node 5: Median survival time = 14.000000

Node 1: ph1 > 7.3344730 or NA

12.1 Proportional hazards12 NONRANDOMIZED TREATMENTS: RHC DATA

```
Node 3: resp1 <= 38.500000 or NA
Node 6: paco21 <= 29.498050
Node 12: Median survival time = 69.000000
Node 6: paco21 > 29.498050 or NA
Node 13: Median survival time = 239.00000
Node 3: resp1 > 38.500000
Node 7: Median survival time = 164.00000

*****
Predictor means below are means of cases with no missing values.
Regression coefficients are computed from the complete cases.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects
for longitudinal and multiresponse variables", Statistics in Medicine, v.35, 4837-4855.

2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic
effects and post-selection inference", Statistics in Medicine, v.38, 545-557.

3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification",
in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node
A case goes into Node 2 if ph1 <= 7.3344730
ph1 mean = 7.3884135
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value      Minimum      Mean      Maximum
Constant        0.000
swang1.RHC      0.1504       4.494       0.7131E-05   0.000       0.3808    1.000
-----
Node 2: Intermediate node
A case goes into Node 4 if cat2 = "MOSF w/Sepsis", "NA"
cat2 mode = "NA"
-----
Node 4: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value      Minimum      Mean      Maximum
Constant      -0.6181E-01
swang1.RHC     0.4067       6.034       0.2086E-08   0.000       0.4499    1.000
-----
Node 5: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value      Minimum      Mean      Maximum
Constant        0.8005
swang1.RHC     -0.3295      -1.558       0.1223       0.000       0.3558    1.000
```

12.1 Proportional hazards 12 NONRANDOMIZED TREATMENTS: RHC DATA

```

-----
Node 3: Intermediate node
A case goes into Node 6 if resp1 <= 38.500000 or NA
resp1 mean = 28.418652
-----
Node 6: Intermediate node
A case goes into Node 12 if paco21 <= 29.498050
paco21 mean = 36.054906
-----
Node 12: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value      Minimum      Mean      Maximum
Constant       0.3006
swang1.RHC     -0.3237E-01  -0.3424      0.7322        0.000        0.3916      1.000
-----
Node 13: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value      Minimum      Mean      Maximum
Constant      -0.7105E-01
swang1.RHC     0.5937E-02   0.1159      0.9078        0.000        0.3632      1.000
-----
Node 7: Terminal node
Coefficients of log-relative hazard function (relative to baseline hazard):
Regressor      Coefficient  t-stat      p-value      Minimum      Mean      Maximum
Constant      -0.1150E-01
swang1.RHC     0.3555        4.329      0.1651E-04    0.000        0.3316      1.000
-----
Observed and fitted values are stored in surv-gi.fit
LaTeX code for tree is in surv-gi.tex

```

Figure 29 shows the tree and Figure 30 shows the estimated survival curves in its terminal nodes. The R code for making the plots is given below.

```

library(survival)
z0 <- read.table("rhcddata.txt",header=TRUE)
par(mar=c(3,4,3,1),mfrow=c(2,3),cex=1)
leg.txt <- c("NoRHC","RHC"); leg.col <- c("blue","red"); leg.lty <- 2:1
xr <- range(z0$survtime)
zg <- read.table("surv-gi.fit",header=TRUE)
nodes <- zg$node
uniq.gp <- unique(sort(nodes))
ii <- 0
for(g in uniq.gp){
  ii <- ii+1
  gp <- nodes == g

```

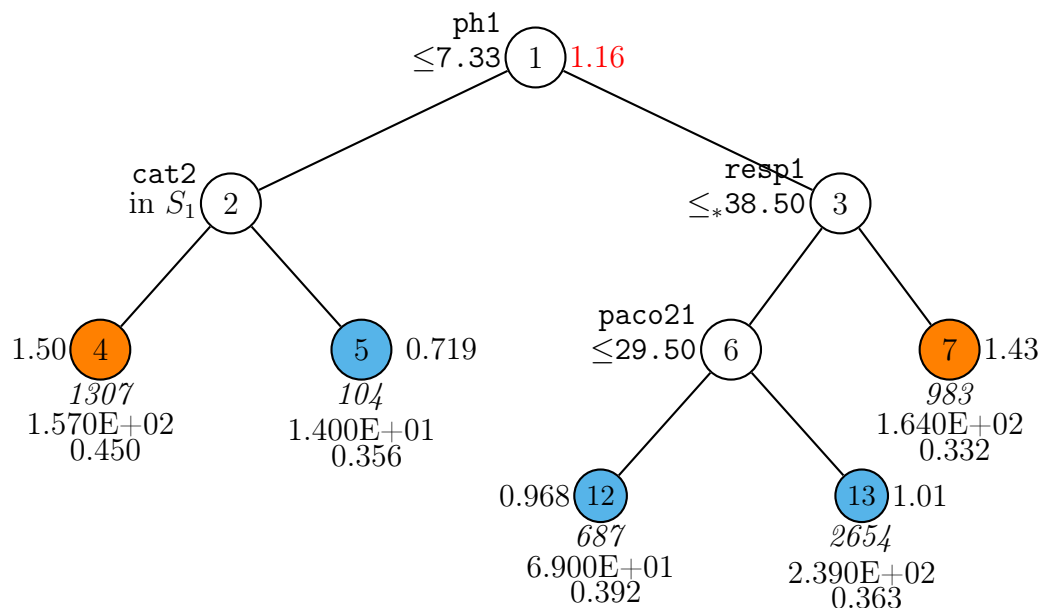



Figure 29: GUIDE v.46.1 0.250-SE proportional hazards regression tree using Gi option for `survtime` and event indicator `death` and treatment variable `swang1` without adjustment for linear prognostic effects. At each split, an observation goes to the left branch if and only if the condition is satisfied. Symbol ' $\leq_*$ ' stands for ' $\leq$ or missing'. $S_1 = \{\text{MOSF w/Sepsis, NA}\}$. Treatment `swang1` hazard ratio of level RHC to level NoRHC beside nodes. Sample size (in *italics*), median survival time, and proportion of `swang1` = RHC printed below nodes. Terminal nodes with treatment hazard ratio above and below 1.162 (ratio at root node) are painted orange and skyblue respectively. Second best split variable at root node is `resp1`.

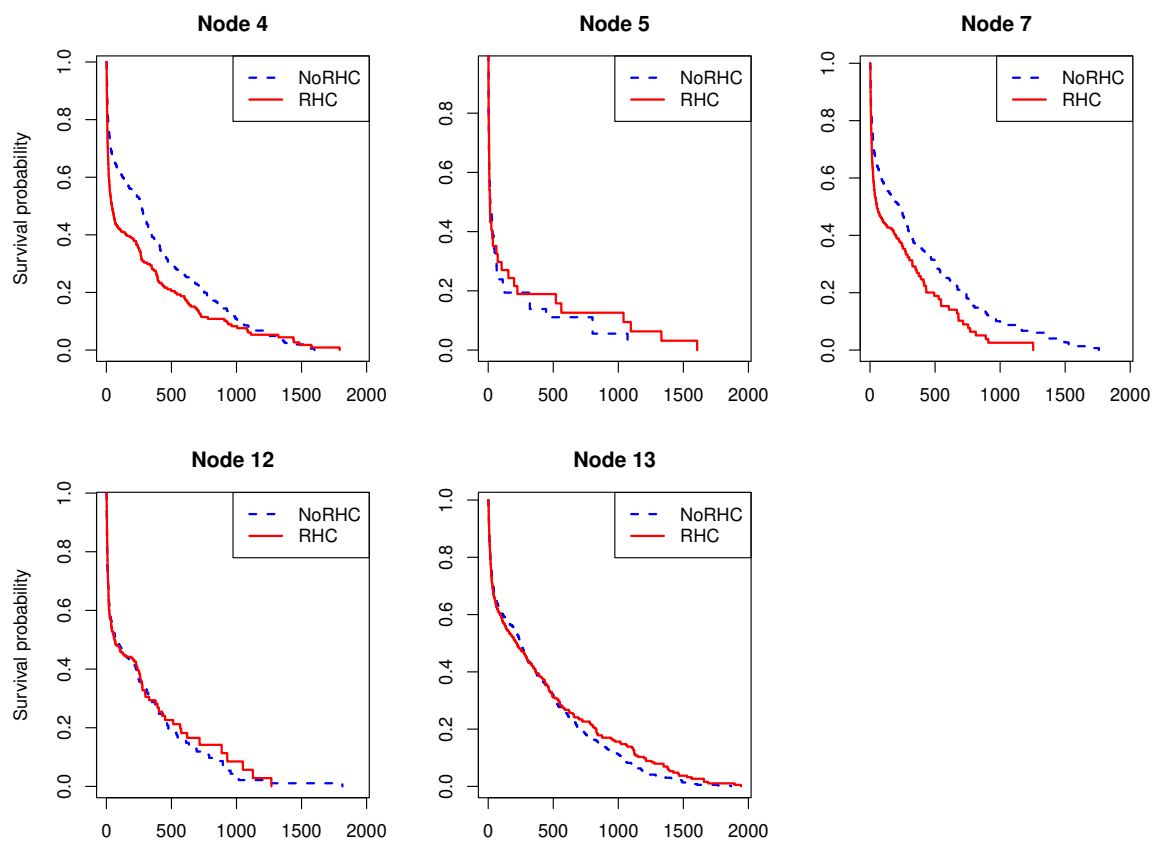


Figure 30: Survival curves for RHC data in nodes of Figure 29

```

y <- z0$survtime[gp]
stat <- z0$death[gp]
treat <- z0$swang1[gp]
fit <- survfit(Surv(y,stat) ~ treat, conf.type="none")
if(g == 4 | g == 12){
  plot(fit,xlim=xr,mark.time=FALSE,xlab="",ylab="Survival probability",
       col=leg.col,lwd=2,lty=leg.lty)
} else {
  plot(fit,xlim=xr,mark.time=FALSE,xlab="",ylab="",col=leg.col,lwd=2,lty=leg.lty)
}
title(paste("Node",g))
legend("topright",legend=leg.txt,lty=leg.lty,col=leg.col,lwd=2)
}

```

Following are the top 3 lines of the file `surv-gi.fit`

train	node	observed	event	logbasecumhaz	survivalprob	mediansurvtime	swang1.RHC
y	13	240.000	n	-0.269165	0.490850	239.000	0.593672E-002
y	4	45.0000	y	-0.757608	0.515901	157.000	0.406690
y	7	317.000	n	-0.633003E-001	0.266047	164.000	0.355517

The column definitions are

train: y if the observation is used for model fitting, n if not.

node: terminal node label of observation.

observed: observed survival time t .

event: y if uncensored (death), n if censored.

logbasecumhaz: log of the estimated baseline cumulative hazard function $\log \Lambda_0(t) = \log \int_0^t \lambda_0(u) du$ at observed time t .

survivalprob: probability that the subject survives up to observed time t . For the first subject, this is

$$\begin{aligned}
\exp\{-\Lambda_0(t) \exp(\beta' \mathbf{x})\} &= \exp\{-\exp(\beta_0 + \text{logbasecumhaz})\} \\
&= \exp(-\exp(-0.242135921383 - 0.3029494)) \\
&= 0.5600147
\end{aligned}$$

where $t = 240$ and $\beta_0 = -0.242135921383$ is the constant term in the node (`surv-gs.r` gives β_0 to higher precision than `surv-gs.out`).

mediansurvtime: median survival time among observations in node estimated from Kaplan-Meier survival function. A trailing plus (+) sign indicates estimate is censored.

swang1.RHC: estimated treatment effect β_1 for level RHC of **swang1**.

12.2 Restricted mean

GUIDE can also construct a tree model such that a restricted mean event time ([Chen and Tsiatis, 2001](#); [Tian et al., 2014](#)) is fitted in each node of the tree.

12.2.1 Gi option

Gi input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: rest-gi.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: rest-gi.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 4
Input 1 for proportional hazards, 2 for restricted mean event time ([1:2], <cr>=1): 2
Choose complexity of model to use at each node:
Choose 0 for stepwise linear regression (recommended for prediction)
Choose 1 for multiple regression
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
0: stepwise linear, 1: multiple linear, 2: best simple polynomial, 3: constant,
4: best simple stepwise ANCOVA ([0:4], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: rhcdsc3.txt
Reading DSC file ...
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
```

```
R variable present
23 N variables changed to S
Warning: model changed to linear in treatment
D variable is death
Reading data file ...
Number of records in data file: 5735
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 31 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Treatment (R) variable is swang1 with values "NoRHC" and "RHC"
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Finished processing 5000 of 5735 observations
Data checks complete
GUIDE will try to create the variables in the DSC file.
If it is unsuccessful, please create the columns yourself...
Number of dummy variables created: 1
Choose a subgroup identification method:
1 = Prognostic priority (Gs)
2 = Predictive priority (Gi)
Input your choice: ([1:2], <cr>=2):
Creating dummy variables ...
Creating missing value indicators ...
Rereading data ...
Largest uncensored and censored survtime by swang1
  "NoRHC"      1867.0000    1243.0000
  "RHC"        1943.0000    1351.0000
Smallest observed uncensored time is 2.0000
Largest observed censored or uncensored time is 1943.0000
Input restriction on event time ([2.00:1943.00], <cr>=622.00):
Proportion of training sample for each level of swang1
  "NoRHC"      0.5993
  "RHC"        0.4007
    Total  #cases w/  #missing
    #cases   miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      5735         0     5157      8      0      0      23
    #P-var  #M-var  #B-var  #C-var  #I-var  #R-var
        0        0        0     30      0      1
```

```

No weight variable in data file
Number of cases used for training: 3763
Number of split variables: 53
Number of dummy variables created: 1
Number of cases excluded due to 0 W or missing D or R variables: 1972
Finished reading data file
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): rest-gi.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: rest-gi.fit
Input 2 to write R function for predicting new cases, 1 otherwise ([1:2], <cr>=2):
Input file name: rest-gi.r
Input rank of top variable to split root node ([1:55], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < rest-gi.in

```

Contents of rest-gi.out

```

Restricted mean event time regression tree
Pruning by cross-validation
DSC file: rhcdsc3.txt
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
23 N variables changed to S
Warning: model changed to linear in treatment
D variable is death
Piecewise linear model
Number of records in data file: 5735
Length of longest entry in data file: 19
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Treatment (R) variable is swang1 with values "NoRHC" and "RHC"
Smallest uncensored survtime: 2.0000
Largest uncensored and censored survtime by swang1

```

swang1	Uncensored	Censored
"NoRHC"	1867.0000	1243.0000
"RHC"	1943.0000	1351.0000

```

Interval for restricted mean event time is from 0 to 622.

```

Proportion of training sample for each level of swang1

```
"NoRHC"    0.5993
  "RHC"     0.4007
```

Summary information for training sample of size 3763 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
2	cat1	c			9	
3	cat2	c			6	2836
4	ca	c			3	
9	death	d	0.000	1.000		
10	cardiohx	c			2	
:						
58	ortho	c			2	
59	adld3p	s	0.000	7.000		3041
60	urin1	s	0.000	9000.		2115
61	race	c			3	
62	income	c			4	
64	survtime	t	2.000	1943.		
===== Constructed variables =====						
65	swang1.RHC	f	0.000	1.000		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
5735	0	5157	8	0	1	23	
#P-var	#M-var	#B-var	#C-var	#I-var	#R-var		
0	0	0	30	0	1		

Number of dummy variables created: 1

No weight variable in data file

Number of cases used for training: 3763

Number of cases excluded due to 0 W or missing D or R variables: 1972

Predictive priority (Gi) using restricted mean event time

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100
 Maximum number of split levels: 8
 Minimum node sample size: 7
 Minimum fraction of cases per treatment at each node: 0.080
 Size and CV MSE and SE of subtrees:

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	17	1.194E+05	3.630E+03	2.243E+03	1.185E+05	1.903E+03
2	16	1.194E+05	3.631E+03	2.232E+03	1.185E+05	1.902E+03
:						
14	3	1.185E+05	3.572E+03	1.952E+03	1.178E+05	1.228E+03
15**	2	1.154E+05	3.251E+03	1.536E+03	1.149E+05	1.343E+03
16	1	1.198E+05	3.143E+03	9.972E+02	1.190E+05	1.421E+03

0-SE tree based on mean is marked with * and has 2 terminal nodes
 0-SE tree based on median is marked with + and has 2 terminal nodes
 Selected-SE tree based on mean using naive SE is marked with **
 Selected-SE tree based on mean using bootstrap SE is marked with --
 Selected-SE tree based on median and bootstrap SE is marked with ++
 * tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is weighted mean of death in the node

Cases fit give the number of cases used to fit node

MSE and R² are based on all cases in node

Node label	Total cases	Cases fit	Matrix rank	Node D-mean	Node MSE	Node R ²	Split variable	Other variables
1	3763	3763	2	2.583E+02	9.489E+04	0.0043	scoma1	
2T	3124	3124	2	2.781E+02	9.938E+04	0.0075	pafi1	
3T	639	639	2	1.333E+02	4.975E+04	0.0016	sod1	

Number of terminal nodes of final tree: 2

Total number of nodes of final tree: 3

Second best split variable (based on curvature test) at root node is ph1

Regression tree:

Node 1: scoma1 <= 49.500000

Node 2: terminal

Node 1: scoma1 > 49.500000 or NA

Node 3: terminal

 Predictor means below are means of cases with no missing values.

Regression coefficients are computed from the complete cases.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

```
Node 1: Intermediate node
A case goes into Node 2 if scoma1 <= 49.500000
scoma1 mean = 20.462797
swang1.RHC    -33.80      -4.020      0.5926E-04    0.000      0.3808      1.000
No truncation of predicted values
-----
Node 2: Terminal node
Coefficients of least squares regression function:
swang1.RHC    -44.75      -4.866      0.1195E-05    0.000      0.3949      1.000
survtime mean = 278.051
No truncation of predicted values
-----
Node 3: Terminal node
Coefficients of least squares regression function:
swang1.RHC    -17.66      -1.003      0.3161      0.000      0.2916      1.000
survtime mean = 133.272
No truncation of predicted values
-----
Number of times Li-Martin approximation used = 51
Observed and fitted values are stored in rest-gi.fit
LaTeX code for tree is in rest-gi.tex
```

Figure 31 shows the Gi restricted mean event time tree.

13 Multiresponse: NMES data

GUIDE has two options for fitting a piecewise-constant regression model to predict two or more dependent variables simultaneously (Loh and Zheng, 2013). The first (named `multiresponse` or option 5 in the input file) requires the number of dependent variables to be the same for each observation. Observations with missing values

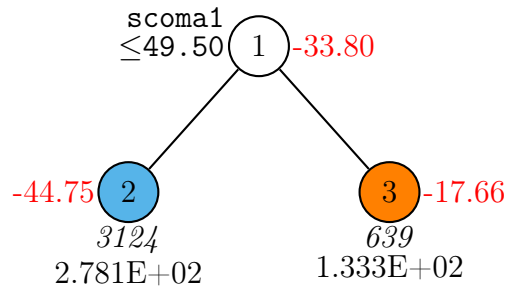


Figure 31: GUIDE v.46.1 0.250-SE regression tree using Gi option for mean `survtime` restricted to less than 622.00 and treatment variable `swang1` without adjustment for linear prognostic effects. At each split, an observation goes to the left branch if and only if the condition is satisfied. Sample size (in *italics*) and mean of `death` printed below nodes. `swang1` effects (in red, relative to reference level NoRHC) of levels RHC (relative to NoRHC) beside nodes. Terminal nodes with treatment effect above and below -33.80 (effect at root node) are painted orange and skyblue respectively. Second best split variable at root node is `ph1`.

in one or more dependent variables are excluded. The second (named `longitudinal data (with T variables)` or option 6 in the input file) requires each dependent variable to be associated with an observation time variable. It fits a model to all observations, including those with missing values in some dependent variables. The observation times are not required to be the same for all subjects, i.e., they may vary from subject to subject, but observations with missing times are excluded from model fitting. We demonstrate the first option in this section. The second option is used in Section 14.

The data file `nmes.txt` contains observations on 4406 subjects from a National Medical Expenditure Survey (NMES) conducted in 1987 and 1988. Table 10 gives the names of the variables and their definitions. The data were previously analyzed in Deb and Trivedi (1997), Cameron and Trivedi (1998, chap. 6), and Zeileis (2006). Here we construct a regression tree to predict the outcomes for the first 6 variables (`ofp`, `ofnp`, `opp`, `opnp`, `emer`, and `hosp`). The contents of the description file `nmes.dsc` follow.

```

nmes.txt
NA
1
1 ofp d
2 ofnp d
3 opp d
4 opnp d

```

Table 10: Definitions of variables in NMES data

ofp	number of physician office visits
ofnp	number of nonphysician office visits
opp	number of physician outpatient visits
opnp	number of nonphysician outpatient visits
emer	number of emergency room visits
hosp	number of hospitalizations
health	self-perceived health (poor, average, or excellent)
numchron	number of chronic conditions
adldiff	has condition that limits daily living (no, yes)
region	region of U.S. (midwest, northeast, west, other)
age	age in years
black	African American (no, yes)
gender	sex (female, male)
married	married (no, yes)
school	number of years of education
faminc	family income in \$10,000
employed	employed (no, yes)
privins	covered by private insurance (no, yes)
medicaid	covered by Medicaid (no, yes)

```

5 emer d
6 hosp d
7 health c
8 numchron n
9 adldiff c
10 region c
11 age n
12 black c
13 gender c
14 married c
15 school n
16 faminc n
17 employed c
18 privins c
19 medicaid c

```

13.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: mult.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1):
Name of batch output file: mult.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
  7=binary logistic regression.
Input choice ([1:7], <cr>=1): 5
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: nmes.dsc
Reading DSC file ...
Training sample file: nmes.txt
Missing value code: NA
Records in data file start on line 1
4 N variables changed to S
Number of D variables: 6
D variables are:
ofp

```

```

ofnp
opp
opnp
emer
hosp
Multivariate or univariate split variable selection:
Choose multivariate if there is an order among the D variables;
choose univariate otherwise or if item response
Input 1 for multivariate, 2 for univariate ([1:2], <cr>=1): 2
D variables can be normalized to have unit variance,
e.g., if they have different scales or units
Input 1 to normalize D variables, 2 for no normalization ([1:2], <cr>=1):

Input 1 for equal, 2 for unequal weighting of D variables ([1:2], <cr>=1):

Reading data file ...
Number of records in data file: 4406
Length of longest entry in data file: 9
Checking for missing values ...
Finished checking
Assigning integer codes to values of 9 categorical variables
Re-checking data ...
Assigning codes to missing values, if any ...
Data checks complete
Normalizing data
Rereading data ...
PCA can be used for variable selection
Do not use PCA if differential item functioning (DIF) scores are wanted
Input 1 to use PCA, 2 otherwise ([1:2], <cr>=2): 1
#cases w/ miss. D = number of cases with some D values missing
      Total #cases w/ #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      4406         0         0        0        0        0        4
      #P-var  #M-var  #B-var  #C-var  #I-var  #R-var
           0         0         0         9         0         1
Number of cases used for training: 4406
Number of split variables: 13
Finished reading data file
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): mult.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save node IDs of observations, 1 otherwise ([1:2], <cr>=2):

```

```

Input name of file to store terminal node ID of each case: mult.nid
Input 2 to save fitted values at each terminal node; 1 otherwise ([1:2], <cr>=2):
Input name of file to store node fitted values: mult.fit
Input 2 to write R function for predicting new cases, 1 otherwise ([1:2], <cr>=2):
Input file name: mult.r
Input rank of top variable to split root node ([1:13], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < mult.in

```

13.2 Contents of *mult.out*

```

Multi-response or longitudinal data without T variables
Pruning by cross-validation
DSC file: nmes.dsc
Training sample file: nmes.txt
Missing value code: NA
Records in data file start on line 1
4 N variables changed to S
Number of D variables: 6
Univariate split variable selection method
Mean-squared errors (MSE) are calculated from normalized D variables
D variables equally weighted
Piecewise constant model
Number of records in data file: 4406
Length of longest entry in data file: 9
Model fitted to subset of observations with complete D values
PCA on dependent variables for split selection

Summary information for training sample of size 4406
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight

```

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	ofp	d	0.000	89.00		
2	ofnp	d	0.000	104.0		
3	opp	d	0.000	141.0		
4	opnp	d	0.000	155.0		
5	emer	d	0.000	12.00		
6	hosp	d	0.000	8.000		
7	health	c			3	
8	numchron	s	0.000	8.000		

9	adldiff	c			2
10	region	c			4
11	age	s	6.600	10.90	
12	black	c			2
13	gender	c			2
14	married	c			2
15	school	s	0.000	18.00	
16	faminc	s	-1.012	54.84	
17	employed	c			2
18	privins	c			2
19	medicaid	c			2

#cases w/ miss. D = number of cases with some D values missing
 Numbers of N, F, S, and P below exclude constant variables

Total #cases	#cases w/ miss. D	#missing ord. vals	#X-var	#N-var	#F-var	#S-var
4406	0	0	0	0	0	4
#P-var	#M-var	#B-var	#C-var	#I-var		
0	0	0	9	0		

Number of cases used for training: 4406

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 44

Ranks of variables and their 1-df chi-squared values at root node

1	0.5227E+03	numchron
2	0.3280E+03	health
3	0.1718E+03	adldiff
4	0.6060E+02	privins
5	0.5697E+02	region
6	0.5290E+02	age
7	0.5172E+02	medicaid
8	0.3676E+02	school
9	0.2714E+02	gender
10	0.2577E+02	black
11	0.1036E+02	faminc
12	0.9199E+01	married
13	0.7753E+01	employed

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	19	1.639E+00	1.305E-01	1.434E-01	1.428E+00	1.081E-01
2	18	1.620E+00	1.304E-01	1.461E-01	1.428E+00	1.073E-01
3	17	1.617E+00	1.304E-01	1.464E-01	1.428E+00	1.087E-01
4	16	1.592E+00	1.304E-01	1.471E-01	1.428E+00	8.733E-02
5	15	1.572E+00	1.304E-01	1.477E-01	1.428E+00	6.736E-02
6	13	1.528E+00	1.304E-01	1.542E-01	1.339E+00	9.682E-02
7	11	1.497E+00	1.304E-01	1.567E-01	1.301E+00	8.848E-02
8	10	1.468E+00	1.304E-01	1.589E-01	1.295E+00	6.558E-02
9	8	1.417E+00	1.304E-01	1.443E-01	1.268E+00	7.670E-02
10	7	1.362E+00	1.304E-01	1.525E-01	1.225E+00	9.328E-02
11	6	1.122E+00	1.296E-01	1.491E-01	9.523E-01	9.815E-02
12++	4	1.050E+00	1.294E-01	1.475E-01	8.972E-01	1.046E-01
13**	3	1.063E+00	1.293E-01	1.422E-01	9.711E-01	7.747E-02
14	2	1.259E+00	1.296E-01	1.461E-01	1.068E+00	9.920E-02
15	1	1.635E+00	1.308E-01	1.448E-01	1.421E+00	1.078E-01

0-SE tree based on mean is marked with * and has 4 terminal nodes

0-SE tree based on median is marked with + and has 4 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree same as + tree

** tree same as -- tree

+ tree same as ++ tree

* tree same as ++ tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

Cases fit give the number of cases used to fit node

MSE is residual sum of squares divided by number of cases in node

Node	Total	Cases	Node	Split
label	cases	fit	MSE	variable
1	4406	4406	1.000E+00	numchron
2T	2523	2523	5.688E-01	numchron
3	1883	1883	1.528E+00	health
6T	426	426	2.282E+00	medicaid
7T	1457	1457	1.277E+00	region

Number of terminal nodes of final tree: 3

Total number of nodes of final tree: 5

Second best split variable (based on curvature test) at root node is health

Regression tree for multi-response data:

For categorical variable splits, values not in training data go to the right

```
Node 1: numchron <= 1.5000000
  Node 2: Mean cost = 0.56857139
Node 1: numchron > 1.5000000 or NA
  Node 3: health = "poor"
    Node 6: Mean cost = 2.2768607
  Node 3: health /= "poor"
    Node 7: Mean cost = 1.2765754
```

```
Node 1: Intermediate node
A case goes into Node 2 if numchron <= 1.5000000
numchron mean = 1.5419882
Means of ofp, ofnp, opp, opnp, emer, and hosp
  5.7744E+00  1.6180E+00  7.5079E-01  5.3609E-01  2.6350E-01
  2.9596E-01
-----
Node 2: Terminal node
Means of ofp, ofnp, opp, opnp, emer, and hosp
  4.4392E+00  1.4491E+00  4.6968E-01  3.9516E-01  1.6488E-01
  1.6647E-01
-----
Node 3: Intermediate node
A case goes into Node 6 if health = "poor"
health mode = "average"
-----
Node 6: Terminal node
Means of ofp, ofnp, opp, opnp, emer, and hosp
  9.4319E+00  1.5000E+00  1.5282E+00  6.8310E-01  7.2066E-01
  7.9108E-01
-----
Node 7: Terminal node
Means of ofp, ofnp, opp, opnp, emer, and hosp
  7.0172E+00  1.9451E+00  1.0103E+00  7.3713E-01  3.0062E-01
  3.7543E-01
-----
Case and node IDs are in file: mult.nid
Node fitted values are in file: mult.fit
LaTeX code for tree is in mult.tex
```

The tree is shown in Figure 32. The file *mult.fit* saves the mean values of the dependent variables in each terminal node:

node	ofp	ofnp	opp	opnp	emer	hosp
------	-----	------	-----	------	------	------

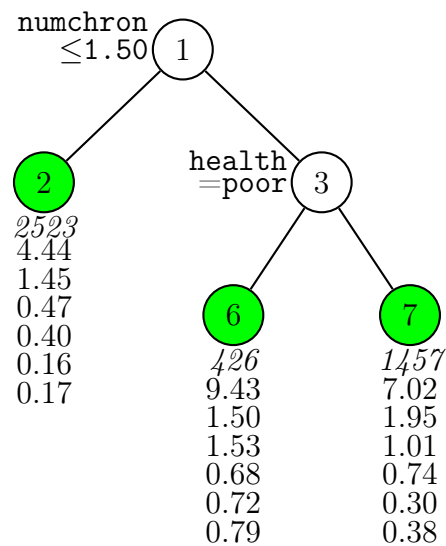


Figure 32: GUIDE v.46.1 0.250-SE regression tree for predicting response variables `ofp`, `ofnp`, `opp`, `opnp`, `emer`, and `hosp`, using PCA at each node. At each split, an observation goes to the left branch if and only if the condition is satisfied. Sample size (in *italics*) and predicted values of `ofp`, `ofnp`, `opp`, `opnp`, `emer`, and `hosp` printed below nodes. Second best split variable at root node is `health`.

```
2 0.44392E+01 0.14491E+01 0.46968E+00 0.39516E+00 0.16488E+00 0.16647E+00
6 0.94319E+01 0.15000E+01 0.15282E+01 0.68310E+00 0.72066E+00 0.79108E+00
7 0.70172E+01 0.19451E+01 0.10103E+01 0.73713E+00 0.30062E+00 0.37543E+00
```

The file `mult.nid` gives the terminal node number for each observation, including those that are not used to construct the tree (indicated by the letter “n” in the `train` column of the file).

14 Longitudinal response with varying times

The data come from a longitudinal study on the hourly wage of 888 male high-school dropouts (246 black, 204 Hispanic, 438 white), where the observation time points as well as their number (1–13) varied across individuals (Murnane et al., 1999; Singer and Willett, 2003). An earlier version of GUIDE was used to analyze the data in Loh and Zheng (2013).

The response variable is hourly wage (in 1990 dollars) and the predictor variables are `hgc` (highest grade completed; 6–12), `exper` (years in labor force; 0.001–12.7 yrs), and `race` (Black, Hispanic, and White). The data file `wagedat.txt` is in **wide format**, where each record refers to one individual. The DSC file `wagedsc.txt` is given below. Observation time points are indicated by `t`. The `d` and `t` variable columns may appear anywhere in the data, but the first `d` must be associated with the first `t`, second `d` with the second `t`, and so on. The number of `d` and `t` variables must be the same. Missing `d` values are permitted to allow for observations with unequal numbers of observation times. Observations with missing values in one or more `t` variable are excluded from model fitting.

```
wagedat.txt
NA
1
1 id x
2 hgc n
3 exper1 t
4 exper2 t
5 exper3 t
6 exper4 t
7 exper5 t
8 exper6 t
9 exper7 t
10 exper8 t
11 exper9 t
```

12 exper10 t
13 exper11 t
14 exper12 t
15 exper13 t
16 postexp1 x
17 postexp2 x
18 postexp3 x
19 postexp4 x
20 postexp5 x
21 postexp6 x
22 postexp7 x
23 postexp8 x
24 postexp9 x
25 postexp10 x
26 postexp11 x
27 postexp12 x
28 postexp13 x
29 wage1 d
30 wage2 d
31 wage3 d
32 wage4 d
33 wage5 d
34 wage6 d
35 wage7 d
36 wage8 d
37 wage9 d
38 wage10 d
39 wage11 d
40 wage12 d
41 wage13 d
42 ged1 x
43 ged2 x
44 ged3 x
45 ged4 x
46 ged5 x
47 ged6 x
48 ged7 x
49 ged8 x
50 ged9 x
51 ged10 x
52 ged11 x
53 ged12 x
54 ged13 x
55 uerate1 x
56 uerate2 x
57 uerate3 x

```

58 uerate4 x
59 uerate5 x
60 uerate6 x
61 uerate7 x
62 uerate8 x
63 uerate9 x
64 uerate10 x
65 uerate11 x
66 uerate12 x
67 uerate13 x
68 race c

```

14.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: wage.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1):
Name of batch output file: wage.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
  7=binary logistic regression.
Input choice ([1:7], <cr>=1): 6
Input 1 for lowess smoothing, 2 for spline smoothing ([1:2], <cr>=1):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: wagedsc.txt
Reading DSC file ...
Training sample file: wagedat.txt
Missing value code: NA
Records in data file start on line 1
One N variable changed to S
Number of D variables: 13
D variables are:
wage1
wage2
wage3
wage4

```

```

wage5
wage6
wage7
wage8
wage9
wage10
wage11
wage12
wage13
T variables are:
exper1
exper2
exper3
exper4
exper5
exper6
exper7
exper8
exper9
exper10
exper11
exper12
exper13
D variables can be grouped into segments to look for patterns
Input 1 for equal-sized groups, 2 for custom groups ([1:2], <cr>=1):
Input number of roughly equal-sized groups ([2:9], <cr>=3):
Input number of interpolating points for prediction ([10:100], <cr>=31):
Reading data file ...
Number of records in data file: 888
Length of longest entry in data file: 16
Checking for missing values ...
Finished checking
Missing values found in D variables
Assigning integer codes to values of 1 categorical variables
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
#cases w/ miss. D = number of cases with all D values missing
  Total  #cases w/  #missing
#cases   miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
   888         0       0      40       0       0       1
#P-var  #M-var  #B-var  #C-var  #I-var
   0       0       0       1       0

```

```

Number of cases used for training: 888
Number of split variables: 2
Number of cases excluded due to 0 W or missing D variable: 0
Finished reading data file
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): wage.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=3):
Input file name: wage.var
Input 2 to save node IDs of observations, 1 otherwise ([1:2], <cr>=2):
Input name of file to store terminal node ID of each case: wage.nid
Input 2 to save fitted values at each terminal node; 1 otherwise ([1:2], <cr>=2):
Input name of file to store node fitted values: wage.fit
Input 2 to write R function for predicting new cases, 1 otherwise ([1:2], <cr>=2):
Input file name: wage.r
Input rank of top variable to split root node ([1:2], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < wage.in

```

14.2 Contents of wage.out

```

Longitudinal data with T variables
Lowess smoothing
Pruning by cross-validation
DSC file: wagedsc.txt
Training sample file: wagedat.txt
Missing value code: NA
Records in data file start on line 1
One N variable changed to S
Number of D variables: 13
Number of D variables: 13
D variables are:
wage1
wage2
wage3
wage4
wage5
wage6
wage7
wage8
wage9

```

wage10

wage11

wage12

wage13

T variables are:

exper1

exper2

exper3

exper4

exper5

exper6

exper7

exper8

exper9

exper10

exper11

exper12

exper13

Number of records in data file: 888

Length of longest entry in data file: 16

Missing values found in D variables

Model fitted to subset of observations with complete D values

Summary information for training sample of size 888

d=dependent, b=split and fit cat variable using indicator variables,

c=split-only categorical, i=fit-only categorical (via indicators),

s=split-only numerical, n=split and fit numerical, f=fit-only numerical,

m=missing-value flag variable, p=periodic variable, w=weight

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
2	hgc	s	6.000	12.00		
3	exper1	t	0.1000E-02	5.637		
4	exper2	t	0.000	7.584		38
5	exper3	t	0.000	9.777		77
6	exper4	t	0.000	10.81		124
7	exper5	t	0.000	11.78		159
8	exper6	t	0.000	10.59		233
9	exper7	t	0.000	11.28		325
10	exper8	t	0.000	10.58		428
11	exper9	t	0.000	11.62		551
12	exper10	t	0.000	12.26		678
13	exper11	t	0.000	11.98		791
14	exper12	t	0.000	12.56		856
15	exper13	t	0.000	12.70		882
29	wage1	d	2.030	68.65		

30	wage2	d	2.069	50.40	38
31	wage3	d	2.046	34.50	77
32	wage4	d	2.117	33.15	124
33	wage5	d	2.104	49.30	159
34	wage6	d	2.208	74.00	233
35	wage7	d	2.104	47.28	325
36	wage8	d	2.316	37.71	428
37	wage9	d	2.529	46.11	551
38	wage10	d	2.998	56.54	678
39	wage11	d	4.084	22.20	791
40	wage12	d	3.432	46.20	856
41	wage13	d	4.563	7.776	882
68	race	c			3

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/	#missing					
#cases	miss.	D	ord. vals	#X-var	#N-var	#F-var	#S-var
888		0	0	40	0	0	1
#P-var	#M-var	#B-var	#C-var	#I-var			
0	0	0	1	0			

Number of cases used for training: 888

Number of cases excluded due to 0 W or missing D variable: 0

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 10

Ranks of variables and their 1-df chi-squared values at root node

1	0.1235E+02	hgc
2	0.6915E+01	race

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	17	1.283E+02	1.051E+01	9.475E+00	1.261E+02	1.045E+01
2	16	1.283E+02	1.051E+01	9.475E+00	1.261E+02	1.045E+01
3	15	1.253E+02	1.048E+01	9.841E+00	1.218E+02	1.118E+01
4	13	1.253E+02	1.048E+01	9.841E+00	1.217E+02	1.118E+01
5	10	1.254E+02	1.058E+01	9.880E+00	1.224E+02	1.126E+01
6*	8	1.240E+02	1.055E+01	9.851E+00	1.205E+02	1.108E+01
7+	7	1.243E+02	1.064E+01	1.002E+01	1.205E+02	1.125E+01
8**	1	1.244E+02	1.065E+01	1.011E+01	1.210E+02	1.171E+01

0-SE tree based on mean is marked with * and has 8 terminal nodes
 0-SE tree based on median is marked with + and has 7 terminal nodes
 Selected-SE tree based on mean using naive SE is marked with **
 Selected-SE tree based on mean using bootstrap SE is marked with --
 Selected-SE tree based on median and bootstrap SE is marked with ++
 ** tree same as ++ tree
 ** tree same as -- tree
 ++ tree same as -- tree

WARNING: tree based on mean CV estimate of error has no splits
 Choosing smallest nontrivial tree with no larger CV error estimate

Structure of final tree. Each terminal node is marked with a T.

Cases fit give the number of cases used to fit node
 MSE is residual sum of squares divided by number of cases in node

Node label	Total cases	Cases fit	Node MSE	Split variable
1	888	888	1.222E+02	hgc
2	873	873	1.215E+02	hgc
4T	32	32	9.598E+01	race
5	841	841	1.226E+02	race
10T	233	233	1.159E+02	hgc
11	608	608	1.248E+02	race
22	192	192	1.251E+02	hgc
44T	118	118	1.017E+02	hgc
45T	74	74	1.569E+02	hgc
23	416	416	1.247E+02	hgc
46T	284	284	1.004E+02	hgc
47T	132	132	1.718E+02	hgc
3T	15	15	1.565E+02	-

Number of terminal nodes of final tree: 7

Total number of nodes of final tree: 13

Second best split variable (based on curvature test) at root node is race

Regression tree for longitudinal data:

For categorical variable splits, values not in training data go to the right

Node 1: hgc <= 11.500000

Node 2: hgc <= 6.5000000

Node 4: Mean cost = 92.977093

Node 2: hgc > 6.5000000 or NA

Node 5: race = "black"

Node 10: Mean cost = 115.39395

```

Node 5: race /= "black"
Node 11: race = "hispanic"
Node 22: hgc <= 9.5000000
Node 44: Mean cost = 100.87229
Node 22: hgc > 9.5000000 or NA
Node 45: Mean cost = 154.79614
Node 11: race /= "hispanic"
Node 23: hgc <= 9.5000000
Node 46: Mean cost = 100.02686
Node 23: hgc > 9.5000000 or NA
Node 47: Mean cost = 170.51645
Node 1: hgc > 11.500000 or NA
Node 3: Mean cost = 146.10583

```

```

*****

```

```

Node 1: Intermediate node
A case goes into Node 2 if hgc <= 11.500000
hgc mean = 8.9166667
-----
Node 2: Intermediate node
A case goes into Node 4 if hgc <= 6.5000000
hgc mean = 8.8636884
-----
Node 4: Terminal node
-----
Node 5: Intermediate node
A case goes into Node 10 if race = "black"
race mode = "white"
-----
Node 10: Terminal node
-----
Node 11: Intermediate node
A case goes into Node 22 if race = "hispanic"
race mode = "white"
-----
Node 22: Intermediate node
A case goes into Node 44 if hgc <= 9.5000000
hgc mean = 9.0000000
-----
Node 44: Terminal node
-----
Node 45: Terminal node
-----
Node 23: Intermediate node
A case goes into Node 46 if hgc <= 9.5000000

```

```

hgc mean = 8.9495192
-----
Node 46: Terminal node
-----
Node 47: Terminal node
-----
Node 3: Terminal node
-----
Case and node IDs are in file: wage.nid
Node fitted values are in file: wage.fit
LaTeX code for tree is in wage.tex
Split and fit variable names are stored in wage.var

```

Figure 33 shows the tree and Figure 34 plots lowess-smoothed curves of mean wage in the two terminal nodes. The figure is produced by the following R code.

```

z <- read.table("wagedat.txt",header=FALSE)
names(z) <- c("id","hgc","exper1","exper2","exper3","exper4","exper5","exper6",
             "exper7","exper8","exper9","exper10","exper11","exper12","exper13",
             "postexp1","postexp2","postexp3","postexp4","postexp5","postexp6",
             "postexp7","postexp8","postexp9","postexp10","postexp11","postexp12",
             "postexp13","wage1","wage2","wage3","wage4","wage5","wage6","wage7",
             "wage8","wage9","wage10","wage11","wage12","wage13","ged1","ged2",
             "ged3","ged4","ged5","ged6","ged7","ged8","ged9","ged10","ged11",
             "ged12","ged13","uerate1","uerate2","uerate3","uerate4","uerate5",
             "uerate6","uerate7","uerate8","uerate9","uerate10","uerate11",
             "uerate12","uerate13","race")
exper <- c(z$exper1,z$exper2,z$exper3,z$exper4,z$exper5,z$exper6,z$exper7,
          z$exper8,z$exper9,z$exper10,z$exper11,z$exper12,z$exper13)
wage <- c(z$wage1,z$wage2,z$wage3,z$wage4,z$wage5,z$wage6,z$wage7,z$wage8,
          z$wage9,z$wage10,z$wage11,z$wage12,z$wage13)
xr <- range(exper,na.rm=TRUE)
yr <- range(wage,na.rm=TRUE)

guide.fit <- read.table("wage.fit",header=TRUE)
g.node <- guide.fit$node
g.start <- guide.fit$t.start
g.end <- guide.fit$t.end
n <- length(g.node)
m <- dim(guide.fit)[2]
npts <- m-3 # number of time points for plotting

xvals <- guide.fit[,2:3]
xvals <- as.numeric(unlist(xvals))
yvals <- guide.fit[,4:m]

```

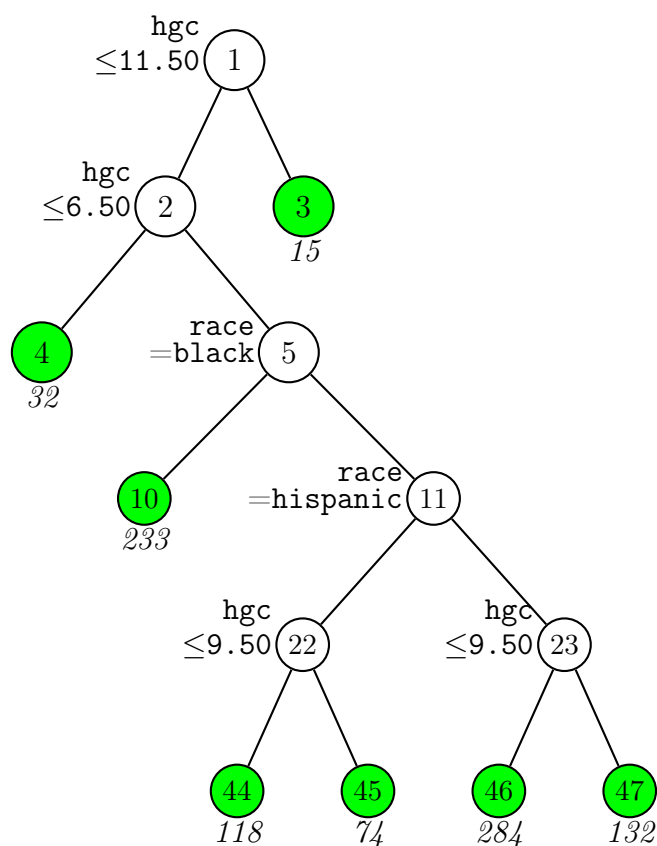


Figure 33: GUIDE v.46.1 0.038-SE (0.250-SE has no splits) regression tree for predicting longitudinal variables `wage1`, `wage2`, etc. At each split, an observation goes to the left branch if and only if the condition is satisfied. Sample size (*in italics*) printed below nodes. Second best split variable at root node is **race**.

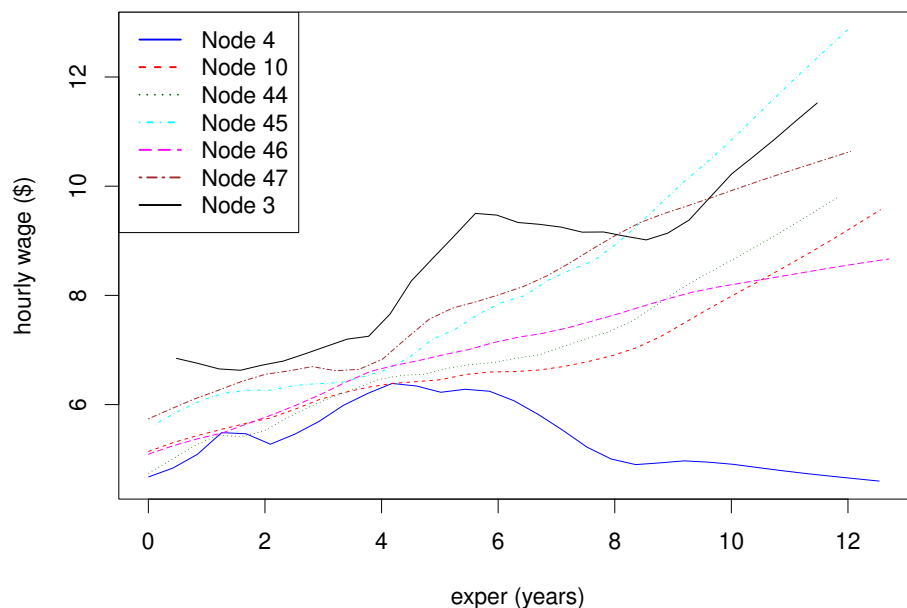


Figure 34: Lowess-smoothed mean wage curves in the terminal nodes of Figure 33.

```

yvals <- as.numeric(unlist(yvals))
plot(range(xvals),range(yvals),type="n",xlab="exper (years)",ylab="hourly wage ($)")
leg.col <- c("blue","red","darkgreen","cyan","magenta","brown","black")
leg.lty <- 1:n
for(i in 1:n){
  node <- g.node[i]
  start <- g.start[i]
  end <- g.end[i]
  gap <- (end-start)/(npts-1)
  x <- start+(0:(npts-1))*gap
  y <- as.numeric(guide.fit[i,4:m])
  lines(x,y,col=leg.col[i],lty=leg.lty[i])
}
leg.txt <- paste("Node",g.node)
legend("topleft",legend=leg.txt,lty=leg.lty,col=leg.col,lwd=2)

```

The plotting values are obtained from the result file `wage.fit` whose contents are given below. The first column gives the node number and the next two columns the start and end of the times at which fitted values are computed. The other columns give the fitted values equally spaced between the start and end times.

node	t.start	t.end	fitted1	fitted2	fitted3	fitted4	fitted5
1	0	12	5.5	6.5	7.5	8.5	9.5
2	0	12	5.5	6.5	7.5	8.5	9.5
3	0	12	6.5	7.5	8.5	9.5	10.5
4	0	12	5.5	6.5	7.5	8.5	9.5
5	0	12	5.5	6.5	7.5	8.5	9.5
6	0	12	5.5	6.5	7.5	8.5	9.5
7	0	12	5.5	6.5	7.5	8.5	9.5
8	0	12	5.5	6.5	7.5	8.5	9.5
9	0	12	5.5	6.5	7.5	8.5	9.5
10	0	12	5.5	6.5	7.5	8.5	9.5
11	0	12	5.5	6.5	7.5	8.5	9.5
12	0	12	5.5	6.5	7.5	8.5	9.5
13	0	12	5.5	6.5	7.5	8.5	9.5
14	0	12	5.5	6.5	7.5	8.5	9.5
15	0	12	5.5	6.5	7.5	8.5	9.5
16	0	12	5.5	6.5	7.5	8.5	9.5
17	0	12	5.5	6.5	7.5	8.5	9.5
18	0	12	5.5	6.5	7.5	8.5	9.5
19	0	12	5.5	6.5	7.5	8.5	9.5
20	0	12	5.5	6.5	7.5	8.5	9.5
21	0	12	5.5	6.5	7.5	8.5	9.5
22	0	12	5.5	6.5	7.5	8.5	9.5
23	0	12	5.5	6.5	7.5	8.5	9.5
24	0	12	5.5	6.5	7.5	8.5	9.5
25	0	12	5.5	6.5	7.5	8.5	9.5
26	0	12	5.5	6.5	7.5	8.5	9.5
27	0	12	5.5	6.5	7.5	8.5	9.5
28	0	12	5.5	6.5	7.5	8.5	9.5
29	0	12	5.5	6.5	7.5	8.5	9.5
30	0	12	5.5	6.5	7.5	8.5	9.5
31	0	12	5.5	6.5	7.5	8.5	9.5
32	0	12	5.5	6.5	7.5	8.5	9.5
33	0	12	5.5	6.5	7.5	8.5	9.5
34	0	12	5.5	6.5	7.5	8.5	9.5
35	0	12	5.5	6.5	7.5	8.5	9.5
36	0	12	5.5	6.5	7.5	8.5	9.5
37	0	12	5.5	6.5	7.5	8.5	9.5
38	0	12	5.5	6.5	7.5	8.5	9.5
39	0	12	5.5	6.5	7.5	8.5	9.5
40	0	12	5.5	6.5	7.5	8.5	9.5
41	0	12	5.5	6.5	7.5	8.5	9.5
42	0	12	5.5	6.5	7.5	8.5	9.5
43	0	12	5.5	6.5	7.5	8.5	9.5
44	0	12	5.5	6.5	7.5	8.5	9.5
45	0	12	5.5	6.5	7.5	8.5	9.5
46	0	12	5.5	6.5	7.5	8.5	9.5
47	0	12	5.5	6.5	7.5	8.5	9.5
48	0	12	5.5	6.5	7.5	8.5	9.5
49	0	12	5.5	6.5	7.5	8.5	9.5
50	0	12	5.5	6.5	7.5	8.5	9.5
51	0	12	5.5	6.5	7.5	8.5	9.5
52	0	12	5.5	6.5	7.5	8.5	9.5
53	0	12	5.5	6.5	7.5	8.5	9.5
54	0	12	5.5	6.5	7.5	8.5	9.5
55	0	12	5.5	6.5	7.5	8.5	9.5
56	0	12	5.5	6.5	7.5	8.5	9.5
57	0	12	5.5	6.5	7.5	8.5	9.5
58	0	12	5.5	6.5	7.5	8.5	9.5
59	0	12	5.5	6.5	7.5	8.5	9.5
60	0	12	5.5	6.5	7.5	8.5	9.5
61	0	12	5.5	6.5	7.5	8.5	9.5
62	0	12	5.5	6.5	7.5	8.5	9.5
63	0	12	5.5	6.5	7.5	8.5	9.5
64	0	12	5.5	6.5	7.5	8.5	9.5
65	0	12	5.5	6.5	7.5	8.5	9.5
66	0	12	5.5	6.5	7.5	8.5	9.5
67	0	12	5.5	6.5	7.5	8.5	9.5
68	0	12	5.5	6.5	7.5	8.5	9.5
69	0	12	5.5	6.5	7.5	8.5	9.5
70	0	12	5.5	6.5	7.5	8.5	9.5
71	0	12	5.5	6.5	7.5	8.5	9.5
72	0	12	5.5	6.5	7.5	8.5	9.5
73	0	12	5.5	6.5	7.5	8.5	9.5
74	0	12	5.5	6.5	7.5	8.5	9.5
75	0	12	5.5	6.5	7.5	8.5	9.5
76	0	12	5.5	6.5	7.5	8.5	9.5
77	0	12	5.5	6.5	7.5	8.5	9.5
78	0	12	5.5	6.5	7.5	8.5	9.5
79	0	12	5.5	6.5	7.5	8.5	9.5
80	0	12	5.5	6.5	7.5	8.5	9.5
81	0	12	5.5	6.5	7.5	8.5	9.5
82	0	12	5.5	6.5	7.5	8.5	9.5
83	0	12	5.5	6.5	7.5	8.5	9.5
84	0	12	5.5	6.5	7.5	8.5	9.5
85	0	12	5.5	6.5	7.5	8.5	9.5
86	0	12	5.5	6.5	7.5	8.5	9.5
87	0	12	5.5	6.5	7.5	8.5	9.5
88	0	12	5.5	6.5	7.5	8.5	9.5
89	0	12	5.5	6.5	7.5	8.5	9.5
90	0	12	5.5	6.5	7.5	8.5	9.5
91	0	12	5.5	6.5	7.5	8.5	9.5
92	0	12	5.5	6.5	7.5	8.5	9.5
93	0	12	5.5	6.5	7.5	8.5	9.5
94	0	12	5.5	6.5	7.5	8.5	9.5
95	0	12	5.5	6.5	7.5	8.5	9.5
96	0	12	5.5	6.5	7.5	8.5	9.5
97	0	12	5.5	6.5	7.5	8.5	9.5
98	0	12	5.5	6.5	7.5	8.5	9.5
99	0	12	5.5	6.5	7.5	8.5	9.5
100	0	12	5.5	6.5	7.5	8.5	9.5

4	0.50000E-02	0.12535E+02	0.46759E+01	0.48382E+01	0.50885E+01	0.54866E+01	0.54655E+01	0.54655E+01
10	0.40000E-02	0.12558E+02	0.51416E+01	0.53033E+01	0.54330E+01	0.55485E+01	0.56582E+01	0.56582E+01
44	0.60000E-02	0.11831E+02	0.47453E+01	0.49937E+01	0.52497E+01	0.54392E+01	0.54149E+01	0.54149E+01
45	0.12200E+00	0.11990E+02	0.56433E+01	0.58784E+01	0.60783E+01	0.62114E+01	0.62619E+01	0.62619E+01
46	0.10000E-02	0.12700E+02	0.50925E+01	0.52457E+01	0.53753E+01	0.54807E+01	0.56485E+01	0.56485E+01
47	0.20000E-02	0.12045E+02	0.57397E+01	0.59284E+01	0.61090E+01	0.62643E+01	0.64272E+01	0.64272E+01
3	0.48000E+00	0.11473E+02	0.68471E+01	0.67557E+01	0.66532E+01	0.66289E+01	0.67211E+01	0.67211E+01

The contents of the file `wage.var` are given below. The 1st column gives the node number. The 2nd column is a letter, with `t` indicating that the node is terminal and `c`, `s`, or `n` indicating an intermediate node split on a `c`, `n` or `s` variable. The 3rd column gives the name of the variable used to split the node; the name `NONE` is used if a terminal node cannot be split by any variable. The 4th column gives the name of the interacting variable if there is one; otherwise the name of the split variable is repeated. If the node is terminal, the 5th column contains the letter “`t`”; otherwise if it is non-terminal, the 5th column is an integer indicating the number of split values to follow (a split on a `c` variable may have more than one value). In the example below, node 1 is split on `s` variable `hgc` at value 9.50. Nodes 2 and 3 are terminal nodes; each would be split on `race` if they were not terminal.

```

1 s hgc hgc          1    0.9500000000E+01
2 t race race      t
3 t race race      t

```

15 Logistic regression

If the dependent variable Y takes values 0 and 1, GUIDE can construct a tree model such that a simple or multiple linear logistic regression model is fitted in each node. The tree model may be more efficient (in terms of size and prediction accuracy) if a preliminary estimate of $p = P(Y = 1)$ is available. The preliminary estimate of p is not necessary, but it may be easily obtained by fitting a GUIDE forest or kernel discriminant model to the data. If a variable containing the estimated p values are included in the data, it should be specified as an “`e`” variable in the description file (see Section 3.1). Missing values in the predictor variables used in the logistic regression node models are imputed with node means; see Loh (2021) for more details.

We use the NHTSA data to demonstrate this, with $Y = \text{HIC2}$, which takes value 1 if $\text{HIC} > 999$ and 0 otherwise. The DSC file is `nhtsadsc2.txt`. The “`e`” variable is `estHIC2` which is a column of estimated values of $p = P(Y = 1)$ obtained from GUIDE forest.

15.1 Piecewise constant

15.1.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: logitc.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: logitc.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
    1=linear, 2=quantile, 3=Poisson, 4=censored response,
    5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
    7=binary logistic regression.
Input choice ([1:7], <cr>=1): 7
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple polynomial, 3: constant ([1:3], <cr>=3):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: nhtsadsc2.txt
Reading DSC file ...
Training sample file: nhtsadatam.txt
Missing value code: NA
Records in data file start on line 2
48 N variables changed to S
Warning: B variables changed to C
D variable is HIC2
Reading data file ...
Number of records in data file: 3310
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 13 categorical variables
Finished assigning codes to 10 categorical variables
Associating missing values of N, P and S variables with M variable codes ...

```



```

Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      3310      34      3310      2      0      0      48
      #P-var  #M-var  #B-var  #C-var  #I-var
      6      42      0      13      0
Number of cases used for training: 3276
Number of cases excluded due to 0 W or missing D variable: 34
Proportion of ones in HIC2 variable: 8.4554334554334559E-002
Finished reading data file
Default max. number of split levels: 15
Minimum number of D=0 and D=1 in each node: 9
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): logitc.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: logitc.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: logitc.r
Input rank of top variable to split root node ([1:67], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < logitc.in

```

15.1.2 Contents of logitc.out

```

Binary logistic regression tree
Pruning by cross-validation
DSC file: nhtsadsc2.txt
Training sample file: nhtsadatam.txt
Missing value code: NA
Records in data file start on line 2
48 N variables changed to S

```

Warning: B variables changed to C
D variable is HIC2
Piecewise constant model
Number of records in data file: 3310
Length of longest entry in data file: 19
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables

Summary information for training sample of size 3276 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight,
e=estimated success probability
Levels of M variables are for missing values in associated variables

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	BARRIG	c			3	
2	BARSHP	c			21	
3	BARANG	p	0.000	330.0	360	14
4	BARDIA	s	191.0	1000.		2807
5	OCCWT	s	72.00	83.00		3265
6	OCCWT_	m			2	
:						
108	CARANG	p	0.000	99.00	360	991
109	VEHOR	p	0.000	90.00	360	995
110	RSTFRT	c			3	
111	HIC2	d	0.000	1.000		
112	estHIC2	e	0.000	0.8455		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	miss. D	#missing ord. vals	#X-var	#N-var	#F-var	#S-var
3310	34		3310	2	0	0	48
#P-var	#M-var	#B-var	#C-var	#I-var			
6	42	0	13	0			

Number of cases used for training: 3276

Number of cases excluded due to 0 W or missing D variable: 34

Proportion of ones in HIC2 variable: 0.084554

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Nodewise interaction tests on all variables

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 32

Minimum number of D=0 and D=1 in each node: 9

Ranks of variables and their 1-df chi-squared values at root node

1	0.1218E+04	COLMEC
2	0.9001E+03	YEAR
3	0.8714E+03	MODEL
:		
65	0.2537E+00	CB
66	0.1349E+00	KB
67	0.4871E-01	HB

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1+	11	4.562E-01	2.044E-02	8.368E-03	4.472E-01	8.232E-03
2	10	4.545E-01	2.020E-02	7.273E-03	4.484E-01	8.265E-03
3**	9	4.529E-01	2.009E-02	6.781E-03	4.484E-01	5.719E-03
4	8	4.590E-01	2.024E-02	6.010E-03	4.516E-01	5.748E-03
5	6	4.580E-01	2.012E-02	6.699E-03	4.516E-01	7.863E-03
6	5	4.580E-01	2.012E-02	6.699E-03	4.516E-01	7.863E-03
7	4	4.580E-01	2.012E-02	6.699E-03	4.516E-01	7.863E-03
8	3	4.580E-01	2.012E-02	6.699E-03	4.516E-01	7.863E-03
9	2	4.580E-01	2.012E-02	6.699E-03	4.516E-01	7.863E-03
10	1	5.795E-01	2.316E-02	2.216E-03	5.834E-01	3.465E-03

0-SE tree based on mean is marked with * and has 9 terminal nodes

0-SE tree based on median is marked with + and has 11 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

** tree same as ++ tree

** tree same as -- tree

++ tree same as -- tree

* tree same as ** tree

* tree same as ++ tree

* tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is mean of HIC2 in the node

Cases fit give the number of cases used to fit node

Node deviance is residual deviance divided by residual degrees of freedom

Node label	Total cases	Cases fit	Matrix rank	Node D-mean	Node deviance	Split variable	Other variables
1	3276	3276	1	8.455E-02	5.797E-01	COLMEC	
2	2610	2610	1	2.797E-02	2.553E-01	MODEL	
4	436	436	1	1.468E-01	8.361E-01	IMPANG	
8T	56	56	1	3.214E-01	1.279E+00	-	
9	380	380	1	1.211E-01	7.400E-01	RSTFRT	
18T	54	54	1	5.185E-01	1.411E+00	-	
19	326	326	1	5.521E-02	4.285E-01	MODEL	
38T	46	46	1	1.957E-01	1.011E+00	-	
39T	280	280	1	3.214E-02	2.852E-01	-	
5T	2174	2174	1	4.140E-03	5.372E-02	-	
3	666	666	1	3.063E-01	1.234E+00	MODEL	
6	213	213	1	7.089E-01	1.212E+00	MODEL	
12T	113	113	1	5.310E-01	1.395E+00	CLSSPD	
13T	100	100	1	9.100E-01	6.112E-01	-	
7	453	453	1	1.170E-01	7.234E-01	MODEL	
14T	116	116	1	3.276E-01	1.276E+00	VEHSPD	
15T	337	337	1	4.451E-02	3.651E-01	-	

Number of terminal nodes of final tree: 9

Total number of nodes of final tree: 17

Second best split variable (based on curvature test) at root node is YEAR

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: COLMEC = "BWU", "CYL", "NA", "NAP", "UNK"

Node 2: MODEL = "4RUNNER", "9-5", "ACHIEVA", "AEROSTAR", "ASTRO",
 "BEAUVILLE", "BEETLE", "CLUBWAGON MPV", "DAKOTA", "DE VILLE",
 "DEFORMABLE IMPACTOR", "DURANGO", "ELANTRA", "EUROVAN",
 "FIVE HUNDRED", "FORESTER", "FRONTIER", "GALANT", "GRAND AM",
 "INSIGHT", "ION", "L200", "MAXIMA", "MAZDA3", "METRO", "MPV",
 "MUSTANG", "NEON", "PASEO", "PATHFINDER", "PT CRUISER", "QUEST",
 "RAM", "RAM WAGON VAN", "RODEO", "S-10", "S10 BLAZER", "S80",
 "SIDEKICK", "SILVERADO2500", "SPACECAB", "SPORTVAN", "TACOMA",
 "TEMPO", "TERCEL", "TRACKER", "TRAILBLAZER", "TROOPER II",
 "VERONA", "VUE"

Node 4: IMPANG in [269, 287]

Node 8: HIC2 proportion of 1s = 0.32142857

Node 4: IMPANG not in [269, 287] or NA

Node 9: RSTFRT = "0"

Node 18: HIC2 proportion of 1s = 0.51851852

```

Node 9: RSTFRT /= "0"
Node 19: MODEL = "BEAUVILLE", "DE VILLE", "PATHFINDER", "PT CRUISER",
        "RAM WAGON VAN", "S-10", "TEMPO", "TRAILBLAZER", "VERONA"
Node 38: HIC2 proportion of 1s = 0.19565217
Node 19: MODEL /= "BEAUVILLE", "DE VILLE", "PATHFINDER", "PT CRUISER",
        "RAM WAGON VAN", "S-10", "TEMPO", "TRAILBLAZER", "VERONA"
Node 39: HIC2 proportion of 1s = 0.32142857E-1
Node 2: MODEL /= "4RUNNER", "9-5", "ACHIEVA", "AEROSTAR", "ASTRO",
        "BEAUVILLE", "BEETLE", "CLUBWAGON MPV", "DAKOTA", "DE VILLE",
        "DEFORMABLE IMPACTOR", "DURANGO", "ELANTRA", "EUROVAN",
        "FIVE HUNDRED", "FORESTER", "FRONTIER", "GALANT", "GRAND AM",
        "INSIGHT", "ION", "L200", "MAXIMA", "MAZDA3", "METRO", "MPV",
        "MUSTANG", "NEON", "PASEO", "PATHFINDER", "PT CRUISER", "QUEST",
        "RAM", "RAM WAGON VAN", "RODEO", "S-10", "S10 BLAZER", "S80",
        "SIDEKICK", "SILVERADO2500", "SPACECAB", "SPORTVAN", "TACOMA",
        "TEMPO", "TERCEL", "TRACKER", "TRAILBLAZER", "TROOPER II",
        "VERONA", "VUE"
Node 5: HIC2 proportion of 1s = 0.41398344E-2
Node 1: COLMEC /= "BWU", "CYL", "NA", "NAP", "UNK"
Node 3: MODEL = "18", "200", "200 SX", "2000", "210", "240", "244",
        "300 ZX", "318", "325 I", "4000", "4RUNNER", "5000", "504", "505",
        "604", "626", "ALLIANCE", "ASTRO", "AXXESS", "B2000 PICKUP",
        "BLAZER", "CAPRI", "CAPRICE", "CAR ELECTRIC", "CHAMP",
        "CLUBWAGON MPV", "COLT PICKUP", "COLT VISTA", "CONQUEST",
        "COURIER", "DAKOTA", "E100 VAN", "EL CAMINO", "ELECTRA",
        "ELECTREK", "ELECTRICA", "EXCEL GLS", "EXP", "F150 PICKUP",
        "FAIRMONT", "FESTIVA", "FUEGO", "FURY", "GF", "GL", "GLC", "GV",
        "I-MARK", "IMPULSE", "KING CAB PICKUP", "LANCER", "LE BARON",
        "LE CAR", "LEGACY", "LEOPARD", "LIBERTY", "LUMINA", "LUV",
        "MALIBU", "MARQUIS", "MEDALLION", "MERKUR", "METRO", "MONTERO",
        "MONZA", "MPV", "NEW YORKER", "NOVA", "PASSAT", "PICKUP",
        "PREVIA", "PRIZM", "PULSAR", "QUANTUM", "S-10", "SABLE",
        "SCIROCCO", "SHADOW", "SIDEKICK", "SOMERSET", "SONATA",
        "SPECTRUM", "SPIRIT", "SPORTSWAGON", "SPRINT", "ST. REGIS",
        "STANZA", "STARLET", "SUBURBAN", "T1000", "TEMPO", "TORINO",
        "TRANSIT CONNECT", "TREDIA", "TREKKER", "TROOPER II", "VANAGON",
        "WAGON", "WRANGLER"
Node 6: MODEL = "18", "200", "244", "300 ZX", "626", "ALLIANCE", "BLAZER",
        "CAPRI", "CAPRICE", "COLT VISTA", "CONQUEST", "COURIER",
        "DAKOTA", "ELECTREK", "EXCEL GLS", "EXP", "FAIRMONT", "GLC",
        "IMPULSE", "LE BARON", "LEGACY", "LIBERTY", "LUV", "MEDALLION",
        "METRO", "MPV", "NOVA", "PASSAT", "PICKUP", "PREVIA", "S-10",
        "SABLE", "SCIROCCO", "SHADOW", "SONATA", "SPIRIT", "TEMPO",
        "WAGON", "WRANGLER"
Node 12: HIC2 proportion of 1s = 0.53097345
Node 6: MODEL /= "18", "200", "244", "300 ZX", "626", "ALLIANCE", "BLAZER",

```

```

"CAPRI", "CAPRICE", "COLT VISTA", "CONQUEST", "COURIER",
"DAKOTA", "ELECTREK", "EXCEL GLS", "EXP", "FAIRMONT", "GLC",
"IMPULSE", "LE BARON", "LEGACY", "LIBERTY", "LUV", "MEDALLION",
"METRO", "MPV", "NOVA", "PASSAT", "PICKUP", "PREVIA", "S-10",
"SABLE", "SCIROCCO", "SHADOW", "SONATA", "SPIRIT", "TEMPO",
"WAGON", "WRANGLER"
Node 13: HIC2 proportion of 1s = 0.91000000
Node 3: MODEL /= "18", "200", "200 SX", "2000", "210", "240", "244",
"300 ZX", "318", "325 I", "4000", "4RUNNER", "5000", "504", "505",
"604", "626", "ALLIANCE", "ASTRO", "AXXESS", "B2000 PICKUP",
"BLAZER", "CAPRI", "CAPRICE", "CAR ELECTRIC", "CHAMP",
"CLUBWAGON MPV", "COLT PICKUP", "COLT VISTA", "CONQUEST",
"COURIER", "DAKOTA", "E100 VAN", "EL CAMINO", "ELECTRA",
"ELECTREK", "ELECTRICA", "EXCEL GLS", "EXP", "F150 PICKUP",
"FAIRMONT", "FESTIVA", "FUEGO", "FURY", "GF", "GL", "GLC", "GV",
"I-MARK", "IMPULSE", "KING CAB PICKUP", "LANCER", "LE BARON",
"LE CAR", "LEGACY", "LEOPARD", "LIBERTY", "LUMINA", "LUV",
"MALIBU", "MARQUIS", "MEDALLION", "MERKUR", "METRO", "MONTERO",
"MONZA", "MPV", "NEW YORKER", "NOVA", "PASSAT", "PICKUP",
"PREVIA", "PRIZM", "PULSAR", "QUANTUM", "S-10", "SABLE",
"SCIROCCO", "SHADOW", "SIDEKICK", "SOMERSET", "SONATA",
"SPECTRUM", "SPIRIT", "SPORTSWAGON", "SPRINT", "ST. REGIS",
"STANZA", "STARLET", "SUBURBAN", "T1000", "TEMPO", "TORINO",
"TRANSIT CONNECT", "TREDIA", "TREKKER", "TROOPER II", "VANAGON",
"WAGON", "WRANGLER"
Node 7: MODEL = "CELICA", "CHEVETTE", "COLT", "CONCORD", "CORDIA",
"CRESSIDA", "CUTLASS", "DL", "E150 VAN", "EXPLORER", "FOX",
"GRANADA", "JETTA", "MAXIMA", "MOTOR HOME", "PONY EXCEL",
"PRELUDE", "RANGER", "SENTRA", "SPORTSMAN", "SPORTVAN",
"TAURUS", "TERCEL", "THUNDERBIRD", "VAN"
Node 14: HIC2 proportion of 1s = 0.32758621
Node 7: MODEL /= "CELICA", "CHEVETTE", "COLT", "CONCORD", "CORDIA",
"CRESSIDA", "CUTLASS", "DL", "E150 VAN", "EXPLORER", "FOX",
"GRANADA", "JETTA", "MAXIMA", "MOTOR HOME", "PONY EXCEL",
"PRELUDE", "RANGER", "SENTRA", "SPORTSMAN", "SPORTVAN",
"TAURUS", "TERCEL", "THUNDERBIRD", "VAN"
Node 15: HIC2 proportion of 1s = 0.44510386E-1

```

Predictor means below are means of cases with no missing values.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.

2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.

3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

Node 1: Intermediate node

A case goes into Node 2 if COLMEC = "BWU", "CYL", "NA", "NAP", "UNK"

COLMEC mode = "UNK"

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	-2.382	17.39	0.000

Proportion of ones in variable HIC2 = 0.845543E-1

Node 2: Intermediate node

A case goes into Node 4 if MODEL = "4RUNNER", "9-5", "ACHIEVA", "AEROSTAR", "ASTRO", "BEAUVILLE", "BEETLE", "CLUBWAGON MPV", "DAKOTA", "DE VILLE", "DEFORMABLE IMPACTOR", "DURA", "ELANTRA", "EUROVAN", "FIVE HUNDRED", "FORESTER", "FRONTIER", "GALANT", "GRAND AM", "INSIG", "MUSTANG", "NEON", "PASEO", "PATHFINDER", "PT CRUISER", "QUEST", "RAM", "RAM WAGON VAN", "S10 BLAZER", "S80", "SIDEKICK", "SILVERADO2500", "SPACECAB", "SPORTVAN", "TACOMA", "TEMPO", "TRACKER", "TRAILBLAZER", "TROOPER II", "VERONA", "VUE"

MODEL mode = "ACCORD"

Node 4: Intermediate node

A case goes into Node 8 if IMPANG in [269, 287]

IMPANG mean = 35.607798

Node 8: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	-0.7472	5.150	0.3616E-05

Proportion of ones in variable HIC2 = 0.321429

Node 9: Intermediate node

A case goes into Node 18 if RSTFRT = "0"

RSTFRT mode = "1"

Node 18: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	0.7411E-01	7.626	0.3935E-09

Proportion of ones in variable HIC2 = 0.518519

Node 19: Intermediate node

A case goes into Node 38 if MODEL = "BEAUVILLE", "DE VILLE", "PATHFINDER", "PT CRUISER", "RAM WAGON VAN", "S-10", "TEMPO", "TRAILBLAZER", "VERONA"

MODEL mode = "MAXIMA"

Node 38: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	-1.414	3.345	0.1667E-02

Proportion of ones in variable HIC2 = 0.195652

Node 39: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	-3.405	3.049	0.2513E-02

Proportion of ones in variable HIC2 = 0.321429E-1

Node 5: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	-5.483	3.006	0.2675E-02

Proportion of ones in variable HIC2 = 0.413983E-2

Node 3: Intermediate node

A case goes into Node 6 if MODEL = "18", "200", "200 SX", "2000", "210", "240", "244", "300 ZX", "318", "325 I", "4000", "4RUNNER", "5000", "504", "505", "604", "626", "ALLIANCE", "B2000 PICKUP", "BLAZER", "CAPRI", "CAPRICE", "CAR ELECTRIC", "CHAMP", "CLUBWAGON MPV", "COLT PICKUP", "COLT VISTA", "CONQUEST", "COURIER", "DAKOTA", "E100 VAN", "EL CAMINO", "EL ELECTREK", "ELECTRICA", "EXCEL GLS", "EXP", "F150 PICKUP", "FAIRMONT", "FESTIVA", "FUEGO", "IMPULSE", "KING CAB PICKUP", "LANCER", "LE BARON", "LE CAR", "LEGACY", "LEOPARD", "LIBERTY", "MARQUIS", "MEDALLION", "MERKUR", "METRO", "MONTERO", "MONZA", "MPV", "NEW YORKER", "NOVA", "QUANTUM", "S-10", "SABLE", "SCIROCCO", "SHADOW", "SIDEKICK", "SOMERSET", "SONATA", "SPECT", "SPORTSWAGON", "SPRINT", "ST. REGIS", "STANZA", "STARLET", "SUBURBAN", "T1000", "TEMPO", "TRANSIT CONNECT", "TREDIA", "TREKKER", "TROOPER II", "VANAGON", "WAGON", "WRANGLER"

MODEL mode = "CIVIC"

Node 6: Intermediate node

A case goes into Node 12 if MODEL = "18", "200", "244", "300 ZX", "626", "ALLIANCE", "BLAZER", "CAPRI", "CAPRICE", "COLT VISTA", "CONQUEST", "COURIER", "DAKOTA", "ELECTREK", "EXCEL GLS", "FAIRMONT", "GLC", "IMPULSE", "LE BARON", "LEGACY", "LIBERTY", "LUV", "MEDALLION", "METRO", "SCIROCCO", "SHADOW", "SONATA", "SPIRIT", "TEMPO", "WAGON", "WRANGLER"

MODEL mode = "PICKUP"

Node 12: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	0.1241	11.31	0.5471E-15

Proportion of ones in variable HIC2 = 0.530973

Node 13: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	2.314	31.80	0.000

Proportion of ones in variable HIC2 = 0.910000

Node 7: Intermediate node

A case goes into Node 14 if MODEL = "CELICA", "CHEVETTE", "COLT", "CONCORD", "CORDIA", "CRESSIDA", "CUTLASS", "DL", "E150 VAN", "EXPLORER", "FOX", "GRANADA", "JETTA", "MAXIMA", "MOTOR HOME", "PONY EXCEL", "PRELUDE", "RANGER", "SENTRA", "SPORTSMAN", "SPORTVAN", "TAURUS", "THUNDERBIRD", "VAN"

MODEL mode = "CIVIC"

Node 14: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	-0.7191	7.518	0.000

Proportion of ones in variable HIC2 = 0.327586

Node 15: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value
Constant	-3.067	3.962	0.9074E-04

Proportion of ones in variable HIC2 = 0.445104E-1

Observed and fitted values are stored in logitc.fit
LaTeX code for tree is in logitc.tex

The logistic regression tree is shown in Figure 35.

15.2 Simple linear

We can also construct a logistic regression tree with a simple linear logistic regression model fitted to each node.

15.2.1 Input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: logits.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: logits.out
```

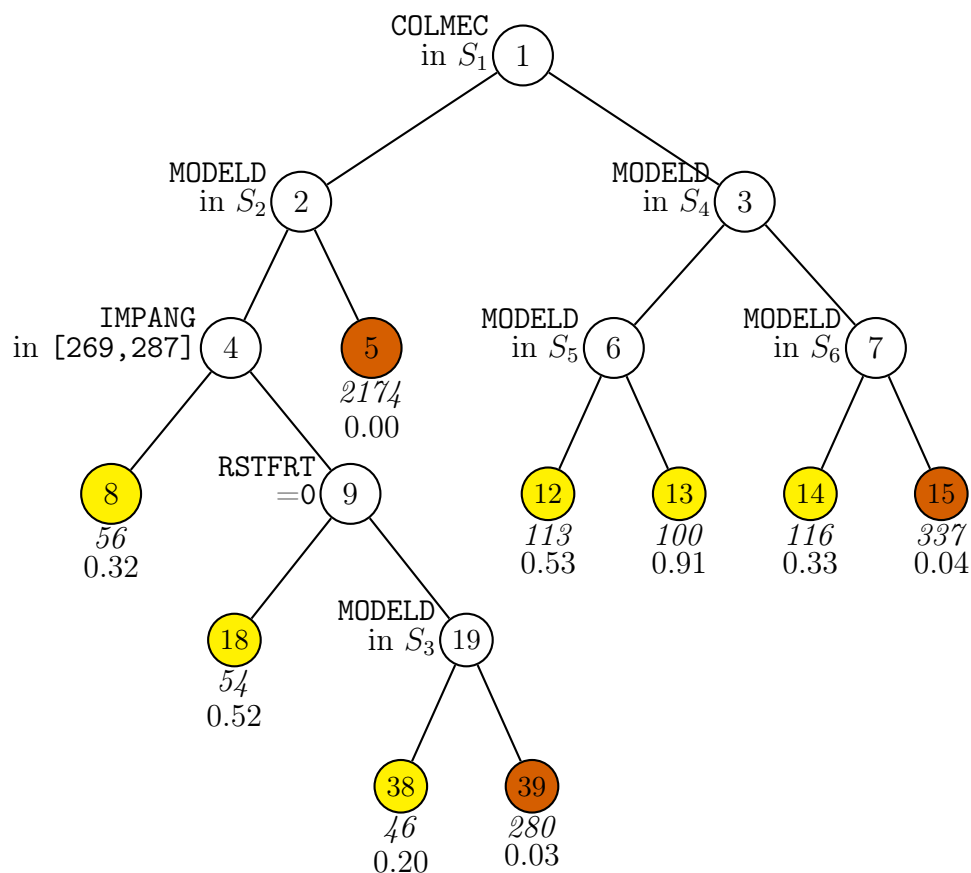


Figure 35: GUIDE v.46.1 0.250-SE piecewise-constant logistic regression tree for predicting $P(\text{HIC2}=1)$. At each split, an observation goes to the left branch if and only if the condition is satisfied. $S_1 = \{\text{BWU, CYL, NA, NAP, UNK}\}$. $S_2 = \{4\text{RUNNER, 9-5, ACHIEVA, AEROSTAR, ASTRO, BEAUVILLE, BEETLE, CLUBWAGON MPV, DAKOTA, DE VILLE, DEFORMABLE IMPACTOR, DURANGO, ELANTRA, EUROVAN, FIVE HUNDRED, FORESTER, FRONTIER, GALANT, GRAND AM, INSIGHT, ION, L200, MAXIMA, MAZDA3, METRO, MPV, MUSTANG, NEON, PASEO, PATHFINDER, PT CRUISER, QUEST, RAM, RAM WAGON VAN, RODEO, S-10, S10 BLAZER, S80, SIDEKICK, SILVERADO2500, SPACECAB, SPORTVAN, TACOMA, TEMPO, TERCEL, TRACKER, TRAILBLAZER, TROOPER II, VERONA, VUE}\}$. $S_3 = \{\text{BEAUVILLE, DE VILLE, PATHFINDER, PT CRUISER, RAM WAGON VAN, S-10, TEMPO, TRAILBLAZER, VERONA}\}$. $S_4 = \{18, 200, 200 \text{ SX}, 2000, 210, 240, 244, 300 \text{ ZX}, 318, 325 \text{ I}, 4000, 4\text{RUNNER}, 5000, 504, 505, 604, 626, \text{ALLIANCE, ASTRO, AXCESS, B2000 PICKUP, BLAZER, CAPRI, CAPRICE, CAR ELECTRIC, CHAMP, CLUBWAGON MPV, COLT PICKUP, COLT VISTA, CONQUEST, COURIER, DAKOTA, E100 VAN, EL CAMINO, ELECTRA, ELECTREK, ELECTRICA, EXCEL GLS, EXP, F150 PICKUP, FAIRMONT, FESTIVA, FUEGO, FURY, GF, GL, GLC, GV, I-MARK, IMPULSE, KING CAB PICKUP, LANCER, LE BARON, LE CAR, LEGACY, LEOPARD, LIBERTY, LUMINA, LUV, MALIBU, MARQUIS, MEDALLION, MERKUR, METRO, MONTERO, MONZA, MPV, NEW YORKER, NOVA, PASSAT, PICKUP, PREVIA, PRIZM, PULSAR, QUANTUM, S-10, SABLE, SCIROCCO, SHADOW, SIDEKICK, SOMERSET, SONATA, SPECTRUM, SPIRIT, SPORTSWAGON, SPRINT, ST. REGIS, STANZA, STARLET, SUBURBAN, T1000, TEMPO, TORINO, TRANSIT CONNECT, TREDIA, TREKKER, TROOPER II, VANAGON, WAGON, WRANGLER}\}$. $S_5 = \{18, 200, 244, 300 \text{ ZX}, 626, \text{ALLIANCE, BLAZER, CAPRI, CAPRICE, COLT VISTA, CONQUEST, COURIER, DAKOTA, ELECTREK, EXCEL GLS, EXP, FAIRMONT, GLC, IMPULSE, LE BARON, LEGACY, LIBERTY, LUV, MEDALLION, METRO, MPV, NOVA, PASSAT, PICKUP, PREVIA, S-10, SABLE, SCIROCCO, SHADOW, SONATA, SPIRIT, TEMPO, WAGON, WRANGLER}\}$. $S_6 = \{\text{CELICA, CHEVETTE, COLT, CONCORD, CORDIA, CRESSIDA, CUTLASS, DL, E150 VAN, EXPLORER, FOX, GRANADA, JETTA, MAXIMA, MOTOR HOME, PONY EXCEL, PRELUDE, RANGER, SENTRA, SPORTSMAN, SPORTVAN, TAURUS, TERCEL, THUNDERBIRD, VAN}\}$. Sample size (in *italics*) and proportion of 1s in HIC2 printed below nodes. Terminal nodes with proportions of 1s above and below value of 0.08 at root node are painted yellow and vermillion respectively. Second best split variable at root node is YEAR.

```

Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables),
  7=binary logistic regression.
Input choice ([1:7], <cr>=1): 7
Choose complexity of model to use at each node:
Choose 1 for multiple regression (recommended for prediction)
Choose 2 for best simple polynomial in one N or F variable
Choose 3 for constant fit (recommended for interpretability or if there is an R variable)
1: multiple linear, 2: best simple polynomial, 3: constant ([1:3], <cr>=3): 2
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: nhtsadsc2.txt
Reading DSC file ...
Training sample file: nhtsadatam.txt
Missing value code: NA
Records in data file start on line 2
Warning: B variables changed to C
D variable is HIC2
Reading data file ...
Number of records in data file: 3310
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 13 categorical variables
Finished assigning codes to 10 categorical variables
Associating missing values of N, P and S variables with M variable codes ...
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Numbers of N, F, S, and P below exclude constant variables

```

Total	#cases w/	#missing					
#cases	miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var	
3310	34	3310	2	48	0	0	
#P-var	#M-var	#B-var	#C-var	#I-var			
6	42	0	13	0			

```

Number of cases used for training: 3276
Number of cases excluded due to 0 W or missing D variable: 34
Proportion of ones in HIC2 variable: 8.4554334554334559E-002
Finished reading data file
Constant fitted to cases with missing values in regressor variables
Default max. number of split levels: 15
Minimum number of D=0 and D=1 in each node: 9
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): logits.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save regressor names in a file, 1 otherwise ([1:2], <cr>=2):
Input file name: logits.reg
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: logits.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: logits.r
Input rank of top variable to split root node ([1:67], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < logits.in

```

15.2.2 Contents of logits.out

```

Binary logistic regression tree
Pruning by cross-validation
DSC file: nhtsadsc2.txt
Training sample file: nhtsadatam.txt
Missing value code: NA
Records in data file start on line 2
Warning: B variables changed to C
D variable is HIC2
Piecewise simple linear logistic model
Number of records in data file: 3310
Length of longest entry in data file: 19
Missing values found in D variable
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables

```

Summary information for training sample of size 3276 (excluding observations with non-positive weight or missing values in d, e, t, r or z variables)
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight,
e=estimated success probability

Levels of M variables are for missing values in associated variables

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	BARRIG	c			3	
2	BARSHP	c			21	
3	BARANG	p	0.000	330.0	360	14
4	BARDIA	n	1.9100E+02	1000.		2807
:						
109	VEHOR	p	0.000	90.00	360	995
110	RSTFRT	c			3	
111	HIC2	d	0.000	1.000		
112	estHIC2	e	0.000	0.8455		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
3310	34	3310	2	48	0	0	
#P-var	#M-var	#B-var	#C-var	#I-var			
6	42	0	13	0			

Number of cases used for training: 3276

Number of cases excluded due to 0 W or missing D variable: 34

Proportion of ones in HIC2 variable: 0.084554

Constant fitted to cases with missing values in regressor variables

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

Nodewise interaction tests on all variables

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 15

Minimum node sample size: 33

Minimum number of D=0 and D=1 in each node: 9

Size and CV Loss and SE of subtrees:

Tree	#Tnodes	Mean Loss	SE(Mean)	BSE(Mean)	Median Loss	BSE(Median)
1	15	7.144E-01	4.578E-02	7.517E-02	6.485E-01	1.081E-01
2	14	7.121E-01	4.579E-02	7.570E-02	6.485E-01	1.116E-01

3	13	7.121E-01	4.574E-02	7.593E-02	6.437E-01	1.127E-01
4	10	6.540E-01	2.923E-02	6.352E-02	6.137E-01	7.424E-02
5	8	6.712E-01	2.906E-02	6.403E-02	6.092E-01	9.406E-02
6	7	6.597E-01	2.885E-02	6.506E-02	5.613E-01	9.848E-02
7	6	6.721E-01	2.097E-02	6.801E-02	5.613E-01	1.205E-01
8	5	6.735E-01	2.102E-02	6.844E-02	5.613E-01	1.225E-01
9	3	7.398E-01	1.931E-02	8.496E-02	6.962E-01	1.514E-01
10	2	4.547E-01	1.932E-02	9.157E-03	4.653E-01	1.100E-02
11**	1	4.547E-01	1.932E-02	9.157E-03	4.653E-01	1.100E-02

0-SE tree based on mean is marked with * and has 1 terminal node

0-SE tree based on median is marked with + and has 1 terminal node

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

* tree, ** tree, + tree, and ++ tree all the same

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is mean of HIC2 in the node

Cases fit give the number of cases used to fit node

Node deviance is residual deviance divided by residual degrees of freedom

Node	Total	Cases	Matrix	Node	Node	Split	Other
label	cases	fit	rank	D-mean	deviance	variable	variables
1T	3276	3276	2	8.455E-02	4.546E-01	RSTFRT	-YEAR

Best split at root node is on RSTFRT

Number of terminal nodes of final tree: 1

Total number of nodes of final tree: 1

Regression tree:

Node 1: HIC2 proportion of 1s = 0.84554335E-1

Predictor means below are means of cases with no missing values.

Regression coefficients are computed from the complete cases.

Node 1: Terminal node

Coefficients of logit function:

Regressor	Coefficient	t-stat	p-value	Minimum	Mean	Maximum
Constant	258.0	17.26	0.3833E-15			
YEAR	-0.1306	-17.38	0.9021E-15	1972.	2000.	2017.

If regressor has missing values, predicted value = 0.84554335E-1

Observed and fitted values are stored in `logits.fit`
Regressor names and coefficients are stored in `logits.reg`
LaTeX code for tree is in `logits.tex`

The results show that the tree has no splits. It fits a simple linear logistic regression model to the whole data set with `YEAR` as linear predictor. If the value of `YEAR` is missing, the predicted value of p is the mean of `HIC2`.

16 Importance scoring

When there are numerous predictor variables, it may be useful to rank them in order of their “importance”. GUIDE has a facility to do this. In addition, it provides thresholds for grouping the variables by their importance—see [Loh and Zhou \(2021\)](#).

16.1 Classification: RHC data

We show here how to obtain the importance scores for predicting `swang1`, the variable that takes values RHC and NoRHC.

16.1.1 Input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: imp.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1): 2
Name of batch output file: imp.out
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: rhcdsc1.txt
Reading DSC file ...
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
23 N variables changed to S
D variable is swang1
Reading data file ...
Number of records in data file: 5735
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Recoding D values to integers
Finished recoding
Number of classes: 2
Assigning integer codes to values of 30 categorical variables
Finished assigning codes to 10 categorical variables
```



```

Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Finished processing 5000 of 5735 observations
Data checks complete
Creating missing value indicators ...
Rereading data ...
Class  #Cases      Proportion
NoRHC   3551      0.61918047
RHC     2184      0.38081953
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      5735      0      5157      10      0      0      23
      #P-var  #M-var  #B-var  #C-var  #I-var
      0      0      0      30      0
Number of cases used for training: 5735
Number of split variables: 53
Number of cases excluded due to 0 W or missing D variable: 0
Finished reading data file
Choose 1 for estimated priors, 2 for equal priors, 3 to input priors from a file
Input 1, 2, or 3 ([1:3], <cr>=1):

Choose 1 for unit misclassification costs, 2 to input costs from a file
Input 1 or 2 ([1:2], <cr>=1):

Default max. number of split levels: 4
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=0):
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
You can create a DSC file with the selected variables included or excluded
Input 2 to create such a file, 1 otherwise ([1:2], <cr>=1):
You can also output the importance scores and variable names to a file
Input 1 to create such a file, 2 otherwise ([1:2], <cr>=1):
Input file name: imp.scr
Input file is created!
Run GUIDE with the command: guide < imp.in

```

16.1.2 Contents of imp.out

The most interesting part of the output file is at the end, as shown below.

Scaled importance scores of predictor variables

Score	Rank	Variable
2.007E+01	1.00	cat1
1.927E+01	2.00	aps1
1.575E+01	3.00	crea1
1.542E+01	4.00	pafi1
1.409E+01	5.00	meanbp1
9.920E+00	6.00	neuro
9.256E+00	7.00	alb1
9.002E+00	8.00	cat2
8.441E+00	9.00	card
8.033E+00	10.00	hema1
7.422E+00	11.00	wtkilo1
6.709E+00	12.00	seps
6.503E+00	13.00	adld3p
5.492E+00	14.00	dnr1
5.408E+00	15.00	resp
5.120E+00	16.00	bili1
4.429E+00	17.00	paco21
4.166E+00	18.00	surv2md1
3.520E+00	19.00	transhx
3.435E+00	20.00	resp1
3.359E+00	21.00	chrpulhx
3.094E+00	22.00	hrt1
2.981E+00	23.00	ninsclas
2.837E+00	24.00	dementhx
2.711E+00	25.00	ph1
1.977E+00	26.00	psychhx
1.933E+00	27.00	gastr
1.913E+00	28.00	das2d3pc
1.825E+00	29.00	renal
1.661E+00	30.00	income
1.597E+00	31.00	cardiohx
1.287E+00	32.00	urin1
1.230E+00	33.00	trauma
1.108E+00	34.00	age
1.103E+00	35.00	sex
1.092E+00	36.00	edu
1.048E+00	37.00	sod1
9.518E-01	38.00	immunhx
9.517E-01	39.00	wblc1
9.030E-01	40.00	ca
8.900E-01	41.00	malighx
7.916E-01	42.00	scoma1
7.817E-01	43.00	amihx
7.000E-01	44.00	chfhx

```

6.124E-01    45.00    gibledhx
4.960E-01    46.00    ortho
4.872E-01    47.00    renalhx
4.450E-01    48.00    pot1
4.402E-01    49.00    liverhx
4.146E-01    50.00    hema
3.704E-01    51.00    temp1
3.674E-01    52.00    meta
2.300E-01    53.00    race
99% threshold is 1.2588
95% threshold is 1.0000
90% threshold is 0.8965
80% threshold is 0.7408
50% threshold is 0.5542
Number of variables above 99% threshold is 32
Number of variables between 95% and 99% thresholds is 5
Number of variables between 90% and 95% thresholds is 3
Number of variables between 80% and 90% thresholds is 3
Number of variables between 50% and 80% thresholds is 2
Number of variables below 50% threshold is 8
Importance scores are stored in imp.scr

```

The variables, sorted according to their importance scores, are divided into 5 groups:

- A. Scores above 99% threshold
- B. Scores above 95% threshold and below 99% threshold
- C. Scores above 90% threshold and below 95% threshold
- D. Scores above 80% threshold and below 90% threshold
- E. Scores above 50% threshold and below 80% threshold
- F. Scores below 50% threshold

The groups and thresholds have the following interpretation. Let H_0 denote the null hypothesis H_0 that the dependent variable is independent of the predictor variables (it is not assumed that the predictor variables are independent of each other). If H_0 is true, there is a 0.01, 0.05, 0.10, 0.20, and 0.50 probability that one or more predictor variables falls into groups A , $A \cup B$, $A \cup B \cup C$, $A \cup B \cup C \cup D$, and $A \cup B \cup C \cup D \cup E$, respectively. The importance scores are normalized so that the 95% threshold is 1.0.

The file `imp.scr` lists the rank, group membership, importance score, number of missing values, and variable name.

Rank	Group	Score	Missing	Variable
1	A	2.007E+01	0	cat1
2	A	1.927E+01	0	aps1
3	A	1.575E+01	0	crea1
4	A	1.542E+01	0	pafi1
5	A	1.409E+01	80	meanbp1
6	A	9.920E+00	0	neuro
7	A	9.256E+00	0	alb1
8	A	9.002E+00	4535	cat2
9	A	8.441E+00	0	card
10	A	8.033E+00	0	hema1
11	A	7.422E+00	515	wtkilo1
12	A	6.709E+00	0	seps
13	A	6.503E+00	4296	adld3p
14	A	5.492E+00	0	dnr1
15	A	5.408E+00	0	resp
16	A	5.120E+00	0	bili1
17	A	4.429E+00	0	paco21
18	A	4.166E+00	0	surv2md1
19	A	3.520E+00	0	transhx
20	A	3.435E+00	136	resp1
21	A	3.359E+00	0	chrpulhx
22	A	3.094E+00	159	hrt1
23	A	2.981E+00	0	ninsclas
24	A	2.837E+00	0	dementhx
25	A	2.711E+00	0	ph1
26	A	1.977E+00	0	psychhx
27	A	1.933E+00	0	gastr
28	A	1.913E+00	0	das2d3pc
29	A	1.825E+00	0	renal
30	A	1.661E+00	0	income
31	A	1.597E+00	0	cardiohx
32	A	1.287E+00	3028	urin1
33	B	1.230E+00	0	trauma
34	B	1.108E+00	0	age
35	B	1.103E+00	0	sex
36	B	1.092E+00	0	edu
37	B	1.048E+00	0	sod1
38	C	9.518E-01	0	immunhx
39	C	9.517E-01	0	wblc1
40	C	9.030E-01	0	ca
41	D	8.900E-01	0	malighx
42	D	7.916E-01	0	scoma1
43	D	7.817E-01	0	amihx

44	E	7.000E-01	0	chfhx
45	E	6.124E-01	0	gibledhx
46	F	4.960E-01	0	ortho
47	F	4.872E-01	0	renalhx
48	F	4.450E-01	0	pot1
49	F	4.402E-01	0	liverhx
50	F	4.146E-01	0	hema
51	F	3.704E-01	0	temp1
52	F	3.674E-01	0	meta
53	F	2.300E-01	0	race

Figure 36 shows a barplot of the scores, produced by the following R code.

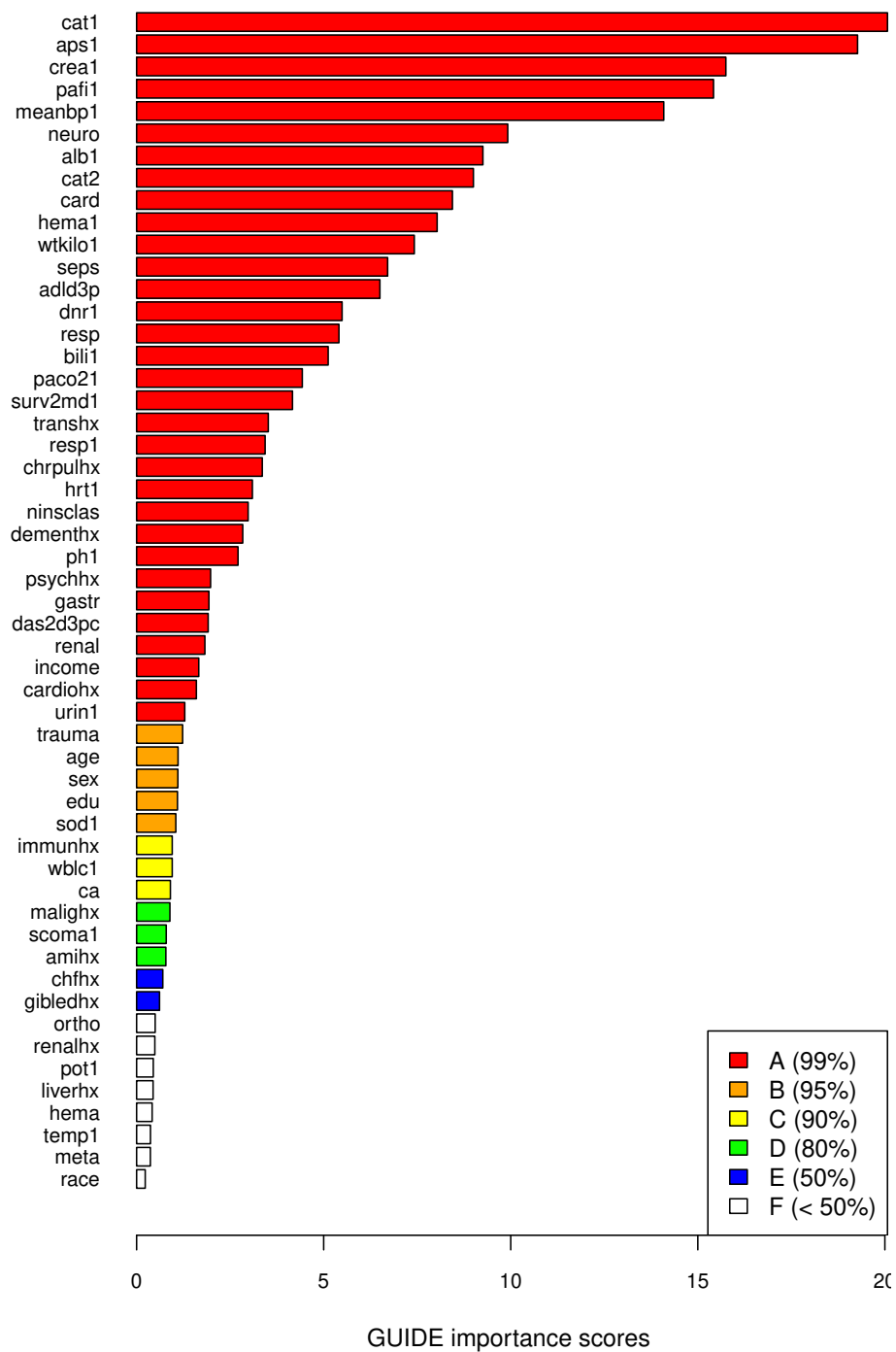
```
par(las=1,mar=c(5,12,4,2),cex.axis=0.8)
leg.col <- c("red","orange","yellow","green","blue","white")
leg.txt <- c("A (99%)","B (95%)","C (90%)","D (80%)","E (50%)","F (< 50%)")
x <- read.table("imp.scr",header=TRUE)
n <- nrow(x)
score <- x$Score
vars <- x$Variable
group <- x$Group
barcol <- rep("white",n)
letrs <- c("A","B","C","D","E","F")
for(i in 1:5){
  barcol[group == letrs[i]] <- leg.col[i]
}
barplot(rev(score[1:n]),names.arg=rev(vars[1:n]),col=rev(barcol[1:n]),horiz=TRUE,
        xlab="GUIDE importance scores")
legend("bottomright",legend=leg.txt,fill=leg.col)
```

16.2 Censored response with R variable

Following is the corresponding scoring procedure for a censored response with a treatment (R) variable (swang1). The R variable is not given a score because it acts as a linear predictor in the nodes of the tree.

16.2.1 Input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: imp_surv.in
```

Figure 36: Scores of important variables for predicting `swang1`

```

Input 1 for model fitting, 2 for importance or DIF scoring,
  3 for data conversion ([1:3], <cr>=1): 2
Name of batch output file: imp_surv.out
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables).
Input choice ([1:6], <cr>=1): 4
Input 1 for proportional hazards, 2 for restricted mean event time ([1:2], <cr>=1):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: rhcdsc3.txt
Reading DSC file ...
Training sample file: rhcdata.txt
Missing value code: NA
Records in data file start on line 2
R variable present
23 N variables changed to S
Warning: model changed to linear in treatment
D variable is death
Reading data file ...
Number of records in data file: 5735
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Assigning integer codes to values of 31 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Treatment (R) variable is swang1 with values "NoRHC" and "RHC"
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Finished processing 5000 of 5735 observations
Data checks complete
Smallest uncensored survtime: 2.0000
Number of complete cases excluding censored T < smallest uncensored T: 5735
Number of cases used to compute baseline hazard: 5735
Number of cases with D=1 and T >= smallest uncensored: 3722
GUIDE will try to create the variables in the description file.
If it is unsuccessful, please create the columns yourself...
Input 1 if randomized trial, 2 if observational study: ([1:2], <cr>=1): 2

```

```

Choose a subgroup identification method:
1 = Prognostic priority (Gs)
2 = Predictive priority (Gi)
Input your choice: ([1:2], <cr>=2):
Creating dummy variables ...
Creating missing value indicators ...
Rereading data ...
Largest uncensored and censored survtime by swang1
  "NoRHC"      1867.0000    1243.0000
  "RHC"        1943.0000    1351.0000
Proportion of training sample for each level of swang1
  "NoRHC"      0.6192
  "RHC"        0.3808
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      5735      0      5157      8      0      1      23
      #P-var  #M-var  #B-var  #C-var  #I-var  #R-var
      0      0      0      30      0      1
Number of dummy variables created: 1
Survival time variable in column: 64
Event indicator variable in column: 9
Proportion uncensored among nonmissing T and D variables: .649
Number of cases used for training: 5735
Number of cases excluded due to 0 W or missing D, T or R variables: 0
Finished reading data file
Default max. number of split levels: 8
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=0):
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
You can create a DSC file with the selected variables included or excluded
Input 2 to create such a file, 1 otherwise ([1:2], <cr>=1):
You can also output the importance scores and variable names to a file
Input 1 to create such a file, 2 otherwise ([1:2], <cr>=1):
Input file name: imp_surv.scr
Input file is created!
Run GUIDE with the command: guide < imp_surv.in

```

16.2.2 Partial contents of imp_surv.out

```

Scaled importance scores of predictor variables
(F, I and R variables are excluded)
  Score    Rank  Variable

```

9.968E-01	1.00	dnr1
8.981E-01	2.00	ph1
7.948E-01	3.00	chrpulhx
7.360E-01	4.00	paco21
7.110E-01	5.00	resp1
4.373E-01	6.00	liverhx
4.306E-01	7.00	pot1
4.148E-01	8.00	gastr
3.964E-01	9.00	cat2
3.826E-01	10.00	age
3.521E-01	11.00	gibledhx
3.428E-01	12.00	pafi1
3.195E-01	13.00	aps1
3.129E-01	14.00	malighx
2.958E-01	15.00	hrt1
2.856E-01	16.00	amihx
2.838E-01	17.00	surv2md1
2.803E-01	18.00	ninsclas
2.576E-01	19.00	income
2.491E-01	20.00	meanbp1
2.469E-01	21.00	edu
2.287E-01	22.00	das2d3pc
2.044E-01	23.00	ortho
2.019E-01	24.00	scoma1
1.790E-01	25.00	sex
1.755E-01	26.00	alb1
1.664E-01	27.00	meta
1.646E-01	28.00	hema1
1.584E-01	29.00	temp1
1.555E-01	30.00	crea1
1.551E-01	31.00	ca
1.528E-01	32.00	renalhx
1.522E-01	33.00	trauma
1.449E-01	34.00	hema
1.401E-01	35.00	psychhx
1.392E-01	36.00	wtkilo1
1.390E-01	37.00	urin1
1.366E-01	38.00	neuro
1.337E-01	39.00	dementhx
1.231E-01	40.00	chfhx
1.204E-01	41.00	wblc1
1.066E-01	42.00	sod1
9.480E-02	43.00	race
9.229E-02	44.00	adld3p
8.731E-02	45.00	seps
8.553E-02	46.00	cardiohx

8.401E-02	47.00	cat1
6.692E-02	48.00	resp
5.832E-02	49.00	card
5.806E-02	50.00	immunhx
5.768E-02	51.00	renal
5.684E-02	52.00	bili1
5.596E-02	53.00	transhx

99% threshold is 1.3762
95% threshold is 1.0000
90% threshold is 0.9475
80% threshold is 0.7654
50% threshold is 0.7235
Number of variables above 99% threshold is 0
Number of variables between 95% and 99% thresholds is 0
Number of variables between 90% and 95% thresholds is 1
Number of variables between 80% and 90% thresholds is 2
Number of variables between 50% and 80% thresholds is 1
Number of variables below 50% threshold (excluding F, I and R variables) is 51

17 Propensity scores

17.1 Causal inference

Propensity scores are often used in causal inference to estimate average treatment effects. Given a treatment variable Z taking values 0 (no treatment) and 1 (treatment), the propensity score for a subject with covariate $X = x$ is $\pi(x) = P(Z = 1 | X = x)$. If n denotes the sample size and Y_i the response of the i th subject, the average treatment effect may be estimated by the *Horvitz-Thompson estimate (HT)*

$$n^{-1} \sum_{i=1}^n \left\{ \frac{Z_i Y_i}{\hat{\pi}(X_i)} - \frac{(1 - Z_i) Y_i}{1 - \hat{\pi}(X_i)} \right\}$$

or the *Hájek inverse probability estimate (IPW)*

$$\frac{\sum_i Z_i Y_i / \hat{\pi}(X_i)}{\sum_i Z_i / \hat{\pi}(X_i)} - \frac{\sum_i (1 - Z_i) Y_i / (1 - \hat{\pi}(X_i))}{\sum_i (1 - Z_i) / (1 - \hat{\pi}(X_i))}$$

where $\hat{\pi}(x)$ is an estimate of $\pi(x)$. Clearly, $\hat{\pi}(x)$ cannot be 0 or 1.

Propensity scores are traditionally estimated by logistic regression, but this approach has difficulties if there are missing values in the covariates or if the number of covariates is large. Random forest has been used, but the version implemented in R is not applicable to data with missing values in predictor variables. Even when there are no missing values, the propensity score estimates from logistic regression and random forest are not easy to interpret.

A logistic regression tree or a piecewise-constant regression tree for estimating $\pi(x)$ is more interpretable than a forest. To prevent any $\hat{\pi}$ from being 0 or 1, the “propensity score” option in GUIDE fits a piecewise-constant regression tree to the Z_i such that no terminal node has all $Z_i = 0$ or all $Z_i = 1$. If this option is used to estimate the propensity scores, the HT and IPW estimates are identical and reduce to the *sample size weighted estimate* $n^{-1} \sum_t n_t \hat{\beta}_t$, where the sum is over the terminal nodes and n_t and $\hat{\beta}_t$ are the sample size and estimated treatment effect in node t .

We demonstrate the propensity score feature with the RHC data. Doctors believe that direct measurement of cardiac function by right heart catheterization for some critically ill patients yields better outcomes. The benefit of RHC has not been demonstrated in a randomized clinical trial due to ethical concerns. In observational studies, the relative risk of death was found to be higher in the elderly and in patients with acute myocardial infarction who received RHC. In such studies, the decision to use RHC is at the discretion of the physician. Therefore treatment assignment is

confounded with patient factors that are also related to outcomes, e.g., patients with low blood pressure are more likely to get RHC, and such patients are also more likely to die. The data consist of observations on more than 60 variables for 5735 patients from 5 medical centers over 5 years (Connors et al., 1996). The treatment variable is `swang1` (RHC or NoRHC) which we define as 1 if RHC and 0 if NoRHC. The resulting data are in the file `propendata.csv` and the DSC file is `propen.dsc` where `swang` is specified as `d` and `death` and `dth30` are specified as `x`.

The next section shows how the input file for propensity score trees is created. After the propensity score option is chosen, the program will ask for a value for the parameter $\delta = \min(p, 1 - p)$, where p denotes a propensity score in any node of the tree, i.e., $\delta < p < 1 - \delta$. The default value $\delta = 0.05$ ensures that no propensity score is less than 0.05 or greater than 0.95.

17.1.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: propen.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: propen.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 3
Input min(p,1-p), where p is propensity score ([0.01:0.49], <cr>=0.05):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: propen.dsc
Reading DSC file ...
Training sample file: propendata.csv
Missing value code: NA
Records in data file start on line 2
35 N variables changed to S
D variable is swang1
Reading data file ...
Number of records in data file: 5735
Length of longest entry in data file: 19
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables

```

```

Assigning integer codes to values of 18 categorical variables
Finished assigning codes to 10 categorical variables
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Finished processing 5000 of 5735 observations
Data checks complete
Creating missing value indicators ...
Rereading data ...
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      5735      0      5157      10      0      0      35
      #P-var  #M-var  #B-var  #C-var  #I-var
      0      0      0      18      0
No weight variable in data file
Number of cases used for training: 5735
Number of cases excluded due to 0 W or missing D variable: 0
Finished reading data file
Default max. number of split levels: 12
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): propen.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: propen.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: propen.r
Input rank of top variable to split root node ([1:53], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < propen.in

```

17.1.2 Contents of propen.out

```

Propensity score tree with  $\min(p, 1-p) = 0.050$  where  $p$  is an estimated propensity score
Least squares regression tree
Pruning by cross-validation
DSC file: propen.dsc
Training sample file: propendata.csv

```

Missing value code: NA
 Records in data file start on line 2
 35 N variables changed to S
 D variable is swang1
 Piecewise constant model
 Number of records in data file: 5735
 Length of longest entry in data file: 19
 Missing values found among categorical variables
 Separate categories will be created for missing categorical variables
 Missing values found among non-categorical variables

Summary information for training sample of size 5735
 d=dependent, b=split and fit cat variable using indicator variables,
 c=split-only categorical, i=fit-only categorical (via indicators),
 s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
 m=missing-value flag variable, p=periodic variable, w=weight

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
2	cat1	c			9	
3	cat2	c			6	4535
4	ca	c			3	
10	cardiohx	s	0.000	1.000		
11	chfhx	s	0.000	1.000		
:						
44	ph1	s	6.579	7.770		
45	swang1	d	0.000	1.000		
46	wtkilo1	s	19.50	244.0		515
47	dnr1	c			2	
48	ninsclas	c			6	
49	resp	c			2	
50	card	c			2	
51	neuro	c			2	
52	gastr	c			2	
53	renal	c			2	
54	meta	c			2	
55	hema	c			2	
56	seps	c			2	
57	trauma	c			2	
58	ortho	c			2	
59	adld3p	s	0.000	7.000		4296
60	urin1	s	0.000	9000.		3028
61	race	c			3	
62	income	c			4	

Numbers of N, F, S, and P below exclude constant variables

```

      Total #cases w/ #missing
#cases  miss. D ord. vals #X-var #N-var #F-var #S-var
    5735      0    5157    10      0      0     35
#P-var  #M-var #B-var #C-var #I-var
      0      0      0    18      0

```

No weight variable in data file

Number of cases used for training: 5735

Number of cases excluded due to 0 W or missing D variable: 0

Pruning by v-fold cross-validation, with v = 10

Selected tree is based on mean of CV estimates

Number of SE's for pruned tree: 0.2500

No nodewise interaction tests

Split values for N and S variables based on exhaustive search

Max number of interaction splits: 100

Maximum number of split levels: 12

Minimum node sample size: 5

Ranks of variables and their 1-df chi-squared values at root node

```

  1  0.3346E+03  cat1
  2  0.2728E+03  aps1
  3  0.2430E+03  crea1
  4  0.2402E+03  meanbp1
  :
 43  0.1052E+01  meta
 44  0.6357E+00  race

```

Size and CV MSE and SE of subtrees:

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	39	2.044E-01	2.778E-03	1.535E-03	2.059E-01	1.812E-03
2	37	2.044E-01	2.778E-03	1.535E-03	2.059E-01	1.812E-03
3	36	2.044E-01	2.778E-03	1.535E-03	2.059E-01	1.821E-03
4*	33	2.041E-01	2.773E-03	1.554E-03	2.048E-01	1.699E-03
5	32	2.042E-01	2.772E-03	1.557E-03	2.050E-01	1.726E-03
6--	31	2.042E-01	2.765E-03	1.589E-03	2.047E-01	2.013E-03
7	30	2.046E-01	2.740E-03	1.493E-03	2.049E-01	1.762E-03
:						
17	18	2.047E-01	2.679E-03	1.405E-03	2.066E-01	2.138E-03
18**	16	2.047E-01	2.647E-03	1.224E-03	2.049E-01	1.498E-03
19++	15	2.048E-01	2.603E-03	9.774E-04	2.050E-01	1.448E-03
20	14	2.050E-01	2.593E-03	1.010E-03	2.055E-01	1.741E-03
21	12	2.052E-01	2.586E-03	9.845E-04	2.056E-01	1.461E-03
22	11	2.053E-01	2.564E-03	8.040E-04	2.056E-01	1.172E-03
23	10	2.057E-01	2.518E-03	9.355E-04	2.059E-01	1.089E-03
24	9	2.062E-01	2.497E-03	1.152E-03	2.060E-01	8.934E-04
25	8	2.062E-01	2.497E-03	1.152E-03	2.060E-01	8.934E-04

26	7	2.070E-01	2.498E-03	1.448E-03	2.060E-01	9.126E-04
27	6	2.093E-01	2.468E-03	1.610E-03	2.092E-01	1.371E-03
28	5	2.100E-01	2.403E-03	1.650E-03	2.106E-01	1.783E-03
29	4	2.121E-01	2.384E-03	1.268E-03	2.123E-01	1.435E-03
30	3	2.192E-01	2.130E-03	1.388E-03	2.194E-01	1.228E-03
31	2	2.284E-01	1.885E-03	1.252E-03	2.281E-01	1.729E-03
32	1	2.358E-01	1.528E-03	6.632E-05	2.357E-01	1.152E-04

0-SE tree based on mean is marked with * and has 33 terminal nodes

0-SE tree based on median is marked with + and has 31 terminal nodes

Selected-SE tree based on mean using naive SE is marked with **

Selected-SE tree based on mean using bootstrap SE is marked with --

Selected-SE tree based on median and bootstrap SE is marked with ++

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is mean of swang1 in the node

Cases fit give the number of cases used to fit node

MSE is residual sum of squares divided by number of cases in node

Node	Total	Cases	Matrix	Node	Node	Split
label	cases	fit	rank	D-mean	MSE	variable
1	5735	5735	1	3.808E-01	2.358E-01	cat1
2	1683	1683	1	5.401E-01	2.485E-01	meanbp1
4	1117	1117	1	6.204E-01	2.357E-01	pafi1
8	655	655	1	6.962E-01	2.118E-01	resp1
16T	197	197	1	8.731E-01	1.114E-01	-
17T	458	458	1	6.201E-01	2.361E-01	adld3p
9	462	462	1	5.130E-01	2.504E-01	ninsclas
18T	218	218	1	3.853E-01	2.379E-01	chfhx
19T	244	244	1	6.270E-01	2.348E-01	-
5T	566	566	1	3.816E-01	2.364E-01	alb1
3	4052	4052	1	3.147E-01	2.157E-01	pafi1
6	1292	1292	1	4.837E-01	2.499E-01	paco21
12	1042	1042	1	5.278E-01	2.495E-01	aps1
24	463	463	1	4.255E-01	2.450E-01	resp1
48T	152	152	1	5.987E-01	2.419E-01	hema1
49T	311	311	1	3.408E-01	2.254E-01	aps1
25	579	579	1	6.097E-01	2.384E-01	resp1
50T	111	111	1	8.198E-01	1.491E-01	-
51T	468	468	1	5.598E-01	2.469E-01	-
13T	250	250	1	3.000E-01	2.108E-01	pafi1
7	2760	2760	1	2.355E-01	1.801E-01	aps1
14	2100	2100	1	1.838E-01	1.501E-01	cat1
28	1326	1326	1	2.353E-01	1.801E-01	wtkilo1

56T	390	390	1	1.333E-01	1.159E-01	-
57T	936	936	1	2.778E-01	2.008E-01	alb1
29T	774	774	1	9.561E-02	8.658E-02	cat1
15	660	660	1	4.000E-01	2.404E-01	crea1
30T	129	129	1	2.171E-01	1.713E-01	-
31	531	531	1	4.444E-01	2.474E-01	wtkilo1
62T	85	85	1	1.765E-01	1.471E-01	-
63T	446	446	1	4.955E-01	2.505E-01	adld3p

Number of terminal nodes of final tree: 16

Total number of nodes of final tree: 31

Second best split variable (based on curvature test) at root node is aps1

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: cat1 = "CHF", "MOSF w/Sepsis"

Node 2: meanbp1 <= 68.500000 or NA

Node 4: pafi1 <= 266.15625

Node 8: resp1 <= 17.000000 or NA

Node 16: swang1-mean = 0.87309645

Node 8: resp1 > 17.000000

Node 17: swang1-mean = 0.62008734

Node 4: pafi1 > 266.15625 or NA

Node 9: ninsclas = "Medicaid", "Medicare", "Medicare & Medicaid"

Node 18: swang1-mean = 0.38532110

Node 9: ninsclas /= "Medicaid", "Medicare", "Medicare & Medicaid"

Node 19: swang1-mean = 0.62704918

Node 2: meanbp1 > 68.500000

Node 5: swang1-mean = 0.38162544

Node 1: cat1 /= "CHF", "MOSF w/Sepsis"

Node 3: pafi1 <= 142.35938

Node 6: paco21 <= 47.046875

Node 12: aps1 <= 56.500000

Node 24: resp1 <= 17.000000

Node 48: swang1-mean = 0.59868421

Node 24: resp1 > 17.000000 or NA

Node 49: swang1-mean = 0.34083601

Node 12: aps1 > 56.500000 or NA

Node 25: resp1 <= 25.500000

Node 50: swang1-mean = 0.81981982

Node 25: resp1 > 25.500000 or NA

Node 51: swang1-mean = 0.55982906

Node 6: paco21 > 47.046875 or NA

Node 13: swang1-mean = 0.30000000

Node 3: pafi1 > 142.35938 or NA

```

Node 7: aps1 <= 62.500000
Node 14: cat1 = "ARF", "Colon Cancer", "MOSF w/Malignancy"
Node 28: wtkilo1 <= 56.949995 or NA
Node 56: swang1-mean = 0.13333333
Node 28: wtkilo1 > 56.949995
Node 57: swang1-mean = 0.27777778
Node 14: cat1 /= "ARF", "Colon Cancer", "MOSF w/Malignancy"
Node 29: swang1-mean = 0.95607235E-1
Node 7: aps1 > 62.500000 or NA
Node 15: crea1 <= 1.3498535
Node 30: swang1-mean = 0.21705426
Node 15: crea1 > 1.3498535 or NA
Node 31: wtkilo1 <= 57.399980
Node 62: swang1-mean = 0.17647059
Node 31: wtkilo1 > 57.399980 or NA
Node 63: swang1-mean = 0.49551570

```

Predictor means below are means of cases with no missing values.

WARNING: p-values below not adjusted for split search. For a bootstrap solution see:

1. Loh et al. (2016), "Identification of subgroups with differential treatment effects for longitudinal and multiresponse variables", *Statistics in Medicine*, v.35, 4837-4855.
2. Loh et al. (2019), "Subgroups from regression trees with adjustment for prognostic effects and post-selection inference", *Statistics in Medicine*, v.38, 545-557.
3. Loh and Zhou (2020), "The GUIDE approach to subgroup identification", in "Design and Analysis of Subgroups with Biopharmaceutical Applications", Springer, pp.147-165.

```

Node 1: Intermediate node
A case goes into Node 2 if cat1 = "CHF", "MOSF w/Sepsis"
cat1 mode = "ARF"
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat    p-value
Constant     0.3808      59.39    0.3027-313
-----
Node 2: Intermediate node
A case goes into Node 4 if meanbp1 <= 68.500000 or NA
meanbp1 mean = 72.674985
-----
Node 4: Intermediate node
A case goes into Node 8 if pafi1 <= 266.15625
pafi1 mean = 241.37331
-----

```

Node 8: Intermediate node

A case goes into Node 16 if resp1 <= 17.000000 or NA
resp1 mean = 28.430124

Node 16: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.8731	36.72	0.3027-313

swang1 mean = 0.873096

Node 17: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.6201	27.31	0.3027-313

swang1 mean = 0.620087

Node 9: Intermediate node

A case goes into Node 18 if ninsclas = "Medicaid", "Medicare", "Medicare & Medicaid"
ninsclas mode = "Private"

Node 18: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.3853	11.66	0.3027-313

swang1 mean = 0.385321

Node 19: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.6270	20.21	0.3027-313

swang1 mean = 0.627049

Node 5: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.3816	18.67	0.3027-313

swang1 mean = 0.381625

Node 3: Intermediate node

A case goes into Node 6 if pafi1 <= 142.35938
pafi1 mean = 211.08630

Node 6: Intermediate node

A case goes into Node 12 if paco21 <= 47.046875
paco21 mean = 40.529679

```

Node 12: Intermediate node
A case goes into Node 24 if aps1 <= 56.500000
aps1 mean = 60.471209
-----

Node 24: Intermediate node
A case goes into Node 48 if resp1 <= 17.000000
resp1 mean = 26.771058
-----

Node 48: Terminal node
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat      p-value
Constant     0.5987        15.01       0.3027-313
swang1 mean = 0.598684
-----

Node 49: Terminal node
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat      p-value
Constant     0.3408        12.66       0.3027-313
swang1 mean = 0.340836
-----

Node 25: Intermediate node
A case goes into Node 50 if resp1 <= 25.500000
resp1 mean = 33.916667
-----

Node 50: Terminal node
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat      p-value
Constant     0.8198        22.37       0.3027-313
swang1 mean = 0.819820
-----

Node 51: Terminal node
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat      p-value
Constant     0.5598        24.37       0.3027-313
swang1 mean = 0.559829
-----

Node 13: Terminal node
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat      p-value
Constant     0.3000        10.33       0.3027-313
swang1 mean = 0.300000
-----

Node 7: Intermediate node
A case goes into Node 14 if aps1 <= 62.500000
aps1 mean = 50.276087
-----

```

Node 14: Intermediate node

A case goes into Node 28 if cat1 = "ARF", "Colon Cancer", "MOSF w/Malignancy"
cat1 mode = "ARF"

Node 28: Intermediate node

A case goes into Node 56 if wtkilo1 <= 56.949995 or NA
wtkilo1 mean = 72.028899

Node 56: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.1333	7.736	0.3027-313

swang1 mean = 0.133333

Node 57: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.2778	18.96	0.3027-313

swang1 mean = 0.277778

Node 29: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.9561E-01	9.040	0.3027-313

swang1 mean = 0.956072E-1

Node 15: Intermediate node

A case goes into Node 30 if crea1 <= 1.3498535
crea1 mean = 3.0464749

Node 30: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.2171	5.957	0.3027-313

swang1 mean = 0.217054

Node 31: Intermediate node

A case goes into Node 62 if wtkilo1 <= 57.399980
wtkilo1 mean = 75.515042

Node 62: Terminal node

Coefficients of least squares regression function:

Regressor	Coefficient	t-stat	p-value
Constant	0.1765	4.243	0.3027-313

swang1 mean = 0.176471

```

Node 63: Terminal node
Coefficients of least squares regression function:
Regressor    Coefficient  t-stat    p-value
Constant     0.4955      20.91     0.3027-313
swang1 mean = 0.495516
-----
Proportion of variance (R-squared) explained by tree model: 0.1805

Observed and fitted values are stored in propen.fit
LaTeX code for tree is in propen.tex
R code is stored in propen.r

```

The propensity score tree is shown in Figure 37. The two numbers below each terminal node are the sample size (in italics) and the estimate of $P(\text{swang1} = \text{RHC})$.

The file `propen.fit` gives the estimated propensity scores in the last column. Below are the top 7 rows of the file:

train	node	observed	predicted
y	48	0.00000	0.598684
y	17	1.00000	0.620087
y	63	1.00000	0.495516
y	56	0.00000	0.133333
y	18	1.00000	0.385321
y	29	0.00000	0.956072E-001
y	56	0.00000	0.133333

The following R code may be used to compute the Horvitz-Thompson and inverse probability weighted (IPW) estimates of average treatment effect for probability of death.

```

data <- read.csv("propendata.csv",header=TRUE)
y <- 1-data$dth30
fit <- read.table("propen.fit",header=TRUE)
train <- fit$train == "y"
z <- fit$observed[train]
p <- fit$predicted[train]
node <- fit$node[train]
nodenum <- unique(sort(node))
n <- sum(train)
horvitz <- (sum(z*y/p)-sum((1-z)*y/(1-p)))/n
ipw <- sum(z*y/p)/sum(z/p) - sum((1-z)*y/(1-p))/sum((1-z)/(1-p))

```

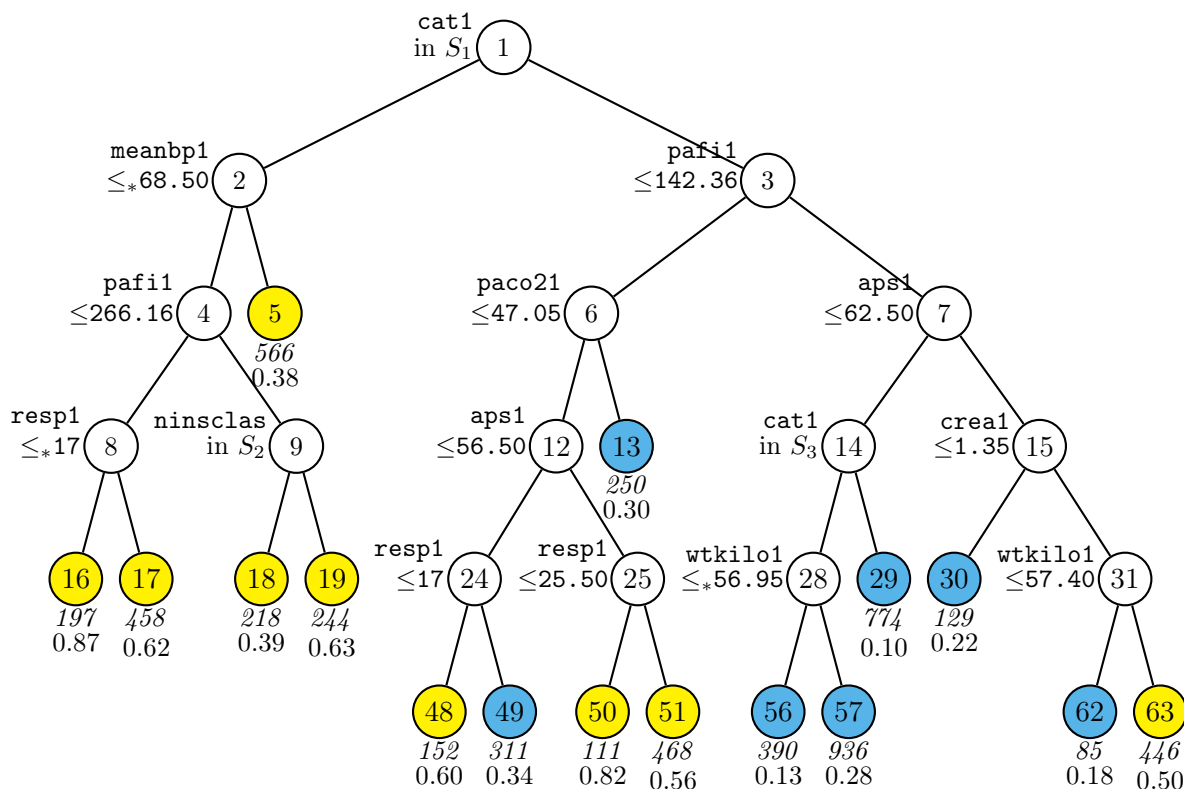


Figure 37: GUIDE v.46.1 0.250-SE tree for estimating propensity scores of `swang1`. At each split, an observation goes to the left branch if and only if the condition is satisfied. Symbol ' $\leq_{*}$ ' stands for ' $\leq$ or missing'. $S_1 = \{\text{CHF, MOSF w/Sepsis}\}$. $S_2 = \{\text{Medicaid, Medicare, Medicare \& Medicaid}\}$. $S_3 = \{\text{ARF, Colon Cancer, MOSF w/Malignancy}\}$. Sample size (in *italics*) and estimated propensity score printed below nodes. Terminal nodes with propensity scores above and below value of 0.381 at root node are painted yellow and skyblue respectively. Second best split variable at root node is `aps1`.

17.2 Missing-value imputation

Section 6.1.3 showed how to use a regression tree to impute missing values in a response variable to estimate the population mean of INTRDVX in the BLS data. Another common imputation method is hot-deck via propensity scores. In this method, propensity scores are used to partition the sample space into “imputation cells” and missing values in a cell are imputed with random draws from the observed responses in the cell.

To construct a propensity score tree and the imputation cells, we first replace the values of INTRDVX in `ce2021.txt` by a nonmissing indicator variable (`INTRDVXnonmiss`, say) that takes value 1 if INTRDVX is nonmissing and 0 otherwise, and save the new file as `ce2021miss.txt`, as shown by the following R code.

```
z <- read.table("ce2021.txt",header=TRUE)
y <- z$INTRDVX
INTRDVX <- rep(1,nrow(z))
INTRDVX[is.na(y)] <- 0
z$INTRDVX <- INTRDVX
names(z)[which(names(z) == "INTRDVX")] <- "INTRDVXnonmiss"
write.table(z,"ce2021miss.txt",row.names=FALSE,col.names=TRUE)
```

Key parts of the corresponding DSC file, which we call `ce2021miss.dsc`, are shown below. The file is the same as `ce2021reg.dsc` except for (i) a change in the first line from “`ce2021.txt`” to “`ce2021miss.txt`”, and (ii) a change in the line for the dependent variable from “`INTRDVX`” to “`INTRDVXnonmiss`”.

```
ce2021miss.txt
NA
2
1 DIRACC n
2 DIRACC_ m
3 AGE_REF n
4 AGE_REF_ m
5 AGE2 n
6 AGE2_ m
:
404 FSMPFRMX n
405 FSMP_RMX m
406 INTRDVXnonmiss d
407 INTRDVX_ x
408 IRAB n
409 IRAB_ m
:
```



```

547 WHLFYR c
548 WHLFYR_ m
549 FFTAXOWE n
550 FSTAXOWE n

```

17.2.1 Input file creation

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: propen.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1):
Name of batch output file: propen.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1):
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 3
Input min(p,1-p), where p is propensity score ([0.01:0.49], <cr>=0.05):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: ce2021miss.dsc
Reading DSC file ...
Training sample file: ce2021miss.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
384 N variables changed to S
D variable is INTRDVXnonmiss
Reading data file ...
Number of records in data file: 3965
Length of longest entry in data file: 11
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Finding number of levels of M variables associated with C variables ...
Assigning integer codes to values of 47 categorical variables
Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Finished assigning codes to 40 categorical variables
Associating missing values of N and S variables with M variable codes ...
Re-checking data ...
Allocating missing value information ...

```

```

Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Smallest positive weight:  1.0725E+03
Largest positive weight:  9.3902E+04
Numbers of N, F, S, and P below exclude constant variables
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      3965      0      3965      1      0      0      384
      #P-var  #M-var  #B-var  #C-var  #I-var
      0      116      0      47      0
Weight variable FINLWT21 in column: 31
Number of cases used for training: 3965
Number of cases excluded due to 0 W or missing D variable: 0
Finished reading data file
Input 1 for unweighted, 2 for weighted error estimates during pruning ([1:2], <cr>=2):
Default max. number of split levels: 15
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=2):
Input file name to store LaTeX code (use .tex as suffix): propen.tex
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
Input 2 to save fitted values and node IDs, 1 otherwise ([1:2], <cr>=2):
Input name of file to store node ID and fitted value of each case: propen.fit
You can output an R function for predicting future observations
Input 1 to skip this step
Input 2 to output an R function that applies to the training data
Input 3 to output an R function that applies to a new data set
Input your choice ([1:3], <cr>=2):
Input filename to write R function: propen.r
Input rank of top variable to split root node ([1:431], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < propen.in

```

17.2.2 Output file

```

Propensity score tree with  $\min(p, 1-p) = 0.050$  where  $p$  is an estimated propensity score
Least squares regression tree
Pruning by cross-validation
DSC file: ce2021miss.dsc
Training sample file: ce2021miss.txt
Missing value code: NA
Records in data file start on line 2

```

Number of M variables associated with C variables: 19
 384 N variables changed to S
 D variable is INTRDVXnonmiss
 Piecewise constant model
 Number of records in data file: 3965
 Length of longest entry in data file: 11
 Missing values found among categorical variables
 Separate categories will be created for missing categorical variables
 Missing values found among non-categorical variables
 Smallest and largest positive weights are 1.0725E+03 and 9.3902E+04

Summary information for training sample of size 3965
 d=dependent, b=split and fit cat variable using indicator variables,
 c=split-only categorical, i=fit-only categorical (via indicators),
 s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
 m=missing-value flag variable, p=periodic variable, w=weight
 Levels of M variables are for missing values in associated variables

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	DIRACC	s	1.000	1.000		170 constant
2	DIRACC_	m			2	
3	AGE_REF	s	18.00	87.00		
4	AGE_REF_	m			0	
5	AGE2	s	21.00	87.00		1734
:						
31	FINLWT21	w	1072.	0.9390E+05		
:						
406	INTRDVXnonmiss	d	0.000	1.000		
:						
547	WHLFYR	c			1	3964
548	WHLFYR_	m			1	
549	FFTAXOWE	s	-0.3368E+05	0.3997E+06		
550	FSTAXOWE	s	-3309.	0.7223E+05		

Numbers of N, F, S, and P below exclude constant variables

Total	#cases w/ #cases	#missing miss. D	ord. vals	#X-var	#N-var	#F-var	#S-var
3965	0	3965	1	0	0	379	
#P-var	#M-var	#B-var	#C-var	#I-var			
0	116	0	47	0			

Number of constant S variables: 5
 Weight variable FINLWT21 in column: 31
 Number of cases used for training: 3965
 Number of cases excluded due to 0 W or missing D variable: 0

Pruning by v-fold cross-validation, with v = 10
 Selected tree is based on mean of CV estimates
 Number of SE's for pruned tree: 0.2500

Weighted error estimates used for pruning
 No nodewise interaction tests
 Split values for N and S variables based on exhaustive search
 Max number of interaction splits: 100
 Maximum number of split levels: 15
 Minimum node sample size: 3
 Ranks of variables and their 1-df chi-squared values at root node

1	0.2268E+03	STATE
2	0.1587E+03	INCLASS2
3	0.1569E+03	ERANKH
:		
416	0.2087E-04	TOTEXPCQ
417	0.2087E-04	TOTEX4CQ
418	0.8284E-07	TAIRFARP

Warning: all zeroes in node 207; shrinking propensity score away from 0
 Size and CV MSE and SE of subtrees:

Tree	#Tnodes	Mean MSE	SE(Mean)	BSE(Mean)	Median MSE	BSE(Median)
1	63	4.881E+03	1.008E+02	5.767E+01	4.799E+03	1.154E+02
2	62	4.880E+03	1.008E+02	5.734E+01	4.799E+03	1.146E+02
:						
28	10	4.770E+03	9.675E+01	7.654E+01	4.675E+03	1.296E+02
29--	9	4.740E+03	9.532E+01	6.872E+01	4.658E+03	1.190E+02
30**	8	4.762E+03	9.353E+01	7.339E+01	4.687E+03	9.692E+01
31	7	4.954E+03	8.175E+01	6.892E+01	4.903E+03	1.144E+02
32	6	4.954E+03	8.175E+01	6.892E+01	4.903E+03	1.144E+02
33	5	4.951E+03	8.137E+01	6.997E+01	4.903E+03	1.144E+02
34	2	5.183E+03	7.727E+01	7.528E+01	5.182E+03	1.099E+02
35	1	5.519E+03	6.287E+01	3.904E+01	5.515E+03	4.372E+01

0-SE tree based on mean is marked with * and has 9 terminal nodes
 0-SE tree based on median is marked with + and has 9 terminal nodes
 Selected-SE tree based on mean using naive SE is marked with **
 Selected-SE tree based on mean using bootstrap SE is marked with --
 Selected-SE tree based on median and bootstrap SE is marked with ++
 * tree same as + tree
 ** tree same as ++ tree
 * tree same as -- tree

Following tree is based on mean CV with naive SE estimate (**)

Structure of final tree. Each terminal node is marked with a T.

D-mean is weighted mean of INTRDVXnonmiss in the node

Cases fit give the number of cases used to fit node

MSE is residual sum of squares divided by number of cases in node

Node label	Total cases	Cases fit	Matrix rank	Node D-mean	Node MSE	Split variable
1	3965	3965	1	6.217E-01	5.518E+03	STATE
2	1425	1425	1	7.722E-01	3.850E+03	INCLASS2
4T	65	65	1	2.990E-01	4.739E+03	BEDROOMQ
5T	1360	1360	1	7.952E-01	3.561E+03	DIVISION
3	2540	2540	1	5.458E-01	6.038E+03	RETSURVX
6	1838	1838	1	5.478E-01	6.186E+03	FINDRETX
12	1164	1164	1	4.692E-01	6.128E+03	ERANKH
24	635	635	1	5.696E-01	5.969E+03	STATE
48T	85	85	1	1.702E-01	2.519E+03	MISCCQ
49T	550	550	1	6.125E-01	6.025E+03	-
25T	529	529	1	3.515E-01	5.690E+03	LIQUIDX
13T	674	674	1	6.781E-01	5.597E+03	POPSIZE
7	702	702	1	5.403E-01	5.659E+03	RETSURVX
14T	129	129	1	1.048E-01	2.294E+03	STATE
15T	573	573	1	6.465E-01	5.131E+03	PSU

Number of terminal nodes of final tree: 8

Total number of nodes of final tree: 15

Second best split variable (based on curvature test) at root node is INCLASS2

Regression tree:

For categorical variable splits, values not in training data go to the right

Node 1: STATE = "2", "6", "10", "11", "21", "24", "25", "27", "31", "40", "41",
"47", "48", "49"

Node 2: INCLASS2 = NA

Node 4: INTRDVXnonmiss-mean = 0.29897738

Node 2: INCLASS2 /= NA

Node 5: INTRDVXnonmiss-mean = 0.79522322

Node 1: STATE /= "2", "6", "10", "11", "21", "24", "25", "27", "31", "40", "41",
"47", "48", "49"

Node 3: RETSURVX = NA & RETS_RVX = "A"

Node 6: FINDRETX <= 391.50000

Node 12: ERANKH <= 0.65269515

Node 24: STATE = "13", "15", "19", "22", "28", "32", "45"

Node 48: INTRDVXnonmiss-mean = 0.17022443

Node 24: STATE /= "13", "15", "19", "22", "28", "32", "45"

Node 49: INTRDVXnonmiss-mean = 0.61251648

Node 12: ERANKH > 0.65269515 or NA

Node 25: INTRDVXnonmiss-mean = 0.35154294

```

Node 6: FINDRETX > 391.50000 or NA
Node 13: INTRDVXnonmiss-mean = 0.67809123
Node 3: not (RETSURVX = NA & RETS_RVX = "A")
Node 7: RETSURVX = NA
Node 14: INTRDVXnonmiss-mean = 0.10477178
Node 7: RETSURVX != NA
Node 15: INTRDVXnonmiss-mean = 0.64648526

```

Figure 38 shows the propensity score tree for INTRDVX being nonmissing. The terminal nodes of the tree can serve as imputation cells for hot-deck imputation of the missing values in INTRDVX to obtain the values of $\hat{y}_i$ in equation (1). The estimated mean is highly random, because hot-deck itself is an intrinsically random process. As a result, it may be necessary to repeat the hot-deck sampling several times, as shown in the following R code.

```

z <- read.table("ce2021.txt",header=TRUE)
w <- z$FINLWT21
y <- z$INTRDVX
prop <- read.table("propen.fit",header=TRUE)
node <- prop$node
ntimes <- 100
est.mean <- 0
totwt <- 0
for(k in 1:ntimes){
  for(i in 1:nrow(z)){
    if(is.na(y[i])){
      responses <- node == node[i] & !is.na(y)
      if(sum(responses) > 0){
        y[i] <- sample(y[responses],1)
      }
    }
  }
  gp <- !is.na(y)
  est.mean <- est.mean+sum(w[gp]*y[gp])/sum(w[gp])
}
est.mean <- est.mean/ntimes
print(est.mean)

```

An alternative method to estimate a population mean is by inverse propensity weighting

$$\left(\sum_{i \in S_1} w_i / \hat{\pi}_i \right)^{-1} \sum_{i \in S_1} w_i y_i / \hat{\pi}_i$$

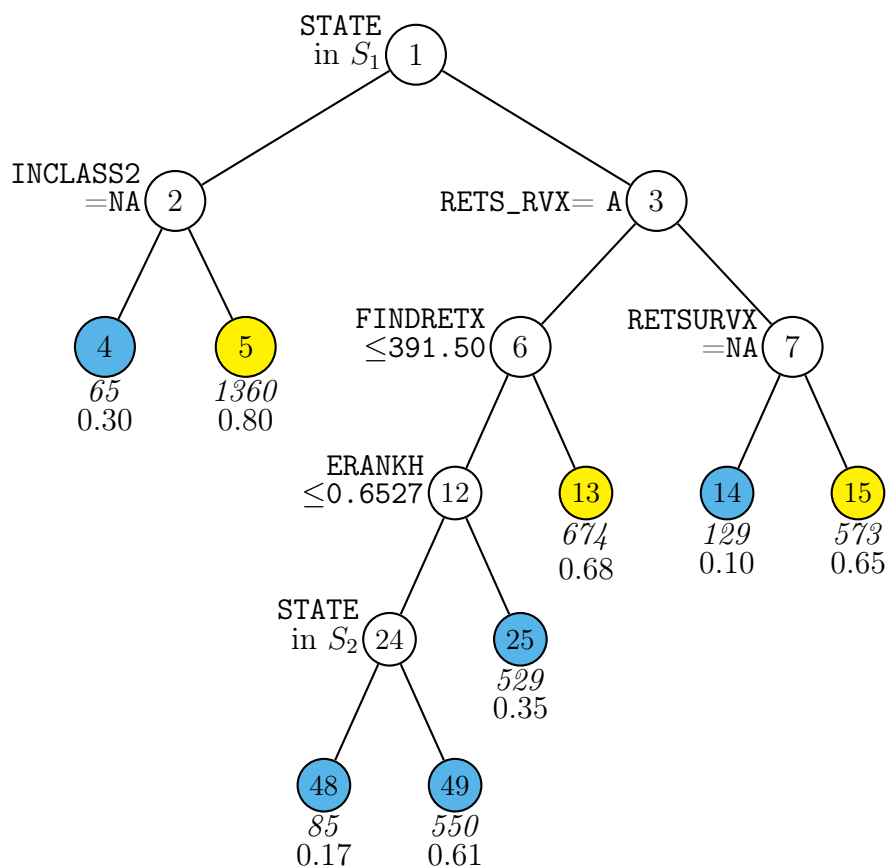


Figure 38: GUIDE v.46.1 0.250-SE tree for estimating propensity scores of INTRDVXnonmiss. At each split, an observation goes to the left branch if and only if the condition is satisfied. $S_1 = \{2, 6, 10, 11, 21, 24, 25, 27, 31, 40, 41, 47, 48, 49\}$. $S_2 = \{13, 15, 19, 22, 28, 32, 45\}$. Sample size (in *italics*) and estimated propensity score printed below nodes. Terminal nodes with propensity scores above and below value of 0.622 at root node are painted yellow and skyblue respectively. Second best split variable at root node is INCLASS2.

where S_1 is the set of observations where INTRDVX is nonmissing and $\hat{\pi}_i$ is its estimated propensity score. This method yields a nonrandomized estimate of 5246.455 as calculated by the following R code.

```
z <- read.table("ce2021.txt",header=TRUE)
w <- z$FINLWT21
y <- z$INTRDVX
prop <- read.table("propen.fit",header=TRUE)
score <- prop$predicted
gp <- !is.na(y)
ipw <- sum(w[gp]*y[gp]/score[gp])/sum(w[gp]/score[gp])
```


18 Differential item functioning: GDS data

GUIDE has an experimental option to identify important predictor variables and items with differential item functioning (DIF) in a data set with two or more item (dependent variable) scores. We illustrate it with a data set from [Broekman et al. \(2011, 2008\)](#) and [Marc et al. \(2008\)](#). It consists of responses from 1978 subjects on 15 items. There are 3 predictor variables (age, education, and gender). The data and DSC files are `GDS.dat` and `GDS.dsc`. Although the item responses in this example are 0-1, GUIDE allows them to be in any ordinal (e.g., Likert) scale. The contents of `GDS.dsc` are:

```
GDS.dat
NA
1
1 rid x
2 satis d
3 drop d
4 empty d
5 bored d
6 spirit d
7 afraid d
8 happy d
9 help d
10 home d
11 memory d
12 alive d
13 worth d
14 energy d
15 hope d
16 better d
17 total x
18 gender c
19 education n
20 age n
21 dxcurrent x
22 sumscore x
```

Here is the session log to create an input file for identifying DIF items and the important predictor variables:

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: dif.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1): 2
```

```
Name of batch output file: dif.out
Input 1 for classification, 2 for regression, 3 for propensity score tree
Input your choice ([1:3], <cr>=1): 2
Choose type of regression model:
  1=linear, 2=quantile, 3=Poisson, 4=censored response,
  5=multiresponse or itemresponse, 6=longitudinal data (with T variables).
Input choice ([1:6], <cr>=1): 5
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: GDS.dsc
Reading DSC file ...
Training sample file: GDS.dat
Missing value code: NA
Records in data file start on line 1
2 N variables changed to S
Number of D variables: 15
D variables are:
satis
drop
empty
bored
spirit
afraid
happy
help
home
memory
alive
worth
energy
hope
better
Multivariate or univariate split variable selection:
Choose multivariate if there is an order among the D variables;
choose univariate otherwise or if item response
Input 1 for multivariate, 2 for univariate ([1:2], <cr>=1): 2
D variables can be normalized to have unit variance,
e.g., if they have different scales or units
Input 1 to normalize D variables, 2 for no normalization ([1:2], <cr>=1): 2
Input 1 for equal, 2 for unequal weighting of D variables ([1:2], <cr>=1):
Reading data file ...
Number of records in data file: 1978
Length of longest entry in data file: 4
Checking for missing values ...
Finished checking
Missing values found in D variables
```

```

Assigning integer codes to values of 1 categorical variables
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Some D variables have missing values
Rereading data ...
PCA can be used for variable selection
Do not use PCA if differential item functioning (DIF) scores are wanted
Input 1 to use PCA, 2 otherwise ([1:2], <cr>=2):
#cases w/ miss. D = number of cases with all D values missing
      Total  #cases w/  #missing
      #cases  miss. D  ord. vals  #X-var  #N-var  #F-var  #S-var
      1978      0      0      4      0      0      2
      #P-var  #M-var  #B-var  #C-var  #I-var
      0      0      0      1      0
Number of cases used for training: 1977
Number of split variables: 3
Number of cases excluded due to 0 W or missing D variable: 1
Finished reading data file
Input 1 to save p-value matrix for differential item functioning (DIF), 2
otherwise ([1:2], <cr>=1):
Input file name to store DIF p-values: dif.pv
Default max. number of split levels: 4
Input 0=skip LaTeX tree, 1=tree without node numbers, 2=with node numbers ([0:2], <cr>=0):
You can store the variables and/or values used to split and fit in a file
Choose 1 to skip this step, 2 to store split and fit variables,
3 to store split variables and their values
Input your choice ([1:3], <cr>=1):
You can create a DSC file with the selected variables included or excluded
Input 2 to create such a file, 1 otherwise ([1:2], <cr>=1):
You can also output the importance scores and variable names to a file
Input 1 to create such a file, 2 otherwise ([1:2], <cr>=1):
Input file name: dif.scr
Input file is created!
Run GUIDE with the command: guide < dif.in

```

The importance scores are in the file `dif.scr`. They show that `age` is most important, followed by `gender` and `education`.

Rank	Score	Variable
1.00	8.89389E+00	age
2.00	5.05776E+00	gender
3.00	3.40934E+00	education

The word ‘yes’ in the last column of `dif.pv` below shows which item has DIF. In this example, only item #10 (`memory`) has DIF.

Item	Itemname	education	age	gender	DIF
1	satis	0.465E-01	0.377E-01	0.102E+00	no
2	drop	0.135E-01	0.218E+00	0.923E+00	no
3	empty	0.180E-02	0.122E+00	0.185E+00	no
4	bored	0.230E-05	0.218E+00	0.300E+00	no
5	spirit	0.973E+00	0.734E+00	0.362E-01	no
6	afraid	0.297E-01	0.411E-03	0.264E-02	no
7	happy	0.734E+00	0.353E+00	0.248E-01	no
8	help	0.397E-01	0.624E+00	0.435E-02	no
9	home	0.359E+00	0.114E+00	0.771E-03	no
10	memory	0.357E+00	0.000E+00	0.205E-01	yes
11	alive	0.172E+00	0.151E+00	0.444E+00	no
12	worth	0.405E+00	0.739E+00	0.698E+00	no
13	energy	0.661E+00	0.657E+00	0.117E-03	no
14	hope	0.607E+00	0.413E+00	0.216E+00	no
15	better	0.518E+00	0.644E+00	0.446E+00	no

19 Bootstrap confidence intervals

Owing to the numerous procedures that are performed during tree construction (such as selection of the variable and the split set to partition each intermediate node), proper statistical inference must account for the multiple testing and estimation issues. Otherwise, the error variance will be underestimated. Suppose, for example, we wish to obtain confidence intervals for the proportion of “RHC” in each terminal node of the tree in Figure ?? . Let n denote the sample size in a node and $\hat{p}$ the proportion of observations in it with the response value RHC. The usual $(1 - \alpha)$ binomial interval is then $\hat{p} \pm z_{1-\alpha/2} \sqrt{\hat{p}(1 - \hat{p})/n}$, where z_α is the α -quantile of the standard normal distribution. This formula yields intervals that are too short because it does not account for the extra variance due to model construction. Bonferroni corrections, which are traditionally used for multiple testing, are inapplicable here because the number of tests are not specified in advance. For example, the number of chi-squared tests at each node depends on the number of variables eligible to split the node and the number of levels of splits depends on the total sample size, extent of pruning, and other parameters such as the minimum sample size in each node.

As with the Bonferroni correction, a natural solution is to change the multiplier $z_{1-\alpha/2}$ to a larger value. The bootstrap method provides one simple solution. Called “bootstrap calibration”, the procedure is described and analyzed in [Loh \(1987, 1991\)](#) in the context of estimating a nonparametric mean; it is extended to subgroup analysis from regression tree models in [Loh et al. \(2016, 2019c\)](#) and [Loh and Zhou \(2020\)](#). The R code below implements the procedure. It can be used by following these steps:

1. Change the name of the data file (`rhcddata.txt` here) to `realdata.txt`.
2. Change the name of the DSC file (`rhcdsc1.txt` here) to `real.dsc`.
3. Change the name of the GUIDE input file (`classin.txt` here) to `real.in`.
4. Change the word “RHC” in line 1 of the R code to the name of the desired class in the data file.
5. In Windows, change the word “system” in lines 32, 32, 74 and 75 to “shell” if necessary.
6. Source the program in R.

```
1 class.name <- "RHC"  ## name of desired class in realdata.txt
2 nboot <- 1000
3 probs <- c(0.80,0.90,0.95,0.98)
4 zstat <- rep(0,nboot)
5 ### write bootstrap DSC file boot.dsc
6 file <- readLines("real.dsc")  ## read real DSC file
7 write("bootdata.txt",file="boot.dsc")
8 len <- length(file)
9 write(file[2:length(file)],"boot.dsc",append=TRUE)
10 write(paste(len-2,"w_w"),"boot.dsc",append=TRUE)
11 ### write bootstrap input file boot.in
12 file <- readLines("real.in")  ## read real input file
13 file2 <- gsub("real.", "boot.",file) ## replace "real." with "boot."
14 write(file2,"boot.in")
15 ### read real data
16 z0 <- read.table("realdata.txt",header=TRUE)
17 nob <- nrow(z0)
18 zt <- cbind(z0,rep(0,nob)) ## add column of weight 0
19 write("Bootstrap_simultaneous_intervals_by_linear_interpolation_of_z",
20       "results.txt")
21 write("trials_z80_z90_z95_z98_bias.err_sd.err",
22       "results.txt", append=TRUE)
23 err.test <- rep(0,nboot) ## misclassification rates
24 bias <- 0
25 for(i in 1:nboot){
26   zb <- z0[sample(nob,nob,replace=TRUE),]
27   zb <- cbind(zb,rep(1,nob)) ## add column of weight 1
28   write.table(zb,"bootdata.txt",col.names=TRUE,row.names=FALSE)
29   write.table(zt,"bootdata.txt",col.names=FALSE,row.names=FALSE,
30             append=TRUE)
31   system("rm_f_log.txt_boot.out_boot.fit")
32   system("guide_<_boot.in_>_log.txt")
33   bfit <- read.table("boot.fit",header=TRUE)  ## read boot results
34   test <- bfit$train == "n"
35   err.test[i] <- sum(bfit$observed[test] != bfit$predicted[test])/nob
36   err.resub <- sum(bfit$observed[!test] != bfit$predicted[!test])/nob
37   bias <- bias+(err.resub-err.test[i])
38   unodes <- unique(sort(bfit$node))
39   for(j in 1:length(unodes)){
40     gp <- bfit$node == unodes[j] & bfit$train == "y" ## training data
41     n0 <- sum(bfit$observed[gp] != class.name)
42     n1 <- sum(bfit$observed[gp] == class.name)
43     ntot <- n0+n1
44     estp <- n1/ntot
45     if(n1 == 0 | n0 == 0){
46       p <- (n1+0.5)/(ntot+1)
```

```

47         sd <- sqrt(p*(1-p)/(ntot+1))
48     } else {
49         sd <- sqrt(estp*(1-estp)/ntot)
50     }
51     gp <- bfit$node == unodes[j] & bfit$train == "n" ## real data
52     n0 <- sum(bfit$observed[gp] != class.name)
53     n1 <- sum(bfit$observed[gp] == class.name)
54     realp <- n1/(n0+n1)
55     zstat[i] <- max(zstat[i],abs(realp-estp)/sd)
56 }
57 if(i %% 100 == 0){
58     sd.err <- sqrt(var(err.test[1:i])) ## linear interpolation
59     q <- quantile(zstat[1:i],probs=probs,type=4)
60     write(c(i,q,bias/i,sd.err),"results.txt",append=TRUE,ncol=7)
61 }
62 }
63 ### find calibrated z.alpha
64 write(paste("No. of bootstraps=",nboot),"results.txt",append=TRUE)
65 write(c("Calibrated z at levels",probs),file="results.txt",ncol=5,
66       append=TRUE)
67 q <- quantile(zstat,probs=probs,type=4) ## linear interpolation
68 write(q,"results.txt",append=TRUE,ncol=4)
69 write(paste("Bootstrap estimate of bias of error rate=",bias/nboot),
70       "results.txt",append=TRUE)
71 write(paste("Bootstrap estimate of SD of error rate=",
72             sqrt(var(err.test))), "results.txt",append=TRUE)
73 ### fit real data
74 system("rm -f log.txt real.out real.fit")
75 system("guide < real.in > log.txt")
76 realfit <- read.table("real.fit",header=TRUE)
77 train <- realfit$train == "y"
78 err.obs <- sum(realfit$observed[train] != realfit$predicted[train])/nobs
79 write(paste("Real data observed error rate=",err.obs),"results.txt",
80       append=TRUE)
81 k <- 3 ## 95% level
82 z0 <- q[k] ## 95% z value
83 write(c("Simultaneous intervals at level",probs[k]),
84       file="results.txt",ncol=2,append=TRUE)
85 write(paste0("Node N P(",class.name,") halfwid left right"),
86       "results.txt", append=TRUE)
87 unodes <- unique(sort(realfit$node))
88 for(j in 1:length(unodes)){
89     gp <- realfit$node == unodes[j] & realfit$train == "y"
90     n0 <- sum(realfit$observed[gp] != class.name)
91     n1 <- sum(realfit$observed[gp] == class.name)
92     ntot <- n0+n1

```

```

93     if(n1 == 0 | n0 == 0){
94         p <- (n1+0.5)/(ntot+1)
95         sd <- sqrt(p*(1-p)/(ntot+1))
96     } else {
97         p <- n1/ntot
98         sd <- sqrt(p*(1-p)/(ntot))
99     }
100     p <- n1/ntot
101     halfwid <- z0*sd
102     left <- p-halfwid
103     right <- p+halfwid
104     write(c(unodes[j],ntot,p,halfwid,left,right),"results.txt",
105           append=TRUE,ncol=6)
106 }
107 ## write(sort(zstat),"zstat.txt",ncol=1) ## output sorted zstat values

```

Figure 39 gives the contents of the file `results.txt`. It shows that the calibrated z -multiplier is 3.961722, 4.325215, 4.690964, or 5.337637 for 80%, 90%, 95%, or 98% simultaneous confidence intervals. For 95% intervals, the left and right end points of the intervals in each terminal node are given in the bottom half of the file.

20 Tree ensembles

A tree ensemble is a collection of trees. GUIDE has two methods of constructing an ensemble.

GUIDE forest. This the preferred method. Similar to Random Forest (Breiman, 2001), it fits *unpruned* trees to bootstrap samples and randomly selects a small subset of variables to search for splits at each node. There are, however, two important differences:

1. GUIDE forest uses the unbiased GUIDE method for split selection; Random Forest uses the biased CART method. One consequence is that GUIDE forest can be very much faster than Random Forest if the dependent variable is a class variable having more than two distinct values and some categorical predictor variables have many categories.
2. GUIDE forest is applicable to data with missing values. The R implementation of Random Forest (Liaw and Wiener, 2002) requires apriori imputation of missing values in the predictor variables.

The default number of trees for GUIDE forest is 1000 if there are fewer than 500 training samples and 100 predictor variables; otherwise, the default is 500.


```

Bootstrap simultaneous intervals by linear interpolation of z
trials  z80    z90    z95    z98    bias.err    sd.err
100 4.036962 4.458809 4.545827 4.922293 -0.03357803 0.005906056
200 4.123996 4.508203 4.777955 5.035208 -0.03335222 0.005670584
300 4.093978 4.513735 4.918732 5.117146 -0.0335048 0.00598086
400 4.108083 4.519645 4.835633 5.28808 -0.03360811 0.005930667
500 4.108083 4.508203 4.826329 5.117146 -0.03377507 0.005887693
600 4.144132 4.548011 4.895352 5.408027 -0.03397879 0.005812075
700 4.123996 4.529434 4.889087 5.408027 -0.03377357 0.005839512
800 4.117319 4.51814 4.845685 5.365021 -0.03369159 0.00588305
900 4.108552 4.50332 4.835633 5.408027 -0.03358888 0.005924705
1000 4.108083 4.495735 4.845685 5.397256 -0.03353304 0.005951228
No. bootstraps = 1000
Calibrated z at levels 0.8 0.9 0.95 0.98
4.108083 4.495735 4.845685 5.397256
Bootstrap estimate of bias of error rate = -0.0335330427201395
Bootstrap estimate of SD of error rate = 0.00595122775778847
Real data observed error rate = 0.296251089799477
Simultaneous intervals at level 0.95
Node N P(RHC) halfwid left right
5 566 0.3816254 0.09894446 0.282681 0.4805699
7 2760 0.2355072 0.03913718 0.1963701 0.2746444
8 655 0.6961832 0.08707675 0.6091065 0.78326
18 244 0.6270492 0.1500158 0.4770334 0.7770649
19 218 0.3853211 0.1597212 0.2255999 0.5450423
25 66 0.3484848 0.2842088 0.06427609 0.6326936
26 110 0.6363636 0.2222518 0.4141119 0.8586154
27 601 0.3627288 0.09503228 0.2676965 0.4577611
48 438 0.6552511 0.1100458 0.5452053 0.7652969
49 77 0.3506494 0.2635033 0.08714608 0.6141526

```

Figure 39: Contents of `results.txt`

Bagged GUIDE. This fits *pruned* GUIDE trees to bootstrap samples of the training data (Breiman, 1996). Each tree is pruned by 5-fold cross-validation. The default number of trees is 200 if there are fewer than 500 training samples and 100 predictor variables; otherwise, the default is 100.

With the default settings, GUIDE forest is typically much faster than bagged GUIDE.

20.1 GUIDE forest: CE data

20.1.1 Input file creation

```
0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: gf.in
Input 1 for model fitting, 2 for importance or DIF scoring, 3 for data
conversion ([1:3], <cr>=1):
Name of batch output file: gf.out
Input 1 for single tree, 2 for ensemble ([1:2], <cr>=1): 2
Input 1 for bagging, 2 for rforest: ([1:2], <cr>=2):
Input 1 for classification, 2 for least-squares regression
Input your choice ([1:2], <cr>=1):
Input 1 for default options, 2 otherwise ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: ce2021class.dsc
Reading DSC file ...
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX_
Reading data file ...
Number of records in data file: 3965
Length of longest entry in data file: 11
Checking for missing values ...
Finished checking
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Recoding D values to integers
Finished recoding
Number of classes: 3
Finding number of levels of M variables associated with C variables ...
Assigning integer codes to values of 48 categorical variables
```

```

Finished assigning codes to 10 categorical variables
Finished assigning codes to 20 categorical variables
Finished assigning codes to 30 categorical variables
Finished assigning codes to 40 categorical variables
Associating missing values of N and S variables with M variable codes ...
Re-checking data ...
Allocating missing value information ...
Assigning codes to missing values, if any ...
Data checks complete
Creating missing value indicators ...
Rereading data ...
Warning: S variable DIRACC is constant
Warning: S variable TOTHVHRP is constant
Warning: S variable TOTHVHRC is constant
Warning: S variable ROTHFLC is constant
Warning: S variable WELFREBX is constant
Smallest positive weight: 1.0725E+03
Largest positive weight: 9.3902E+04
Class #Cases      Proportion
C      1478      0.37276166
D      2431      0.61311475
T        56      0.01412358
  Total #cases w/ #missing
  #cases miss. D ord. vals #X-var #N-var #F-var #S-var
    3965      0    3965      1      0      0    383
  #P-var #M-var #B-var #C-var #I-var
      0    116      0     48      0
Number of cases used for training: 3965
Number of split variables: 431
Number of cases excluded due to 0 W or missing D variable: 0
Finished reading data file
Warning: No linear splits; number of S variables must be < 225
Choose 1 for estimated priors, 2 for equal priors, 3 to input priors from a file
Input 1, 2, or 3 ([1:3], <cr>=1):
Choose 1 for unit misclassification costs, 2 to input costs from a file
Input 1 or 2 ([1:2], <cr>=1):
Warning: All positive weights treated as 1
Default max. number of split levels: 18
Input name of file to store predicted class and probability: gf.pro
Input rank of top variable to split root node ([1:431], <cr>=1):
Input file is created!
Run GUIDE with the command: guide < gf.in

```

20.1.2 Contents of gf.out

Note: Owing to the intrinsic randomness in forests, your results may differ from those shown below. “OOB” stands for “out-of-bag”.

```
Random forest of classification trees
No pruning
DSC file: ce2021class.dsc
Training sample file: ce2021.txt
Missing value code: NA
Records in data file start on line 2
Number of M variables associated with C variables: 19
383 N variables changed to S
D variable is INTRDVX_
Number of records in data file: 3965
Length of longest entry in data file: 11
Missing values found among categorical variables
Separate categories will be created for missing categorical variables
Missing values found among non-categorical variables
Number of classes: 3
Warning: S variable DIRACC is constant
Warning: S variable TOTHVHRP is constant
Warning: S variable TOTHVHRC is constant
Warning: S variable ROTHFLC is constant
Warning: S variable WELFREBX is constant
Smallest and largest positive weights are 1.0725E+03 and 9.3902E+04
Training sample class proportions of D variable INTRDVX_:
Class  #Cases      Proportion
C         1478      0.37276166
D         2431      0.61311475
T          56       0.01412358
```

```
Summary information for training sample of size 3965
d=dependent, b=split and fit cat variable using indicator variables,
c=split-only categorical, i=fit-only categorical (via indicators),
s=split-only numerical, n=split and fit numerical, f=fit-only numerical,
m=missing-value flag variable, p=periodic variable, w=weight
Levels of M variables are for missing values in associated variables
```

Column	Name		Minimum	Maximum	#Codes/ Levels/ Periods	#Missing
1	DIRACC	s	1.000	1.000		170
2	DIRACC_	m			2	
3	AGE_REF	s	18.00	87.00		
4	AGE_REF_	m			0	
5	AGE2	s	21.00	87.00		1734
6	AGE2_	m			1	

```

:
547 WHLFYR      c                      1      3964
548 WHLFYR_     m                      1
549 FFTAXOWE    s  -0.3368E+05  0.3997E+06
550 FSTAXOWE    s   -3309.      0.7223E+05

```

```

Total #cases w/ #missing
#cases miss. D ord. vals #X-var #N-var #F-var #S-var
3965      0      3965      1      0      0      383
#P-var #M-var #B-var #C-var #I-var
0      116      0      48      0

```

Number of cases used for training: 3965

Number of split variables: 431

Number of cases excluded due to 0 W or missing D variable: 0

Number of trees in ensemble: 500

Number of variables used for splitting: 21

Warning: No linear splits; number of S variables must be < 225

Simple node models

Estimated priors

Unit misclassification costs

Warning: All positive weights treated as 1

Univariate split highest priority

No interaction splits

No linear splits

Max number of splits on N and S variables: 1000

Maximum number of split levels: 18

Minimum node sample size: 2

Mean number of terminal nodes: 59.11

Classification matrix for training sample:

Predicted	True class		
class	C	D	T
C	639	31	4
D	839	2400	48
T	0	0	4
Total	1478	2431	56

Number of cases used for tree construction: 3965

Number misclassified: 922

Resubstitution estimate of mean misclassification cost: .2325

Number of OOB cases: 3965

Number OOB misclassified: 1130

OOB estimate of mean misclassification cost: .2850

Mean number of trees per OOB observation: 183.90

Predicted class probabilities are stored in `gf.pro`

Following are the top few rows of the file `gf.pro`, which give the estimated class posterior probabilities and the predicted and observed values of each case in the data.

train	"P(C)"	"P(D)"	"P(T)"	predicted	observed
y	0.59569E+00	0.39907E+00	0.52325E-02	"C"	"C"
y	0.23720E+00	0.75292E+00	0.98837E-02	"D"	"D"
y	0.26304E+00	0.73013E+00	0.68219E-02	"D"	"D"
y	0.31232E+00	0.67988E+00	0.78014E-02	"D"	"D"
y	0.41418E+00	0.57643E+00	0.93887E-02	"D"	"D"
y	0.37058E+00	0.46101E+00	0.16841E+00	"D"	"T"
y	0.35868E+00	0.62997E+00	0.11344E-01	"D"	"D"

20.2 Bagged GUIDE

This option uses an ensemble of **pruned** GUIDE trees. It often takes longer to execute and does not appear to produce more accurate results. It is made available for research purposes.

21 Other features

21.1 Pruning with test samples

GUIDE typically has three pruning options for deciding the size of the final tree: (i) cross-validation, (ii) test sample, and (iii) no pruning. Test-sample pruning is available only when there are no derived variables, such as creation of dummy indicator variables when ‘b’ variables are present. If test-sample pruning is chosen, the program will ask for the name of the file containing the test samples. This file must have the same column format as the training sample file. Pruning with test-samples or no pruning are non-default options.

21.2 Prediction of test samples

GUIDE can produce R code to predict future observations from all except kernel and nearest neighbor classification and ensemble models. This is also a non-default option.

Predictions of the training data for all models can be obtained, however, at the time of tree construction. This feature can be used to obtain predictions on “test samples” (i.e., observations that are not used in tree construction) by adding them to the training sample file. There are two ways to distinguish the test observations from the training observations:

1. Use a *weight* variable (designated as W in the DSC file) that takes value 1 for each training observation and 0 for each test observation.
2. Replace the D values of the test observations with the missing value code.

For tree construction, GUIDE does not use observations in the training sample file that have zero weight.

21.3 GUIDE in R and in simulations

GUIDE can be used in simulations or used repeatedly on bootstrap samples to produce an ensemble of tree models. For the latter,

1. Create a file (with name `data.txt`, say) containing one set of bootstrapped data.
2. Create a DSC file (with name `desc.txt`, say) that refers to `data.txt`.
3. Create an input file (with name `input.txt`, say) that refers to `desc.txt`.
4. Write a batch program (Windows) or a shell script (Linux or Macintosh) that repeatedly:
 - (a) replaces the file `data.txt` with new bootstrapped samples;
 - (b) calls GUIDE with the command: `guide < input.txt`; and
 - (c) reads and processes the results from each GUIDE run.

In R, the command in step 4b depends on the operating system. If the GUIDE program and the files `data.txt` and `input.txt` are in the same folder as the working R directory, the command is:

Linux/Macintosh: `system("guide < input.txt > log.txt")`

Windows: `shell("guide < input.txt > log.txt")`

If the files are not all in the same folder, full path names must be given. Here `log.txt` is a text file that stores messages during execution. If GUIDE does not run successfully, errors are also written to `log.txt`.

21.4 Generation of powers and products

GUIDE allows the creation of certain powers and products of regressor variables on the fly. Specifically, variables of the form $X_1^p X_2^q$, where X_1 and X_2 are numerical predictor variables and p and q are integers, can be created by adding one or more lines of the form

```
0 i p j q a
```

at the end of the DSC file. Here i and j are integers giving the column numbers of variables X_1 and X_2 , respectively, in the data file and a is one of the letters **n**, **s**, or **f** (corresponding to a numerical variable used for both splitting and fitting, splitting only, or fitting only).

To demonstrate, suppose we wish to fit a piecewise quadratic model in the variable `wtgain` in the birthweight data. This is easily done by adding one line to the file `birthwt.dsc`. First we assign the **s** (for splitting only) designator to every numerical predictor except `wtgain`. This will prevent all variables other than `wtgain` from acting as regressors in the piecewise quadratic models. To create the variable `wtgain2`, add the line

```
0 8 2 8 0 f
```

to the end of `birthwt.dsc`. The 8's in the above line refer to the column number of the variables `wtgain` in the data file, and the **f** tells the program to use the variable `wtgain2` for fitting terminal node models only. Note: The line defines `wtgain2` as `wtgain2 × wtgain0`. Since we can equivalently define the variable by `wtgain2 = wtgain1 × wtgain1`, we could also have used the line: “0 8 1 8 1 f”.

The resulting DSC file now looks like this:

```
birthwt.dat
NA
1
1 weight d
2 black c
3 married c
4 boy c
5 age s
6 smoke c
7 cigsper s
8 wtgain n
9 visit c
10 ed c
11 lowbwt x
0 8 2 8 0 f
```


When the program is given this DSC file, the output will show the regression coefficients of `wtgain` and `wtgain`² in each terminal node of the tree.

21.5 Data formatting functions

GUIDE has a utility function for reformatting data files into forms required by some old statistical software packages:

1. R/Splus: Fields are space delimited. Missing values are coded as `NA`. Each record is written on one line. Variable names are given on the first line.
2. SAS: Fields are space delimited. Missing values are coded with periods. Character strings are truncated to eight characters. Spaces within character strings are replaced with underscores (`_`).
3. TEXT: Fields are comma delimited. Empty fields denote missing values. Character strings longer than eight characters are truncated. Each record is written on one line. Variable names are given on the first line.
4. STATISTICA: Fields are comma delimited. Commas in character strings are stripped. Empty fields denote missing values. Each record occupies one line.
5. SYSTAT: Fields are comma delimited. Strings are truncated to eight characters. Missing character values are replaced with spaces, missing numerical values with periods. Each record occupies one line.
6. BMDP: Fields are space delimited. Categorical values are sorted in alphabetic order and then assigned integer codes. Missing values are indicated by asterisks. Variable names longer than eight characters are truncated.
7. DataDesk: Fields are space delimited. Missing categorical values are coded with question marks. Missing numerical values are coded with asterisks. Each record is written on one line. Spaces within categorical values are replaced with underscores. Variable names are given on the first line of the file.
8. MINITAB: Fields are space delimited. Categorical values are sorted in alphabetic order and then assigned integer codes. Missing values are coded with asterisks. Variable names longer than eight characters are truncated.
9. NUMBERS: Same as **TEXT** option except that categorical values are converted to integer codes.

10. C4.5: This is the format required by the C4.5 (Quinlan, 1993) program.
11. ARFF: This is the format required by the WEKA (Witten and Frank, 2000) programs.

Following is a sample session where the NHTSA comma-separated data are re-formatted to tab-delimited for R or Splus.

```

0. Read the warranty disclaimer
1. Create a GUIDE input file
Input your choice: 1
Name of batch input file: format.in
Input 1 for model fitting, 2 for importance or DIF scoring,
    3 for data conversion ([1:3], <cr>=1): 3
Name of batch output file: format.out
Input 1 if D variable is categorical, 2 if real ([1:2], <cr>=1):
Input name of DSC file (max 100 characters);
enclose with matching quotes if it has spaces: nhtsaiclass.dsc
nhtsaiclass.dsc
Reading DSC file ...
Training sample file: nhtsadata.csv
Missing value code: NA
Records in data file start on line 2
Warning: 48 N variables changed to S
Dependent variable is HIC2
Reading data file ...
Number of records in data file: 3310
Length of longest entry in data file: 19
Checking for missing values ...
Total number of cases: 3310
Number of classes: 2

Warning: "x" variables will be excluded
Choose one of the following data formats:
      Field  Miss.val.codes
No. Name  Separ  char.   numer. Remarks
-----
1  R/Splus  space  NA      NA      1 line/case, var names on 1st line
2  SAS      space  .       .       strings trunc., spaces -> '_'
3  TEXT     comma  empty   empty   1 line/case, var names on 1st line
4  STATISTICA comma  empty   empty   1 line/case, commas stripped
                                var names on 1st line
5  SYSTAT   comma  space   .       1 line/case, var names on 1st line
                                strings trunc. to 8 chars
6  BMDP     space          *       strings trunc. to 8 chars
                                cat values -> integers (alph. order)

```


A CE variables

Table 11: Some CE variables and their missing rates (if any)

Name	Definition	Missing
AGE_REF	Age of reference person	
AGE2	Age of spouse	0.44
ALCBEVCQ	Alcoholic beverages this quarter	
ALCBEVPQ	Alcoholic beverages last quarter	
ALLFULCQ	Fuel oil and other fuels this quarter	
ALLFULPQ	Fuel oil and other fuels last quarter	
APPARCQ	Apparel and services this quarter (MENBOYCQ + WOMGRLCQ + CHLDRNCQ + FOOTWRCQ + OTH-APLCQ)	
APPARPQ	Apparel and services last quarter (same composition as APPARCQ)	
AS_COMP1	Number of males age 16 and over in CU	
AS_COMP2	Number of females age 16 and over in CU	
AS_COMP3	Number of males age 2 through 15 in CU	
AS_COMP4	Number of females age 2 through 15 in CU	
AS_COMP5	Number of members under age 2 in CU	
BATHRMQ	Number of complete baths in this unit	
BBYDAYCQ	Babysitting and child day care this quarter	
BBYDAYPQ	Babysitting and child day care last quarter	
BEDROOMQ	Number of bedrooms in CU	0.01
BLS_URBN	Is this CU located in an urban or rural area? (1=urban, 2=rural)	
BUILDING	Which of these descriptions from the list best describes this building? (1–11)	
BUILT	Year property was built	0.23
BUSCREEN	Has household had business expenses that could be reimbursed? (1=yes, 2=no)	<0.01
CARTKNCQ	Cars and trucks, new (net outlay) this quarter	
CARTKNPQ	Cars and trucks, new (net outlay) last quarter	
CARTKUCQ	Cars and trucks, used (net outlay) this quarter	
CARTKUPQ	Cars and trucks, used (net outlay) last quarter	
CASHCOCQ	Cash contributions this quarter	
CASHCOPQ	Cash contributions last quarter	

CHILDAGE	Age of children of reference person (0=no children, 1=all children less than 6, 2=oldest child 6–11 and at least one child less than 6, 3=all children 6–11, 4=oldest child 12–17 and at least one child less than 12, 5=all children 12–17, 6=oldest child greater than 17 and at least one child less than 17, 7=all children greater than 17)	
CREDFINX	What was the total amount paid in finance, late charges, and interest for all cards in last month?	0.82
CREDITB	Could you tell me which range that best reflects the total amount owed on all major credit cards including store cards and gas cards? (1=0–499, 2=500–999, 3=1000–2499, 4=2500–9999, 5=10000–34999, 6=35K and over)	0.99
CREDITBX	Median bracket range of CREDITB	0.99
CREDITX	Total amount owed on all cards	0.81
CREDTYRX	Total amount owed on all cards one year ago today	0.90
CREDYR	Did you have any credit cards including store cards and gas cards one year ago today? (1=yes, 2=no)	0.99
CREDYRB	Range that best reflects the total amount owed on all major credit cards including store cards and gas cards one year ago today (1=0–499, 2=500–999, 3=1000–2499, 4=2500–9999, 5=10000–34999, 6=35K and over)	0.99
CREDYRBX	Median bracket range of CREDYRB	0.99
CUTENURE	Housing tenure (1=homeowner with mortgage, 2=homeowner without mortgage, 3=homeowner, mortgage not reported, 4=rented, 5=occupied without payment of rent, 6=student housing)	
DEFBENRP	Do you have a defined retirement plan, such as a pension, from an employer? (1=yes, 2=no)	0.77
DIRACC	Is access to the quarters direct or through another unit? (1=direct, 2=another)	0.04
DIVISION	Census division (1=New England, 2=Middle Atlantic, 3=East North Central, 4=West North Central, 5=South Atlantic, 6=East South Central, 7=West South Central, 8=Mountain, 9=Pacific)	0.07
DOMSRVCQ	Domestic services this quarter	
DMSXCCCQ	Domestic services excluding child care this quarter	
DMSXCCPQ	Domestic services excluding child care last quarter	
DOMSRVPQ	Domestic services last quarter	

EARNCOMP	Composition of earners (1=reference person only, 2=reference person and spouse, 3=reference person, spouse and others, 4=reference person and others, 5=spouse only, 6=spouse and others, 7=others, 8=no earners)	
ECARTKNC	Outlays for new vehicle purchases this quarter including down payment, principal and interest paid on loans, or if not financed, purchase amount	
ECARTKNP	Outlays for new vehicle purchases last quarter including down payment, principal and interest paid on loans, or if not financed, purchase amount	
ECARTKUC	Outlays for used vehicle purchases this quarter including down payment, principal and interest paid on loans, or if not financed, purchase amount	
ECARTKUP	Outlays for used vehicle purchases last quarter including down payment, principal and interest paid on loans, or if not financed, purchase amount	
EDUC_REF	Education of reference person (10=grades 1–8; 11=grades 9–12, no degree; 12=high school graduate; 13=some college, no degree; 14=Associate's degree in college; Bachelors degree; 16=Masters degree or professional/doctorate degree)	
EDUCA2	Education level of spouse (same levels as EDUC_REF)	0.44
EDUCACQ	Education this quarter	
EDUCAPQ	Education last quarter	
EENTMSCC	Miscellaneous entertainment outlays this quarter including photographic and sports equipment and boat and RV rentals	
EENTMSCP	Miscellaneous entertainment outlays last quarter including photographic and sports equipment and boat and RV rentals	
EENTRMTC	Total entertainment outlays this quarter including sound systems, sports equipment, toys, cameras, and down payments on boats and campers (FEEADMCQ + TVRDIOCQ + PETTOYCQ + EOTHENTC)	
EENTRMTP	Total entertainment outlays last quarter including sound systems, sports equipment, toys, cameras, and down payments on boats and campers (same composition as EENTRMTC)	

EHOUSNGC	Total housing outlays this quarter including maintenance, fuels, public services, household operations, house furnishings, and mortgage (lump sum home equity loan or line of credit home equity loan) principle and interest ($\text{ESHELTRC} + \text{UTILCQ} + \text{HOUSOPCQ} + \text{HOUSEQCQ}$)
EHOUSNGP	Total housing outlays last quarter including maintenance, fuels, public services, household operations, house furnishings, and mortgage (lump sum home equity loan or line of credit home equity loan) principle and interest (same composition as EHOUSNGC)
ELCTRCCQ	Electricity this quarter
ELTRCPQ	Electricity last quarter
EMISCELC	Miscellaneous outlays this quarter including reduction of mortgage principal (lump sum home equity loan) on other property ($\text{MISCPQ} + \text{EMISCMTP}$)
EMISCELP	Miscellaneous outlays last quarter including reduction of mortgage principal (lump sum home equity loan) on other property (same composition as EMISCELC)
EMISCMTC	Mortgage principal outlays this quarter for other property
EMISCMTP	Mortgage principal outlays last quarter for other property
EMRTPNOC	Mortgage principal outlays this quarter for owned home
EMRTPNOP	Mortgage principal outlays last quarter for owned home
EMRTPNVC	Mortgage principal outlays this quarter for owned vacation home
EMRTPNVP	Mortgage principal outlays last quarter for owned vacation home
EMOTRVHC	Outlays for motored recreational vehicles this quarter
EMOTRVHP	Outlays for motored recreational vehicles last quarter
ENOMOTRC	Outlays for non-motored recreational vehicles this quarter
ENOMOTRP	Outlays for non-motored recreational vehicles last quarter
ENTERTCQ	Entertainment this quarter ($\text{FEEADMCCQ} + \text{TVRDIOCQ} + \text{OTHEQPCQ}$)

ENTERTPQ	Entertainment last quarter (same composition as ENTERTCQ)	
EOTHENTC	Outlays for other entertainment supplies this quarter, equipment, and services including down payments on boats and campers (ENOMOTRC + EMOTRVHC + EENTMSCC)	
EOTHENTP	Outlays for other entertainment supplies last quarter, equipment, and services including down payments on boats and campers (same composition as EOTHENTC)	
EOTHLODC	Outlays for other lodging this quarter such as owned vacation home, including mortgage principal and interest, property taxes, maintenance, insurance, and other expenses (OTHLODCQ + EMRTPNVC)	
EOTHLODP	Outlays for other lodging last quarter such as owned vacation home, including mortgage principal and interest, property taxes, maintenance, insurance, and other expenses (same composition as EOTHLODC)	
EOTHVEHP	Outlays for other vehicle purchases last quarter such as motorcycles and airplanes including down payment, principal and interest paid on loans, or if not financed, purchase amount	
EOTHVEHC	Outlays for other vehicle purchases this quarter such as motorcycles and airplanes including down payment, principal and interest paid on loans, or if not financed, purchase amount	
EOTHVEHP	Outlays for other vehicle purchases last quarter such as motorcycles and airplanes including down payment, principal and interest paid on loans, or if not financed, purchase amount	
EOWNDWLC	Owned home outlays this quarter including mortgage principal and interest, property taxes, maintenance, insurance, and other expenses (OWNDWECQ + EMRTPNOC)	
EOWNDWLP	Owned home outlays last quarter including mortgage principal and interest, property taxes, maintenance, insurance, and other expenses (same composition as EOWNDWLC)	
ERANKH	Percent expenditure outlay rank	0.08

ERANKHM	Weighted cumulative percent expenditure outlay ranking of CU to total population
ESHELTRC	Shelter outlays this quarter including mortgage principle and interest for owned home and/or vacation home, rents, insurance, taxes, and maintenance (EOWNDWLC + RENDWECQ + EOTHLODC)
ESHELTRP	Shelter outlays last quarter including mortgage principle and interest for owned home and/or vacation home, rents, insurance, taxes, and maintenance (same composition as ESHELTRC)
ETOTALC	Total outlays this quarter, sum of outlays from all major expenditure categories (FOODCQ + AL-CBEVCQ + EHOUSNGC + APPARCQ + ETRANPTC + HEALTHCQ + EENTRMTC + PERSCACQ + READCQ + EDUCACQ + TOBACCCQ + EMISCELC + CASHCOCQ + PERINSCQ)
ETOTALP	Total outlays last quarter, sum of outlays from all major expenditure categories (same composition as ETOTALC)
ETOTACX4	Adjusted total outlays this quarter, sum of outlays from all major expenditure categories (FOODCQ + AL-CBEVCQ + EHOUSNGC + APPARCQ + ETRANPTC + HEALTHCQ + EENTRMTC + PERSCACQ + READCQ + EDUCACQ + TOBACCCQ + MISC1CQ + 4×MISC2CQ + EMISCMTC + PERINSCQ)
ETOTAPX4	Adjusted total outlays last quarter, sum of outlays from all major expenditure categories (same composition as ETOTACX4)
ETRANPTC	Total outlays for transportation this quarter including down payment, principal and finance charges paid on loans, gasoline and motor oil, maintenance and repairs, insurance, public and other transportation, and vehicle rental licenses and other charges (EVEHPURC + GAS-MOCQ + MAINRPCQ + VEHINSCQ + VRNTLOCQ + PUBTRACQ)

ETRAPNTP	Total outlays for transportation last quarter including down payment, principal and finance charges paid on loans, gasoline and motor oil, maintenance and repairs, insurance, public and other transportation, and vehicle rental licenses and other charges (same composition as ETRAPNTPC)
EVEHPURC	Outlays for vehicle purchases this quarter including down payment, principal and interest paid on loans, or if not financed, purchase amount (ECARTKNC + ECARTKUC + EOTHVEHC)
EVEHPURP	Outlays for vehicle purchases last quarter including down payment, principal and interest paid on loans, or if not financed, purchase amount (same composition as EVEHPURC)
FAM_SIZE	Number of Members in CU
FAM_TYPE	Family type (1-9)
FDAWAYCQ	Food away from home this quarter
FDAWAYPQ	Food away from home last quarter
FDHOMEQC	Food at home this quarter
FDHOMEPC	Food at home last quarter
FDMAPCQ	Meals as pay this quarter
FDMAPPQ	Meals as pay last quarter
FDXMAPCQ	Food away excluding meals as pay this quarter
FDXMAPPC	Food away excluding meals as pay last quarter
FEEADMCC	Fees and admissions this quarter
FEEADMPC	Fees and admissions last quarter
FFTAXOWE	Weighted estimate for Federal tax liabilities for entire CU
FGOVRETM	Amount of government retirement deducted from last pay, annualized for all CU members
FGOVRETX	Amount of government retirement deducted from last pay annualized
FINCBTAX	Total family income before taxes in last 12 months (INTRDVX + INTRDVBX + ROYESTX + ROYESTBX + OTHREGX + OTHREGBX + WELFAREX + WELFREBX + RETSURVX + RETSRVBX + NETRENTX + NETRNTBX + OTHRINCX)
FINDRETX	Money placed in self-employed retirement plan in past year for all CU members

FINLWT21	Sampling weight	
FJSSDEDX	Estimated amount contributed to Social Security by all CU members past 12 mos.	
FLRCVRCQ	Floor coverings this quarter	
FLRCVRPQ	Floor coverings last quarter	
FMLPYRX	Annual value of free meals received as part of pay	0.99
FOODCQ	Total food this quarter	
FOODPQ	Total food last quarter	
FPRIPENM	Amount of private pensions deducted from last pay, annualized, for all CU members	
FPRIPENX	Amount of private pensions	
FRRDEDM	Amount of Railroad Retirement deducted from last pay, annualized for all CU members	
FRRDEDX	Amount of railroad retirement deducted from last pay annualized	
FRRETIRM	Amount of social security and railroad retirement income, prior to deductions for medical insurance and Medicare, received by all CU members in the past 12 months	
FRRETIRX	Social security and railroad retirement income	
FS_MTHI	In how many of the last 12 months were food stamps or EBTs received?	0.98
FSALARYX	Wage and salary income of all members past 12 mos.	
FSMPFRMX	Family level summation for new variable SEMPFRMX and SMPFRMBX	
FSSIX	Amount supplemental security income from all sources received by all CU members in past 12 months	
FSTAXOWE	Weighted estimate for State tax liabilities for entire CU	
FURNTRCQ	Furniture this quarter	
FURNTRPQ	Furniture last quarter	
GASMOCQ	Gasoline and motor oil this quarter	
GASMOPQ	Gasoline and motor oil last quarter	
FULOILCQ	Fuel oil this quarter	
FULOILPQ	Fuel oil last quarter	
FURNTRCQ	Furniture this quarter	
FURNTRPQ	Furniture last quarter	
HEALTHCQ	Health care this quarter (HLTHINCQ + MEDSRVCQ + PREDRGCQ + MEDSUPCQ)	

HEALTHPQ	Health care last quarter (same composition as HEALTHCQ)	
HIGH_EDU	Highest level of education within the CU (0=never attended, 10=1-8 grade, 11=9-12 grade, 12=HS grad, 13=some college, 14=AA degree, 15=Bachelors, 16=Masters/professional/doctorate)	
HISP_REF	Hispanic origin of reference person (1=Hispanic, 2=non-Hispanic)	
HISP2	Hispanic origin of spouse (1=Hispanic, 2=non-Hispanic)	
HH_CU_Q	Count of CUs in household	
HLFBATHQ	How many half bathrooms are there in this unit?	0.01
HLTHINCQ	Health insurance this quarter	
HLTHINPQ	Health insurance last quarter	
HORREF1	Hispanic origin of reference person (1=Mexican, 2=Mexican-American, 3=Chicano, 4=Puerto Rican, 5=Cuban, 6=Other)	0.96
HORREF2	Hispanic origin of spouse (same codes as HORREF1)	0.98
HOUSCQ	Housing this quarter	
HOUSEQCQ	House furnishings and equipment this quarter (TEXTILCQ + FURNTRCQ + FLRCVRCQ + MAJAPPCQ + SMLAPPCQ + MISCEQCQ)	
HOUSEQPQ	House furnishings and equipment last quarter (same composition as HOUSEQCQ)	
HOUSOPCQ	Household operations this quarter	
HOUSPQ	Housing last quarter	
HOUSOPPQ	Household operations last quarter	
INC_HRS1	Number hours worked per week by reference person	0.38
INC_HRS2	Number hours worked per week by spouse	0.66
INC_RANK	Income rank of CU to total population	
INCLASS2	Income class based on INC_RANK (1=0-0.1667, 2=0.1667-0.3333, 3=0.3334-0.4999, 4=0.5000-0.6666, 5=0.6667-0.8333, 6=0.8334-1, 7=incomplete reporting)	
INCNONW1	Reason for not working during past 12 months (1=retired, 2=take care of home, 3=going to school, 4=ill, disabled, unable to work, 5=unable to find work, 6=doing something else)	0.62
INCNONW2	Reason spouse did not work during past 12 months (same codes as INCNONW1)	0.78

INCOMEY1	Employer paying most earnings in past 12 months (1=private company, business or individual, 2=Federal govt, 3=State govt, 4=local govt, 5=self-employed, 6=family business or farm, working without pay)	0.38
INCOMEY2	Employer from which spouse received most earnings during the past 12 months	0.66
INCWEEK1	Weeks worked full or part time in last 12 months	
INCWEEK2	Weeks worked by spouse full or part time last 12 months	0.44
INTRDVX	Amount received in interest or dividend during past 12 mos.	0.37
IRA	Do you have any retirement accounts such as 401(k)s, IRAs, thrift saving plans? (1=yes, 2=no)	0.76
IRAB	Range that best reflects the total value of all retirement accounts such as 401(k)s, IRAs, and thrift savings plans (1=0-1999, 2=2000-9999, 3=10K-49999, 4=50K-199999, 5=200K-449999, 6=450K or more)	0.97
IRAX	Total amount put into retirement accounts past 12 mos.	0.87
IRAYRB	Range which best reflects the total value of all retirement accounts one year ago today (same codes as IRAB)	0.96
IRAYRBX	Median value of bracket range for IRAYRB	0.96
IRAYRX	Total value of retirement accounts one year ago	0.88
JFS_AMT	Annual value of food stamps	
LIFINSCQ	Life and other personal insurance this quarter	
LIFINSPQ	Life and other personal insurance last quarter	
LIQDYRBX	Median value of bracket range for LIQUDYRB	0.96
LIQUDYR	Did you have any checking savings money market accounts, or CDs one year ago? (1=yes, 2=no)	>0.99
LIQUID	Do you have any checking, saving, money market accounts, or CDs? (1=yes, 2=no)	0.76
LIQUIDB	Range that best reflects total value of checking, savings, money market accounts, CDs (1=0-499, 2=500-999, 3=1000-2499, 4=2.5K-9999, 5=10K-34999, 6=35K and over)	0.97
LIQUIDBX	Median value of bracket range LIQUIDB	0.97
LIQUIDX	Total value of all checking, savings, money market, and CD accounts	0.83

LIQUDYRB	Range that best reflects the total value of all checking, savings, money market accounts, and CDs one year ago today (same codes as LIQUIDB)	0.97
LIQUDYRX	Total value of all checking, savings, money market accounts, and CDs one year ago today	0.84
MAINRPCQ	Maintenance and repairs this quarter	
MAINRPPQ	Maintenance and repairs last quarter	
MAJAPPCQ	Major appliances this quarter	
MAJAPPPQ	Major appliances last quarter	
MARITAL1	Marital status of reference person (1=married, 2=widowed, 3=divorced, 4=separated, 5=never married)	
MEALSPAY	Have you received any free meals at work as part of your pay? (1=yes, 2=no)	<0.01
MEDSRVCQ	Medical services this quarter	
MEDSRVPQ	Medical services last quarter	
MEDSUPCQ	Medical supplies this quarter	
MEDSUPPQ	Medical supplies last quarter	
MENBOYCQ	Clothing for men and boys this quarter	
MENSIXCQ	Clothing for men, 16 and over this quarter	
MENSIXPQ	Clothing for men, 16 and over last quarter	
MENBOYPQ	Clothing for men and boys last quarter	
MISC1CQ	Miscellaneous expenditures this quarter	
MISC1PQ	Miscellaneous expenditures last quarter	
MISCEQCQ	Miscellaneous household equipment this quarter	
MISCEQPQ	Miscellaneous household equipment last quarter	
MISCCQ	Miscellaneous expenditures this quarter (MISC1CQ + MISC2CQ)	
MISCPQ	Miscellaneous expenditures last quarter (same composition as MISCCQ)	
MISCTAXX	During past 12 months, what was total amount paid for personal property taxes and other taxes not reported elsewhere by all CU members?	0.99
MISCX4CQ	Adjusted miscellaneous expenditures this quarter (MISC1CQ + 4×MISC2CQ)	
MISCX4PQ	Adjusted miscellaneous expenditures last quarter (same composition as MISCX4CQ)	
MLPAYWKX	About what was the weekly dollar value of these meals?	0.99

MLPYQWKS	For how many weeks did members of your household receive these meals during the past 12 months?	0.99
MRPINSQC	Maintenance, repairs, insurance, and other expenses this quarter	
MRPINSPQ	Maintenance, repairs, insurance, and other expenses last quarter	
MRTINTCQ	Mortgage interest this quarter	
MRTINTPQ	Mortgage interest last quarter	
MRTPRNOC	Outlays on owned vacation home mortgage principle this quarter	
MRTPRNOP	Outlays on owned vacation home mortgage principle last quarter	
NETRENTB	Range that best reflects the total net rental income or loss during the past 12 months (1=0–999, 2=1–2K, 3=2–3K, 4=3–4K, 5=4–5K, 6=5–10K, 7=10–15K, 8=15–20K, 9=20–30K, 10=30–40K, 11=40–50K, 12=50K and over)	0.99
NETRENTX	What was the amount of net rental income or loss?	0.92
NETRNTBX	Median value of bracket range of NETRENTB	0.99
NTLGASCQ	Natural gas this quarter	
NTLGASPQ	Natural gas last quarter	
NO_EARNR	Number of earners	
NONINCMX	Amount of other money receipts excluded from CU income before taxes received by CU in past 12 months	
NUM_AUTO	Total number of owned cars	
NUM_TVAN	Total number of owned trucks and vans	
OCCUCOD1	Highest paid occupation last 12 months (15 coded values)	0.38
OCCUCOD1	Job in which reference person received most earnings during past 12 months (15 coded values)	0.66
OCCUCOD2	Job in which spouse received most earnings during past 12 months (15 coded values)	0.66
OTHAPLCQ	Other apparel products and services this quarter	
OTHAPLPQ	Other apparel products and services last quarter	
OTHAStB	Range which best reflects the total value of these other financial assets (1=0–2K, 2=2–10K, 3=10–50K, 4=50–200K, 5=200–450K, 6=450K and over)	>0.99
OTHAStBX	Median value of bracket range for OTHAStB	>0.99
OTHAStX	Total value of these other financial assets as of today	0.99
OTHENTCQ	Other entertainment this quarter	

OTHENTPQ	Other entertainment last quarter	
OTHEQPCQ	Other equipment and services this quarter (PETTOYCQ + OTHENTCQ)	
OTHEQPPQ	Other equipment and services last quarter (same composition as OTHEQPCQ)	
OTHFINX	Total amount paid in finance, late charges, and interest for all other loans in the last month	0.99
OTHFLSCQ	Other fuels this quarter	
OTHFLSPQ	Other fuels last quarter	
OTHHEXCQ	Other household expenses this quarter	
OTHHEXPQ	Other household expenses last quarter	
OTHLNYP	Did you have any other debt such as medical loans or personal loans one year ago today? (1=yes, 2=no)	>0.99
OTHLNYPB	Range which best reflects the total amount owed on all other loans one year ago today (1=0-499, 2=500-999, 3=1-2.5K, 4=2.5-10K, 5=10-35K, 6=35K and over)	>0.99
OTHLODCQ	Other lodging this quarter	
OTHLODPQ	Other lodging last quarter	
OTHLONX	Total amount owed on all other loans	0.99
OTHLYRBX	Median value of bracket range for OTHLONBX	>0.99
OTHREGB	Range best reflects total amount received in Veteran's Administration (VA) payments, unemployment compensation, child support, or alimony during the past 12 months (1=0-1K, 2=1-2K, 3=2-3K, 4=3-4K, 5=4-5K, 6=5-10K, 7=10-15K, 8=15-20K, 9=20-30K, 10=30-40K, 11=40-50K, 12=50K and over)	0.99
OTHREGBX	Median value of bracket range for OTHREGB	0.99
OTHRINX	Amount received in other income including money from care of foster children, cash scholarships and fellowships, or stipends not based on working	0.97
OTHREGX	Income on a regular basis from any other source such as Veteran's Administration (VA) payments, unemployment compensation, child support, or alimony	0.92
OTHSTYRB	Range which best reflects total value of these other financial assets one year ago today (1=0-2K, 2=2-10K, 3=10-50K, 4=50-200K, 5=200-450K, 6=450K and over)	>0.99
OTHSTYRX	Value of these other financial assets one year ago today	0.99
OTHSYRBX	Median value of bracket range for OTHSTYRB	>0.99

OTHVEHCQ	Other vehicles this quarter	
OTHVEHPQ	Other vehicles last quarter	
OWNDWECQ	Owned dwellings this quarter (MRTINTCQ + PROP- TXCQ + MRPINSCQ)	
OWNDWEPQ	Owned dwellings last quarter (same composition as OWNDWECQ)	
OWNVACC	Expenditures on owned vacation homes this quar- ter including mortgage interest, insurance, taxes, maintenance, and miscellaneous household equipment (VOTHRLOC + VMISCHEC)	
OWNVACP	Expenditures on owned vacation homes last quarter including mortgage interest, insurance, taxes, mainte- nance, and miscellaneous household equipment (same composition as OWNVACC)	
PERINSCQ	Personal insurance and pensions this quarter (LIFINSCQ + RETPENECQ)	
PERINSPQ	Personal insurance and pensions last quarter (same com- position as PERINSCQ)	
PERSCACQ	Personal care this quarter	
PERSCAPQ	Personal care last quarter	
PERSLT18	Number of CU members less than 18	
PERSOT64	Number of CU members over 64	
PETTOYCQ	Pets, toys, and playground equipment this quarter	
PETTOYPQ	Pets, toys, and playground equipment last quarter	
POPSIZE	Population size of the PSU (1=more than 5M, 2=1–5M, 3=0.5–1M, 4=100–500K, 5=less than 100K)	
PREDRGCQ	Prescription drugs this quarter	
PREDRGPQ	Prescription drugs last quarter	
PRINEARN	Member number of principal earner (5 coded values)	
PROPTXCQ	Property taxes this quarter	
PROPTXPQ	Property taxes last quarter	
PSU	Primary sampling unit	0.52
PUBTRACQ	Public and other transportation this quarter (TRNTR- PCQ + TRNOTHCQ)	
PUBTRAPQ	Public and other transportation last quarter (same com- position as PUBTRACQ)	
RACE2	Race of spouse (same codes as REF_RACE)	0.44
READCQ	Reading this quarter	

READPQ	Reading last quarter	
REF_RACE	Race of reference person (1=white, 2=black, 3=native American, 4=Asian, 5=Pacific islander, 6=multi-race)	
REFGEN	Generation of reference person (1=Greatest/Silent: born 1945 or earlier, 3=Baby boomers: 1946–64, 4=Gen X: 1965–80, 5=Millennials: 1981 or later)	
REGION	Region (1=Northeast, 2=Midwest, 3=South, 4=West)	0.01
RELECTRC	Expenditures on electricity for rented vacation homes this quarter	
RELECTRP	Expenditures on electricity for rented vacation homes last quarter	
RENDWECQ	Rented dwelling this quarter (RNTXRPCQ + RNTAPYCQ)	
RENDWEPQ	Rented dwelling last quarter (same composition as RENDWECQ)	
RENTEQVX	Monthly rent if home rented today	0.20
RETPENCQ	Retirement, pensions, social security this quarter	
RETPENPQ	Retirement, pensions, social security last quarter	
RETSRVBX	Median value of bracket range for RETSURVB	0.99
RETSURV	Did you receive income from retirement, survivor, or disability pensions during past 12 months? (1=yes, 2=no)	
RETSURVX	Retirement, survivor, disability pensions received past 12 mos.	0.78
RNATLGAC	Expenditures on natural gas for rented vacation homes this quarter	
RNATLGAP	Expenditures on natural gas for rented vacation homes last quarter	
RNTAPYCQ	Rent as pay this quarter	
RNTAPYPQ	Rent as pay last quarter	
RNTXRPCQ	Rent excluding rent as pay this quarter	
RNTXRPPQ	Rent excluding rent as pay last quarter	
ROOMSQ	Number of rooms in CU living quarters, including finished living areas, excluding all baths	0.01
ROTHRFLC	Expenditures on other fuels for rented vacation homes this quarter	

ROYESTB	Range that best reflects total amount received in royalty income or income from estates and trusts during past 12 months (1=0–1K, 2=1–2K, 3=2–3K, 4=3–4K, 5=4–5K, 6=5–10K, 7=10–15K, 8=15–20K, 9=20–30K, 10=30–40K, 11=40–50K, 12=50K and over)	>0.99
ROYESTBX	Median value of bracket range for ROYESTB	>0.99
ROYESTX	Amount received in royalty income or income from estates and trusts	0.96
RWATERPC	Expenditures on water and public services for rented vacation homes this quarter	
RWATERPP	Expenditures on water and public services for rented vacation homes last quarter	
SEX_REF	Sex of reference person (1=male, 2=female)	
SEX2	Sex of spouse (1=male, 2=female)	0.44
SHELTCQ	Shelter this quarter (OWNDWECQ + RENDWECQ + OTHLODCQ)	
SHELTPQ	Shelter last quarter (same composition as SHELTCQ)	
SMLAPPCQ	Small appliances, miscellaneous housewares this quarter	
SMLAPPPQ	Small appliances, miscellaneous housewares last quarter	
SMSASTAT	Does CU reside inside a Metropolitan Statistical Area (MSA)? (1=yes, 2=no)	
ST_HOUS	Are these living quarters presently used as student housing by a college or university? (1=yes, 2=no)	
STATE	1=AL, 2=AK, 4=AZ, 5=AR, 6=CA, 8=CO, 9=CT, 10=DE, 11=DC, 12=FL, 13=GA, 15=HI, 16=ID, 17=IL, 18=IN, 19=IA, 20=KS, 21=KY, 22=LA, 23=ME, 24=MD, 25=MA, 26=MI, 27=MN, 28=MS, 29=MO, 30=MT, 31=NE, 32=NV, 33=NH, 34=NJ, 36=NY, 37=NC, 39=OH, 40=OK, 41=OR, 42=PA, 44=RI, 45=SC, 46=SD, 47=TN, 48=TX, 49=UT, 51=VA, 53=WA, 54=WV, 55=WI	0.08
STCKYRBX	Median value of bracket range for STOCKYRB	0.98
STDNTYR	Did you have student loans one years ago today? (1=yes, 2=no)	>0.99
STDNTYRB	Range which best reflects the total amount owed on all student loans one year ago today (1=0–499, 2=500–999, 3=1–2.5K, 4=2.5–10K, 5=10–35K, 6=35K and over)	>0.99

STDNTYRX	Total amount owed on all student loans one year ago today	0.97
STDYRNBX	Median value of bracket range for STDNTYRB	>0.99
STOCKB	Range which best reflects total value of all directly-held stocks, bonds, and mutual funds (1=0–2K, 2=2–10K, 3=10–50K, 4=50–200K, 5=200–450K, 6=450K and over)	0.99
STOCKBX	Median value of bracket range for STOCKB	0.99
STOCKX	Value of directly-held stocks, bonds, mutual funds (median=59,950, mean=411,867)	0.93
STOCKYR	Did you have any directly-held stocks, bonds, or mutual funds one year ago? (1=yes, 2=no)	>0.99
STOCKYRB	Range which best reflects total value of all directly-held stocks, bonds, and mutual funds one year ago today (same codes as STOCKB)	0.98
STOCKYRX	Median value of bracket range of STOCKX	0.93
STUDFINX	Total amount paid in finance, late charges, and interest for all student loans in the last month	0.97
STUDNTB	Range which best reflects the total amount owed on all student loans (1=0–499, 2=500–999, 3=1–2.5K, 4=2.5–10K, 5=10–35K, 6=35K and over)	>0.99
STUDNTBX	Median value of bracket range for STUDNTB	>0.99
STUDNTX	Total amount owed on all student loans	0.97
TAIRFARC	Trip expenditures on airfare this quarter	
TAIRFARP	Trip expenditures on airfare last quarter	
TALCBEVC	Total trip expenditures this quarter on alcoholic beverages at restaurants, cafes, and bars	
TALCBEVP	Total trip expenditures last quarter on alcoholic beverages at restaurants, cafes, and bars	
TELEPHCQ	Telephone services this quarter	
TELEPHPQ	Telephone services last quarter	
TENTRMNC	Total trip expenditures on entertainment this quarter including sporting events, movies, and recreational vehicle rentals (TFEESADC + TOTHEMTC)	
TENTRMNP	Total trip expenditures on entertainment last quarter including sporting events, movies, and recreational vehicle rentals (same composition as TENTRMNC)	
TEXTILCQ	Household textiles this quarter	
TEXTILPQ	Household textiles last quarter	

TFAREC	Trip expenditures this quarter on transportation fares including airfare, intercity bus, train, and ship fare (TAIRFARC + TOTHFARC)
TFAREP	Trip expenditures last quarter on transportation fares including airfare, intercity bus, train, and ship fare (same composition as TFAREC)
TFEESADC	Trip expenditures on miscellaneous entertainment this quarter including recreation expenses, participation sport fees, and admission fees to sporting events and movies
TFEESADP	Trip expenditures on miscellaneous entertainment last quarter including recreation expenses, participation sport fees, and admission fees to sporting events and movies
TFOODAWC	Food and non-alcoholic beverages this quarter at restaurants, cafes, and fast food places during out-of-town trips
TFOODAWP	Food and non-alcoholic beverages last quarter at restaurants, cafes, and fast food places during out-of-town trips
TFOODHOC	Food and beverages purchased and prepared by CU this quarter during out-of-town trips
TFOODHOP	Food and beverages purchased and prepared by CU last quarter during out-of-town trips
TFOODTOC	Total trip expenditures on food this quarter including both restaurant food and food prepared by CU (TFOODAWC + TFOODHOC)
TFOODTOP	Total trip expenditures on food last quarter including both restaurant food and food prepared by CU (same composition as TFOODTOC)
TGASMOTC	Trip expenditures on gas and oil this quarter
TGASMOTP	Trip expenditures on gas and oil last quarter
TLOCALTC	Trip expenditures this quarter on local transportation including taxis, buses etc.
TLOCALTP	Trip expenditures last quarter on local transportation including taxis, buses etc.
TOBACCCQ	Tobacco and smoking supplies this quarter
TOBACCPQ	Tobacco and smoking supplies last quarter
TOTEX4CQ	Adjusted total expenditures this quarter (TOTEXPCQ - MISCCQ + MISC1CQ + 4 × MISC2CQ)
TOTEX4PQ	Adjusted total expenditures last quarter (same composition as TOTEX4CQ)

TOTEXPCQ	Total expenditures this quarter (FOODCQ + AL-CBEVCQ + HOUSCQ + APPARCQ + TRANSCQ + HEALTHCQ + ENTERTCQ + PERSCACQ + READCQ + EDUCACQ + TOBACCCQ + MISCCQ + CASHCOCQ + PERINSCQ)
TOTEXPPQ	Total expenditures last quarter (same composition as TOTEXPCQ)
TOTHENTC	Trip expenditures on recreational vehicle rentals this quarter including campers, boats, and other vehicles
TOTHENTP	Trip expenditures on recreational vehicle rentals last quarter including campers, boats, and other vehicles
TOTHFARC	Tip expenditures this quarter on other transportation fares including intercity bus and train fare, and ship fare
TOTHFARP	Tip expenditures last quarter on other transportation fares including intercity bus and train fare, and ship fare
TOTHRLOC	Total trip expenditures on lodging this quarter including rent for vacation home, and motels
TOTHRLOP	Total trip expenditures on lodging last quarter including rent for vacation home, and motels
TOTHTREC	Trip expenditures this quarter for other transportation expenses including parking fees, and tolls
TOTHTREP	Trip expenditures last quarter for other transportation expenses including parking fees, and tolls
TOTHVHRC	Trip expenditures on other vehicle rentals this quarter
TOTHVHRP	Trip expenditures on other vehicle rentals last quarter
TOTXEST	Estimated total taxes paid (FFTAXOWE + FSTAXOWE + MISCTAXX)
TRANSCQ	Transportation this quarter (CARTKNCQ + CARTKUCQ + OTHVEHCQ + GASMOCQ + VEHFINCQ + MAINRPCQ + VEHINSCQ + VRNT-LOCQ + PUBTRACQ)
TRANSPQ	Transportation last quarter (same composition as TRANSCQ)
TRNOTHCQ	Local public transportation, excluding on trips this quarter
TRNOTHPQ	Local public transportation, excluding on trips last quarter
TRNTRPCQ	Public and other transportation on trips this quarter

TRNTRPPQ	Public and other transportation on trips last quarter
TTOTALC	Total of all trip expenditures this quarter (TFOODTOC + TALCBEVC + TOTHRLC + TTRANPRC + TEN-TRMNPC)
TTOTALP	Total of all trip expenditures last quarter (same composition as TTOTALC)
TTRANPRC	Total trip expenditures on transportation this quarter including airfare, local transportation, tolls and parking fees, and car rentals (TGASMOTC + TVRENTLC + TTRNTRIC)
TTRANPRP	Total trip expenditures on transportation last quarter including airfare, local transportation, tolls and parking fees, and car rentals (same composition as TTRANPRC)
TTRNTRIC	Trip expenditures this quarter for public transportation, including airfares (TFAREC + TLOCALTC)
TTRNTRIP	Trip expenditures last quarter for public transportation, including airfares (same composition as TTRNTRIC)
TVRDIOCQ	Televisions, radios, and sound equipment this quarter
TVRDIOPQ	Televisions, radios, and sound equipment last quarter
TVRENTLC	Trip expenditures on vehicle rentals and other fees this quarter (TCARTRKC + TOTHVHRC + TOTHTREC)
TVRENTLP	Trip expenditures on vehicle rentals and other fees last quarter (same composition as TVRENTLC)
UNISTRQ	How many housing units, both occupied and vacant, are in this structure? (1=only other units, 2=mobile home or trailer, 3=one, detached, 4=one, attached, 5=2, 6=3-4, 7=5-9, 8=10-19, 9=20-49, 10=50 or more)
UTILCQ	Utilities, fuels and public services this quarter (NTLGASCQ + ELCTRCCQ + ALLFULCQ + TELEPHCQ + WATRPSCQ)
UTILOWNC	Expenditures on owned vacation home utilities this quarter including water, trash, electricity, and fuels (VFUELOIC + VOTHRFLC + VELECTRC + VNATLGAC + VWATERPC)
UTILOWNP	Expenditures on owned vacation home utilities last quarter including water, trash, electricity, and fuels (same composition as UTILOWNC)
UTILPQ	Utilities, fuels and public services last quarter

UTILRNTC	Expenditures on rented vacation home utilities this quarter including water, trash, electricity, and fuels (RFUELOIC + ROTHFLC + RELECTRC + RNATLGAC + RWATERPC)	
UTILRNTP	Expenditures on rented vacation home utilities last quarter including water, trash, electricity, and fuels (same composition as UTILRNTC)	
VEHFINCQ	Vehicle finance charges this quarter	
VEHFINPQ	Vehicle finance charges last quarter	
VEHICTAX	Personal property taxes for vehicles	0.92
VEHINSCQ	Vehicle insurance this quarter	
VEHINSPQ	Vehicle insurance last quarter	
VEHQ	Total number of owned vehicles	
VEHQL	Total number of leased autos, trucks and vans	
VELECTRC	Expenditures on electricity for owned vacation homes this quarter	
VELECTRP	Expenditures on electricity for owned vacation homes last quarter	
VFUELOIC	Expenditures on fuel oil for owned vacation homes this quarter	
VFUELOIP	Expenditures on fuel oil for owned vacation homes last quarter	
VNATLGAC	Expenditures on natural gas for owned vacation homes this quarter	
VNATLGAP	Expenditures on natural gas for owned vacation homes last quarter	
VOTHFLC	Expenditures on other fuels for owned vacation homes this quarter	
VOTHFLP	Expenditures on other fuels for owned vacation homes last quarter	
VOTHROP	Expenditures on owned vacation homes last quarter including mortgage interest, insurance, taxes, and maintenance	
VOTHROC	Expenditures on owned vacation homes this quarter including mortgage interest, insurance, taxes, and maintenance	
VRNTLOCQ	Vehicle rental, leases, licenses, and other charges this quarter	

VRNTLOPQ	Vehicle rental, leases, licenses, and other charges last quarter	
VWATERPC	Expenditures on water and public services for owned vacation homes this quarter	
VWATERPP	Expenditures on water and public services for owned vacation homes last quarter	
WATRPSCQ	Water and other public services this quarter	
WATRPSPQ	Water and other public services last quarter	
WELFAREX	Amount received from public assistance or welfare including money received from job training grants	0.99
WELFREBX	Median of bracket range of WELFAREB	0.99
WHLFYR	Did you own any whole life insurance or other life insurance policies that can be surrendered for cash or borrowed against prior to the death of the person insured one year ago today? (1=yes, 2=no)	>0.99
WHLFYRB	Range which best reflects total surrender value of these policies one year ago today (1=0–499, 2=500–999, 3=1–2.5K, 4=2.5–10K, 5=10–35K, 6=35K and over)	0.99
WHLFYRBX	Median value of bracket range for WHLFYRB	0.99
WHLFYRX	Total surrender value of these policies one year ago today	0.98
WHOLIFB	Range which best reflects the total surrender value of these policies (same codes as WHLFYRB)	>0.99
WHOLIFBX	Median value of bracket range for WHOLIFB	>0.99
WHOLIFX	Total surrender value of these policies as of today	0.98

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Table 12: PSU codes

S11A	Boston-Cambridge-Newton, MA-NH
S12A	New York-Newark-Jersey City, NY-NJ-PA
S12B	Philadelphia-Camden-Wilmington, PA-NJ-DE-MD
S23A	Chicago-Naperville-Elgin, IL-IN-WI
S23B	Detroit-Warren-Dearborn, MI
S24A	Minneapolis-St. Paul-Bloomington, MN-WI
S24B	St. Louis, MO-IL
S35A	Washington-Arlington-Alexandria, DC-VA-MD-WV
S35B	Miami-Fort Lauderdale-West Palm Beach, FL
S35C	Atlanta-Sandy Springs-Roswell, GA
S35D	Tampa-St. Petersburg-Clearwater, FL
S35E	Baltimore-Columbia-Towson, MD
S37A	Dallas-Fort Worth-Arlington, TX
S37B	Houston-The Woodlands-Sugar Land, TX
S48A	Phoenix-Mesa-Scottsdale, AZ
S48B	Denver-Aurora-Lakewood, CO
S49A	Los Angeles-Long Beach-Anaheim, CA
S49B	San Francisco-Oakland-Hayward, CA
S49C	Riverside-San Bernardino-Ontario, CA
S49D	Seattle-Tacoma-Bellevue, WA
S49E	San Diego-Carlsbad, CA
S49F	Honolulu, HI
S49G	Anchorage, AK

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